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WEB TECHNOLOGIES

(Computer Science (DSE-2F) (Elective-B))

As Per (O.U) CBCS 2016-17 Syllabus for B.Sc III-Year VI-Semester

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PREFACE

ABOUT THE SUBJECT

Web Technologies is the programming concept that allows users to create interactive web pages. These pages can be created by writing code using various web programming languages like XML, HTML, JavaScript, VBscript, Perl and PHP. A web document can be created using client-side and server-side programming languages and are later embedded within HTML documents. In client-side programming, the execution of program/script is done on client machine. Some of the client-side programming languages are HTML, JavaScript, CSS etc.

Cascading Style Sheets (CSS) is one of the most interesting concepts of HTML as it specifies the rules for organizing elements of a given web document. CSS are used to determine the style and layout of web pages. JavaScript is a programming language with direct support to object-oriented methodologies. It facilitates inclusion of certain executable data along with it. Hence, it can be said that with the usage of JavaScript, a given web page will no longer remain a store house of static data but it can also maintain dynamic data which can interact with the users, it can dynamically create HTML context and also control the browser actions.

ABOUT THE BOOK

This book provides theoretical and practical knowledge to the student about "**Web Technologies**" for B.Sc III-Year VI-Sem. It covers the complete syllabus of the subject prescribed by O.U. The content written in this book is presented in a consistently readable and student friendly format so that student can prepare well for their both mid and final exams. This book is beneficial because it gives complete up-to-date information about each topic and the questions to be asked in the exams. This helps the student in getting a clear idea about the important questions in each topic. This book has been prepared keeping student's views, ideas and suggestions in mind. The main motivation behind the publication of this book is to help the student to gain good marks and knowledge in the subject.

According to the examination pattern of B.Sc students, this book provides the following features:

- **List of Programs** are provided before the units for easy reference.
- Every unit is structured into two main sections viz. **Short Questions and Essay Questions** with solutions along with Learning Objectives and Introduction.
- **Objective Type** along with **Very Short Answers** are also given at the end of each unit.
- **Three Model Papers** are provided in order to help students to understand the paper pattern in the end examination.
- **Lab Practical with Solutions** are also provided at the end of units.
- **Important Questions** are included to help the students to prepare for Internal and External Assessment.

The table below gives complete idea about the subject, which will be helpful to plan and score good marks in their exams.

Unit No.	Unit Name	Description
1.	Structuring Documents for the Web, Links and Navigation, Images, Audio and Video, Tables, Forms, Frames	This unit includes topics like Structuring Documents for the Web: Introducing HTML and XHTML, Basic Text Formatting, Presentational Elements, Phrase Elements, Lists, Editing Text, Core Elements and Attributes, Attribute Groups. Links and Navigation: Basic Links, Creating Links with the <a> Element, Advanced E-mail Links. Images, Audio and Video: Adding Images Using the <img> Element, Using Images as Links, Image Maps, Choosing the Right Image Format, Adding Flash, Video and Audio to Your Web Pages, Table: Introducing Tables, Grouping Section of a Table, Nested Tables, Accessing Tables, Forms: Introducing Forms, Form Controls, Sending Form Data to the Server, Frames: Introducing Frameset, <frame> Element, Creating Links Between Frames, Setting a Default Target Frame Using <base> Element, Nested Framesets, Inline or Floating Frames with <iframe>.
2.	Cascading Style Sheets	This unit includes topics like Cascading Style Sheets: Introducing CSS, Where you Can Add CSS Rules, CSS Properties: Controlling Text, Text Formatting, Text Pseudo Classes, Selectors, Lengths, Introducing the Box Model, More Cascading Style Sheets: Links, Lists, Tables, Outlines, The :focus and :activate Pseudo Classes Generated Contents, Miscellaneous Properties, Additional Rules, Positioning and Layout with CSS, Page Layout: Understanding the Site's Audience, Page Size, Designing Pages, Coding Your Design, Developing for Mobile Devices, Design Issues: Typography, Navigation, Tables, Forms.
3.	Learning and Working with JavaScript, Putting Your Site on the Web	This unit includes topics like Learning JavaScript: How to Add Script to Your Pages, The Document Object Model, Variables, Operators, Functions, Control Statements, Looping, Events, Built-in Objects, Working with JavaScript: Practical Tips for Writing Scripts, Form Validation, Form Enhancements, JavaScript Libraries. Putting Your Site on the Web: Meta Tags, Testing Your Site, Talking the Leap to Live, Telling the World about Your Site, Understanding Your Visitors.

It is sincerely hoped that this book will satisfy the expectations of students and at the same time helps them to score maximum marks in exams.

Suggestions for improvement of the book from our esteemed readers will be highly appreciated and incorporated in our forthcoming editions.

WEB TECHNOLOGIES

B.Sc III-Year VI-Semester (OU)

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SYLLABUS

UNIT-I

Structuring Documents for the Web: Introducing HTML and XHTML, Basic Text Formatting, Presentational Elements, Phrase Elements, Lists, Editing Text, Core Elements and Attributes, Attribute Groups.

Links and Navigation: Basic Links, Creating Links with the <a> Element, Advanced E-mail Links.

Images, Audio and Video: Adding Images Using the <img> Element, Using Images as Links, Image Maps, Choosing the Right Image Format, Adding Flash, Video and Audio to Your Web Pages.

Tables: Introducing Tables, Grouping Section of a Table, Nested Tables, Accessing Tables.

Forms: Introducing Forms, Form Controls, Sending Form Data to the Server.

Frames: Introducing Frameset, <frame> Element, Creating Links Between Frames, Setting a Default Target Frame Using <base> Element, Nested Framesets, Inline or Floating Frames with <iframe>.

UNIT-II

Cascading Style Sheets: Introducing CSS, Where you Can Add CSS Rules.

CSS Properties: Controlling Text, Text Formatting, Text Pseudo Classes, Selectors, Lengths, Introducing the Box Model.

More Cascading Style Sheets: Links, Lists, Tables, Outlines, The :focus and :activate Pseudo Classes Generated Content, Miscellaneous Properties, Additional Rules, Positioning and Layout with CSS.

Page Layout: Understanding the Site's Audience, Page Size, Designing Pages, Coding Your Design, Developing for Mobile Devices.

Design Issues: Typography, Navigation, Tables, Forms.

UNIT-III

Learning JavaScript: How to Add Script to Your Pages, The Document Object Model, Variables, Operators, Functions, Control Statements, Looping, Events, Built-in Objects.

Working with JavaScript: Practical Tips for Writing Scripts, Form Validation, Form Enhancements, JavaScript Libraries.

Putting Your Site on the Web: Meta Tags, Testing Your Site, Talking the Leap to Live, Telling the World about Your Site, Understanding Your Visitors.

MODEL PAPER 1

FACULTY OF SCIENCE

B.Sc. (CBCS) VI-Semester Examinations

Subject: COMPUTER SCIENCE

DSE-2F: Web Technologies

Paper-VIII (B)

Time: 2 Hours | Max. Marks: 60

SECTION - A (5 × 3 = 15 Marks)

Answer any *Five* of the following Eight questions. Each carries *Three* marks.

1. Write about various controls of forms. **(Unit-I, Page No. 2, Q2)**
2. Write about the tags used for creating table. **(Unit-I, Page No. 3, Q4)**
3. Write about style rule. **(Unit-II, Page No. 80, Q2)**
4. What is a selector? List the types of selectors in CSS. **(Unit-II, Page No. 81, Q5)**
5. Discuss in brief about tracking. **(Unit-II, Page No. 81, Q8)**
6. What is a variable in java script? How to declare a variable **(Unit-III, Page No. 174, Q3)**
7. Write a short note on meta tags. **(Unit-III, Page No. 175, Q9)**
8. What are head, title, body tags in HTML? **(Unit-I, Page No. 4, Q7)**

SECTION - B (3 × 15 = 45 Marks)

Answer all of the following *Three* questions. Each carries *Fifteen* marks.

9. (a) Explain in detail about HTML. **(Unit-I, Page No. 5, Q11)**

OR

(b) Explain how basic table is created in xhtml. **(Unit-I, Page No. 46, Q48)**

10. (a) How style sheets are specified into HTML document? **(Unit-II, Page No. 82, Q12)**

OR

(b) Explain in detail about background properties of CSS3 with an example. **(Unit-II, Page No. 117, Q29)**

11. (a) How a script is added to the webpage. **(Unit-III, Page No. 176, Q13)**

OR

(b) Write short notes on following,

(i) Focusing on first form item

(ii) Auto tabbing between fields. **(Unit-III, Page No. 231, Q40)**

MODEL PAPER 2

FACULTY OF SCIENCE

B.Sc. (CBCS) VI-Semester Examinations

Subject: COMPUTER SCIENCE

DSE-2F: Web Technologies

Paper-VIII (B)

Time: 2 Hours Max. Marks: 60

SECTION - A (5 × 3 = 15 Marks)

Answer any *Five* of the following Eight questions. Each carries *Three* marks.

1. Write about types of lists. **(Unit-I, Page No. 3, Q3)**
2. What is a frame in HTML? Give an example. **(Unit-I, Page No. 3, Q5)**
3. What is CSS? **(Unit-II, Page No. 80, Q1)**
4. Write about vertical-align property. **(Unit-II, Page No. 81, Q6)**
5. Write a short note on,
   (i) list-style-position property
   (ii) list-style-image property. **(Unit-II, Page No. 81, Q9)**
6. What is an array in java script? **(Unit-III, Page No. 175, Q7)**
7. List the different points used to make people aware of the site. **(Unit-III, Page No. 175, Q11)**
8. What is hyperlink? Explain the anchor tag with syntax. **(Unit-I, Page No. 4, Q9)**

SECTION - B (3 × 15 = 45 Marks)

Answer all of the following *Three* questions. Each carries *Fifteen* marks.

9. (a) Discuss about presentational elements. **(Unit-I, Page No. 15, Q20)**

OR

(b) What are different types of Form controls in xhtml? **(Unit-I, Page No. 56, Q55)**

10. (a) Explain how the fonts can be controlled. List and explain each of the properties that directly affect the font and its appearance. **(Unit-II, Page No. 89, Q18)**

OR

(b) Discuss in brief about the composition and grids. **(Unit-II, Page No. 142, Q44)**

11. (a) What is an operator? What are the different types of operators? **(Unit-III, Page No. 188, Q21)**

OR

(b) What are the two different states of testing? Explain about validating HTML, XHTML and CSS. **(Unit-III, Page No. 243, Q51)**

MODEL PAPER 3

FACULTY OF SCIENCE

B.Sc. (CBCS) VI-Semester Examinations

Subject: COMPUTER SCIENCE

DSE-2F: Web Technologies

Paper-VIII (B)

Time: 2 Hours Max. Marks: 60

SECTION - A (5 × 3 = 15 Marks)

Answer any *Five* of the following Eight questions. Each carries *Three* marks.

1. Write about different formatting tags. **(Unit-I, Page No. 2, Q1)**
2. Explain <img> tag in HTML. **(Unit-I, Page No. 4, Q6)**
3. List the properties that influence the appearance of the font. **(Unit-II, Page No. 80, Q4)**
4. What are the different properties that are used to position the items on the page. **(Unit-II, Page No. 81, Q7)**
5. List the advantages of fixed-width design and liquid design. **(Unit-II, Page No. 81, Q10)**
6. Define functions in javascripts. **(Unit-III, Page No. 174, Q4)**
7. Write a short note on domain name. **(Unit-III, Page No. 175, Q12)**
8. Define with syntax <FONT> tag. **(Unit-I, Page No. 4, Q10)**

SECTION - B (3 × 15 = 45 Marks)

Answer all of the following *Three* questions. Each carries *Fifteen* marks.

9. (a) Explain about images and supported attributes in HTML. **(Unit-I, Page No. 30, Q31)**

OR

(b) Discuss about frameset. **(Unit-I, Page No. 69, Q64)**

10. (a) Discuss in detail about the dimensions of a box. **(Unit-II, Page No. 111, Q27)**

OR

(b) Write in detail about layout of forms. **(Unit-II, Page No. 165, Q66)**

11. (a) Write about form validation. **(Unit-III, Page No. 219 , Q33)**

OR

(b) What are the different pointers used to make more people aware of the site? **(Unit-III, Page No. 250, Q57)**

UNIT 1

STRUCTURING DOCUMENTS FOR THE WEB, LINKS AND NAVIGATION, IMAGES, AUDIO AND VIDEO, TABLES, FORMS, FRAMES

Marketed by:
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LEARNING OBJECTIVES

After studying this unit, a student will have thorough knowledge about the following key concepts,

- Introduction to HTML and XHTML
- Text formatting tags, Presentational elements and Phrase elements
- Basic links and usage of <a> element
- <img> element and linking image maps
- Adding flash, video and audio to the web pages
- Tables and Nested tables
- Forms and Form controls
- Frameset and <frame> element
- Links between frames and <iframe> element.

INTRODUCTION

HTML is a markup language immensely used to develop web pages. It creates a web page with the help of in-built tags, which are the special keywords supported by HTML. The most commonly used HTML tags include <HTML>---</HTML>, <HEAD>---</HEAD>, <BODY>---</BODY>, <TITLE>--- </TITLE> etc. HTML supports formatting tags which are used for designing and coloring the text. A hyperlink creates a link. This link directs the users to another HTML page or to any specific part of a web page or the same webpage. An image tag or <img> tag inserts an image in HTML page.

In HTML, data can also be represented in the form of a table consisting of rows and columns. The HTML table is created using <TABLE>…</TABLE> tags. Form is a web page which contains elements like text fields, check boxes. Usually forms are created by inserting<FORM>---</FORM> tags in the HTML code. The output screen of every HTML application is referred as a frame which is defined as the region of the browser that displays certain output.

PART-A

SHORT QUESTIONS WITH SOLUTIONS

Q1. Write about different formatting tags.

Answer : Model Paper-III, Q1

Formatting the text/content present in a web page can be done through formatting tags. Various formatting tags in HTML are as follows,

1. Paragraph

The tag <p> allows the text to be written in a paragraph (i.e., block-wise).

2. Center Tag

The tag allows the text to be written in center of a page.

3. Break Tag

The tag
 shows a line break.

4. Bold Text

The tag <b> displays the text in bold format.

5. Italic Text

The tag <i> displays the text in italicized format.

6. Marked Text

The tag <mark> displays the text by highlighting or marking it.

7. Underlined Text

The tag <u> displays the text in underlined format.

8. Subscript

The tag <sub> displays the text in the subscripted format.

9. Superscript

The tag <sup> displays the text in the superscripted format.

Q2. Write about various controls of forms.

Answer : Model Paper-I, Q1

The various form controls are as follows,

(i) The BUTTON Element

In HTML, button is generally used to trigger appropriate code of javascript (or) to perform some action.

Syntax

<INPUT TYPE = "button" NAME = "Name of button" VALUE = "button label">

Example

<INPUT TYPE = "button", NAME = "b1" VALUE = "Add">

(ii) The TEXT Element

In HTML, Text elements are considered as data entry fields. These fields generally accept single line of text.

Syntax

<INPUT TYPE = "text" Name = "name of text field" VALUE = "default value">

Example

<INPUT TYPE = "text" NAME = "L1" VALUE = "JOHN">

(iii) The PASSWORD Element

It is similar to text filed but accepts confidential information like password, account number and personnel identification number etc. The password element are echoed as '*'.

Syntax

<INPUT TYPE = "Password" NAME = "name of the password" VALUE = "default">

Example

<INPUT TYPE = "password" NAME = "emp_id" VALUE = "default">

(iv) The Submit Button

In HTML, the submit button sends the data entered into the form to the web server.

Syntax

<INPUT TYPE = "submit" NAME = "name of the submit button" VALUE = "submit button label">

Example

<INPUT TYPE = "SUBMIT" NAME = "employee1" VALUE = "VALIDATE">

Q3. Write about types of lists.

Answer : Model Paper-II, Q1

List are displayed in bulleted format by using bullets or specific sequence. The different types of lists are,

1. Unordered Lists

Unordered lists displays certain text with a bullet mark. These lists are used when the user does not want to follow any specific order.

Here, usually the entire list item is enclosed under the tags **<UL> text to be displayed </UL>** with each list item sentence initiating with **<LI>** and closing with **</LI>**.

Syntax

 . . .

 . . .

Example

- o item
- o item

2. Ordered Lists

Ordered list is used to display list items with order. Ordered list is usually initiated using the tags **<OL type = "value">** text to be displayed **</OL>**. These lists are used when user is willing to display lists sequentially each list item preceeded with <li> . . . , default sequence is number. It can be customized using type attributes <ol type = "1|A|a|i|I"> value is the first characters of sequence.

Syntax

 . . .
 . . .

Example

1. item
2. item

3. Definition Lists

It is a special type of list that will list the terms and their definitions. It is created using (define list) <dl> tag. A term in a definition list is specified using <dt> (define term) tag and its definition is specified using <dd> (define definition) tag.

Syntax

<dl>
<dt> . . . </dt>
<dd> . . . </dd>
</dl>

Q4. Write about the tags used for creating table.

Answer : Model Paper-I, Q2

In HTML, data can also be represented in table format. In this approach, data is usually organized in the form of rows and columns. The tags and attributes that helps in creating HTML table are as follows.

1. <TABLE>...</TABLE>
2. <TR>...</TR>
3. <TH>...</TH>
4. <TD>...</TD>
5. <THEAD>...</THEAD>
6. <CAPTION>...</CAPTION>
7. <TBODY>...</TBODY>
8. <TFOOT>...</TFOOT>

Q5. What is a frame in HTML? Give an example.

Answer : Model Paper-II, Q2

Frame

The output screen of every HTML application is nothing but a **frame**. Hence, frame can be define as the part of the browser that displays a particular output. Here, each independent frame can display an independent "HTML" file. Hence within a single window, various files can be viewed at one glance.

Example

Frame.html

```
<HTML>
<HEAD>
<TITLE> Frame </TITLE>
</HEAD>
<BODY BGCOLOR = "pink">
<CENTER> <STRONG> <H1> FRAME example </H1> </STRONG> </CENTER>
</BODY>
</HTML>
```

Mainframe.html

```
<HTML>
<HEAD>
<TITLE> inserting frame</TITLE>
<FRAMESET ROW = "25%">
<FRAME SRC = "frame.html">
</FRAMESET>
</HEAD>
</HTML>
```

Q6. Explain <img> tag in HTML.

Answer : Model Paper-III, Q2

Image tag is used to add an image to the page. This is done by providing suitable image source and image file format in the tag.

Syntax

<img Src = "location of the image"/name of the image">

Attributes supported by the image tag are,

(a) src
(b) Alt
(c) Align
(d) Border
(e) Hight /Width
(f) Hspace/Vspace
(g) Longdesc
(h) Name.

Q7. What are head, title, body tags in HTML?

Answer : Model Paper-I, Q8

Head Tag

<HEAD> ……</HEAD> it holds the head information of the HTML code i.e., title, link, meta are same of the tags that appears in the head region of the html document.

Title Tag

<TITLE> ……</TITLE> the name of the document which is displayed at the title bar of the html document is provided using <TITLE> tag. It is an optional tag.

Body Tag

<BODY> ……</BODY> it is the major tag which holds the data and elements on a web page. It holds eight attributes. They are : background, bgcolor, bottom margin, left margin, tag margin, right margin, text, scroll respectively. These attributes can make the representation of elements depending on the requirement.

Q8. Define the structure of html program.

Answer :

The structure of a basic HTML document is given below,

```
<HTML>
    <HEAD>
        <!— A HTML comment—>
        <TITLE> A HTML Document </TITLE>
    </HEAD>
    <BODY>
        ...
        ...
    </BODY>
</HTML>
```

Q9. What is hyperlink? Explain the anchor tag with syntax.

Answer : Model Paper-II, Q8

Hyperlink

A hyperlink directs the users to another HTML page or to any specific part of a web page or the same webpage. A hyperlink allows users to navigate between various parts of a web page or different web pages. It allows users to create hyperlinks using text/images available on webpage.

Anchor Tag

In HTML, a link is defined using <a> tag. <a> is an anchor tag. The code for creating a link is embedded between the tags <a> ... .

The syntax of creating HTML tag is as follows,

Text

Q10. Define with syntax <FONT> tag.

Answer : Model Paper-III, Q8

By using font tag, the style and size of the text can be changed. The attributes of the font tag are size and color which are shown below,

<font size = "[t/ –]n" color = "# rrggbb" >

This tag sets the font size to the text. It has highest priority than the default value or any size set by <basefont> i,e., it changes the font size to a particular size by modifying the <basefont> size or default size.

The size attributes uses the numbers from 1 to 7 for absolutes font sizes but for relative sizes it uses t/ – 1 to 7.

For example: <font size = "+5" Font

It displays font with size 5.

PART-B

ESSAY QUESTIONS WITH SOLUTIONS

1.1 STRUCTURING DOCUMENTS FOR THE WEB

1.1.1 Introducing HTML and XHTML

Q11. Explain in detail about HTML.

Answer : Model Paper-I, Q9(a)

HTML

HTML stands for HyperText Markup Language. It is markup language immensely used in developing web pages. In similar way, the Hyper Text Markup Language highlights different views of a given web page in terms of headers, titles, plain text, images, sounds etc.

HTML creates a web page with the help of numerous in built **tags**. A tag is nothing but a special keyword and instruction supported by browser.

Considerations to be taken while creating a HTML page,

- ❖ Every HTML code should begin with **<HTML>** tag and end with **</HTML>** tag.
- ❖ Every HTML code should be saved by **.HTML** or **.HTM** extension.
- ❖ There is no specific **editor** devised for writing **HTML** codes, i.e., a given HTML code can be written on a simple notepad.
- ❖ The **output** of a given HTML document can be visualized by typing the path (where a given HTML document has been saved) along with .HTML extension on the address bar of any **web browser** (internet explorer, netscape navigator etc.)
- ❖ If the browser is not displaying expected output then, it means that the HTML code contains certain **errors** which should be sufficiently **debugged**.
- ❖ It has to be noted that the HTML language is not **case sensitive**, hence writing the code either in **upper case** or **lower case** will make no difference.

Steps for Writing a HTML Program

Step 1

To write a HTML program, open a suitable notepad. This can be done by clicking the start button and typing "notepad" at the command prompt.

An untitled notepad will be displayed as shown below.

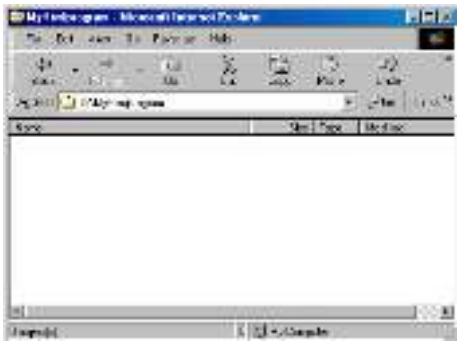




Type the following Code First.HTML

```
<HTML>
<HEAD>
<TITLE> MY FIRST HTML PROGRAM</TITLE>
</HEAD>
<BODY>
<P>SIA Publishers And Distributors Private Limited </P>
</BODY>
</HTML>
```

Now, save the above code as first.html (for convenience create a new directory in C:\ MyHtmlprogram and save the program in it).



Step 2

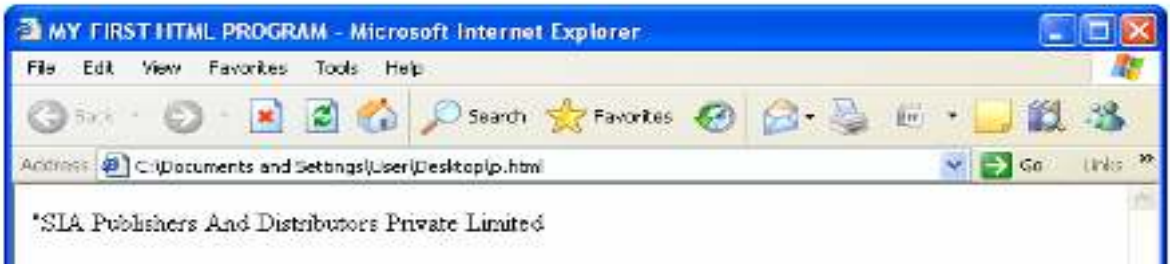
After saving, an internet explorer icon will be displayed with the file name as shown below.



first. html

Double click to execute it. The output will be available in the following manner.

Output



Q12. Write about XHTML.

Answer :

XHTML is the lastest version of HTML. It has become a W3C recommendation in January 2000. XHTML is a enhanced version of HTML by using XML that changes the direction and using the HTML in the way it must to be. In short, rules does matter. Earlier, the browser must be intimated about anything and it would render. Inorder to form a valid XHTML document the following rules must be followed,

- ❖ A doctype indicator which should conform must be available to its rules.
- ❖ The tags such as <html>, <head> and <body> (or a <frameset> containing a <body> inside of a <no frames> tag)are required.
- ❖ <title> must be first in the <head> element.
- ❖ All the attributes, even simple ones such as must be quoted
- ❖ Tags must be nested properly.
- ❖ The tags such as and are required.
- ❖ Empty tags need to be closed, so tags such as <hr> become <hr/>.
- ❖ All tags and attribute names must be in lower case.
- ❖ XHTML is case sensitive.
- ❖ XHTML is sensitive to a single white space character.
- ❖ XHTML follows a content model.

Q13. Discuss various tags of HTML.

Answer :

	Commonly Used HTML Tags	Description
1.	<HTML> ……</HTML>	◆ It is placed at the start as well as at the end of every HTML page. The code which resides between these tags forms the head, title and body of the HTML program. It is compulsory that every HTML program should have <HTML> …… </HTML> at the start as well as at the end of the document.
2.	<HEAD> ……</HEAD>	◆ It holds the head information of the HTML code i.e., title, link, meta are some of the tags that appears in the head region of the html document.
3.	<TITLE> ……</TITLE>	◆ The title which is displayed at the title of the html document is provided using <TITLE> tag. It is an optional tag which displays in title bar.
4.	<BODY> ……</BODY>	◆ It is the major tag which holds the contents appearing on a web page. It holds eight attributes. They are : background, bgcolor, bottom margin, left margin, tag margin, right margin, text and scroll. These attributes can make the representation of elements depending on the requirement.
5.	Headers tag (<H1><H2><H3>……<H6>)	◆ These are frequently used tags which display data in bold font with different sizes. <H1> causes data to be displayed in biggest font whereas <H6> in the smallest font.
6.	<!……comment…>	◆ It provides comments to be displayed in the html document. They make the document elucidating.
7.	Formatting text tags	
◆	BOLD (<B> ……</B>)	◆ It makes the text font to bold.
◆	ITALIC (<I> ……</I>)	◆ It makes the text to appear in italic form.
◆	UNDERLINE (<U> ……</U>)	◆ It underlines the text.
◆	LINE BREAK (… </BR>) and NOLINE BREAK (<NOBR> ……</NOBR>)	◆ It discriminates between two lines or prevents addition of gap between two lines.

◆	PARAGRAPH (<P> ……</P>)	◆ It makes the text to appear on the screen, by leaving certain space towards it's left, for the first line.
◆	CENTER (<CENTER> … </CENTER>)	◆ It aligns the text at the center of the screen.
8.	LISTS (<UL> …</UL>) or (<OL> ……</OL>)	◆ When a text (consisting several lines) is embedded in these tags, a bullet mark is inserted at the beginning of every line or a number is added to each line.
9.	Background <body bgcolor = " " >	◆ It is used to set of background to the web page. The background can be any color(or) an image
10.	Text styles	
◆	Base font <basefont>	◆ It changes the font of the entire text in a page.
◆	font <font>	◆ It changes the font of only specified text.
11.	Combining styles <u>.....</u>	◆ It combines the style tags for the text Example: Bold and underline together.
12.	Preformatted Text <pre> ..... </pre>	◆ It does not substitute the consecutive white spaces by any other tag or text.
13.	Strike through text <del>	◆ It is used for displaying a strike through text. This enhances the users interest. It specifically used in advertising.
14.	HTML symbols & name;	◆ It is used to insert symbol on the webpage. They are called as entities.

Q14. Write in brief about document head and document body.

Answer :

Document Head

The head tag in the document contains the control information that is called in body tag. The <title> tag is the only tag used in head section. It displays the title of the document at the top of the browser. Title of the document should be placed within the opening and closing tags of title as shown below,

Document Body

The document body contains the text and elements that msut be displayed on the web page. The formatting of the text can be controlled by HTML documents are divided into blocks of text. The body of the HTML documents generally include paragraph and heading. Paragraph and heading are considered as two major blocks in HTML documents in which each block is formatted independently and contains all the text and images of a document. Lists and tables are exceptional.

Paragraph contains the major part of the information and should be tagged explicitly within and .

Example

```
<html>
<head>
<title> Document Head </title>
</head>
<body>
<p>This is a document head tag  </p> //Document body
</body>
</html>
```

Output



Q15. What is hyperlink? Also, explain about anchor tag with example program.

Answer :

Hyperlink

A hyperlink directs the users from one web page to another HTML page or to any specific part of it or the same webpage. A hyperlink allows users to navigate between various web pages. It allows users to create hyperlinks using text/images available on webpage.

Anchor Tag

In HTML, a link is defined using <a> tag. <a> is a anchor tag. The code for creating a link is embedded between the tags <a> ... .

The syntax of creating HTML tag is as follows,

Text

1. Href

This attribute defines the destination address. It is usually a URL path.

2. Text

This attribute embeds text/image or any other HTML element on web page.

3. Target

This attribute specifies the location to open a link document. Its values are blank, self, parent top and target frame.

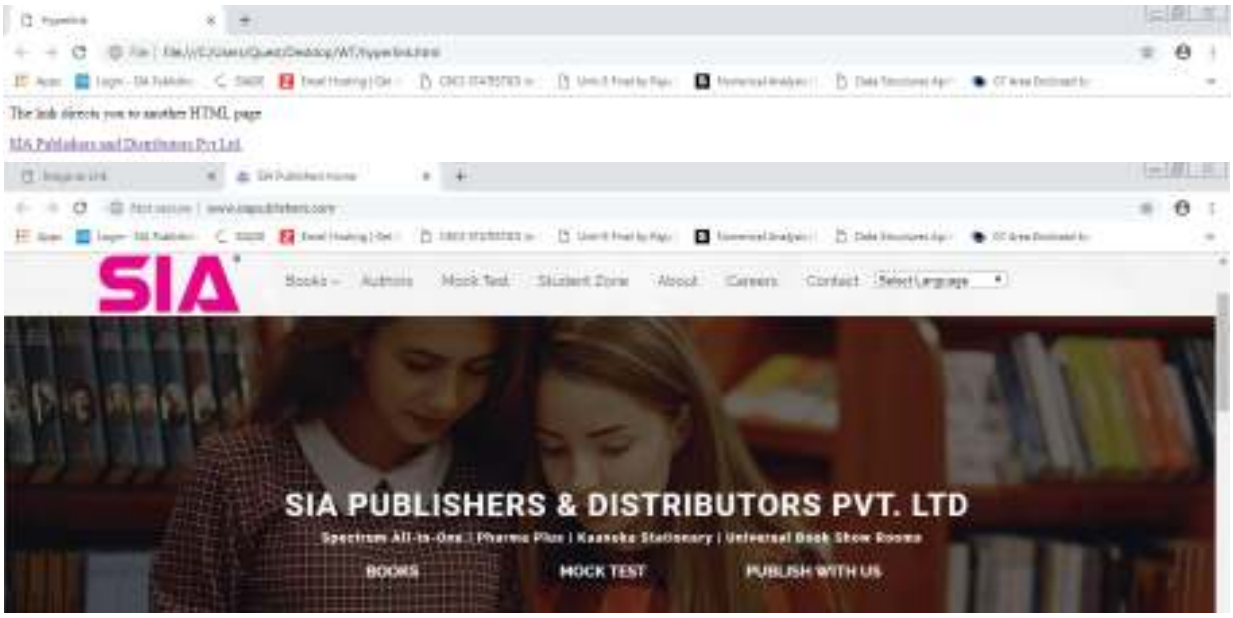
4. Base Path

This attribute is used in head region of HTML tag. It provides the base path for the link.

Program

```
<html>
<head>
<title> Hyperlink </title>
</head>
<body>
<p> The link directs you to another HTML page </p>
<a href = "http://www.siapublishers.com/" target = "self"> SIA Publishers and Distributors Pvt Ltd. </a>
</body>
</html>
```

Output



1.1.2 Basic Text Formatting

Q16. What are Basic Text formattings? Write about White-Space property in XHTML.

Answer :

Basic Text Formatting

It is a process of changing the original state of the document such as text color back ground color using textformatting tags. Every document contains some type text that is represented by using the formatting elements such as <h1>, <h2>, <h3>, <h4>, <h5>, <h6>, <p>,
 and <pre>.

White-space Property

The white-space property is used to control the amount of space between each word. The possible values of white-space property are as follows,

(i) Pre

The "pre" value preserves the white spaces and does not break the line until the
 element is used explicitly.

(ii) Nowrap

The "nowrap" value does not preserve the white spaces instead it collapses the sequence of white spaces into a single space. It does not breaks the line until the
 element is used explicitly.

(iii) Normal

The "normal" is the default value. It does not preserves the white spaces instead it collapses the sequence of white spaces into a single space. It breaks the line if it exceeds the available line space of the browser screen.

Example

```
<html>
<head>
<style type= "text/css">
p.pre {white-space:pre;}
p.nowrap{white-space:nowrap;}
```

```
p.normal{white-space:normal;}
</style>
</head>
<body>
<h3>Effect of “pre” value</h3>
<p class= “pre”>This paragraph demonstrates the use of pre value of the white space property, this paragraph demonstrates the use of pre value of the white space property, this paragraph demonstrates the use of pre value of the white space property..</p>
<h3>Effect of “Nowrap” Value</h3>
<p class= “nowrap”>This paragraph demonstrates the use of nowrap value of the white space property..This paragraph demonstrates the use of nowrap value of the white space property..This paragraph demonstrates the use of nowrap value of the white space property..</p>
<h3>Effect of “Normal” Value</h3>
<p class= “normal”>This paragraph demonstrates the use of normal value of the white space property..This paragraph demonstrates the use of normal value of the white space property..This paragraph demonstrates the use of normal value of the white space property..</p>
</body>
</html>
```

Output



Q17. Discuss in detail about Headings.

Answer :

Heading tags are used to displays the text in predefined text size and bold face. These tags are used for defining HTML heading with different levels. Html supports six types of headings.

Where,

- <h1> defines the text embedded between these tag with Largest bold font.
- <h2> defines the text with font lesser than <h1>
- <h3> defines the text with font lesser than <h2>
- <h4> define the text with font lesser than <h3>
- <h5> defines the text with font lesser than <h4>
- <h6> defines the text with smallest bold font lesser than <h5>.

Example

```
<HTML>
<HEAD>
<TITLE> HEADER TAG </TITLE>
</HEAD>
<BODY>
<h1> Web Programming </h1>
<h2> Web Programming </h2>
<h3> Web Programming </h3>
<h4> Web Programming </h4>
<h5> Web Programming </h5>
<h6> Web Programming </h6>
</BODY>
</HTML>
```

Output



Q18. Write in detail about text align attribute.

Answer :

text-align Property

The text-align property places the text at left, right or at center of browser's window according to the specified value. The attributes of text-align property are as follows,

(i) left

Positions the text at the left side of the browser's window.

(ii) right

Positions the text at the right side of the browser's window.

(iii) center

Positions the text at the middle of the browser's window.

Example

```
<html>
<title>Text Align</title>
<body>
<h1 style= “text-align:left”>Left</h1>
<h2 style= “text-align:center”>Center</h2>
<h3 style= “text-align:right”>Left</h3>
</body>
</html>
```

Output



Q19. Discuss about the following,

(i) <p> element

**(ii)
 element**

(iii) <pre> element.

Answer :

(i) <p> Element

The text can be written in a paragraph (i.e., block-wise) by using <p> tag.

<p> element contains universal attributes and deprecated align attribute such as align, class, id, style, title, chir, long.

Example

The following example illustrates the use of <p> tag.,

```
<html>
<head>
<title> Paragraph Tag </title>
</head>
<body>
<p> This is First Paragraph </p>
</body>
</html>
```

Output



**(ii)
 Element**

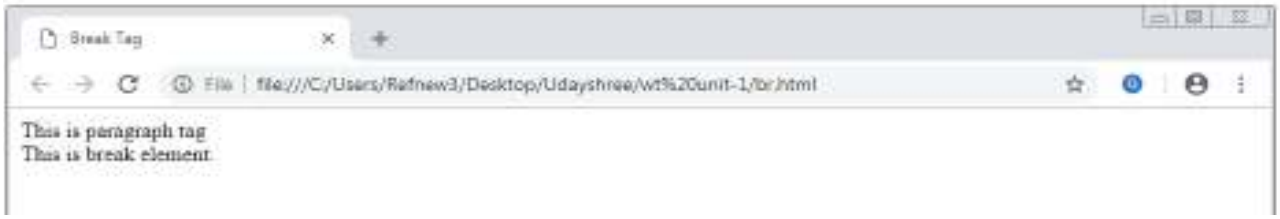
This tag/element is used to break the current line of text. It displays the remaining text in the a next line. This tag does not contain opening and closing tags, since it is an empty element. Multiple breaks (
) can be used in between paragraphs to display the text in different lines. This tag element contains the attributes such as clear, class, id, style, title.

Example

Consider below example that illustrates the use of break
 element,

```
<html>
<head>
<title> Break Tag </title>
</head>
<body>
<p> This is paragraph tag <br/> This is break element.</p>
</body>
</html>
```

Output



(iii) <pre> Element

<pre> element block is used to preserve the specific format of the source code. It contains both opening and closing tag. Mostly, this element is used to show the computer source code. It is mostly used to display the special characters such as /, %, <, > etc.

Example

```
<html>
<head>
<title>Pre tag</title>
</head>
<body>
<pre>
This  element block is used to preserve the format for the source code
</pre>
</body>
</html>
```

Output



1.1.3 Presentational Elements, Phrase Elements

Q20. Discuss about presentational elements.

Answer : Model Paper-II, Q9(a)

Presentational Elements

The text can be displayed in different styles or presentation styles using the following tags,

1. element
2. element
3. <u> element
4. <s> (or) <strike> element
5. <tt> element
6. <sup> element
7. <sub> element
8. <big> element
9. <small> element
10. <hr> element.

1. Element

The tag displays the text in bold form.

Example

<P> Web Programming </P>

2. element

The tag displays the text in italic form.

Example

<P> Web Programming </P>

3. <u> element

The tag <u> displays the text in underline form.

Example

<P> <u> Web Programming <u> </P>

4. <s> (or) <strike> element

The tag <strike> displays the text with a strike through it.

Example

<P> <strike> Web Programming </strike> </P>

5. <tt> element

The tag <tt> displays the text in monospared font. Here tt stands for teletype.

Example

<P> <tt> Monospared </tt> font </P>

6. <sup> element

The tag <sup> displays the text in the superscript format.

Example

<P> Web Programming is an ^{easy} language </p>

7. <sub> element

The tag <sub> displays the text in the subscripted format.

Example

<P> Web Programming is an _{easy} language </P>

8. <big> element

The tag <big> displays the text in the large size.

Example

<P> <big> Web Technologies </big> </P>

9. <small> element

The tag <small> displays the text in small size.

Example

<P> < small> Web Programming </small> </P>

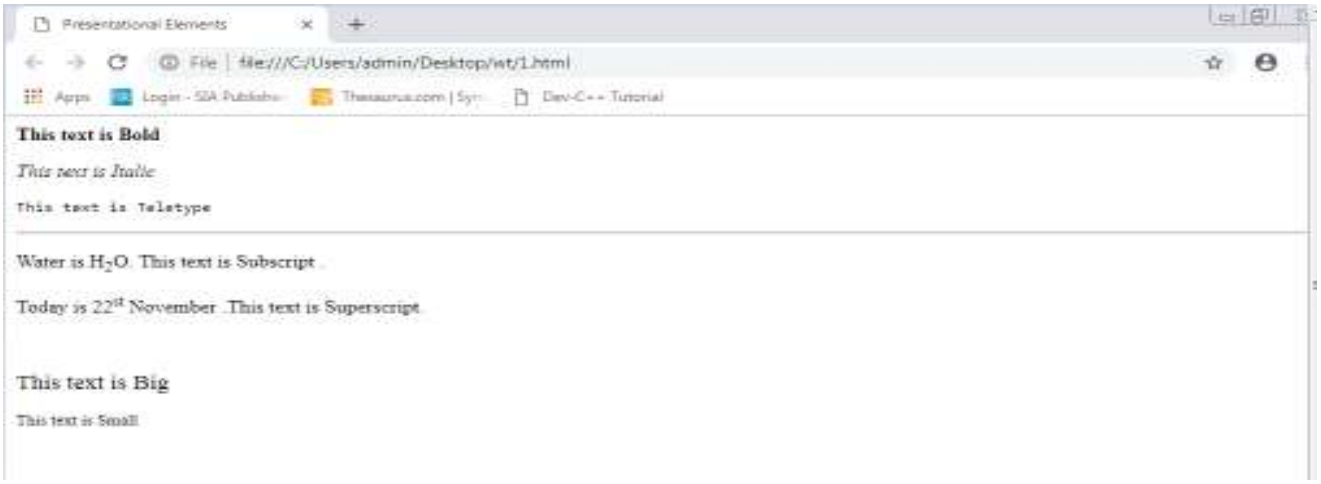
10. <hr> element

The tag <hr/> is used for creating the horizontal rule across the page. It is an empty element similar to
 element.

Example

```
<html>
<head>
<title> Presentational Elements </title>
</head>
<body>
<p><b>This text is Bold</b></p>
<p><i>This text is Italic</i></p>
<p><tt>This text is Teletype</tt></p>
<hr />
<p> Water is H<sub>2</sub>O. This text is Subscript .</p>
<p>Today is 22<sup>st </sup> November .This text is Superscript. </p>
<p><sup>This text is Superscript</sup></p>
<p><big>This text is Big</big></p>
<p><small>This text is Small</small></p>
</body>
</html>
```

Output



Q21. Discuss about Phrase Elements.

Answer :

Phrase Elements

The following are phrase elements in XHTML,

1. Element

The tag displays the text in an emphasized form.

Example

<P> Web Programming </P>

2. Element

The tag displays the text same as bold and with strong emphasis.

Example

<P> Web Programming </P>

<P> Hello   World </Strong> </P>

3. <address> Element

The tag <address> displays the text of addresses.

Example

<address> SIA PUBLISHERS & DISTRIBUTORS, HYD </address>

4. <abbr> Element

The tag <abbr> is used for abbreviations of text displayed in between opening and closing tags.

Example

This is an phrase element <abbr title = "Web Technolgoies"> WT</abbr>

5. <acronym> Element

The tag <acronym> is used for acronyms of text displayed in between opening and closing tags.

Example

<acronym title = "Hyper Text Markup Language"> HTML </acroynm>

6. <dfn> Element

The tag <dfn> is used for defining a special or a key form in an italicized format.

Example

This is an element of phrase element <dfn> WT </dfn>

7. <blockquote> Element

The tag <blockquote> is used for quoting the text from other document.

Example

<P> This is a paragraph </P>

<blockquote> Web Technologies </blockquote>

Cite Attribute with <blockquote> Element

Cite attribute is used with the <blockquote> element for defining the source or path of the quote. It contains an URL.

Example

<blockquote cite = "http://www.tutorialpoint.com"> </blockquote>

8. <q> Element

The tag <q> is used for short quotations i.e., a quote is added to a word.

Example

<P> <q> It's raining </q> </P>

9. <cite> Element

It defines the work title for example, A painting, BOOK, TV show, program, school, sports etc. It does not support any specific attributes except global attributes, event attributes etc.

10. <code> Element

The tag <code> is used for including the codes of any programming language between opening and closing <Code> tags.

Example

<P> <code> <h2> This is phrase element &.lt; /h2> </code> </P>

11. <kbd> Element

The tag <kbd> is used for displaying the text entered by a keyboard.

Example

<P> <kbd> SIA </kbd> <kbd> PUBLISHERS </kbd> and <kbd> DISTRIBUTORS </kbd>.<P>.

12. <var> Element

The tag <var> is used to display the programming variables. This is combined with <P> and <code> elements.

Example

<P> <code> document.write ("<var> Password </var>") </code> </P>

13. <Samp> Element

The tag <Samp> is used for displaying the program output.

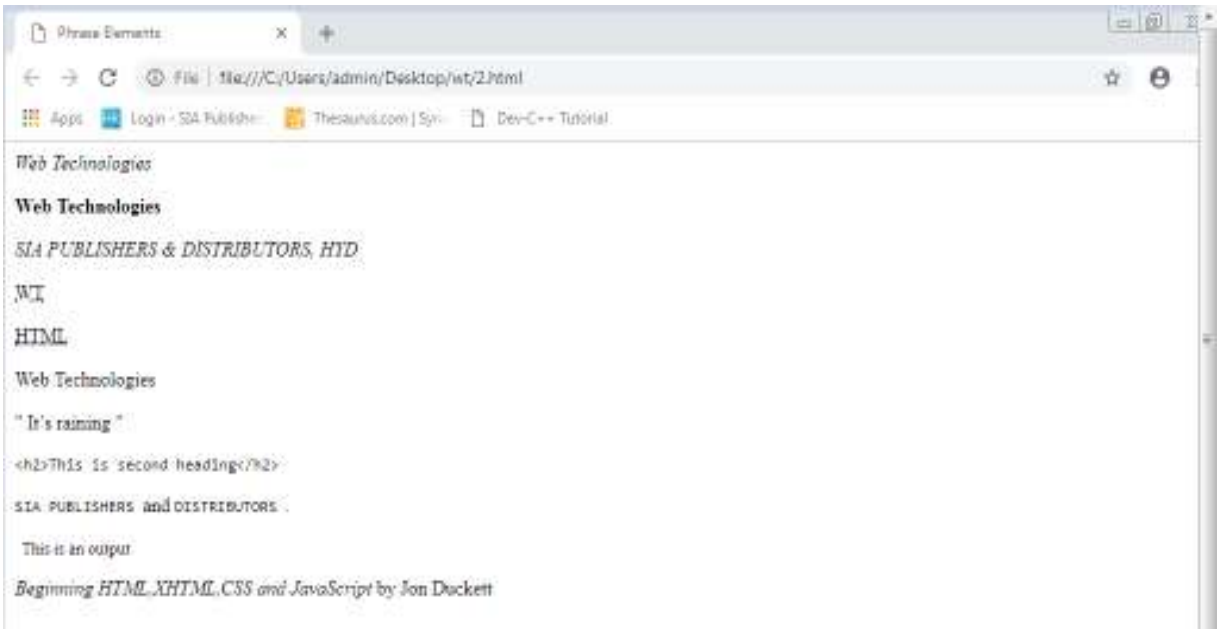
Example

<P> <Samp> This is an output </Samp> </P>.

Program

```
<HTML>
<HEAD>
<TITLE> Phrase Elements </TITLE>
</HEAD>
<BODY>
<P> <em> Web Technologies</em> </P>
<P> <strong> Web Technologies </strong> </P>
<P>  <address> SIA PUBLISHERS & DISTRIBUTORS, HYD </address></P>
<P>  <abbr title = "Web Technologies"> WT</abbr></P>
<P>  <acronym title = "Hyper Text Markup Language"> HTML </acroynm></P>
<P> <blackquote> Web Technologies </blockquote> </P>
<P> <q> It's raining </q> </P>
<P> <code> &lt;h2&gt;This is second heading&lt;/h2&gt;</code> </P>
<P> <kbd> SIA </kbd> <kbd> PUBLISHERS </kbd> and <kbd> DISTRIBUTORS </kbd>.<P>
<P> <Samp> This is an output </Samp> </P>
<P> <cite> Beginning HTML,XHTML,CSS and JavaScript</cite> by Jon Duckett </P>
</BODY>
</HTML>
```

Output



1.1.4 Lists, Editing Text

Q22. What are lists? Explain different types of lists with examples.

Answer :

List

"List" refers to a record that displays the text in arranged in sequential format or bullet format. Examples in this aspect can be a shopping list, list of names of employees in an organization, list of names of places.

Types of Lists

XHTML supports three types of lists. They are,

1. Unordered list
2. Ordered list
3. Definition list.

1. Unordered List

Unordered list display certain text on the console, with each independent sentence being initiated with a bullet mark.

Here, usually the entire text is enclosed under the tags **<UL> text to be displayed </UL>** with each independent sentence initiating with **<LI>** and closing with **</LI>**.

Unordered list does not account for different forms of displaying text like, sentences initiating with large roman numbers, small roman numbers, capital letters, small letters etc. Hence, to achieve applications displaying text in the form of above mentioned formats, the concept of ordered lists is used.

2. Ordered Lists

It is used to display list items with specific order. Ordered list displays the text containing large roman numbers, small roman numbers, capital letters etc. Ordered list is usually initiated using the tags **<OL type = "value">** text to be displayed **</OL>**. Here the value 'attribute' can accommodate any of the following values.

Value			Independent Sentences Initiates with
1.	A	1.	Capital alphabets
		E.g:	A. All men are blue
			B. All women are red
			C. All children are purple.
2.	a	2.	Small alphabets
		E.g:	a. All men are blue
			b. All women are red
			c. All children are purple.
3.	I	3.	Large roman numerals
		E.g:	I. All men are blue
			II. All women are red
			III. All children are purple.
4.	i	4.	Small roman numerals
		E.g:	i. All men are blue
			ii. All women are red
			iii. All children are purple.
5.	1	5.	Numerals usually the positive integers starting from 1
		E.g:	1. All men are blue
			2. All women are red
			3. All children are purple.

If no value is specified, HTML prints the text containing the (+ve) integer numerals as default case.

3. Definition List

It is a special type of list that lists terms and their definitions. It is created using (define list) <dl> tag. Every term in a definition list is defined within <dt> (define term) tag and its definition is defined within <dd> (define definition) tag.

Lists can be Nested or Embedded

When one list type is included in other list type then such type of succession is usually referred to as Miscellaneous or Nested Lists.

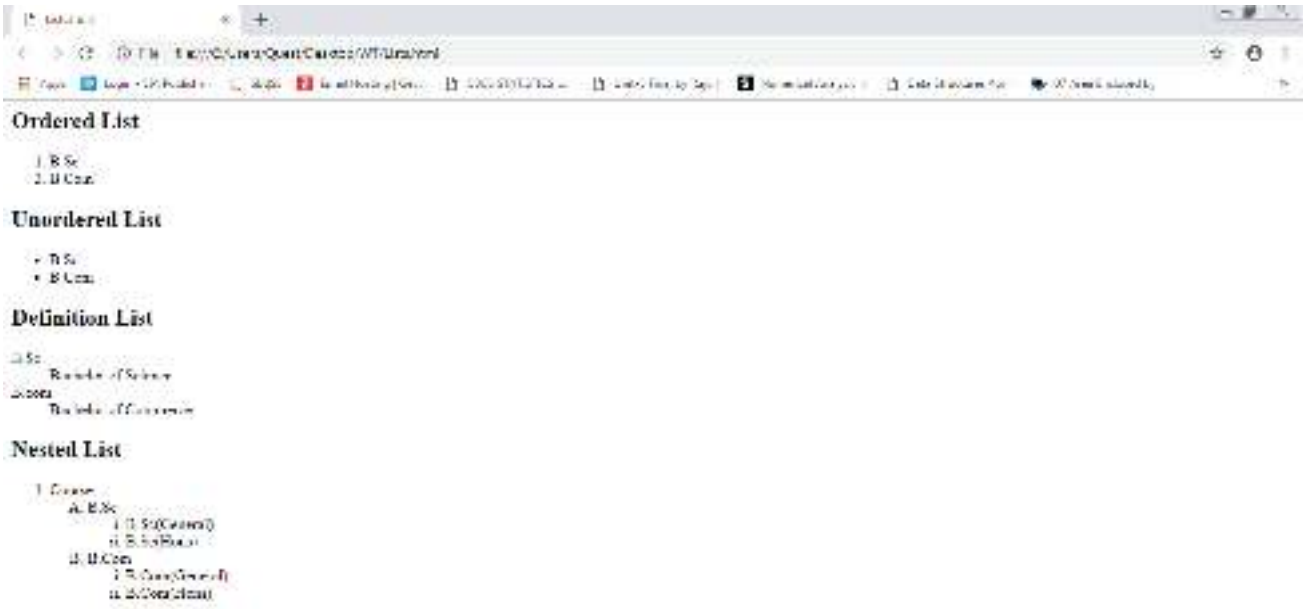
Program

An example to illustrate the types of lists is given below,

```
<html>
<body>
<h2>Ordered List</h2>
<ol>
<li>B.Sc</li>
```

```
<li>B.Com</li>
</ol>
<h2>Unordered List</h2>
<ul>
<li>B.Sc</li>
<li>B.Com</li>
</ul>
<h2>Definition List</h2>
<dl>
<dt>B.Sc</dt>
<dd>Bachelor of Science</dd>
<dt>B.com</dt>
<dd>Bachelor of Commerce</dd>
</dl>
<h2>Nested List</h2>
<ol TYPE=”1”>
<li>Course</li>
<ol TYPE=”A”>
<li>B.Sc
<ol TYPE=”i”>
<li>B.Sc(General)</li>
<li>B.Sc(Hons)</li>
</ol>
<li>B.Com
<ol TYPE=”i”>
<li>B.Com(General)</li>
<li>B.Com(Hons)</li>
</ol>
</li>
</ol>
</body>
</html>
```

Output



Q23. Write about the elements used for editing text in XHTML.

Answer :

The elements used for editing the text in XHTML are as follows,

(i) <ins> Element

(ii) <del> Element.

(i) <ins> Element

The tag <ins> displays the text that is to be inserted or added to XHTML.

Example

<P> Web Programming <ins> is </ins> an easy subject </P>

(ii) <del> Element

The tag <del> displays the text that is to be delete/removed from XHTML.

Example

<P> Web Programming is <del> also </del> an easy subject </P>.

Example

```
<html>
<body>
<p>My favorite color is <del>Blue</del> <ins>Red</ins>!</p>
</body>
</html>
```

Output



1.1.5 Core Elements and Attributes, Attribute Groups

Q24. What are the core elements and attributes of an XHTML document.

Answer :

The core elements and attributes of an XHTML document are given below,

(i) <html>

(ii) <head>

(iii) <little> and

(iv) <body>.

(i) <html> Element

XHTML document contains <html> element, under an optional XML declaration and a DOCTYPE declaration. An opening <html> tag includes namespace identifier which specifies the markup language to which the element belongs. For an XHTML document, a namespace identifier is used to specify that markup language belongs to XHTML 1.0 namespace in the document. It contains id, dir, lang, xml long attributes.

Example

<html xmlns = “http://www.tutorialspoint/Xhtml”>

(ii) <head> Element

<head> element contains the header elements. It is written after the opening tag of html in the document. <title> element is used to specify or declare the title of the document and is written in the block of <head> element.

The following are the attributes used in an opening<head>tag

id, dir, lang, xml: lang, profile

(iii) <title> Element

The title is displayed at the title bar of an xhtml document by using <title> element.

<title> Element can be used in the following ways,

(a) It is used at the top of the browser window.

(b) It is used as a default name for a bookmarks in any browsers.

(c) It is used by search engines for index pages.

Example

<title> Beginning HTML, XHTML, CSS and Javascript by Jon Duckett </title>

The following are the attributes used in <title> element id, dir, lang, xml: lang

(iv) <body> Element

<body> element contains the content of a web page. It uses all attributes available in the attribute group. The background, bgcolor, alink, vlink, link.text are the deprecated attributes used in this <body> element. The browser specific attributes such as language, top margin, bottom margin, left margin, right margin, scroll, bg properties, margin height, margin width are also used in <body> element.

Example

<html xmlns="http://www.w3.org/1999/xhtml">

<head>

<title>Web Technologies</title>

</head>

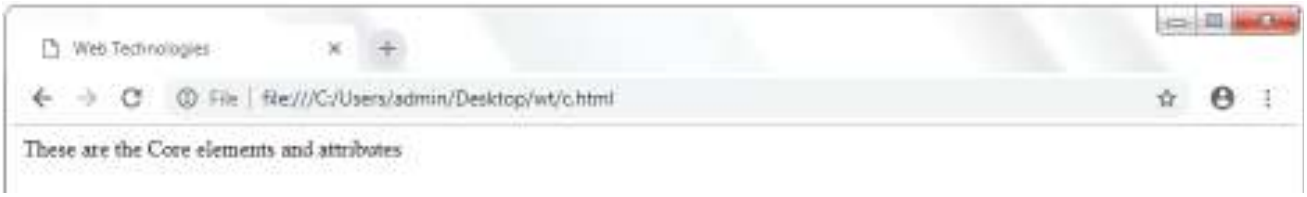
<body>

These are the Core elements and attributes

</body>

</html>

Output



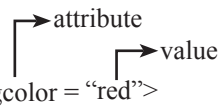
Q25. What are attribute groups? Explain.

Answer :

Attribute Groups

Attribute groups are embeded within opening tag. They provide extra information about an element. These attributes contain name as well as value. A name is the property or a feature of an element. A value is the value of an attribute assigned to that property.

Example



Types of Attribute Groups

Attribute groups are of three types. They are,

(i) Core attributes

(ii) Internationalization attributes

(iii) UI events.

(i) Core Attributes

Core attributes used in an XHTML document are id, title, class and style

(a) id Attribute

id attribute is used for identifying an element in a page uniquely.

Syntax

id = "string"

id attribute should start with any alphabet, digits, hyphens, underscores and colons. In an XHTML page, the attribute values should be different or unique.

(b) class Attribute

class attribute specifies that an element belongs to a class of elements.

Syntax

Class = "ClassName"

Example

<P class = "colors"> List of colors </P>

(c) title Attribute

title attribute provides the title for an element.

Syntax

title = "string"

(d) style Attribute

style attribute specifies the CSS rules in the block of element.

Example

<P style = "color : pink;"> Text </P>

Example

```
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
</head>
<body>
These are the Core elements and attributes
<p id = "string"> Hello </p>
<p class = "colors"> World </p>
<p title = "string" ></p>
<p style = "font-family:arial;color:#FF0000;">Hello World </p>
</body>
</html>
```

Output



(ii) Internationalization Attributes

Internationalization attributes are used for helping the users to write pages of different languages and character sets. The following are three types of internationalization attributes,

(a) dir

(b) lang

(c) xml:lang.

(a) dir Attribute

dir attribute is used to specify the direction in which the text must be displayed in the browser i.e., from left to right or from right to left.

dir attribute contain the following values,

- **ltr**

  ltr refers to the value from left to right and is considered as default value.

- **rtl**

  rtl refers to the value from right to left.

(b) lang Attribute

lang attribute specifies the language that must be used in an xhtml document. This is used by search engines, screen readers and other applications. It provides language – specific display to the users.

(c) xml:lang Attribute

xml.lang attribute is the xhtml attribute that is replaced for lang attribute. The value for this attribute must be a country code ISO-639.

Program

```
<html >
<head>
<title>Internationalization Attributes</title>
</head>
<body>
<p dir =“rtl” >This is dir attribute.</p>
<p lang=“en”>This is lang attribute.</p>
<p xml:lang=“us”>This is xml:lang attribute.</p>
</body>
</html>
```

Output



(iii) UI Events

UI event attributes are used to link the elements with the event. Keypress or mouse movement are the examples of UI events.

There are ten common events that are given as follows,

Onclick, Ondoubleclick, Onmouseup, Onmousemove, Onmousedown, Onmouseover, Onmouseout, Onkeypress, Onkeydown and Onkeyup.

The <body> and <frame set> elements contain onload and onunload events whenever a page is opened or closed.

1.2 LINKS AND NAVIGATION

1.2.1 Basic Links

Q26. Explain the following,

(i) How to link to other web-pages?

(ii) How to link to e-mail addresses?

Answer :

(i) Linking to other Web-Pages

A page can be linked to other web-pages by using the <a> element with the href attribute specifying the URL of the other web-page. If the URL described by the href attribute is any other complete web-page then this URL is known as qualified URL. This can be shown with the help of following example,

Example

```
<html>
<head>
<title>main page</title>
</head>
<h1>one of the most popular engine is
<a href = "http://www.Google.com">Google</a></h1>
</body>
</html>
```

Output



When the phrase "yahoo" from the site.html page is clicked, the http://www.yahoo.com site appeared in the same window.

In addition to the href attribute, the anchor element can contain the title attribute which describes about the other page the link navigates to the value of the title attribute is displayed in a tooltip when the user hovers the mouse over the link. This can be shown with the help of following example,

Example

```
<html>
<head>
<title> main page</title>
<h1><a href = "http://www.Google.com/" title = "searches the info with google search engine">Google</a> is a very
popular search engine</h1>
</body>
</html>
```

Output



(ii) Linking to e-mail Addresses

A link is created to an e-mail address by using anyone of the following syntax.

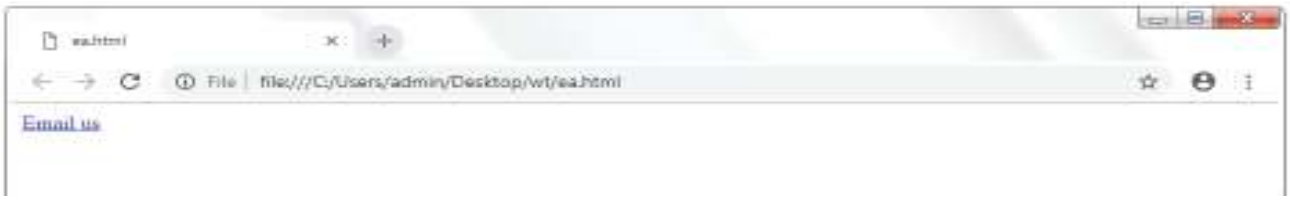
 abc@emp.com

 Email us

Example

```
<html>
<body>
 <a href = "mailto:pqr@exmp.com"> Email us </a>
</body>
</html>
```

Output



1.2.2 Creating Links with the <a> Element

Q27. How source anchor is created to link?

Answer :

A document can be linked to other documents by using the <a> element. It contains an attribute "href" which specifies the location link of the document that is to be linked. The text inside the anchor element should be clearly mentioned about the page that is going to be linked there should not be any space between the <a> element because it renders all its contents including the spaces as links. This is shown in the following example,

Example

home.html

```
<html>
<head>
<title>Home page</title>
</head>
<body>
<h1>Navigate to <a href = "Lists.html">Course page</a></h1>
</body>
<html>
```

Output



When user clicks on the "lists page of the home page, the index.html page gets loaded into the same window by replacing the home.html page. This can be shown as follows,

For remaining answer refer Unit-I, Page No. 19, Q.No. 22, Topic: Program.

Q28. Explain how the specific parts of a page can be linked.

Answer :

The page can contain links to specific parts of it when the length of page is long or when the page does not fit in the browser's window or when the user has to scroll the page to move to the specific location. It is also called internal links.

Some examples of the linking to the specific parts to a page on web pages are as follows,

- ❖ Long page contains the links "Back of top" at the bottom of the page to navigate.
- ❖ Page contains the lists of contents that will direct to the relevant part of the page.
- ❖ Page contains the links in the text to navigate to definitions or to the footnotes.

It is possible by two attributes,

1. Name (or) id attribute
2. Source attribute.

The linking to the specific parts of a page can be achieved by using, source anchors and destination anchors.

(a) Source Anchors

The source anchors specify the links to the specific parts of same page. The href attribute in the source anchor should contain the value same as the value of id attribute in the destination anchor except that it should be preceeded by a hash sign (#) (also known as pond sign).

(b) Destination Anchors

The destination anchors contain particular parts in the page that can be linked by source anchor. The destination anchor contains an attribute, "id", or name attribute which provides the identification to a particular part of the page. This identification is used as the value (preceeded by pond sign #) of the href attribute in the source anchor.

Example

```
<html>
<body>
<a name="top">Context</a>
<a href="#top">top</a>
</body>
</html>
```

Output



Q29. What are the other attributes of <a> element?

Answer :

Attributes of <a> Element

<a> element can contain the attributes like universal attributes, UI event attributes and other attributes such as access key, charset, coords, href, hreflong, rel, rev, shape, style, tabindex, target and type.

1. access key Attribute

access key attribute is used for creating a keyboard shortcut to enable a link.

For example, A value is assigned to the access key attribute such as alphabet 'P'. When the key "P" is pressed with alt or ctrl the link is activated.

The access key attribute must be specified on the source anchor.

Example

```
<a id = "bottom" access key = "P"> Back to top </a>
```

In the above example, the link at the top of the page can be accessed using the access key attribute by pressing on the keyboard.

2. charset Attribute

charset attribute is used for indicating the character encoding of the document to which the URL is pointed. This can be used to link a page in different languages with different character set. The value of charset attribute must be string i.e., UTF-8 or ISO-8859–1.

Example

 yahoo Japan

In the above example, a document is linked with the japanese character set.

3. coords Attribute

coords attribute is used by source anchor that has image in it. This creates an image map i.e., same image link is used at different places. The value of coords attribute is provided by x and y coordinates such that a particular part of image has link.

4. hreflang Attribute

hreflang attribute is used to specify the language of the page that is linked. The value of hreflang attribute is provided by two letter language code.

Example

 India

5. rel Attribute

rel attribute is used to specify the relationship between current document and resource provided by href attribute on the source anchor. It does not support major browsers.

Example

 next

This attribute can have the following values.

(a) toc (or) contents

toc refers to the document that contains table of contents of the current webpage.

(b) index

index refers to the document the contains index of the current webpage.

(c) glossary

glossary refers to the document that contains glossary of terms related to the current webpage.

(d) copyright

copyright refers to the document that contains a copyright statement of the current document.

(e) start

start refers to the document that is at the top position in a collection of ordered documents.

(f) next

next refers to the document that is at the next position in a collection of ordered documents.

(g) prev (or) previous

prev refers to the document that is at the previous position in a total of ordered documents.

(h) help

help refers to the document that is user understandable and can be navigated from one page or site to other.

(i) chapter

chapter refers to the document that represents as a chapter from the total documents.

(j) section

section refers to the document that represents as a section in a total documents.

(k) subsection

subsection refers to the document that represents as a subsection in a total documents.

(l) appendix

appendix refers to the document that is represented as a appendix in a total documents.

6. rev Attribute

rev attribute is same as rel attribute used on destination anchor that specifies the relation between the destination anchor and the source anchor. It does not support major browsers.

7. shape Attribute

shape attribute is used to specify the shape of an area while creating an image map that becomes a clickable hotspot.

8. target Attribute

This attribute specifies the location to open a link document. Its values are blank, self, parent top and target frame.

Example

```
<a href \ "Pgm.html/" target = "_self"> Program </a>
```

9. title Attribute

The title attribute is used on any links that has images. Additional information is provided to the users that uses visual text tooltip in any browsers.

10. type Attribute

type attribute is used to specify the MIME type of a link. These MIME types are compared with file extensions.

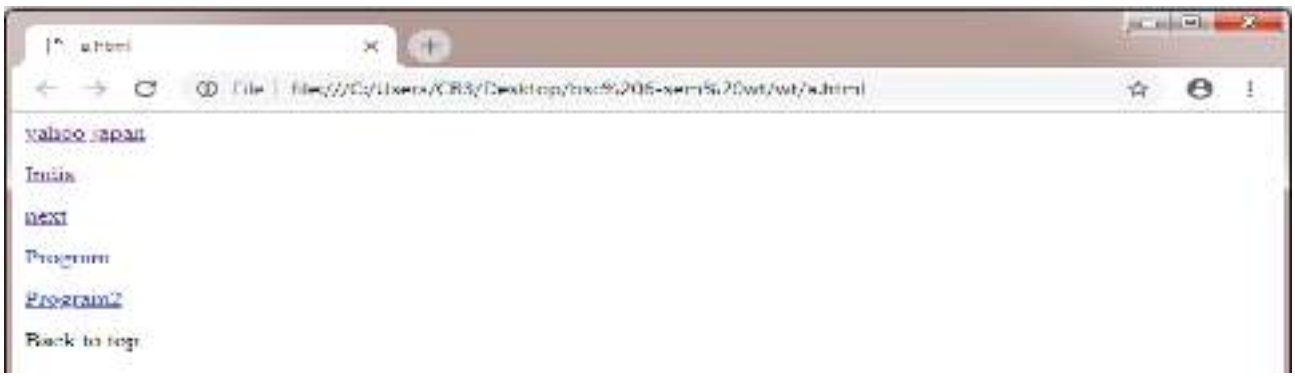
Example

```
<a href = "Pgmz.html" type = "text/html"> Program2 </a>
```

Program

```
<html>
<body>
<p><a href = "http://www.yahoo.co.jp/"  charset = "ISO-2022-JP"> yahoo japan </a> </p>
<p><a href = "http://www.google.co.in/" hre?ang = "In"> India </a></p>
<p><a href = "#next" rel = "next"> next </a></p>
<p><a href = "Pgm1.htm/" target = "_self"> Program </a></p>
<p><a href = "Pgm2.html" type = "text/html"> Program2 </a></p>
<a id = "bottom" access key = "P"> Back to top </a>
</body>
</html>
```

Output



1.2.3 Advanced E-mail Links

Q30. Discuss about advanced E-mail links.

Answer :

Advanced E-mail Links

A link is created to the users default e-mail editor by providing an e-mail address as shown in below,

```
<a href = "mailto: abcd@exam.org"> abcd@exam.org </a>
```

Other parts of the message can also be specified by using C c'd or BCC'd for subject, body and e-mail addresses. A question mark is placed after the e-mail address and name|value separated by an equal sign.

The properties are described below,

(i) subject

This property is used for adding subject line to an e-mail.

(ii) body

This property is used for adding a message into the body of the e-mail.

(iii) cc

This property is used for sending a carbon copy of a mail to the given CC'd address. The value provided should be a valid e-mail address. By using an ampersand, multiple addresses can be sent.

(iv) bcc

This property is used for sending a carbon copy of a mail secretly to the given BCC'd address without any recipient. The value provided should be a valid e-mail address. Ampress and is used for providing multiple addresses.

A space between words in the subject line can be provided by adding % 20 in between words. A line break in a body of message is created by adding %0D % 0A.

Example

```
<html>
<body>
<p><a href="mailto:ab@email.address?subject=Information">Home</a></p>
<p><a href="mailto:cd@email.address?cc=ef@email.address,gh@email.address">About us</a></p>
<p><a href="mailto:stu@email.address?bcc=vw@email.address,xyz@email.address">Contact us</a></p>
<a href="mailto:abcd@email.address?body=One paragraph.">Details</a>
</body>
</html>
```

Output



1.3 IMAGES, AUDIO AND VIDEO

1.3.1 Adding Images Using the <img> Element, Using Images as Links

Q31. Explain about images and supported attributes in HTML.

Answer : **Model Paper-III, Q9(a)**

Images (<img>)

This tag adds images to the web pages. They increase the visibility of web pages. These images are stored in computers by following several strict formats, most common among them are BMP, JPEG, GIF, PMP etc. However '.GIF' are most popular because of its clarity and support for a wide range of colors. The following is the image tag,

```
<img src = "pic.gif"  alt = "picture"/>
```

This forms the primary tag to insert images onto the web page.

Supported Attributes

The following are the support attributes of <img> tag,

1. SRC

It specifies the URL (Uniform Resource Locator) i.e., where the image is actually stored in the computer or server. For example, if the image is stored in the 'C' drive, images folder, with name rose.gif, then SRC (Source) can supply URL address as SCR = "C:\images\rose.gif". It is mandatory to include SRC attribute within the <IMG> tag.

```
<img src = "c:/images/rose.gif"/>
```

2. ALT

It is used to display alternative text if the browser fails to display the required image due to several underlying reasons. This can be done using ALT attribute.

3. ALIGN

It aligns the given web page. Its values can be "TOP", "BOTTOM", "LEFT", "RIGHT", "MIDDLE" respectively.

4. BORDER

It specifies the width of the border for the image can be specified. It is usually measured in terms of pixels and its default value is zero.

<img src = "url" border = "Value in pixels">

5. HEIGHT/WIDTH

Height and width of the image can be specified using these attributes. Often height and width of an image are measured in terms of pixels.

6. HSPACE/VSPACE

The gaps or white spaces between the text and images can be controlled using these attributes

7. LONGDESC

It usually takes certain URL as argument, where further information related to a given image is stored.

<img src = "url" Width = "Value in Pixel" Height = "Value in Pixel" Longdesc = "file1.html">

8. NAME

This attribute is used to specify name of the image.

9. Ismap and Use map

These attributes are used with image maps.

Following is an application supporting insertion of images on the given web pages,

Program

```
<HTML>
<HEAD>
<TITLE>IMAGES</TITLE>
</HEAD>
<BODY>
<IMG SRC= "IMG1.PNG" ALT= "ERROR" ALIGN= "CENTER" NAME= "IMAGE1" HEIGHT= "200" WIDTH= "200" HSPACE= "0" VSPACE= "1" LONGDESC= "HTTP://EXAMPLE.COM/DESC#IMAGE1">
</BODY>
</HTML>
```

Output



Q32. How an image is used as link?

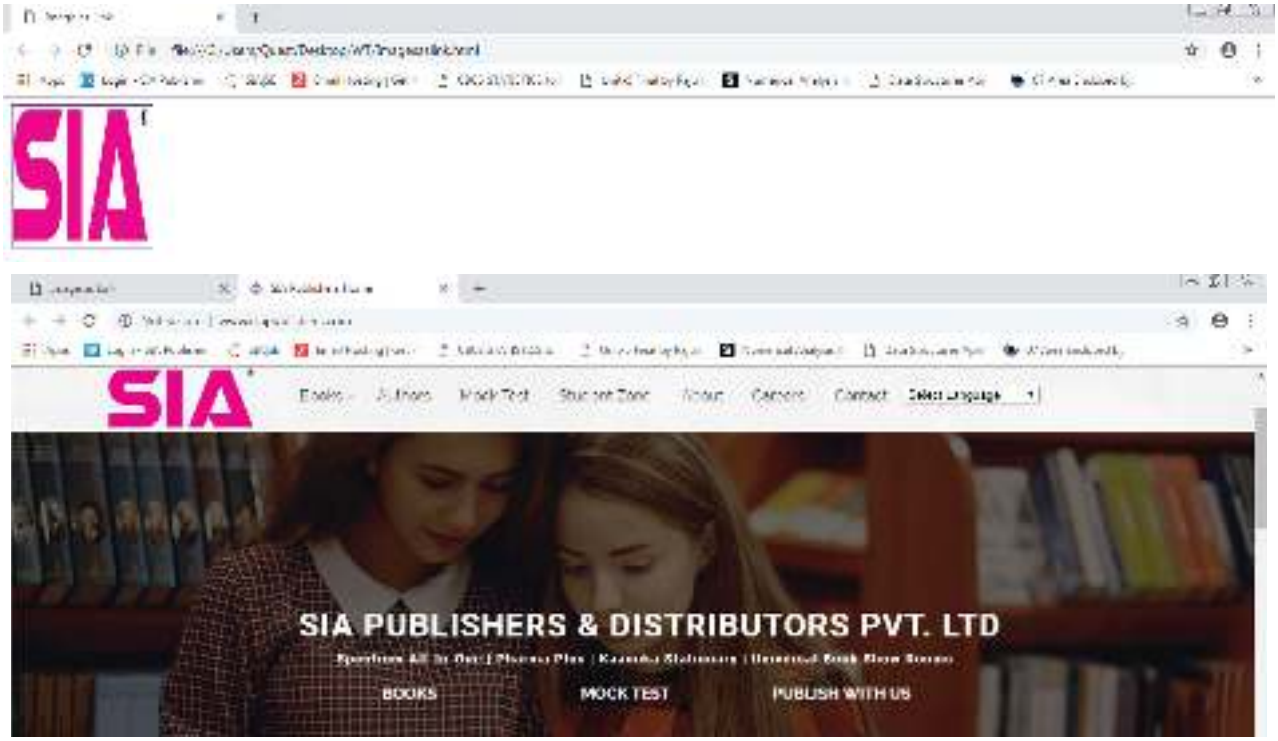
Answer :

Using Images Links

An image is inserted into the link using <a> element shown below,

```
<html>
<head>
<title> Image as Link </title>
</head>
<body>
<a href = "http://www.siapublishers.com/" target = "self">
<img src = "logo.png" align = "bottom" alt = "Error" border = "1" width = "150" height = "150"/> </a>
</body>
</html>
```

Output



1.3.2 Image Maps

Q33. What are image maps? List the different types of image maps.

Answer :

Image Maps

Image maps are used for creating different links within an image so that, the user can navigate to different pages when clicked on different parts of the image. The places where the user clicks in an image map are called as clickable hotspots. Image maps are helpful in dividing the image into irregular partitions such as maps. The two different types of maps are,

(i) Server-side image maps

(ii) Client-side image maps.

(i) Server-side Image Maps

The server side image maps allow the browser to collect the coordinates where the user clicks on the image and sends it to the server. These coordinates are processed by a script present on the server side and determines the page to be sent to the client.

(ii) Client-side Image Maps

The client-side image maps allow the browser to represent the page based on the location where the user clicks on the image.

Q34. Explain about client-side image maps.

Answer :

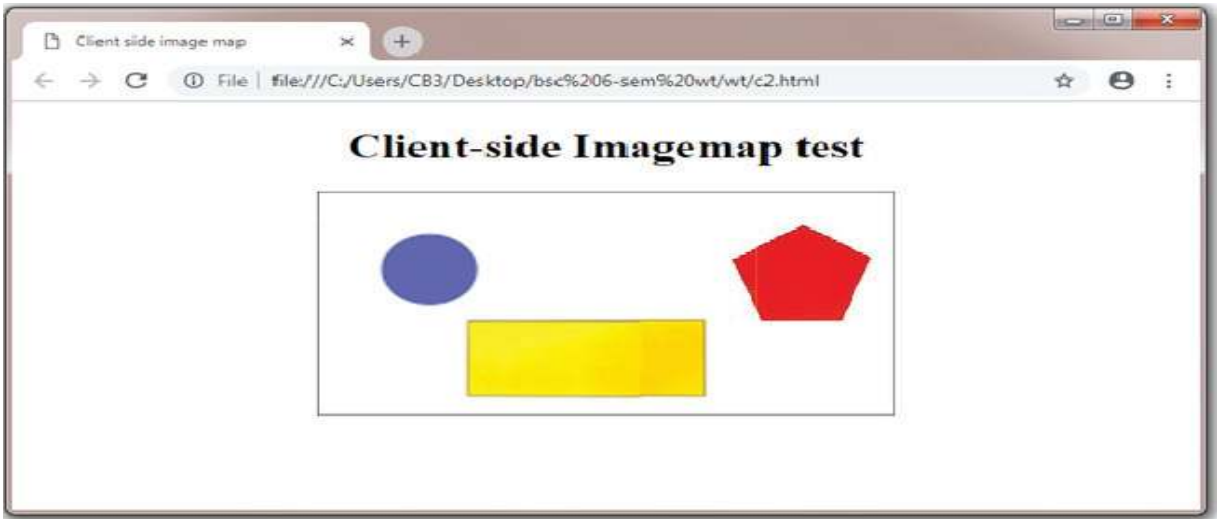
Client-side image maps are written in XHTML code. It depends on the browser to specify the part of the image that should be clicked to navigate to the respective web page. The client-side image maps can be created by using the <map> and <area> elements within the <img> element.

- ❖ The <img> element specifies the image that must be inserted as map in the document. It includes an extra attribute i.e., usemap whose value is similar to the value of the name attribute in the <map> element except that it is proceeded by a pound or hash sign (#).
- ❖ The <map> element must be followed by the <img> element. It includes an attribute, "name" which specifies the identity of the map.
- ❖ The <area> element must be contained within the <map> element. It defines the clickable hotspots within the map. It also defines the shape and the coordinates of the clickable hotspots. It defines the clickable sections of the image.

Example

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtmll/DTD/xhtmll-transitional.dtd">
<HTML xmlns="http://www.w3.org/1999/xhtml" lang="en">
<head>
<title>Client side image map </title>
<meta http-equiv="content-type" content="text/html"; charset=I80-8859-1"/>
</head>
<BODY>
<h1 align="center">Client-side Imagemap test</h1>
<div align="center">
<img src="C:\Users\CB3\Desktop\1.gif" usemap="#shapes" alt="shapes" boder="0" width="400" height="200"/>
</div>
<MAP Name="shapes" id="shapes">
<AREA Shape="circle" Coords="70,84,51" Href="https://kb.iu.edu/d/abyt" target="blank" alt="CIRCLE">
<AREA Shape="rect" Coords="25,180,125,280" Href="https://www.cs.uct.ac.za/mit_notes/web_programming/html/ch18s03.html" target="blank" alt="RECTANGLE">
<AREA Shape="poly" Coords="153,106,186,225,340,193,315,81,304,167" Href="http://www.columbia.edu/itc/visualarts/r4100/html_example_pages/g_image_map.htm" target="blank" alt="POLYGON">
<AREA Shape="default" Coords="0,0,500,300" Href="https://www.december.com/html/demo/imagemap.html" target="blank" alt="">
</MAP>
</BODY>
</HTML>
```

Output



Attributes of <area> Element

The following are some of the attributes used by the <area> element,

(a) The shape Attribute

The shape attribute specifies the shape of the portions in the map. If the value of the shape is not provided then the area is assumed to be rectangle. The following are the possible values that the shape attributes can accept,

default → No shape is created

rectangle or rect → Rectangle shape is created

polygon or poly → Polygon shape is created

Circle or circ → Circle shape is created

(b) The coords Attribute

The coords attribute specifies the coordinates of the area that can be selected. The specification of the number of coordinates depends upon the value of the 'shape' attribute for example:

rect → It should specify four coordinates.
The first two indicate the top-left and the next two indicate the botton-right of the rectangle.

Circle → It should specify three coordinates. The first two indicates the centre of the circle where as the third coordinate indicates the radius in pixels.

polygon → It should specify two coordinates for each of its vertex. So for the triangle it should specify six coordinates, for the pentagon it should specify ten coordinates.

(c) The href and nohref Attributes

The href attribute provides the URL of the image. If href is not used nohref attribute is used with the value nohref.

<area shape = "rect" coords = "212, 123, 323, 250" nohref = "nohref"/>

(d) The alt Attribute

The alt attribute specifies the text which is displayed in place of image if the image does not get displayed due to certain reasons.

(e) The target Attribute

The target attribute specifies the frame/window in which the page can be loaded.

(f) tabindex Attribute

The tabindex attribute specifies the order in which the users has to tab items on a page. This value ranges from 1 to 32767.

Q35. Write about client-side image maps using <object> element.

Answer :

Client-Side Image Maps Using the <Object> Element

<Object> element is used to add image maps to documents. This concept is supported by the major browsers such as IE8, Firefox 3, safari 3.

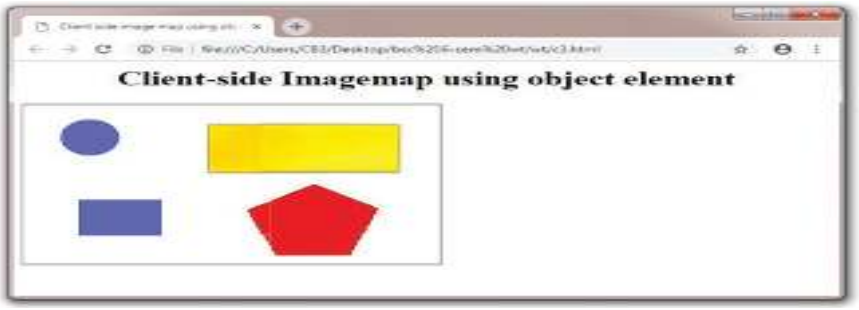
<map> element with name attribute is used in the <object> element. <a> element is used instead of <area> element in the <object> element.

Example

```
<HTML xmlns="http://www.w3.org/1999/xhtml" lang="en">
<head>
<title>Client side image map using object </title>
<meta http-equiv="content-type" content="text/html"; charset=I80-8859-1"/>
</head>
<BODY>
<h1 align="center">Client-side Imagemap using object element</h1>
<object data="C:\Users\CB3\Desktop\1.gif" type="image/gif" alt="shapes info" width="400" height="250" border="0" usemap="#shapes"/>
<map name="shapes">
<a shape="rect" href="http://speed.eik.bme.hu/help/html/Special_Edition-Using_CGI/ch9.htm" coords="0,0,118,28" target="_self" >Link1</a> |
<a shape="rect" href="https://docs.oracle.com/javaee/7/tutorial/jsf-custom002.htm" coords="118,0,184,28" target="_self">Link2</a> |
```

```
<a shape="circ" href=" http://www.howtocreate.co.uk/tutorials/html/imagemaps "  coords="184,200,60"  target="_self"
>Link3</a> |
 <a shape="poly" href=" http://www.htmlhelp.com/reference/html40/special/object.html" coords="276,0,373,28,50,50,276,0"
target="_self"  >Link4</a>
</map>
</body>
</html>
```

Output



Q36. Explain about server side image maps.

Answer :

In the server-side image maps, the ismap attribute is used the <img> element wthin the <a> element. The ismap instructs the browser to send the coordinates (that have selected by the user while clicking on the image maps) to the server. The sent coordinates are then processed by the script (written in PHG, GGI, JSP, Asp.net etc) at the server side. The ismap attribute doesnot need any value in HTML but, it requires the value in XHTML hence its own name can be used as its value to make this attribute accessible. This can be shown as follows,

<imp src = "/images/countries/country.gif alt" = "India map" border = "0" ismap = "ismap"/>

When the user selects any part of the map with the coordinates 85 pixels from the right of top-left corner and 75 pixels from the down of the same corner, the browser appends these coordinates at the end of the URL and sends it to the server as follows,

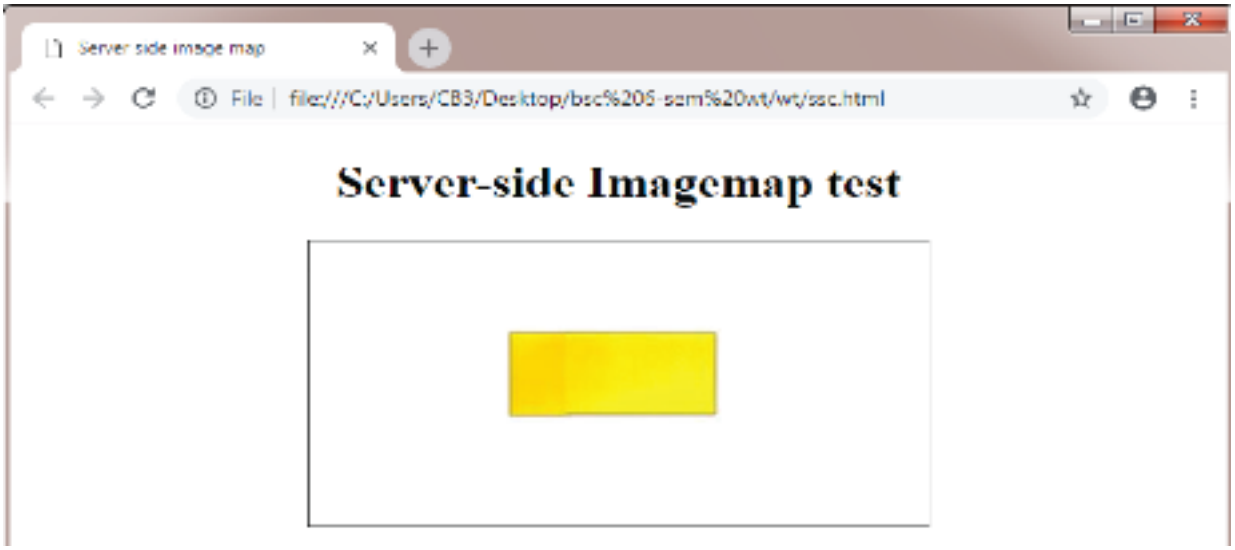
http://www.mapsexample.org/location/servermap.aspx?85,75.

Thses coordinates are then processed by the script present at the serverside to decide page that is to be send to the client.

Program

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 4.01 Transitional//EN" "http://www.w3.org/TR/html4/loose.dtd">
<HTML >
<head>
<title>Server side image map </title>
<meta http-equiv="content-type" content="text/html"; charset=I80-8859-1"/>
</head>
<BODY>
<h1 align="center">Server-side Imagemap test</h1>
<div align="center">
<a href="https://www.google.com/" shape="rect" coords="0,0,118,28">
<img src="C:\Users\CB3\Desktop\1.gif"  alt="shapes" boder="0" width="400" height="200" ismaps="ismap"/></a>
</div>
</BODY>
</HTML>
```

Output



1.3.3 Choosing the Right Image Format

Q37. Explain about bitmapped graphics. Write about reducing file size.

Answer :

Bitmapped Graphics

The bitmapped category of image formats such as : JPEG (Joint Photographic Experts Groups Format), GIF (Graphics Interchange Format), TIFF, PNG (Portable Network Graphics) and BMP divides an image into number of pixels and defines the color of each pixel similar to the color of pixels defined on the screen by the computer. These type of image formats are suitable for photographs and the complicated groups of shade and color.

The resolution can be defined as the number of pixels per square inch. The web stores the image at the resolution of 72 pixels per square inch. The printers often prints the image at 300 dots per inch (in printer the word dot is used in place of pixels) hence the file, it prints will be of the larger size.

Various types of bitmap graphics formats are as follows,

1. GIF Image

The GIF (Graphic Interchange Format) images uses a palette consisting of 256 colors. Each pixel of a GIF image carry one of these colors. There are several palettes of 256 colors made by using the group of 16 million colors. Every GIF image can use any of these different palettes.

The GIF image stores the information about the palette of 256 colors in a lookup table. Each pixel of the GIF image references to a color in the look up table.

The GIF image doesnot repeat the color information in the lookup table if many of the pixels in it references to a same color. This type of storing an image is known as “index color formatting”.

When GIF requires more than 256 colors then this technique used to save GIF images that help to represent extra colors. Two or more colors are used in adjacent pixels for creating an effect of a third color. GIFs are used for specifying one or more colors in a GIF that represents a transparent background. LZW compression is a technique used for compressing the GIF files to small. This is used for scanning the rows of an image for consecutive pixels with same color. This technique is also referred as lossless compression since there is no loss of data and quality.

Animated GIFS

GIF images are used for storing the frames more than one within a file. GIF are allowed to rotate in between versions or frames to create a simple animation. Its working process is same as a flip-book animation. In this flip book animation, the drawing on a single page of a book is changed slightly compared with the previous page. Whenever a user flips the pages it looks like as the images are moving in a book. When an animated image has a larger areas of a flat colour then this is used very well. As the image is compressed after the recording of initial frame, the pixels are changed and required to store with a single subsequent frame.

2. JPEG Image

The JPEG image format is used to store and compress the image. This format can be used to control the compression of the image. When the image is compressed the color data related to it gets lost and the compressed image cannot be converted back to the original version of the image. This type of compression is known as "lossy compression".

The size of the file depends upon the compression that is performed on it. The quality of the image can be either represented in percentages or by using the words such as excellent, very good, good and so on.

3. PNG Images

The PNG (Portable Network Graphics) format was created in the late 1990's. It provides image same as GIF formats and reduces the drawbacks associated with them. The two types of PNG was developed are,

(a) The 8 bit PNG has the disadvantages as the 8-bit GIF formats such as,

- ❖ It also uses only 256 colors and
- ❖ In case of transparency each pixel is either in the on or off state.

(b) The 24 bit PNG format possesses the following advantages:

- ❖ It uses unrestricted colors without loosing any data
- ❖ It provides the different level of transparency for each pixel by using the maps (such as lookup tables).
- ❖ It displays and image 8 times faster than a GIF image by replacing the approach of sampling one in eight line with two- dimensional sample.

Reducing File Size

An image is required to be download or saved on the desktop in the format that compresses the size such that the page can loaded quickly. The JPEG format can be used for photo-realistic pictures that contains heavy or subtle shades that need to be preserved. The GIF or PNG format can be used for images with flat color, hard edges like diagrams, texts and logos.

Consider the two images saved in different formats and file size shown as below,

Image	JPEG	GIF
Image 1	50.15k	110.1k
Image 2	8.16k	7.012k

The small size image file are known as thumbnails and found in the image gallery or on some pages. An image is created in the smaller version and it is scaled in on image editing program. Users download the full-size of image even it is of smaller version. So, the width and height attributes of an image or object must not be changed. A special thumbnail is created for a smaller image that can be used to load the page faster.

1.3.4 Adding Flash, Video and Audio to Your Web Pages

Q38. How Flash, Video and Audio are added to web pages?

Answer :

Adding Flash, Video and Audio to Web Pages

Rich Media known to be the addition of flash animations, video and audio to web pages. The following are the two reasons that make displaying of video or audio difficult on the site.

1. Flash format is easy since it has only one format for flash animation whereas a video contains various formats such as AVI, Flash video, MPEG, Quicktime, Windows Media etc and in audio MP3, WAV.
2. Browsers are built and used for displaying text and images. The various formats of rich media are not shown automatically by browsers. The browsers depends on additional programs such as players or plug ins.

   An <object> element is used to convey the following things to the browser.

   (a) The name of the file and where it has to be find

   (b) The format of the file

   (c) The type of plug in required for playing the file.

<param> element is used to send additional information to the player present in the <object> element. The <embed> element also be used in the <object> element.

1. **Adding Flash Movies to Web Site**
   For answer refer Unit-I, Page No. 38, Q.No. 39.
2. **Adding Videos to Website**
   For answer refer Unit-I, Page No. 40, Q.No. 42.
3. **Adding Audio to Web Site**
   For answer refer Unit-I, Page No. 42, Q.No. 43.

Q39. How flash movies are added to web site?

Answer :

Adding Flash Movies to Website

1. Flash files are created by using flash authoring environment.
2. The flash project is saved with the extension of .fla.
3. These files are exported into web in the .swf format.
4. After finishing the exporting of files, they are referred as SWF files or flash movies.
5. A flash player is a free plugin used to play SWF files incase of browsers that does not support SWF files.

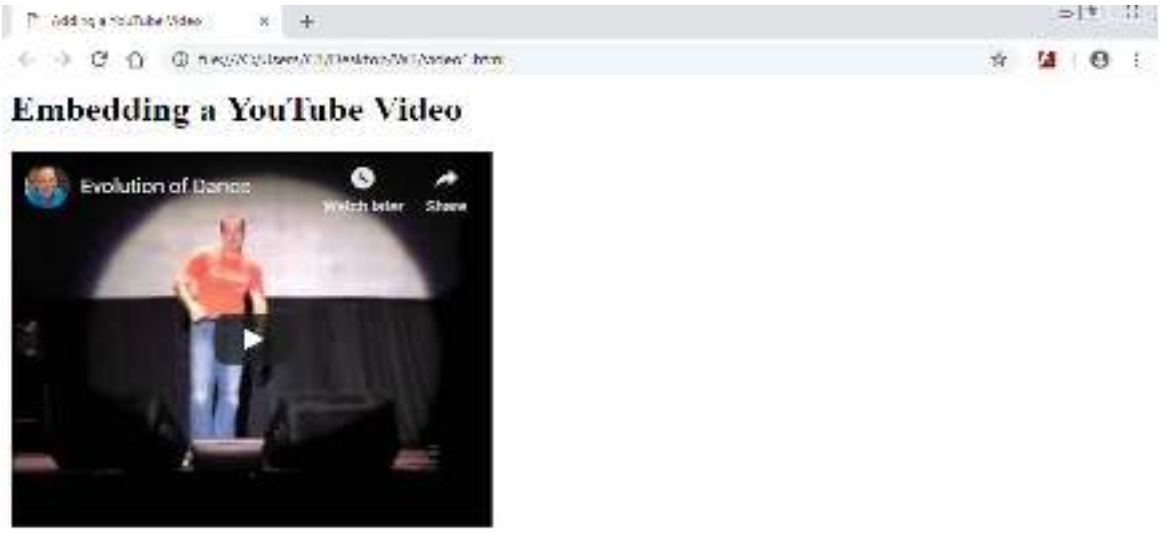
Example: Adding the Youtube Flash Video Player to a Page

Youtube is easy for users to embed the youtube player on their own pages and this youtube player is written in flash. A line of a code or a link is copied from the youtube site and is pasted on any web page.

Program

```
<html>
<head>
<title>Adding a YouTube Video</title>
</head>
<body>
<h1>Embedding a YouTube Video</h1>
<object width="425" height="344">
<param name="movie" value="http://www.youtube.com/v/dMH0bHeiRNg&hl=en&fs=1"></param>
<param name="allowFullScreen" value="true"></param>
<param name="allowscriptaccess" value="always"></param>
<embed src="http://www.youtube.com/v/dMH0bHeiRNg&hl=en&fs=1" type="application/x-shockwave-flash"
allowscriptaccess="always" allowfullscreen="true" width="425" height="344"></embed>
</object>
</body>
</html>
```

Output



In the above code, the <param> element is used to specify the name as movie with value given as URL. It specifies whther the player should be in full-screen mode or not. In also allows to play a flash movie from other sites.

Q40. How flash is added to a web page using <object> and <embed> element?

Answer :

<Object> and <Embed> Element

Flash is added to the webpage by using <object> and <embed> elements. A basic flash animation can be added to a web page by using a different technique called Flash B. When the FIA file exported for creating SWF file. The code created by the flash adds the SWF file to a web page.

Program

```
<html>
<head>
<title>Adding a flash movie with the &lt;object&gt; element</title>
</head>
<body>
<p>
<object classid="clsid:D27CDB6E-AE6D-11cf-96B8-444553540000" width="300" height="200" codebabse="http://download.macromedia.com/pub/shockwave/cabs/flash/swflash.cab">
<param name="movie" value="flash/flash_sample.swf">
<param name="play" value="true">
<param name="loop" value="false">
<embed SRC="flash/flash_sample.swf" WIDTH="300" HEIGHT="200" PLAY="true" LOOP="false" QUALITY="best" MENU="false" TYPE="application/x-shockwave-flash" PLUGINSPAGE="http://www.macromedia.com/shockwave/download/index.cgi?P1_Prod_Version=ShockwaveFlash"></embed>
</object>
</p>
</body>
</html>
```

Output



(a) Classid attribute is used for specifying the type of application to be played for a file.

(b) Codebase attribute is used for specifying the plugin or a program at the place where this file is required to be played.

Q41. How flash is added to a web page using SWF object?

Answer :

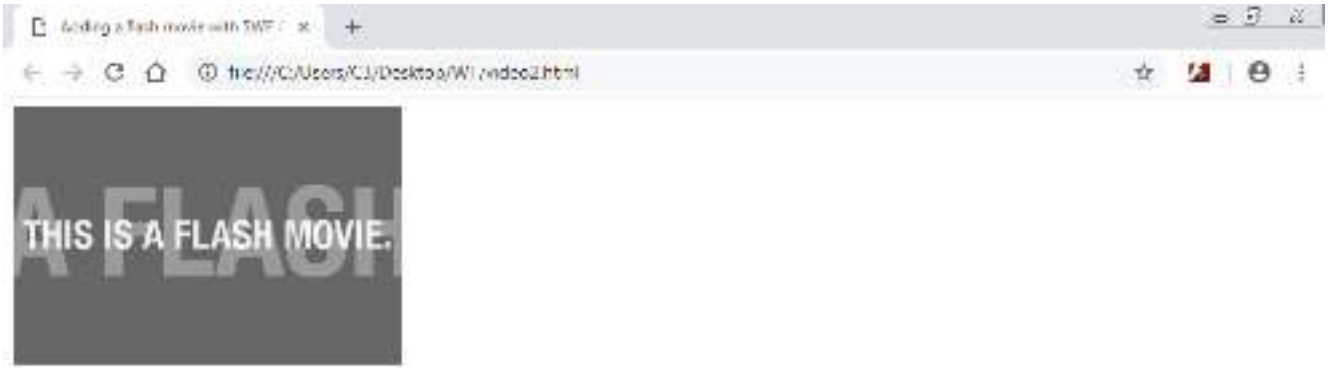
SWF Object

Flash is added to a web page using SWF object. In this technique, a page is created by using element that holds the flash video. CSS is used to ensure that element has the width and height same as the SWF file needs element has an id attribute that specifies uniquely the identity for an element. Images and texts can be used inside element. A javascript is added to the page that checks for the installed flash player in the browser. When a flash player is installed then the javascript is used to replace the content of element with SWF file. SWF object can be download from the site http://code.google.com/p/swfobject/.

Program

```
<html>
<head>
<title>Adding a flash movie with SWF Object</title>
<script type="text/javascript" src="swfobject.js"></script>
<script type="text/javascript">
swfobject.embedSWF("flash/flash_sample.swf", "flash_movie", "300", "200", "8.0.0");
</script>
</head>
<body>
<div id="flash_movie">*Install Flash player</div>
</body>
</html>
```

Output



Q42. How videos are added to website?

Answer :

Adding Videos to Website

A video can be uploaded on the website in a easy way. The following are the things to be considered while playing a video on the web.

(i) File Format

Selection of user desired format for the video must be done. Video can be in any formats such as AVI, Flash video, MPEG, Quicktime, Windows Media. Based on the selected format for the video, the user can download it with the specified size and quality of the picture.

(ii) Plug in is Required for Playing the File

Various formats uses different plugins such as flash player, Quicktime players, Real player, Windows Media player for playing the file.

The different kind or a type of video required for playing the file can be conveyed to a browser by using type attribute in the <object> element. Its value is MIME type.

The type of plug in to be used by a browser can be specified by class id attribute. Its value is group of numbers/letters used for identifying the plugin.

The class id value of the Quicktime Player Plug-in is,

02BF25D5 – 8C17 – 4B23 – BC80 – R3488 ABDDC6B

Commonly used video formats on the web with its file extension, MIME type, default player on PC and Mac are given below,

Video File Format	File Extension	MIME Type	Default Player on PC	Default Player on MAC
Flash video player	.swf	Application/x-shock-wave flash	Flash player	Flash player
AVI	.avi	video/x-ms video	Windows media player	Windows media components are required to be download-ed by user
MPEG	.mpg or .mpeg	video/mpeg	Windows media player	Quick time player
Quicktime movie	.mov	video/quicktime	Plug in should be downloaded by the user	Quick time player
Windows media video	.wmv	video/x-ms-wmv	Windows media player	Windows media components should be downloaded by the user
Real media	.rm	Application/vnd.rn-realmedia	Real player	Real player

Example: Adding Videos Using Flash Video

The following are the reasons to describe that the flash video is popularly used to add videos on websites.

(i) Flash player must be installed on computer connected to the internet server or the web and different video formats are also installed such that users can view FLV files.

(ii) Flash is used for controlling the appearance of the player so that users can create branded players.

(iii) FLV files are compared such that these are smaller than MPEG-1 format with some loss in quality.

(iv) The videos are faster for loading and playing.

(v) Its cost is low for running a website.

A user does not upload the video to anyone of the site then the following things are to be done.

(a) The video has to be converted from one format to the required FLV format.

(b) FLV player is written or downloaded for playing FLV files.

The video can be converted from one formats into the required FLV format in two following ways,

(i) Adobe Software

Adobe software is used to convert video format to FLV provided by flash authoring environment.

(ii) Third-Party Software

Different companies have developed softwares for converting various video formats into FLV files.

Different flash players are available that can be downloaded even though it does have any flash authoring tool as given below,

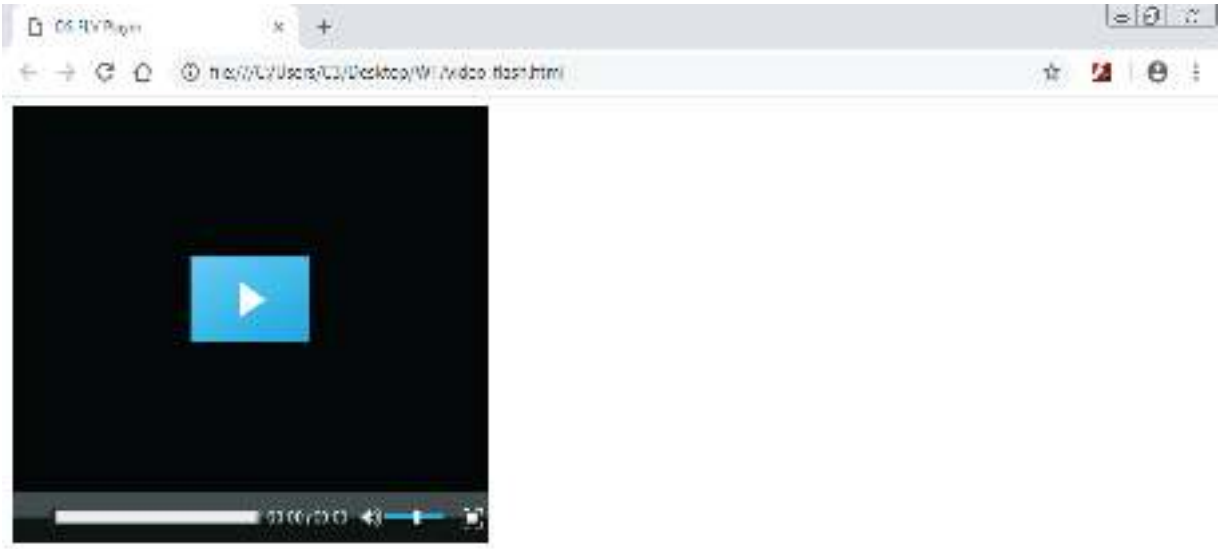
(a) OS FLV player
(b) JW FLV player
(c) Flow player
(d) Slideshow pro.

The above players have their own way to specify how to find FLV file.

Program

```
<html>
<head>
<title>OS FLV Player</title>
<meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1">
<script type="text/javascript" src="swfobject.js"></script>
<script type="text/javascript">
swfobject.embedSWF("flash/os_flv_player.swf?movie=video.flv", "flash_video", "400", "340", "8.0.0");
</script>
</head>
<body>
<div id="flash_video">*Install Flash player</div>
</body>
</html>
```

Output



Video Size

The size of a video must be known when it is downloaded the larger video has more file size. These big files can be considered as a problem due to following reasons,

(i) If the size of a file is less or small then it is downloaded very quickly. Users loose interest to watch videos which consume more time for downloading. User's might pause them for a while.

(ii) If the size of the file is big then it consumes more bandwidth. Some companies that contains websites may charge for bandwidth.

The long video can be divided into sections for the users to watched.

Q43. How audio is added to web site?

Answer :

Adding Audio to Website

An audio can be added to web site by considering the following things.

(i) File Format

Various types of compression techniques are used to store various types of audio files that effect the quality and size of file recording.

(ii) Plug in Required for Playing the File

The player is required by the user for playing audio using plug-in.

File Types

The following are the file types used popularly.

(a) MP3

The MP3 is the audio file of the MPEG format. MP3 is popular format for music and songs that are of good quality. Files can be saved with .mp3 extension and are played in most of the browsers. MIME type for MP3 is given by audio/mpeg.

(b) WAVE

Microsoft and IBM developed the WAVE format for audio files that can be played in most of the browsers. These wave files can be saved with the .wav file extension. The quality of wave is better compared to MP3. The file size of wave is more compared to MP3 file. The MIME type for wave file is given by audio/x – wav.

The following are the different formats used on the web that require plug-ins.

(i) Windows Media Audio

It meeds the plug-in of windows media.

(ii) Quicktime

It needs the plug-in of Quicktime.

(iii) Real Audio

It needs the plug-in of Real Media.

Adding Audio using Default Media Players

An extra plugin is not required to download in some browser for playing on MP3 or WAVE files.

An <object> element is used to add the audio file in the page. The type of plug in for playing in audio is determined automatically by the browser and its value is specified by type attribute.

Program

```
<html>
<head>
 <title>Adding Audio using Default Media Players</title>
</head>
<body>
<h1>Adding Audio using Default Media Players</h1>
<object width="250" height="100" type="audio/mpeg">
<param name="src" value="Sleep Away.mp3" />
<param name="autoplay" value="true" />
<param name="autostart" value="1" />
<embed src="Sleep Aay.mp3" width="250" height="100" ></embed>
</object>
</body>
</html>
```

Output



Q44. How MP3s are added to web page using flash?

Answer :

Adding MP3s to Web-Pages Using Flash

Flash is used to add the MP3s to the web pages. They are used to create playlists that contain multiple tracks. The code for an MP3 player can be written from scratch using flash authoring tool and the code will be available in the following sites.

(i) http://musicplayer.sourceforge.net/
(ii) http://www.wimpyplayer.com/
(iii) http://code.google.com/p/mp3player/
(iv) http://developer.yahoo.com/mediaplayer/

Example

```
<html>
<head>
<title>Adding an MP3 player using Flash</title>
</head>
<body>
<h2>Flash MP3 Player Using &lt;object&gt; element</h2>
<object type="application/x-shockwave-flash" width="400" height="200"
data="flash/xspf_player.swf?playlist_url=flash/playlist.xspf">
<param name="movie" value="flash/xspf_player.swf?playlist_url=flash/playlist.xspf" />
</object>
</body>
</html>
```

Output



Q45. How the audio or video is played automatically when an page is loaded?

Answer :

There are several techniques that are used to add the audio or video to the web page. Consider the following issues while playing audio or video automatically when a page is loaded.

(i) Users expect to watch or listen to the file when a page is loaded. Some users will skip the audio or video file when the page is loaded and some of the users watch it.

(ii) A pause or clear or mute button must be provided for an audio or video so that user can turn off the sound when a page is loaded.

(iii) The options should be provided to avoid the repetitions of the track i.e. audio or video.

(iv) Some of the audio and video files are charged for the bandwidth of the files if they are hosted on the server. Whenever a file starts playing automatically when an page is loaded then the user has to pay for watching or listening the file.

Q46. What are the attributes of an <object> element?

Answer :

<Object> Element

HTML uses an object element to insert the multimedia objects directly into the page. This object tag replaces the <img> tag using the object element is a good approach, because it allows to embed any type of data.

Commonly used HTML tags, devoted for inserting objects on webpages are listed below.

<object classid = "URL" data = "URL" [codebase = "URL] type = "string"

[standby = "string"] height = "n" width = "n" [title = "string"] > ----- < /object>

The attributes of an <object> element are as follows,

1. archive Attributes

Archive attribute is used specifically by Java based applications. Its value is the list of URLS separated by spaces from the resources.

2. border Attribute

Border attribute creates a border around an object. Its value is provided in the form of pixels.

3. class id Attribute

This attribute is mandatory for every object to specify its URL of the object. If classid specifies only a filename, then assume that the object is present in the same directory as the HTML page.

Example

class id = "clsid1:02CG36E6 – 9D28 – 2D41 – DE91 – B2398 BCDDB9C"

4. codebase Attribute

This attribute specifies the URL of identify the directory in which the object is present. Codebase attribute is placed in square brackets which means it is optional. If this attribute is not specified, the full URL of the classid is placed.

Example

codebase = "http://www.yahoo.org/java/"

5. codetype Atrribute

Codetype attribute specifies the MIME type of an application. These are used to skip unsupported media types in the browsers.

Example

<object data = "details.xls" code type = "application/vnd.ms-excel">

6. declare Attribute

declare attribute is used for declaring an object that is not instantiated.

Example

declare = "declare"

7. data Attribute

data attribute specifies the URL of the resource to be used by the object.

Example

data = "http://www.yahoo.com/mp4/song.mp4"

8. height and width Attributes

height and width attribute specifies the height and width of an object respectively. The values for these attributes are provided in pixels or percentages.

9. hspace and vspace Attributes

hspace attributes defines the whitespaces on left side and right side of an object. The vspace attribute defines the whitespaces on the objects top and bottom.

10. name Attribute

name attribute specifies the name for referring an object in scripts. This attribute is replaced with an id attribute in xhtml.

11. standby Attribute

This attribute displays alternative text while the object is loading from server.

Example

standby = "The page is loading"

12. tabindex Attribute

tabindex attribute specifies the tabindex of an object in a page.

13. usemap Attribute

usemap attribute specifies the object in an image map with hyperlinks. The value used with an object is map file.

Q47. What is <param> element? Explain the attributes of it.

Answer :

<param> element

<param> element is used for passing parameters to an object.

This tag is used whenever an object requires command-line parameters. Syntax of the param tag is

<param name = "string" value = "string" type = "string"

valuetype = ["ref"/"object"/"data"] >

The attributes of an <param> element are as follows,

1. name Attribute

It specifies a name for every parameter.

Example

<param name = "src" value = "show.mov"/>

<param name = "autoplay" value = "true"/>

2. value Attribute

This attribute specifies the value that is to be is passed to the object. The value passed need not be numerical or textual but can be any of other type.

Example

<param name = "src" value = "onlinesite.com"/>

<param name = "autoplay" value = "true"/>

3. valuetype Attribute

This attribute specifies the format of the parameter of the browser. It may be original data or the URL (ref) or another object (object).

The values that are possible for valuetype attribute are as follows.

(i) data

This parameter value is a string and used as default.

(ii) ref

This parameter value is a URL.

(iii) Object

This parameter value is an other object.

4. type Attribute

This attribute specifies the MIME type of the object.

Example

<param name = "src" value = "show.mid" valuetype = "ref" type = "audio/mid">

1.4 TABLES

1.4.1 Introducing Tables

Q48. Explain how basic table is created in xhtml.

Answer : Model Paper-I, Q9(b)

In HTML, data can also be represented in table format. In this approach a data is usually organized in terms of rows and columns. The tags and attributes that helps in creating HTML table are as follows,

1. <TABLE>...</TABLE>

Entire code corresponding to a given table should be inscribed within these tags.

Relative Attributes

(i) ALIGN

This attribute allows a user to direct the placement of a given table in the web browser window. Here alignment is usually performed horizontally by assigning values LEFT/RIGHT/CENTER respectively.

(ii) BACKGROUND

This attribute enhances the visualization of the table by setting an image or other graphics as background. It usually carries a URL path corresponding to the given image or graphics.

(iii) BGCOLOR

This attributes sets the colour of the window in which the table is currently being displayed.

(iv) BORDERCOLOR

This attribute specifies the colour for the given table border.

(v) BORDER

This attribute sets the width of the table border. Hence a value '0' provided to it, displays a table with no border.

(vi) COLS

This attribute specifies the number of columns that the given table should contain.

(vii) HEIGHT

This attribute specifies the height of the given table.

(viii) WIDTH

This attribute specifies the width of a given table.

2. <TR>...</TR>

It corresponds to a single row of a given table. Data supplied within these tags is usually displayed in a particular row of that table. These tags may have <TH> and <TD> tags within them.

Relative Attributes

(i) ALIGN

It directs the alignment of the text in a given row of a table. It's value can be LEFT/CENTER/RIGHT.

(ii) BGCOLOR

It is used to set the background colour of the blocks of a table.

(iii) BORDERCOLOR

It sets the border colour for a given row of the table.

(iv) VALIGN

It directs vertical placement of data corresponding to a given row of the table. It can have values as TOP/BOTTOM/MIDDLE.

3. <TH>...</TH>

This tag is specifically used to set the headings of the available columns. As the data is inserted into the table in row-wise format, hence <TH> tag usually is followed by <TR> tag. Major difference between <TH> and <TD> tags is that, <TH> makes the data bold and places it at the centre of the table's block. On the other hand <TD> inserts data into the block (of table) in normal font and its positioning depends on its attributes.

Relative Attributes

(i) ALIGN

It directs alignment of data in the header column of the table. Its value can be LEFT/RIGHT/CENTER.

(ii) BGCOLOR

It is used to set the back ground colour for the table headers.

(iii) BORDERCOLOR

It is used to set the border color for the corresponding table headers.

(iv) VALIGN

It directs vertical alignment of data in the table header row.

(v) WIDTH

It sets the width of the table header.

4. <TD>...</TD>

It inserts data into each block of the table. Its supported attributes are **ALIGN/BACKGROUND/BGCOLOR/BORDERCOLOR/HEIGHT/SCOPE/VALIGN/WIDTH** respectively.

5. <CAPTION>...</CAPTION>

It specifies title to the table. This tag can be placed at the bottom or above the table. Its relative attributes can be **ALIGN/VALIGN** respectively.

6. <THEAD>...</THEAD>

It groups certain headers and provides them with single name. This can be accomplished by means of <THEAD> tag. Its relative attributes are **ALIGN/VALIGN/BGCOLOR** respectively.

7. <TBODY>...</TBODY>

It is same as <THEAD>. The only difference here is, it groups the specified rows to frame the table body. Its relative attributes are **ALIGN/VALIGN/BGCOLOR** respectively.

8. <TFOOT>...</TFOOT>

It specifies the table footer by combining its columns. Its relative attributes are **ALIGN/ BGCOLOR/VALIGN**.

Program

```
<HTML>
    <HEAD>
        <TITLE>Table displaying the details of various computer courses. </TITLE>
    </HEAD>
        <BODY>
        <CENTER>
        <FONT SIZE=7 COLOR=BLUE><H1>KNZ INSTITUTE OF COMPUTER SCIENCES
        </H1></FONT>
        <FONT SIZE=7 COLOR=RED><P><P><H2>MALAKPET HYDERABAD</H2></P></P></FONT>
        <TABLE BORDER=3 ALIGN="CENTER" BGCOLOR="CYAN">
        <CAPTION><H3>COURSE DETAILS</H3></CAPTION>
    <TR>
        <TH>COURSES</TH>
        <TH>DURATION</TH>
        <TH>FACULTY</TH>
        <TH>FEES</TH>
    </TR>
    <TR>
        <TD>C++</TD>
        <TD>45 Days</TD>
        <TD>Mr.NADEEM</TD>
        <TD> Rs:1200</TD>
    </TR>
    <TR>
        <TD>JAVA AND J2EE</TD>
        <TD>3 MONTHS</TD>
        <TD>Mr.KHAJA NAZIMUDDIN</TD>
        <TD>Rs:8000</TD>
    </TR>
     <TR>
        <TD>C PROGRAMMING</TD>
        <TD>45 Days</TD>
        <TD>Mr.JANI</TD>
        <TD>Rs:1200</TD>
    </TR>
    </TABLE>
    </BODY>
</HTML>
```

Output



1.4.2 Grouping Section of a Table

Q49. What are the techniques used to group cells, rows and columns of a table? Explain.

Answer :

The techniques used to group cells, rows and columns of a table are as follows.

1. Spanning Columns Using the Colspan Attribute

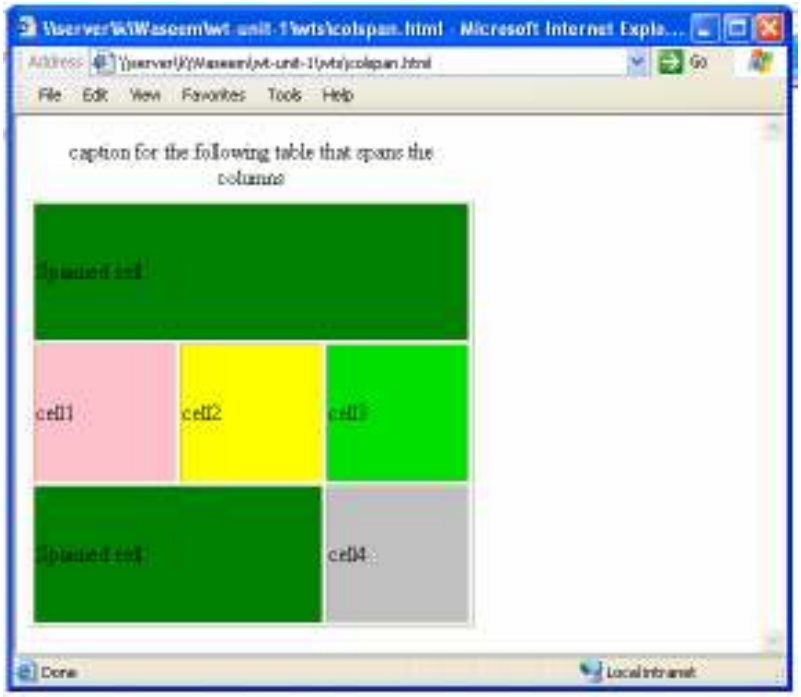
The colspan attribute can be used to stretch a cell horizontally across more than one column. The following example shows the effects of colspan and <caption> elements on the table.

Example

Colspan.html

```
<html>
<head></head>
<body>
<table border = "1">
<caption>caption for the following table that spans the columns</caption>
<tr>
<td colspan = "3" bgcolor = "pink" height = "100"> Spanned cell</td>
</tr>
<tr>
<td bgcolor = "pink" width = "100" height = "100">cell1</td>
<td bgcolor = "yellow" width = "100" height = "100">cell2</td>
<td bgcolor = "grey" width = "100" height = "100">cell3</td>
</tr>
<tr>
<td bgcolor = "green" colspan = "2">Spanned cell</td>
<td bgcolor = "silver" width = "100" height = "100">cell4</td>
</tr>
</body>
</table>
</html>
```

Output



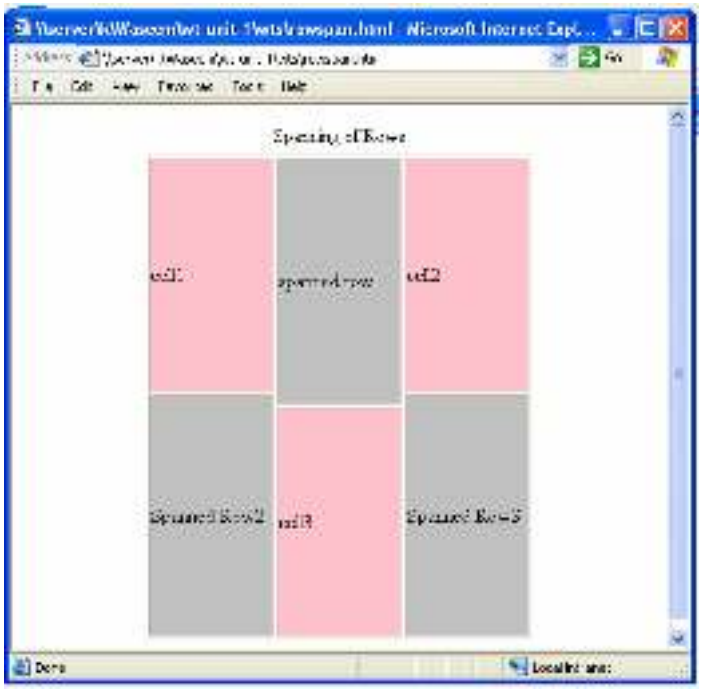
2. Spanning Rows Using the Rowspan Tag

The rowspan attribute can be used to stretch a cell vertically through multiple rows. The following example shows the effect of rowspan attribute on the table.

Example

rowspan.html

```
<html>
<head></head>
<body>
<center>
<table border = "1">
<caption>Spanning of Rows</caption>
<tr>
<td bgcolor = "pink" width = "100" height = "100">cell1</td>
<td rowspan = "2" bgcolor = "silver" width = "100" height = "200">Spanned Row1</td>
<td bgcolor = "pink" width = "100" height = "100">cell2</td>
</tr>
<tr>
<td rowspan = "2" bgcolor = "silver" width = "100" height = "200">Spanned Row2</td>
<td rowspan = "2" bgcolor = "silver" width = "100" height = "200">Spanned Row3</td>
</tr>
<tr>
<td bgcolor = "pink" width = "100" height = "100">cell3</td>
</tr>
</table>
</center>
<body>
</html>
```



3. Splitting up Tables Using a Head, Body and Foot

The following three elements are used in order to define the header, body and foot of a table,

- ❖ <thead>
- ❖ <tbody>
- ❖ <tfoot>

(a) <thead>: The <thead> element defines the head of a table.

(b) <tbody>: The <tbody> element defines the body of a table.

(c) <tfoot>: The <tfoot> element defines the foot of a table. It should always appear before the <tbody> element.

The above elements can be used to provide rich formatting to tables. These elements are not supported by majority of the browsers.

4. Grouping Columns Usign the <colgroup> Element

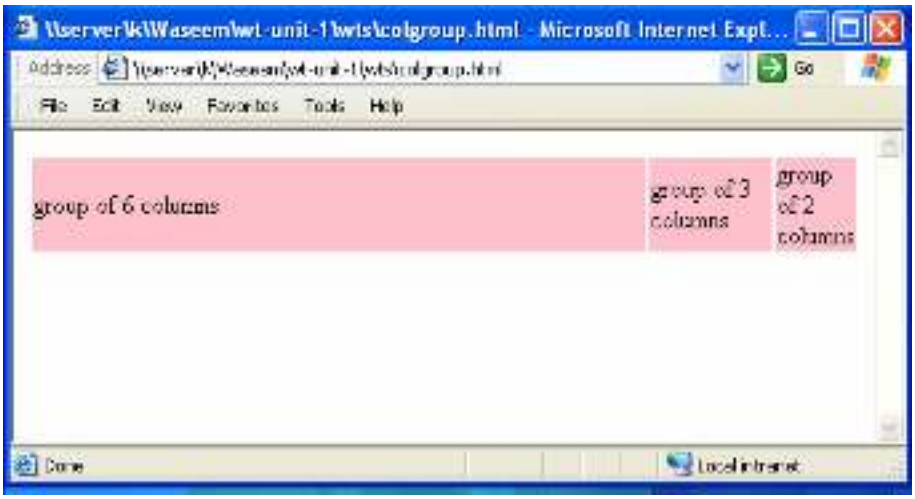
The <colgroup> element can be used to group multiple columns together. This allows us to provide the formatting to the group of columns. The <colgroup> element should be used after the <table> element.

Example

colgroup.html

```
<html>
<head></head>
<body>
<table>
<colgroup span = “6” bgcolor = “pink” width = “60%” height = “500”/>
<colgroup span = “3” bgcolor = “pink” width = “30%” height = “500”/>
<colgroup span = “2” bgcolor = “pink” width = “20%” height = “500”/>
<tr>
<td>group of 6 columns</td>
<td>group of 3 columns</td>
<td>group of 2 columns</td>
</table>
</body>
</html>
```

Output



5. Columns Sharing Styles Using the <col> Element

The col element is used to format or group the table columns in the colgroup element.

colgroup

The colgroup element is used to group the columns of a table and add semantic context to the column data.

Example

```
<html>
<body>
<table>
<caption>Student Information </caption>
<col Span = "1">
<colgroup>
<col span = "1">
<col span = "1">
</colgroup>
<thead>
<tr>
<th scope = "col">Student_name </th>
<th scope = "col"> Roll.no </th>
<th scope = "column"> location </th>
</tr>
</thead>
<tbody>
<tr>
<td> nymisha </td>
<td> 101 </td>
<td> Hyderabad </td>
</tr>
<tr>
<td> Satya </td>
<td> 102 </td>
<td> Vizag</td>
</tr>
</tbody>
```

```
<tfoot>
<tr>
<td> Copyritht@</td>
<td>Inc. of all taxes </td>
</tr>
</tfoot>
</table>
</body>
</html>
```

Output



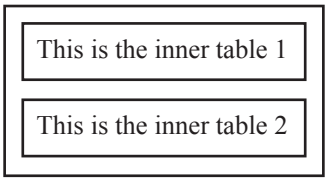
1.4.3 Nested Tables

Q50. Write about Nested tables.

Answer :

Nested Tables

The table that is embedded inside another table is known as nested table.




```
<HTML>
<HEAD>
<BODY>
<H1 align="center"> Displaying Marks of the student </H1>
<TABLE BORDER = 3 align="center">
<TR>
<TH> NAME </TH>
<TH> ROLL NO</TH>
<TH> MARKS </TH>
</TR>
<TR>
<TD> XYZ </TD>
<TD> 12 </TD >
<TD>
<TABLE BORDER = 3 ALIGN = "CENTER" >
<TR>
```

```
<TH> TEL </TH>
<TH> HIN </TH>
<TH> ENG </TH>
<TH>MATH </TH>
<TH> SCI </TH>
<TH> SOC</TH>
</TR>
<TR>
<TD> 89 </TD>
<TD> 95 </TD>
<TD> 91 </TD>
<TD> 81 </TD>
<TD> 85 </TD>
<TD> 99 </TD>
</TR>
</TABLE>
</TD>
</TR>
</TABLE>
</BODY>
</HTML>
```



1.4.4 Accessing Tables

Q51. How tables are accessible?

Answer :

Tables have huge data and they are scanned through rows, columns to find or compare a value. The content of the table is not read to find out about situation or the work. The tables are difficult to the users who use voice browsers or a screen reader to listen pages.

The following are the things to be remembered to understand the table.

1. Caption is added to the table using <caption> element. It is easily understood by a user who read a table using screen reader.
2. <th> element must be used to specify a table heading.
3. Headings must be provided in first row and first column.
4. Nested tables should be avoided as user will find difficult to understand when using screen reader.
5. A user with a screen reader finds it difficult to use rowspan and colspan attributes. So it is better to avoid them.
6. The usage of voice browser or screen reader should be understood properly by a user to read a table.
7. By using scope and headers attributes, the users of screen readers can understand easily and retrieve the headings for a specific cell.

Q52. How Tables can be linearized?

Answer :

Tables can be linearized by using a screen reader for reading a table. Linearization is the process of reading a table by a reader starting from the first row of a table and the cells are read from left to right in a row. Each and every cell in a row is read.

Example

```
<html>
<body>
<table border = "1">
<tr>
<td> Column 1, Row 1 </td>
<td> Column 2, Row 1 </td>
</tr>
<tr>
<td> Column 1, Row 2 </td>
<td> Column 2, Row 2</td>
</tr>
</table>
</body>
</html>
```

Output



Q53. How Id, scope and headers attributes are used to make a table more accessible?

Answer :

Id, scope and headers attributes are used for storing the table structure and make it to be more accessible.

When a cell is made as the heading by adding the scope attribute to <th> element, then this will specify to know the cells under that heading. If the value is given as row then it specifies that the given element/cell is the header for that row. If the value is given as column then it specifies that given element is the header for that column. If the value is given as rowgroup then it specifies that cell is the header for that rowgroup. If the value is given as colgroup then it specifies that cell is the header for colgroup.

Headers attribute is used on <td> element to specify header for respective cells. Its value is the space-separated list of header cell's attribute value.

Example

```
headers = "income p1"
```

In the above example, the headers have the id attributes with the values income and P1.

Header attribute is used for voice browsers. The record of the table is difficult when a table is read so this is used to keep the record about the rows and columns of the table and cell to which row and column belongs to.

1.5 FORMS

1.5.1 Introducing Forms

Q54. Write about forms. Also, explain about different types of form controls.

Answer :

Forms

Form is a web page that contains elements like text fields, check boxes, buttons etc which are used to takes input from the user. These web pages look like normal white sheet of paper consisting of significant spaces information to be entered by users. The forms can be used for different purposes such as online registrations, order entry, subscription etc.

Form Controls

The following are different types of form controls.

1. **Text Input**
   It is a simple text box where a text is entered to search.
2. **Submit Button**
   It is a button that submits the form when clicked on it.
3. **Select Boxes**
   It is a drop-down box containing multiple options.
4. **Radio Buttons**
   Radio buttons have two options (Yes or No). Only one option is selected when a group of radio buttons are available.
5. **Check Boxes**
   It is a small box that can be checked and used during selection.
6. **Text Area**
   It is a large box that can contain text upto a paragraph.

1.5.2 Form Controls

Q55. What are different types of Form controls in xhtml?

Answer : **Model Paper-II, Q9(b)**

The different types of form controls in xhtml are as follows,

1. **Text Input**
   For answer refer Unit-I, Page No. 56, Q.No. 56.
2. **Buttons**
   For answer refer Unit-I, Page No. 58, Q.No. 57.
3. **Check Boxes**
   For answer refer Unit-I, Page No. 61, Q.No. 58(i).
4. **Radio Buttons**
   For answer refer Unit-I, Page No. 62, Q.No. 58(ii).
5. **Select Boxes**
   For answer refer Unit-I, Page No. 63, Q.No. 59.
6. **File Select Boxes**
   For answer refer Unit-I, Page No. 67, Q.No. 62(i).
7. **Hidden Controls**
   For answer refer Unit-I, Page No. 67, Q.No. 62(ii).

Q56. Explain about text input controls.

Answer :

Text Input

It is a simple text box in which certain text is entered.

The following are three of the text input control.

1. **Single Line Text Input Control**

Text elements are data entry fields used in HTML forms. Text fields accept a single line of text entry.

Properties

NAME, VALUE

Syntax

<INPUT TYPE = “text” NAME = “name of the text field” VALUE = “default value”>

Example

<INPUT TYPE = “text” NAME = “t1” VALUE = “Vaseem”>

(double quotes for optional values)

This places a text field (i.e., a single line text edit area) within an HTML form which can be referred by the name t1. The text field displays a default value of Vaseem that can be changed by the user later.

Attributes

The following are the attributes of <input element> used for text input control.

(i) type Attribute

Type attribute in the text input control is used to indicate the type. The value for type attribute must be text.

(ii) name Attribute

name attribute in the text input control is used to provide a name along with the value to forward to the server.

(iii) value Attribute

value attribute in the text input control is used to assign a intial value to the text input field which is displayed when a page is loaded.

(iv) size Attribute

size attribute in the text input control is used for specifying the width of the text input field. This is measured in terms of characters.

(v) max Length

max length attribute in the text input control is used for specifying the maximum value of an entered text by a user.

Example

```
<html>
<body>
<form action= ““method=”get”>
First name:<input type=”text” name=”first_name” />
<input type= “submit” value= “submit” />
</form>
</body>
</html>
```

Output



2. Password-Input Control

For answer refer Unit-I, Page No. 3, Q.No. 2, Topic: The PASSWORD Element.

Example

```
<html>
<body>
<form action="" method="get">
Login : <input type="text" name="login" />
<br>
Password: <input type="Password" name="password" />
<input type="submit" value="submit" />
</form>
</body>
</html>
```

Output



3. Multiple-line Text Input Control

Multiple line text input control is created by using <textarea> element for creating multiple lines of text.

The TEXTAREA Element

The TEXTAREA form element provides a way to create a customized and multiple line text entry form element that can be placed in an HTML form.

Syntax

```
<TEXTAREA  NAME = "text area name"  ROWS= "no of rows" COLS = "no of columns">
some text
</TEXTAREA>
```

Attributes

The following are the attributes of an <textarea> element.

(a) name Attribute

name attribute is used to specify the name of a control that is transmitted to the server.

(b) rows Attribute

rows attribute is used for specifying the number of rows of text present in the <textarea> element.

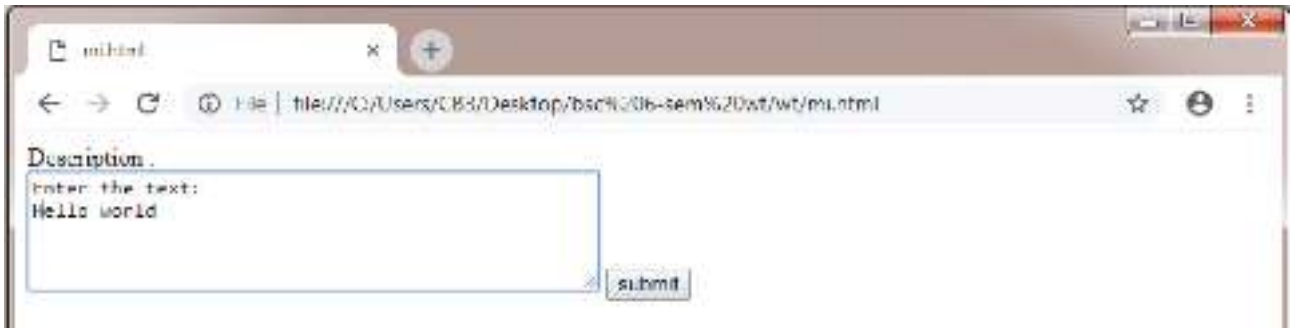
(c) cols Attribute

cols attribute is used for specifying the size and number of columns of text present in the <textarea> element.

Example

```
<html>
<body>
<form action="" method="get">
Description : <br />
<textarea rows="5" cols="50" name="description">
Enter the text:
</textarea>
<input type="submit" value="submit"/>
</form>
</body>
</html>
```

Output



Q57. Discuss in detail about Buttons.

Answer :

Buttons

Buttons are form elements that are used for various purposes such as submitting a form, clearing or resetting a form and triggering the client-side scripts. Buttons are created using the following three ways.

1. An <input> Element with a Type Attribute Whose Values are Submit, Reset or Button

An <input> element is used for creating a button and its type is specified by a type attribute. Type attribute has three different values such as,

(i) Submit

It is a button that submits the form when clicked on it.

(ii) reset

It is a button that resets the form into its original state.

(iii) button

It is a button that triggers a client-side script whenever a user clicks on it.

Attributes

The following are the attributes used by buttons.

(i) Type

It is used to specify the type of button a user creates and the values for this type attribute can be submit, reset button.

(ii) name

It is used to provide a name for the button.

(iii) Value

It is used for specifying the text value for a button.

(iv) onclick

It is used for triggering a script whenever a user clicks on this button.

Example

```
<html>
<body>
<form action="" method="get">
Username: <input type="text" name="Username"><br>
Password: <input type="Password" name="Password" maxlength="10"><br>
<input type="reset" value="Reset">
<input type="submit" value="Submit">
<input type="button" value="Click" onclick="msg()">
</form>
<script>
function msg()
{
alert("Hello!");
}
</script>
</body>
</html>
```

Output



2. An <input> Element with a Type Attribute Whose Value is Image

A button is created with the value image for the type attribute as shown below,

<input type = "image" src = "submit.jpg" alt = "submit" name = "button4"/>

Attributes

The following are the attributes used for an image button.

(i) src

It is used to specify the source path of an image file.

(ii) alt

It is used to provide an alternative text for an image.

Example

```
<html>
<body>
<form method="get" action="">
<input type="image" src="D:image.jpg" alt="Submit" width="100" height="150">
</form>
</body>
</html>
```

Output



<button> Element

Element is used for specifying the text provided between opening and closing tag of <button>.

Example

```
<!DOCTYPE html>
<html>
<body>
<p>button element</p>
<form>
<input type="button" value="Click" onclick="msg()">
</form>
<script>
function msg() {
alert("welcome!");
}
</script>
</body>
</html>
```

Output



Q58. Write about the following,

(i) Checkboxes

(ii) Radiobuttons.

Answer :

(i) Checkboxes

It is a small box that can be checked and used for multiple selection/multiple input.

When the user is left with two multiple options only (like Yes or No), we prefer to use check boxes. Checkbox behaves as a toggle switch (toggles between two options). This means a check box can be in either one of the multiple states, either checked(true state) or unchecked(false state). A checkbox returns either single or multiple value to the Web server, either 'T' or 'F' (or '1' or '0') depending on the checkbox whether checked or unchecked. Basing on the value returned from the HTML form, a Web server script does further processing. Using checkboxes the user can give multiple input to the server.

Properties

NAME, VALUE, CHECKED

Syntax

< INPUT TYPE = "checkbox" NAME = "name of the checkbox" VALUE = "a string that indicates what for the checkbox is meant" CHECKED(optional)>

Example

<INPUT TYPE = "checkbox" NAME = "box1" VALUE = "Male" CHECKED>

This places a checkbox on the HTML form, which can be referenced by the name box1. The VALUE attribute assigns a meaning to the check box. This is the value that is returned if the check box is 'checked'.

Attributes

The following are the attributes used for checkboxes,

(a) type

It is used to specify the checkbox.

(b) name

It is used to provide the name for checkbox.

(c) value

It is used to specify the value that is sent to server when a checkbox is selected.

(d) checked

It is used to select a checkbox when a page is loaded.

Example

```
<html>
<body>
<form action = "" method = "get" >
Select any option:
<br/>
<input type = "checkbox" name = "skill1" value = "html"/> html <br/>
<input type = "checkbox" name = "skill2" value = "xhtml"/> xhtml <br/>
<input type = "checkbox" name = "skill3" value "css"/>css <br/>
<input type="submit" value="Submit">
</form>
</body>
</html>
```

Output



(ii) Radio Buttons

The radio button element has two states and can toggle between them. When several radio buttons are combined into a group, only one radio button can be selected at any given time. Providing the same name to all the radio buttons will place them in the same radio group.

The occurrence of the CHECKED attribute in the INPUT element will preselect the radio button. Only one radio button can be selected at any time in a group.

Syntax

<INPUT TYPE = "radio" NAME = "RadioGroupName" VALUE = "What for the radio button is meant">

Attributes

The following are the attributes for an <input> element that contains the value radio for type attribute.

(a) type

It is used to specify radio button.

(b) name

It is used to specify the name for a radiobutton.

(c) value

It is used to specify the value that is sent to the server when this option is selected.

(d) checked

It is used to specify that this option must be choosen by default whenever a page is loaded.

(e) size

It is used to specify the size of an radio button in terms of pixel.

Example

```
<html>
<body>
<form action="">
<input type="radio" name="language" value="css"> CSS<br>
<input type="radio" name="language" value="xhtml"> XHTML<br>
<input type="radio" name="language" value="javascript"> JAVASCRIPT<br>
<input type="submit" value="Submit">
</form>
</body>
</html>
```

Output



Q59. Write about select boxes.

Answer :

Select Boxes

It is a drop-down select box in which only one item is selected from a list of options. A drop down select box contains <select> element and <option> element.

The following are the elements of select box.,

(a) <select> element

<select> element contains the following attributes,

(i) name

It is used to specify the name.

(ii) size

It is used to present a scrolling list box.

(iii) multiple

It is used to select multiple items from the drop down menu.

<select> element should contain at least one <option> element.

(b) <option> element

Atleast one <option> should be used in <select> element. The following are the attributes of <option> element.

(i) value

It is used to specify the value that is sent to the server whenever this option is choosen.

(ii) selected

It is used to specify the option that must be selected initially whenever a page is loaded.

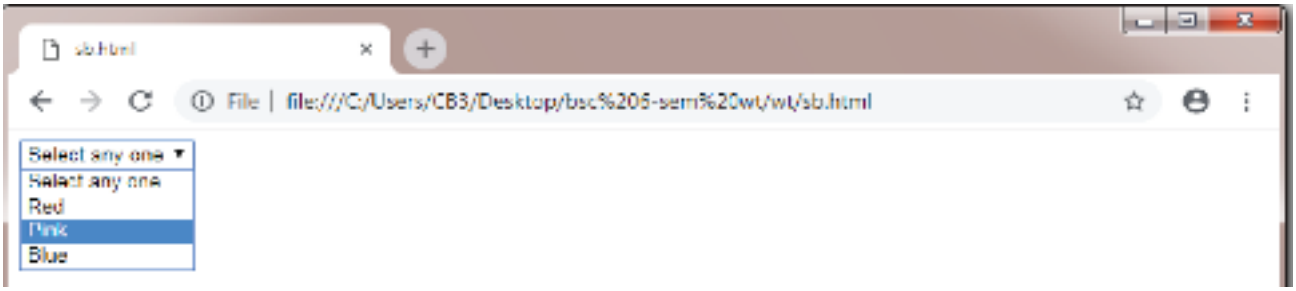
(iii) lable

It is used to label an option.

Example

```
<html>
<body>
<form action="" >
<select name = "colors">
<option selected = "selected" value = " "> Select any one </option>
<option value = "red"> Red </option>
<option value = "pink"> Pink </option>
<option value = "blue"> Blue</option>
</select>
</form>
</body>
</html>
```

Output



Q60. How scrolling select boxes are created?

Answer :

Scrolling Select Boxes

Scrolling select boxes are created by using the <select> element and size attribute is added to it. The value of size attribute is the number of options that are displayed only one time.

Example

```
<html>
<body>
<form action = “” name = “Days” method = “get”>
Select any one:
<br/>
<select size = “7”>
<option value = “Mon”> Monday </option>
<option value = “Tue”> Tuesday </option>
<option value = “Wed”> Wednesday </option>
<option value = “Thu”> Thursday </option>
<option value = “Fri”> Friday </option>
<option value = “Sat”> Saturday </option>
<option value = “Sun”> Sunday </option>
</select>
<br/> <br/> <input type = “submit” value = “submit”/>
</form>
</body>
</html>
```

Output



Multiple Options using Multiple Attribute

Multiple attribute is used for selecting more than one item from a selectbox.

Example

```
<html>
<body>
<form action = " " method = "get" name = "Days">
Select more than one : <br/>
<select name = "select day" multiple = multiple">
<option value = "Mon"> Monday </option>
<option value = "Tue"> Tuesday </option>
<option value = "Wed"> Wednesday </option>
<option value = "Thu"> Thursday </option>
<option value = "Fri"> Friday </option>
<option value = "Sat"> Saturday </option>
<option value = "Sun"> Sunday </option>
</select>
<br/> <br/> <input type = "submit" value = "submit">
</form>
</body>
</html>
```

Output



Q61. How the options are grouped in the select box?

Answer :

The options are grouped using <optgroup> element in the selectbox. <optgroup> element has a label attribute and its value is provided as label.

Example

```
<html>
<body>
<form action = "" method = "get" name = "Info">
select any product:<br/>
<select name = "select Info">
<optgroup label = "Hardware">
<option value = "Desktop"> Desktop computers</option>
<option value = "Laptop"> Laptop computers </option>
</optgroup>
<optgroup label = "software">
<option value = "office software"> office software </option>
<option value = "Games"> Games </option>
</optgroup>
</select>
<br/> <br/> <input type = "submit" value = "submit"/> </form>
</body>
</html>
```

Output



<option> element

An <option> element with disabled attribute is used for grouping the options.

Example

```
<html>
<body>
<form action = “” method = “get” name = “Info”>
select any one product:<br/>
<select name = “selectInfor”>
<option disabled = “disabled” value = “ ”> Hardware </option>
<option value = “Desktop”> Desktop computers</option>
<option value = “Laptop”> Laptop computers </option>
<option disabled = “disabled” value = “ ”> Software </option>
<option value = “officesoftware”> Office Software </option>
<option value = “games”> games </option>
</select>
<br/> <input type = “submit” value = “submit”/>
</form>
</body>
</html>
```

Output



Attributes for Select Boxes

The <select> element contains the following attributes name, size, multiple, disabled, tabindex all universal attributes and UI event attributes.

The <option> element contains the following attributes label, disabled, all universal attributes and UI event attributes.

Q62. Write about the following,

(i) File select boxes

(ii) Hidden controls.

Answer :

(i) File Select Boxes

The file select box or the file upload box is used to include a file from user's computer to the website. This can be done by using the value of the "type" attribute (of the <input> element) as "file" when the file select box is used the 'method' attribute of the <form> element must have the value "post".

When the file upload box is used, the other two attributes that must be used are as follows,

(a) The Enctype Attribute

The enctype attribute is used in the <form> tag to send form controls to the server. It takes the value of multipart form-data.

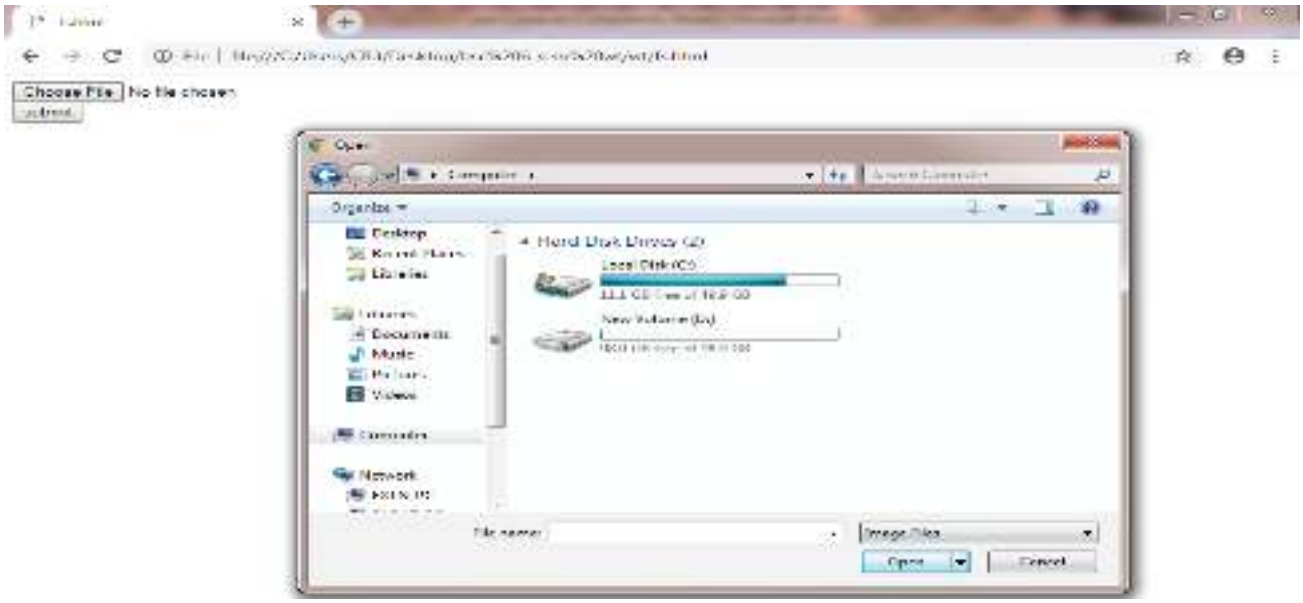
(b) The Accept Attribute

The accept attribute is used in the <form> tag to specify the MIME type of the files that are uploaded through file upload box. Any type of the image can be included along with wildcard character (*).

Example

```
<html>
<body>
<form action = "" method = "post" name = "FromImage" entype = "multipart/form-data">
<input type = "file" name = "file upload" accept = "image/*"/> <br/>
<input type = "submit" value = "submit"/>
</form>
</body>
</html>
```

Output



(ii) Hidden Controls

This tag <INPUT TYPE = HIDDEN>, stores hidden data. The data is not visible unless the user views the Page Source from View menu of the browser.

Syntax

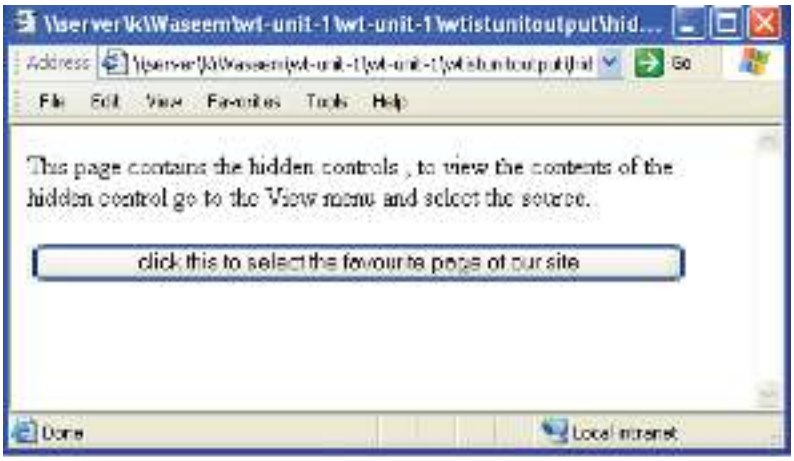
<INPUT TYPE= "HIDDEN" NAME= "stdpassword" value = "MyWord">

The data **MyWord** is forwarded to the Web server along with other data users. The advantage of this tag is, the data MyWord is not displayed in the browser. Through these fields, the programmer or user can pass secret data like standard password or actual sales figures etc., to the Web server. This is because they are not displayed in the browser and they are not known to others.

Example

```
<html>
<head></head>
<body>
<form action = "http://www. siaedugroup.com/ vote.aspx" method = "get" name = "form1">
<p>This page contains the hidden controls, to view the contents of the hidden control go to the View menu and select the source..</p>
<input type = "hidden" name = "hiddencontrol" value = "main page"/>
<input type = "submit" value = "click this to select the favourite page of our site"/>
</form>
</body>
</html>
```

Output



In addition to these controls there is also another control called object control.

Object are embedded in the <object> element within the <form> element of HTML page. For example to use the object for activating graphical inter action and the value must be stored with its object name.

1.5.3 Sending Form Data to the Server

Q63. How the form data is sent to the server?

Answer :

HTTP is used when a browser requests for a webpage and when the server sends this page to the browser. HTTP get and HTTP post are the methods used by the browser to send form data to the server.

(a) HTTP Get

HTTP get method is used to access form data from the server. The form data is appended with the URL written in action attribute of an <form> element by using a question mark in between them. The name/value pairs for each form control is separated using ampersand (&).

Example

<form action = "http://www.google.com/login.aspx" method = "get">

Usename:

<input type = "text" name = "Username" value = " " size = "20" maxlength = "20">

Password:

<input type = "password" name = "password" value = " " size = "10" maxlength = "10">

<input type = "submit"/>

</form>

After clicking on the submit button, the username and password are appended to the URL as shown below,

http://www.google.com/login.aspx?Username = harrys&password = ph123

The above query is known to be query string.

An advantage of sending form data to the server is that it can be bookmarked,

(b) HTTP Post

HTTP Post method is used to send data from a form to the server.

The login form that is sent to the server using post method can be represented in the given HTTP headers.

User_agent : MSIE 7

Content_Type : application/x-www_form_urlencoded

Content_length : 35

..... other headers go here....

Username = harry & password = h123.

1.6 FRAMES

1.6.1 Introducing Frameset, <frame> Element, Creating Links Between Frames

Q64. Discuss about frameset.

Answer : Model Paper-III, Q9(b)

<frameset>

The entire code corresponding to frame(s) is usually embedded between <FRAMESET> ... </FRAMESET> tag. This tag must be included in <HEAD> region of code not in <BODY> region of XHTML code.

Attributes of <frameset>

The <frameset> tag supports the following attributes.

(a) COLS

It displays the total number of columns and the size of frameset. It is measured in pixels.

(b) ROWS

It displays the total number of rows and the size of frameset. It is measured in pixels.

Program

The following code creates a multiple frames on screen. This frames are created using <frameset> tag.

Frame1.HTML

<HTML>

<HEAD>

<TITLE> PROGRAM 1 </TITLE>

<BODY BGCOLOR = "pink">

<CENTER> <STRONG> <H1> This is program1 </H1> </STRONG> </CENTER>

</BODY>

</HTML>

Frame2.HTML

```
<HTML>
<HEAD>
<TITLE> PROGRAM2</TITLE>
</HEAD>
<BODY BGCOLOR = “Cyan”>
<CENTER> <STRONG> <H1> This is progam2 </H1> </STRONG></CENTER>
</BODY>
</HTML>
```

Frame3.HTML

```
<HTML>
<HEAD>
<TITLE> PROGRAM3 </TITLE>
</HEAD>
<BODY BGCOLOR = “Blue”>
<CENTER> <STRONG> <H1> This is program3 </H1> </STRONG> </CENTER>
</BODY>
</HTML>
```

Mainframe.HTML

```
<HTML>
<HEAD>
<TITLE> Program to create multiple frames </TITLE>
<FRAMESET ROWS = “25%, 25%, 25%”>
<FRAME SRC = “Frame1.HTML”>
<FRAME SRC = “Frame2.HTML”>
<FRAME SRC = “Frame3.HTML”>
</FRAMESET>
</HEAD>
</HTML>
```

Output



Q65. Write about <frame> element.

Answer :

<frame> element

The <frame> element should contained within the <frameset>element. It specifies one or more frames within the frameset. It is an empty element. It should always carry an attribute called "src" which indicates the location of the html document to be loaded in the specific frame.

Attributes

(i) src Attribute

It is used to specify the file that must be displayed in the frame.

Example

src = "program.html"

(ii) name Attribute

It is used to provide the name for a frame. This name is used when the links are created in one frame and used as a target in other frame for loading the pages.

Example

name = "program"

(iii) frameborder Attribute

It is used to define the borders around the frames. The values for frameborder attribute is given below.

(a) 1 or yes value

It is the default value that specifies borders that must be shown.

(b) 0 or no value

It is used to specify borders that must not be shown.

(iv) marginwidth and marginheight Attribute

This attribute sets left and right margin size of a frame.

This attribute sets the top and bottom margin of a frame. It is usually measured in pixel.

Example

marginwidth = "15" marginheight = "15"

(v) noresize Attribute

It is used to prevent the resize of a frame. Its value is noresize.

Example

noresize = "noresize"

(vi) scrolling Attribute

The scrolling attribute is used within the <frame> element to provide the scrolling bar if the content of the frame does not fit into the available size of the frame. It can take three possible values they are,

(a) Yes

It specifies that the frame should always have the scroll bar.

(b) No

It specifies that the frame should never have the scroll bar.

(c) Auto

It specifies that the browser can automatically create the scrolling bar if the content of the frame does not fit in the size of the frame.

(vi) longdesc Attribute

This attribute displays the pages that contain long content description

Its value must be URL that points the description.

longdesc = "framedescription.html"

Program

Frame.html

```
<!DOCTYPE html>
<html>
<frameset cols="80%,50%">
<frame src="frame1.html">
<frame frameborder="1">
<frame  marginheight="12" >
<frame  noresize="noresize">
<frame  name="frame1">
<frame  marginwidth="15">
<frame  scrolling="yes">
</frameset>
</html>
```

Frame1.html

```
<HTML>
<HEAD>
<TITLE> PROGRAM 1 </TITLE>
<BODY BGCOLOR = "pink">
<CENTER> <STRONG> <H1> This is program1 </H1> </STRONG> </CENTER>
</BODY>
</HTML>
```

Output



Q66. How the links can be created between frames?

Answer :

The linking between frames is provided in the three conditions,

- ❖ When the main document is lengthy, the navigation bar is used to provide the shortcuts to the parts of the main document.
- ❖ When a user does not want to reload the whole page
- ❖ When the size of the navigation bar is too large, the usage of frames prevents the user to reload the navigation bar each time they view the new page.

The linking between frames can be provided by using the two main attributes they are,

The name attribute and the target attribute.

The name attribute specifies the name of the frame in which the new page can be loaded. The value of the target attribute can be the name of the frame in which the new page is going to be loaded. The difference between name and target attribute is that, the name attribute is used in the <frame> element where as the target attribute is used in the <a> element. The other possible values that can be accepted by the "target" attribute are,

- ❖ **Self**

  It loads the page into the current frame.
- ❖ **Blank**

  It loads the page into the new window.
- ❖ **Parent**

  It loads the page into the parent window i.e., in the case of a single set of frames, the new page will replace all the frames. In case of nested frames, the new page will replace the frame containing the sets of frames.

❖ **Top**

It loads the page into the current window by replacing any current frame.

This concept of frame can be explained with the help of the following program,

Program

frames.html

```
<html>
<frameset cols = 30%, 70%">
<frame src = "links.html"/>
<frame src = "linksinfo.html" name = "main_frame"/>
</frameset>
</html>
```

In the above example of (frames.html), 2 frames are created, one frame loads the page (links.html) where as the other frame initializes it with the page (linksinfo.html)

The code for (linksinfo.html) is as follows,

```
<html>
<head></head>
<body>
<h1>Main page<h1>
<h3>This page provide you the information about books present in the navigation frame. Just click any one them and obtain the relevant information here</h3>
</body>
</html>
```

Output



It is the step to link the above two frames. This can be done as follows,

Consider the code for (links.html).

```
<html>
<head></head>
<body>
<h1>About Books</h1>
<u1>
<li><a href = "b1.html" target = "main_frame">Web Technologies</a></li>
<li><a href = "b2.html" target = "main_frame">J2ME<a></li>
<li><a href = "b3.html" target = "main_frame">Software Engineering<a></li>
</u1>
</body>
</html>
```

In the above example program (links.html), when the link "Web Technologies" is clicked, the contents of the page "b1. html" gets loaded into the target frame named "main_frame". Similarly when the other links are clicked their associated pages "b2.html" and "b3.html" gets loaded into the target-frame. This can be shown as follows,



1.6.2 Setting a Default Target Frame Using <base> Element, Nested Framesets, Inline or Floating Frames with <iframe>

Q67. (a) How <base> element is used to set as a default target frame.

(b) Write about nested framesets.

Answer :

(a) <base> Element

The <base> element is used to set the default target frame in any webpage. The links are provided in other frame. It has target attribute with the value as name for the frame. LinksNav.html is added in the code as a default frame target.

```
<head>
<base target = "main_page"/>
</head>
```

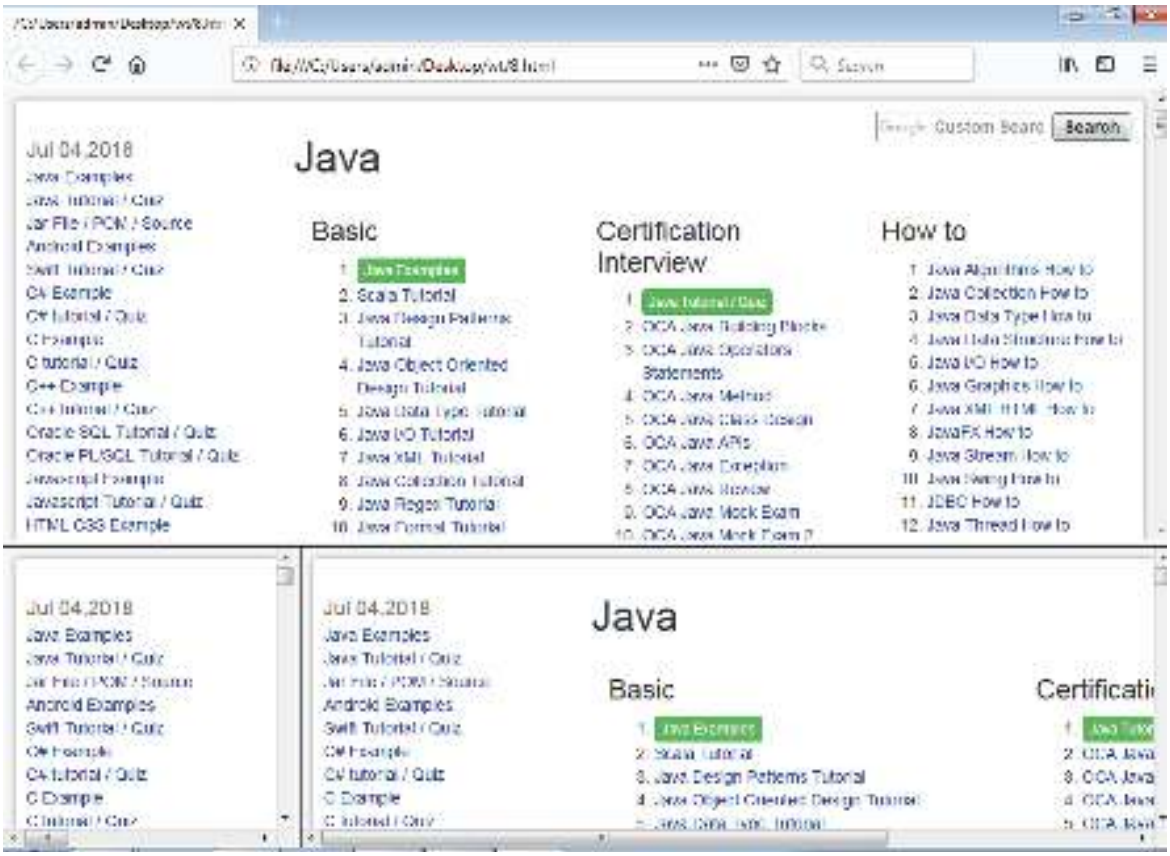
(b) Nested Framesets

A frameset specifies the grid-like structure for displaying rows and columns. A nested frameset is created by a placing the second<frameset> element instead of one of the <frame> element.

Example

```
<!DOCTYPE HTML>
<html>
<frameset rows="60%,40%">
<frame src="http://java2s.com">
<frameset cols="25%,75%">
<frame src="http://java2s.com">
<frame src="http://java2s.com">
</frameset>
</frameset>
</html>
```

Output



Q68. Explain <iframe> element.

Answer :

<iframe> Element

It is also called as inline or floating frames. This tag helps in defining an inline frame. An inline frame embeds one HTML document in another HTML document.

Attributes

The following are the attributes of <iframe> element.

1. src Attribute

It defines a URL path (address) to the iframe.

2. align Attribute

It directs the alignment of <iframe>. Its values are top, bottom, middle, left and right.

The values for the align attribute are as follows,

(a) left: The frame is set at the left margin of the page and text is displayed at the right side.

(b) right: The frame is set at the right margin of the page and text is displayed at the left side.

(c) top: The frame is set at the top and is inlined with the text.

(d) middle: The frame is set at the middle and is inlined with the text.

(e) bottom: The frame is set at the bottom and is inlined with the text.

3. height and width Attributes

(a) height: It defines the height of <iframe>.
Example: height = "120"

(b) width: It defines the width of <iframe>.
Example: width = "400"

4. frameborder Attribute

It sets a border around <iframe>. The value for a frameborder must be number of pixels. The value 0 represents no border.
Example: frameborder = "0"

5. longdesc Attribute

The longdesc attribute is used for specifying indicating a link to another page and a description is given in the form of text. This includes images, charts and graphs in the frame.
Example: longdesc = "../textDescription/iframepgm.html"

6. marginheight and marginwidth Attributes

(a) marginheight Attribute: It sets left and right margin around iframe content. It is used to provide the distance between border and content of the frame interms of pixels.
Example: marginheight = "5"

(b) marginwidth Attribute: It sets left and right margins around iframe content. It provides the distance between border and content of the frame interms of pixels.

7. scrolling Attribute

scrolling attribute is used to specify whether the frame must contain scrollbars.

Example

```
<iframe> element.
<!DOCTYPE html>
<html>
<body>
<h2>iframe element</h2>
<iframe height="300px" width="100%" src="frame1.html" name="iframe element"></iframe>
<p><a href="https://www.google.com/" target="iframe element">Google page</a></p>
</body>
</html>
```

Output



INTERNAL ASSESSMENT

OBJECTIVE TYPE

I. Multiple Choice

1. HTML code should begin with _____ tag and ends with _____ tag. [ ]

   (a) </html>, <html> (b) <html>, </html>

   (c) </html>, </html> (d) <html>, <html>

2. Which of the following is a type of list in HTML? [ ]

   (a) Order list (b) Unordered list

   (c) Nested lists (d) All of the above

3. _____ aligns the text at the centre of the screen. [ ]

   (a) <CENTER> . . . </CENTER> (b) <CENTER> . . . </CENTER>

   (c) </CENTER> . . . <CENTER> (d) </CENTER> . . . </CENTER>

4. _____ inserts a button in the HTML page. [ ]

   (a) <button> (b) Insert button

   (c) ins_button (d) None of the above

5. _____ attribute provides the URL for an image. [ ]

   (a) id (b) href

   (c) src (d) sc

6. _____ attribute is used to specify the number of columns in a givne page. [ ]

   (a) columns (b) clmn

   (c) col (d) cols

7. Which of the following is not a formatting tag? [ ]

   (a)
 (b) <i>

   (c) <p> (d) None of the above

8. _____ tag is used to create a link from one page to another page. [ ]

   (a) Anchor (b) Llink

   (c) Blink (d) Linker

9. The vertical alignment of the data in a table can be done by using _____ attribute. [ ]

   (a) veralign (b) vertical_align

   (c) valign (d) None of the above

10. Tags are classified into _____ types. [ ]

    (a) 3 (b) 2

    (c) 5 (d) None of the above

II. Fill in the Blanks

1. _____ tag is used to add the images to the web page.
2. _____ tag allows the text to be written in paragraph.
3. _____ attribute provide the color to the background of a page.
4. href is used to provide the _____ for a link.
5. The output screen of every HTML page is a _____.
6. A table can be created by using _____ tags.
7. Tables can be linearized by using a _____ for reading a table.
8. In HTML, forms can be created by inserting _____ tags.
9. Break tag in xhtml is given as _____.
10. The process of placing one table inside another table is known as _____.

KEY

I. Multiple Choice

1. (b) 2. (d) 3. (a) 4. (a) 5. (c)
6. (d) 7. (d) 8. (a) 9. (c) 10. (b)

II. Fill in the Blanks

1. <img>
2. <P>
3. bgcolor
4. URL/web address
5. Frame
6. <table> . . . </table>
7. Screen reader
8. <FORM> . . . </FORM>
9.
10. Nested table

III. Very Short Questions and Answers

Q1. Write a short note on HTML.

Answer :

HTML stands for HyperText Markup Language. It is a markup language immensely used in developing web pages. Here the word "markup" in real sense refers to highlighted instance of a given aspect i.e., an author can highlight a particular word of his written document to grab attention of readers. In similar way, the Hyper Text Markup Language highlights different views of a given web page in terms of headers, titles, plain text, images, sounds etc.

Q2. List any five formatting tags.

Answer :

Formatting tags in HTML are as follows,

1. Paragraph Tag <P>
2. Center Tag
3. Break Tag
4. Bold Tag <b>
5. Italic Tag <i>

Q3. Write about the text element in HTML.

Answer :

Text Element

In HTML, Text elements are considered as data entry fields. These fields generally accept single line of text.

Syntax

<INPUT TYPE = "text" Name = "name of text field" VALUE = "default value">

Example

<INPUT TYPE = "text" NAME = "L1" VALUE = "JOHN">

Q4. Define form.

Answer :

Form is a web page which carry elements like text fields, check boxes, buttons etc. These web pages look like normal white sheet of paper consisting of significant spaces for information to be entered by users. Online forms can be used for different purposes. For example it is used, for online registrations, order entry, subscription etc.

Q5. Define list in HTML.

Answer :

"List" refers to a record that gives small amount of information arranged in specific format. Examples in this aspect can be a shopping list, list of names of employees in an organization, list of names of places etc. In HTML, concepts of list is used for displaying information in specific order.

UNIT 2

CASCADING STYLE SHEETS

Marketed by:
SIA GROUP

LEARNING OBJECTIVES

After studying this unit, a student will have thorough knowledge about the following key concepts,

- Cascading Style Sheets
- Various CSS properties
- More Cascading Style Sheets like Links, List, Tables and Outlines
- Pseudo classes like :focus and :active
- Page layout, Designing pages, Coding your design and developing for mobile devices.
- Various design issues.

INTRODUCTION

Cascading Style Sheets (CSS) is one of the most interesting concepts of HTML as it specifies the rules for styling elements of a web document. It not only extends it features in controlling colors and sizes of fonts, but also controls spaces between various elements, the color and width of a given line etc. Thus, CSS are used to determine the style and layout of web pages. There are four different types of column-layouts available for structuring the pages i.e., two-column layout, three-column layout, four-column layout and six-column layout. In CSS syntax value should end with semicolon.

Typography refers to an art and technique used in type design which consists of decisions related to type faces, point size, line length, letter spacing, etc. It means that this keeps a lot of efforts to make the page easy to read, navigate and to provide the messages clearly. Form is a web page which contains elements like text fields, check boxes etc. Usually forms are created by inserting <FORM>---</FORM> tags in the HTML code.

PART-A
SHORT QUESTIONS WITH SOLUTIONS

Q1. What is CSS?

Answer : Model Paper-II, Q3

Cascading Style Sheets (CSS), forms one of the most interesting concepts of HTML as it specifies the rules for organizing elements of a given web document. It not only extends its features in controlling colors and sizes of fonts, but also controls spaces between various elements, the color and width of a given line etc. Thus, CSS are used to determine the style and layout of web pages. Generally, style defines the way in which the HTML elements are shown on the web pages. A style can be used in the following format,

selector{property : value;}

Q2. Write about style rule.

Answer : Model Paper-I, Q3

Style Rule

Style rules specify the appearance of a document. A style rule is composed of two units,

(a) Selector

It represents the elements to which the rule applies. (e.g. h1, h2, p etc.)

(b) Declaration

It represents the properties of an element(i.e., color, font-style, etc) and its value.

Syntax

Selector{Declaration;}

Selector{Property:value;}

Example

h1{Text-align:center;}

Q3. Write a short note on <link> element.

Answer :

The <link> element is an empty element. It is used to link two documents i.e., it is used to link the stylesheets with the document in which it is present. It not only links a stylesheet but also links an RSS feed that correlate with the web page.

The <link> element has 3 attributes when it is used for associating a stylesheet with the document. They are type, rel and href. In addition to these core attributes the <link> element can have other attributes they are, charset, dir, href, hreflang, media, rel, rev, style, target, type.

Q4. List the properties that influence the appearance of the font.

Answer : Model Paper-III, Q3

The properties that directly influence the appearance of the font are as follows,

1. font
2. font-family
3. font-size
4. font-weight
5. font-style
6. font-variant
7. font-stretch
8. font-size-adjust.

Q5. What is a selector? List the types of selectors in CSS.

Answer : Model Paper-I, Q4

Selector

Selector is an element to which the style rule is applied. It provides a way to select the element (s) for applying the style rules. The elements can be selected in several ways by using the different types of selectors.

Types of Selectors in CSS

The following are eight different types of selectors in CSS,

1. Universal selector
2. The Type selector
3. The Class selector
4. The ID selector
5. The child selector
6. The descendent selector
7. The general sibling selector
8. The adjacent sibling selector
9. The attribute selector.

Q6. Write about vertical-align property.

Answer : Model Paper-II, Q4

Vertical-align Property

This property is used to control the vertical position of text or an image.

Syntax

h1 {vertical-align : value;}

It takes different values. They are,

1. baseline
2. sub
3. super
4. middle
5. top
6. text-top
7. bottom
8. text-bottom.

Q7. What are the different properties that are used to position the items on the page.

Answer : Model Paper-III, Q4

The different properties that are used to position the items on the page are as follows,

1. The XHTML Property

It is used to group the items of the XHTML pages to chunks or blocks.

2. The CSS margin Property

It is used to set margin of the element or property.

3. The CSS width property

It is used to set the width of the box.

4. The CSS float Property

It is used to align the elements to the left or right of the page.

Q8. Discuss in brief about tracking.

Answer : Model Paper-I, Q5

Tracking

It is also called as letter spacing. It referred to as the space between the letters. In CSS, the letter-spacing property is used to control the space between the items on the page. The letter spacing is required when dealing with the headings that is in uppercase letters because reading the headings is little bit difficult than reading a sentence.

Example: h1{letterspacing : 0.1 em;}

Q9. Write a short note on,

(i) list-style-position property

(ii) list-style-image property.

Answer : Model Paper-II, Q5

(i) list-style-position Property

The list-style property specifies the bullet points for multiple lines of text by aligning the second line with the start of the first line or aligning it underneath the start of the marker (bullets or numbering). This property has 2 values, they are,

(a) inside : The second line is aligned underneath the start of the marker

(b) outside : The second line is aligned with the start of the first line.

(ii) list-style-image Property

The list-style-image property is used to specify the images as the marker rather than a bullet point or number. It requires the URL of the image as its value.

Q10. List the advantages of fixed-width design and liquid design.

Answer : Model Paper-III, Q5

Advantages of Fixed-width Design

1. The line length of the text can be controlled without bothering about the size of user's window.
2. The page size remains the same.
3. The designer has full control on the position and appearance of items on the page.
4. The width and position of elements can be controlled exactly by pixel values.

Advantages of Liquid Design

1. The page expands and fills the browser window without leaving any gaps around it.
2. The users are allowed to set font size lager than intended because page layout is stretchable.
3. The page can even fit in a small size window, there is no need for scrolling it.

PART-B

ESSAY QUESTIONS WITH SOLUTIONS

2.1 CASCADING STYLE SHEETS

2.1.1 Introducing CSS

Q11. What is CSS? Give an example.

Answer :

Cascading Style Sheet (CSS)

For answer refer Unit-II, Page No. 80, Q.No. 1.

Example

```
<html>
<head>
<title>HTML CSS EXAMPLE</title>
</head>
<body>
<h3 style = "color:green;font-size:24px;font-style:normal;">Hello SIA GROUP!</h3>
</body>
</html>
```

Output



Q12. How style sheets are specified into HTML document?

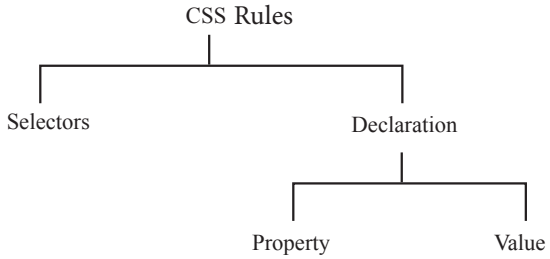
Answer : Model Paper-I, Q10(a)

Rule

A rule is considered as a guide or principle that helps in conducting an action.

CSS Rule

A CSS rule can be associated with the element of the html document. They specify the styles for the content of the elements.



A CSS rule consist of two parts,

1. Selector
2. Declaration.

1. Selector

Selector indicates the elements to be styled. There are three types of selectors namely HTML selector, class selector and ID selector.

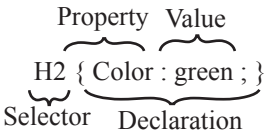
2. Declaration

The declaration indicates the type of the styles that can be applied to an element. The declaration part is again divided into two parts namely a property and value.

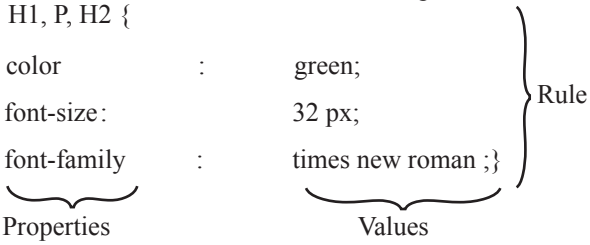
(a) Property: It indicates the property of the attributes of the selected element(s).

(b) Value: Value acts as the specification for the property.

Example



A CSS rule can be applied to more than one element. This can be explained with the help of following example.



The above example shows a rule which is applied to <H1> <P> and <H2> elements of the document. These elements are seperated by comma. The rule also consists of property-value pair in the curly braces{ }. The property and value is separated by a colon(:). Whereas each of the property-value pair separated by semicolon.

An xhtml Code and a CSS Containing Rules

The following is an Xhtml code, to which, the CSS containing different rules are applied. The XHTML code is saved as "fruits.html".

The CSS should be saved with the file extension ·css. Here, the stylesheet that used is "style.css". The stylesheets should be created in the editors such as windows notepad or text editor on the Mac.

fruits.html

```
<html>
<body>
<link rel= "stylesheet" type= "text/css" href= "style.css">
<h1>This is h1 text</h1>
<h2>This is h2 text</h2>
<h3>This is h3 text</h3>
</body>
</html>
```

style.css

```
body {
color: #000000;
background-color: #ffffff;
font-family: arial; }
h1 {
background-color: #efefef ;
border-style: solid;
border-width: 2px;
border-color: #999999;}
h2 {
background-color: #cccccc;
font-weight: bold;
padding: 6px;
}
h3 { padding: 6px;}
```

Output

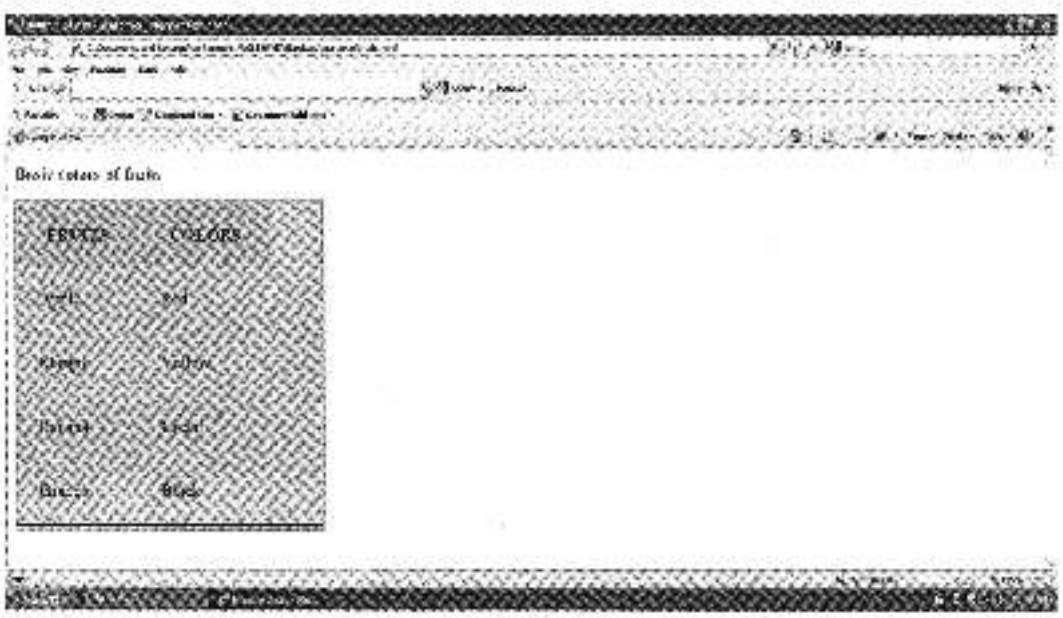


Style.css

The following is the css file for the above xhtml document.

```
/*Rule 1*/
    body {
        color: #000000;
        background-color: #ffffff;
        font-family: arial; }
/*Rule 2*/
     h1 {
        background-color: #efefef ;
        border-style: solid;
        border-width: 2px;
        border-color: #999999;}
/*Rule 3*/
    h2 {
        background-color: #cccccc;
        font-weight: bold;
        padding: 6px;
        }
/*Rule 4*/
    h3 { padding: 6px;}
```

Output



Effect of each of the CSS Rule on the xhtml Document's Elements

The Rule 1 in the styleapply.css is prepared to apply the default color to any text and lines used in the document. The default color of the text and lines of the document is taken as black. The background of the page is specified as white. The type face is arial.

The Rule 2 is used to affect the appearance of the h1. The background-color of the table is specified as light gray and the border's color is specified as dark gray with broadness of 1-pixel.

The Rule 3 affects the appearance of the table h2. The background-color of the table header is taken as medium gray. The distance between the edge of the cell and the text known as padding is taken as 6px. The weight of the font is taken as bold.

The Rule 4 affects the h3 where the distance between the edge of the cell and the text known as padding is taken as 6px.

Q13. Explain the concept of inheritance in CSS with an example program.

Answer :

In cascading style sheets, the element enclosed in another element inherits the styles of its outer element. The outer element is the parent element and the inner element is child element. The child element inherits the styles of parent element. If the inner element, the (child element) defines its own styles(that exist in the parent element), then the parent's styles are overridden.

Cascading means overriding the global style sheet with local style sheet and overriding the local style sheet by a particular element with its own style. Cascading effects allows a chain of precedence in overriding – even a document creator's style by a reader style.

Example

To make use of cascading effect.

File Name

Cascading.html

```
<HEAD>
    <STYLE>
        BODY{color: red; font-size: 15pt;}
    P{color: green;}
    .own{color: blue;}
</STYLE>
</HEAD>
<BODY>
This is in body and is in red color.
<P> This is in paragraph tag and is in green color.
<P> This is in span attribute and is in <SPAN CLASS = own>blue color.<SPAN>
</BODY>
```

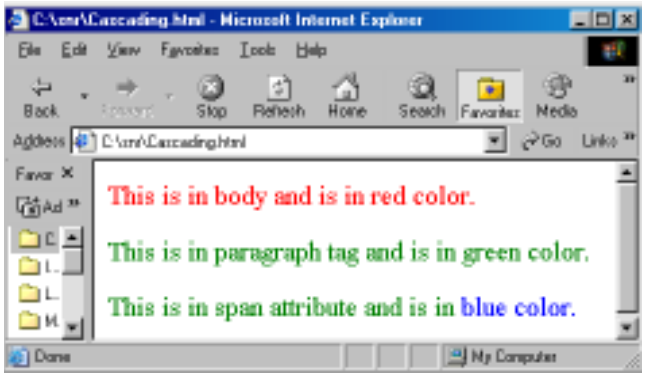


Figure: Output of Cascading.html

In the above program, the body style(red color) is overridden by P element style(green color) and P element style is overridden by SPAN style(blue color).

2.1.2 Where you can Add CSS Rules

Q14. What are the three different ways of adding CSS ruless into documents?

OR

Explain the types of CSS.

Answer :

CSS rules can be added into documents in three ways. They are,

1. External style sheet
2. Internal style sheet
3. Inline style sheet.

1. External Style Sheet

External style sheets allow users to create application with multiple pages. In external style sheets, CSS rule multiple is written in single document and it saved under file name with CSS extension. Hence if any web document wants to be adhered to these rules, then a simple link to filename .css is made using <LINK> tag.

Program

Rule.CSS

```
body{color: BLACK; BACKGROUND-COLOR: lime; FONT-FAMILY:SANS-SERIF; ALIGN: CENTER;}
h1{FONT-SIZE: 20PT ;}
h2{BACKGROUND-COLOR: ORANGE; BORDER-WIDTH: 2PX; BORDER-COLOR: BLACK;}
h3{FONT-WEIGHT: ITALIC; COLOR: BLACK; PADDING: 2PX;}
h4{PADDING: 5PX;}
```

external.html

```
<html>
<head>
<title>External style sheet example</title>
<link rel = “stylesheet” Type=”Text/css” href = “rule.css”/>
</head>
<body>
<h1>This is h1<h1>
<h2>This is h2<h2>
<h3>This is h3<h3>
<h4>This is h4<h4>
</body>
</html>
```

Output



2. Internal Style Sheet

In internal style sheets (also called embedded style sheet) the properties (depending on requirement) are declared in <HEAD> section of HTML code along with their selectors. Whereas the elements are declared in <BODY> region of HTML code along with their selectors. While displaying the output, the browser considers the selectors declared in <BODY> region of code and accordingly searches for their equivalent selector in <HEAD> region and applies the effect as required.

Program

```
<HTML>
<HEAD>
<TITLE> Embedded style sheet example </TITLE>
<style type = “text/CSS”>
H1{Font-family: SANS-SERIF; COLOR:GREEN; TEXT-ALIGN:CENTER; FONT-SIZE: 30PT;}
P{FONT-FAMILY: SANS-SERIF; TEXT-ALIGN: LEFT; TEXT-INDENT: 0.5; FONT-SIZE: 15PT;}
BODY{Background: Yellow; Margin-left: 1.5 In; Margin-right: 1.5 IN;}
</style>
</HEAD>
<BODY>
Welcome to Universal Book Mall!
```

<P> This passage gets effected to CSS properties accertained to P selector in head region</P>

<H1> This passage gets affected to CSS properties accertained to H1 selector in head region</H1>

</BODY>

</HTML>

Output



3. Inline Style Sheet

In inline style sheets, the configuration of current web document (with respect to CSS) is declared in the body region of HTML code and the rules are applied to only single element where given configuration is declared. Moreover, the configuration is rendered by using 'STYLE' attribute as an attribute of selector.

Syntax

<element style = "property : value;"> Text </element>

Example

<html>

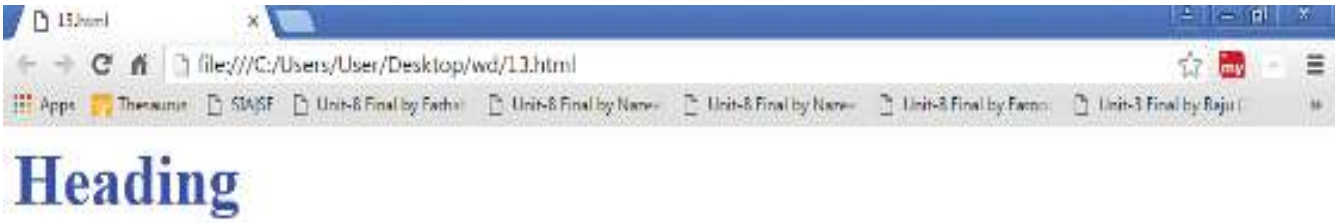
<body>

<h1 style = "color : blue;">Heading</h1>

</body>

</html>

Output



Q15. Explain about <link> and <style> elements.

Answer :

The <link> Element

The <link> element is an empty element. It is used to link two documents i.e., it is used to link the stylesheets to the document in which it is present. It not only links a stylesheet but also links an RSS feed that correlate with the web page.

The <link> element has 3 attributes when it is used for associating a stylesheet with the document. They are type, rel and href. In addition to these core attributes the <link> element can have other attributes they are, charset, dir, href, hreflang, media, rel, rev, style, target, type.

1. The Type Attribute

The type attribute indicates the MIME type of the document which a user need to link. Since the document is css, its MIME type is "text/css".

type = "text/css"

2. The rel Attribute

The rel attribute is used to indicate the relationship between the document in which the link is present and the document which has to be linked with it. The key value which is provided to rel attribute is "stylesheet".

rel = "stylesheet"

3. The href Attribute

The href attribute indicates the URL of the stylesheet. The URL can either be absolute or relative.

href = "../stylesheets/interface.css"

4. The hreflang Attribute

The hreflang attribute indicates the language in which the document to be linked is written.

hreflang = "en – US".

The <style> Element

When a user need to use an inline type of stylesheet, the <style> element is used. This element is used within the <head> element of the document. The style rules are enclosed within the <style> element.

The media Attribute

The media attribute indicates the media/output device that is optimized for the document.

media = "value";

Here, print, screen, tv, tty, projection, handled, aural, all, embossed, Braille are the different possible values of this attribute.

Example

```
<head>
<title>Style Element </title>
<style type = "text/css">
P{color : pink ;}
</style>
</head>
```

dir, title, type, lang and media are the attributes that are used by the <style> element.

Q16. What are the advantages of external stylesheets?

Answer :

The following are the advantages of external stylesheets,

1. External stylesheets can be reused for each page of the website.
2. External stylesheets can be used to avoid the use of stylistic markup in each page of the website.
3. A stylesheet can import the rules from other stylesheets to make the development modular and to obtain good reusability.
4. The site can be made more accessible by removing the stylesheet.
5. The stylesheets can be used as a template by different authors to obtain the same type in their documents.
6. If the first document downloads the stylesheet, the subsequent documents are not needed to download it again because, the browser retains a copy of stylesheet rather than each individual page.
7. Several stylesheets can be associated with a single document.

2.2 CSS PROPERTIES

Q17. List all the main properties available in CSS1 and CSS2.

Answer :

The following table shows all the main properties available in CSS1 and CSS2.

Category	Properties
FONT	font, font-family, font-size, font-size-adjust, font-stretch, font-style, font-variant, font-weight.
TEXT	color, direction, letter-spacing, text-align, text-decoration, text-indent, text-shadow, text-transform, unicode-bidi, white-space, word-spacing
BACKGROUND	background, background-attachment, background-color, background-image, back ground-position, background-repeat.
BORDER	border, border-style, border-width, border-bottom, border-bottom-color, border-bottom-style, border-bottom-width, border-color, border-left, border-left-color, border-left-style, border-left-width, border-right, border-right-color, border-right-style, border-right-width, border-style, border-top, border-top-color, border-top-style, border-top-width, border-width.
CLASSIFICATION	clear, cursor, display, float, position, visibility.
GENERATED CONTENT	content, counter-increment, counter-reset, quotes.
LIST and MARKER	list-style, list-style-image, list-style-position, list-style-type, marker-offset.
TABLE	border-collapse, border-spacing, caption-side, empty-cells, table-layout.
OUTLINES	outline, outline-width, outline-style, outline-color.
POSITIONING	bottom, clip, z-index, vertical-align, left, overflow, top, right.
DIMENSIONS	height, line-height, Max-height, width, min-width, max-width, min-height.
PADDING	padding, padding-bottom, padding-top, padding-right, padding-left.
MARGIN	margin, margin-top, margin-bottom, margin-left, margin-right.

2.2.1 Controlling Text

Q18. Explain how the fonts can be controlled. List and explain each of the properties that directly affect the font and its appearance.

Answer : Model Paper-II, Q10(a)

Controlling Fonts

The fonts can be controlled by using the different available font properties. These properties are categorized into two groups,

(a) The properties that directly influence the appearance of the font.

(b) The properties that have other formatting influences upon the font.

(a) Properties that Influence the Appearance of the Font

The properties that directly influence the appearance of the font are as follows,

1. font
2. font-family
3. font-size
4. font-weight
5. font-style
6. font-variant
7. font-stretch
8. font-size-adjust.

1. font

Font is used to combine several properties into one. A font is different from a typeface. A typeface is a family of fonts where as a font is the member of typeface.

The example of typeface is, Arial

The example of font is, Arial 12-point bold.

Example

```
<html>
<head>
<title> Font Example</title>
</head>
<body>
<p style="font: 15px arial, sans-serif;">first paragraph</p>
<p style="font: italic bold 12px/30px Georgia, serif;">second paragraph</p>
</body>
</html>
```

Output



2. font-family

The font-family property is used to specify the typeface for the text. Multiple font-family properties can be specified so that if the first choice of the value is not available in the system, then the browser can select the next font in the list. This can be shown with the help of following example html code. (program.html) which is applied with stylesheet (s2.css)

Example

```
<html>
<head>
<title> Font-family Example</title>
</head>
<body>
<p style= "font-family: verdana, arial, sans-serif ;" >first paragraph</p>
</body>
</html>
```

Output



The comma-separated list of font should end with one of the following generic font names. These generic font names allow the computer to select its default font type when none of the specified fonts are available in the system. The generic font names are serif, sans-serif, monospace, cursive, fantasy.

3. font-size

The font-size property is used to specify the size of the font. The value of font-size property can be specified in several ways. They are,

- **Absolute** xx-small, x-small, Small, medium, large, x-large, xx-large
- **Relative size** smaller, larger
- **Length** px , em, ex, pt, in, cm, pc, mm
- **Percentage (in relation to parent element)**
  2% , 10% , 25% , 50% , 100%.

Program

```
<html>
<head>
<title> Font-size Example</title>
</head>
<body>
<p style= "font-size: 35px;" >first paragraph</p>
</body>
</html>
```

Output



4. font-weight

The font-weight property represents the text either in thick or light form. The possible values of font-weight property are normal, lighter, bold, bolder, 100, 200, 300, 400, 500, 600, 700, 800, 900.

Example

```
<html>
<head>
<title> Font-weight Example</title>
</head>
<body>
<p style= “font-weight: bold;” >First paragraph</p>
<p style= “font-weight: lighter;” >Second paragraph</p>
</body>
</html>
```

Output



5. font-style

The font-style property is used to specify the style of the font. The styles are, normal, italic, oblique etc,. That can be provided as the values of font-style property.

Example

```
<html>
<head>
<title> Font-Style Example</title>
</head>
<body>
<p style= “font-style: italic;” >First paragraph</p>
<p style= “font-style: normal;” >Second paragraph</p>
</body>
</html>
```

Output



6. font-variant

The font-variant property can take two possible values namely normal and small caps. The font-variant property specifies that the font can be either in normal or small caps. The value small caps of the font-variant allows the appearance of the font appear like a smaller version of the uppercase letter set. This can be shown with an example.

Example

```
<html>
<head>
<title>Font-variant Example </title>
</head>
<body>
<p style= “font-variant: normal”> Normal font <span style= “font-variant: small-caps;”>
is represented in small caps </span> in the middle </p>
</body>
</html>
```

Output



7. font-stretch

The font-stretch property is used to control the width of text in a font but not the spaces between them. The values that it can take is of two types,

1. Relative-normal, wider, narrower
2. Fixed: ultra-condensed, extra-condensed, semi-condensed, semi-expanded, expanded, extra-expanded, ultra-expanded following is a stylesheet (stretch.css) which is applied to the html document (program.html).

Program

```
<html>
<head>
<title> Font-Strech Example</title>
</head>
<body>
<p style= “font-family: verdana, arial, sans-serif; font-stretch: semi-expanded;” >First paragraph</p>
<p style= “font-family: “Arial Black”; font-stretch: condensed;” >Second paragraph</p>
</body>
</html>
```

Output



8. font-size-adjust

The font-size-adjust attribute is used to modify the font’s aspect value. The font’s aspect value is a ratio. This ratio is between the height of the lowercase letter ‘x’ in the font and the height of the font.

Example

```
<html>
<head>
<title> Font-size-adjust Example</title>
</head>
<body>
<p style=“font-family: Times, serif; font-size: 35px;”>SIA</p>
<p style=“font-family: Verdana, sans-serif; font-size: 35px;”>SIA</p>
<p style=“font-family: Times, sans-serif; font-size-adjust: 0.58; font-size: 35px;”>SIA</p>
</body>
</html>
```

Output



(b) Properties that have other Formatting Influences upon the Font

For answer refer Unit-II, Page No. 93, Q.No. 19.

2.2.2 Text Formatting

Q19. List and explain each of the properties that have other formatting affects on the text.

Answer :

The following are the properties that have other formatting influences upon the font,

1. color
2. text-align
3. text-decoration
4. text-indent
5. text shadow
6. text-transform
7. letter-spacing
8. word-spacing
9. white-spacing
10. direction
11. vertical-align.

1. The color Property

The color property is used to apply the color to the text. The values assigned to it can be either a hexadecimal code or a color name.

Example

color.html

```
<html>
<head>
<style type="text/css">
p { color : red; font-size : 28pt ; }
</style>
</head>
<body>
<p> This paragraph shows the effect of color property. it is red in color </p>
</body>
</html>
```

Output



2. The text-align Property

The text-align property places the text in the browser's window based on the specified value. These values can be specified by text-align property as left, right, center and justify.

(i) left: Positions the text at the left side of the browser's window

(ii) right: Positions the text at the right side of the browser's window

(iii) center: Positions the text at the middle of the browser's window

(iv) justify: Spreads the text across the whole width of the browser's window.

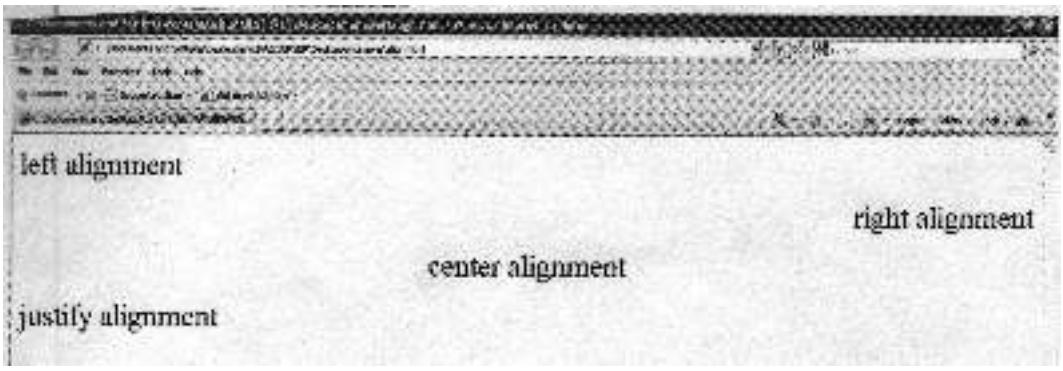
Example

align.html

```
<html>
<head>
<style type= "text /css">
p.id1 {text-align : left ; font-size : 28pt ; }
p.id2 {text-align : right ; font-size : 28pt ; }
```

```
p.id3 {text-align : center ; font-size : 28pt ; }
p.id4 {text-align : justify ; font-size : 28pt ; }
</style>
</head>
<body>
<p class= "id1"> left alignment</p>
<p class= "id2"> right alignment</p>
<p class= "id3"> center alignment</p>
<p class= "id4"> justify alignment</p>
</body>
</html>
```

Output



3. The text-decoration Property

The text-decoration property will apply the effects such as underline, overline , strike through or blinking to the text. The values taken by text-decoration property are,

(i) underline : It draws a line under the content

(ii) overline : It draws a line over the top of the content

(iii) line-through : It draws a line in the middle of the content to show the deletion of text

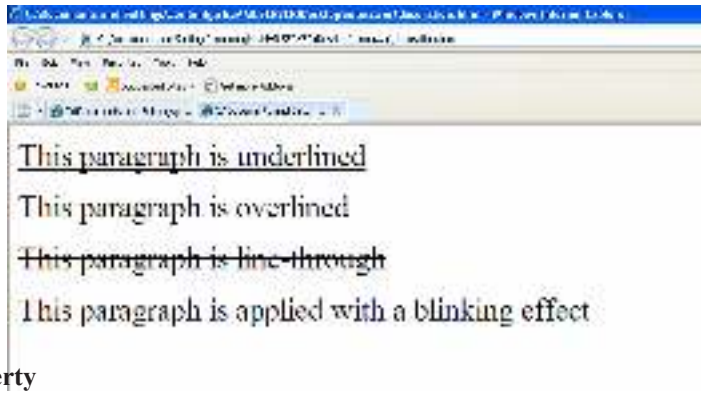
(iv) blink : It creates blinking effect on the text

Example

decoration.html

```
<html>
<head>
<style type= "text/CSS">
p.id1 {text-decoration : underline ; font-size : 28pt ; }
p.id2 {text-decoration : overline ; font-size : 28pt ; }
p.id3 {text-decoration : line-through ; font-size : 28pt ; }
p.id4 {text-decoration : blink ; font-size : 28pt ; }
</style>
</head>
<body>
<p class= "id1"> This paragraph is underlined </p>
<p class= "id2"> This paragraph is overlined </p>
<p class= "id3"> This paragraph is line-through </p>
<p class= "id4"> This paragraph is applied with a blinking effect </p>
</body>
</html>
```

Output



4. The text-indent property

The text-indent property is used to set the indent for first line of the text.

Example

indent.html

```
<html>
<head>
<style type= “text/css”>
p.indent {text-indent : 5em ; font-size : 28pt ; }
p.noindent {font-size : 28pt ; }
</style>
</head>
<body>
<p class= “noindent”>The first line of the this paragraph is intendent
</p>
<p class= “indent”> The first line of the this paragraph is intendent</p>
</body>
</html>
```

Output



5. The text-shadow Property

The text-shadow property is used to apply the drop shadowing effect to the text. This property is often used in print media. Its property includes three lengths , optionally followed by a color as its value. Hence the value is complicated. This can be shown as follows,

drop shadow { text-shadow : 0.4em 0.4em 0.6em green }

In the above example , first two values of text-shadow indicates the x and y coordinates of the offset of the drop shadow. The third value followed by a color indicates the blur effect.

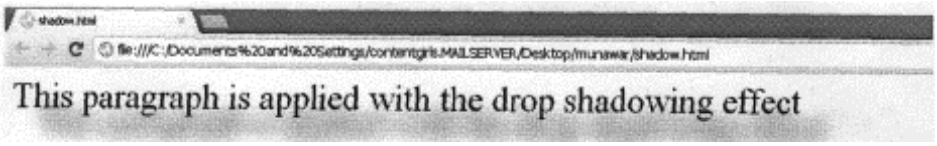
This property is not supported by IEZ and firefox.

shadow.html

```
<html>
<head>
<style type= “text/css”>
```

```
p.shadow { text-shadow : 0.7 em  0.7em  0.9em red; font-size : 28pt; }
</style>
</head>
<body>
<p class= “shadow”> This paragraph is applied with the drop shadowing effect</p>
</body>
</html>
```

Output



6. The text-transform Property

The text-transform property is used to apply the cases such as the lowercase or uppercase to the text.

The possible values taken by text-transform property are,

(i) none: Applies no effect

(ii) Capitalize: Capitalizes the first letter of every word.

(iii) Uppercase: Converts the entire content of an element into uppercase

(iv) Lowercase: Converts the entire content of an elements into lowercase.

Example

```
<html>
<head>
<style type= “text/css”>
</style>
</head>
<body>
<p style=“text-transform : none ; font-size : 20pt ;”> no effect </p>
<p style=“text-transform : capitalize ; font-size : 20pt ;”> capitalized effect </p>
<p style=“text-transform : uppercase ; font-size : 20pt ;”> uppercase effect </p>
<p style=“text-transform : lowercase ; font-size : 20pt ;”> lowercase effect </p>
</body>
</html>
```

Output



7. The letter-spacing Property

The letter-spacing property is used to track the number of spaces between letters. The tracking of space is of three types,

(i) Loose tracking indicates a lot of space between the letters.

(ii) No tracking indicates the normal space between the letters.

(iii) Tight tracking indicates that the letters are squeezed together. The value it accepts is a unit of length.

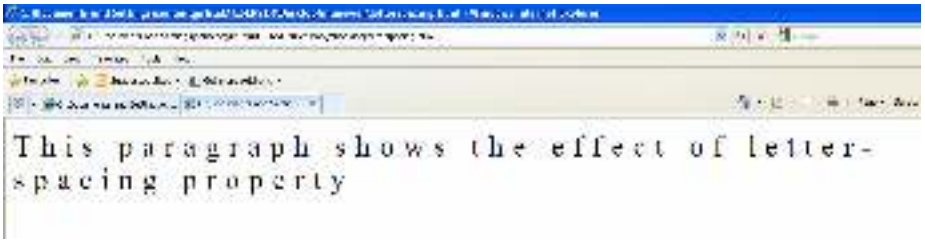
Example

letterspacing.html

```
<html>
<head>
<style type= “text/css”>
p { letter-spacing : 12px ; font-size : 28pt ; }
</style>
```

```
</head>
<body>
<p> This  paragraph shows the effect of letter-spacing property </p>
</body>
</html>
```

Output



8. The word-spacing Property

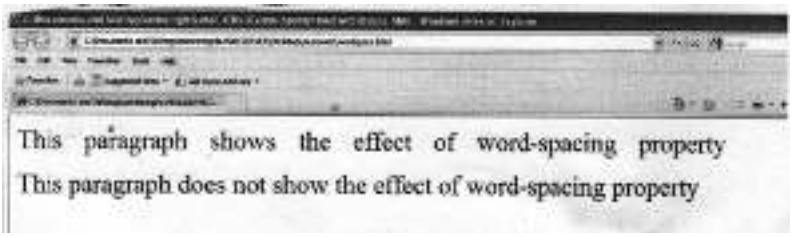
The word-spacing property is used to provide the space between words. The value it accepts is a unit of length.

Example

wordspace.html

```
<html>
<head>
<style type= “text/css”>
p.wordspace {word-spacing : 22px ; font-size : 28pt ; }
p.nospace {font-size : 28pt ; }
</style>
</head>
<body>
<p class= “wordspace”> This paragraph shows the effect of word-spacing property </p>
<p class= “no space”> This paragraph does not show the effect of word-spacing property </p>
</body>
</html>
```

Output



9. The white-space Property

The white-space property is used to control the number of spaces between any two words in the document. The possible values of white-space property are normal , pre and nowrap

(i) Pre: The “pre” value preserves the white spaces and does not break the line until the
 element is used explicitly

(ii) Nowrap: The “nowrap” value does not preserve the white spaces. Instead it collapses the sequence of white spaces into a single space. It does not break the line until the
 element is used explicitly.

(iii) Normal: The “normal” is the default value. It does not preserves the white spaces. Instead it collapses the sequence of white spaces into a single space. It breaks the line if it exceeds the available line space of the browser screen.

Example

```
<html>
<head>
<style type= “text/css”>
p.pre {white-space:pre;}
p.nowrap{white-space:nowrap;}
p.normal{white-space:normal;}
</style>
</head>
```

```
<body>
<h1>EFFECT OF PRE VALUE</h1>
<p class= “pre”> This paragraph demonstrates the use of pre value of the white space property, this paragraph demonstrates the use of pre value of the white space property, this paragraph demonstrates the use of pre value of the white space property..</p>.
<h1>EFFECT OF NOWRAP VALUE</h1>
<p class= “nowrap”> This paragraph demonstrates the use of nowrap value of the white space property, this paragraph demonstrates the use of nowrap value of the white space property, this paragraph demonstrates the use of nowrap value of the white space property. </p>
<h1>EFFECT OF NORMAL VALUE</h1>
<p class= “normal”> This paragraph demonstrates the use of normal value of the white space property, this paragraph demonstrates the use of normal value of the white space property, this paragraph demonstrates the use of normal value of the white space property. </p>
</body>
</html>
```

Output



10. The direction Property

The direction property is used to specifies the direction in which the text must flow such as left direction or right direction. The possible values accepted by direction property are,

(i) ltr: Allows the text to flow from left to right direction

(ii) rtl: Allows the text to flow from right to left direction

(iii) inherit: allows to flow the text in the direction similar to the direction of its parent element.

Example

```
<html>
<head>
<style type= “text/css”>
p.ltr {direction : ltr ; font-size : 28pt ; }
p.rtl {direction : rtl ; font-size : 28pt ; }
p.inh {direction : inherit ; font-size : 28pt ; }
</style>
</head>
<body>
<p class= “ltr”>left to right direction </p>
<p class= “rtl”>right to left direction </p>
<p class= “inh”>direction similar to the parent’s direction </p>
</body>
</head>
```

Output



11. The vertical-align Property

This property is used to control the vertical positioning of text or an image.

Syntax

```
p.a{
    vertical-align: value;
}
```

It takes different values. They are,

(i) **baseline:** It is the default value. It aligns the elements on the baseline.

(ii) **sub:** It aligns the element as subscript of the baseline element.

(iii) **super:** It aligns the element as superscript of the baseline element.

(iv) **middle:** It aligns the element with the midpoint of the parent element.

(v) **top:** It aligns the top of the text and the top of the image at the top of the tallest element on the line.

(vi) **text-top:** It aligns the top of the text and the top of the image at the top of the text on the line.

(vii) **bottom:** It aligns the bottom of the text and the bottom of the image at the bottom of the lowest element on the line.

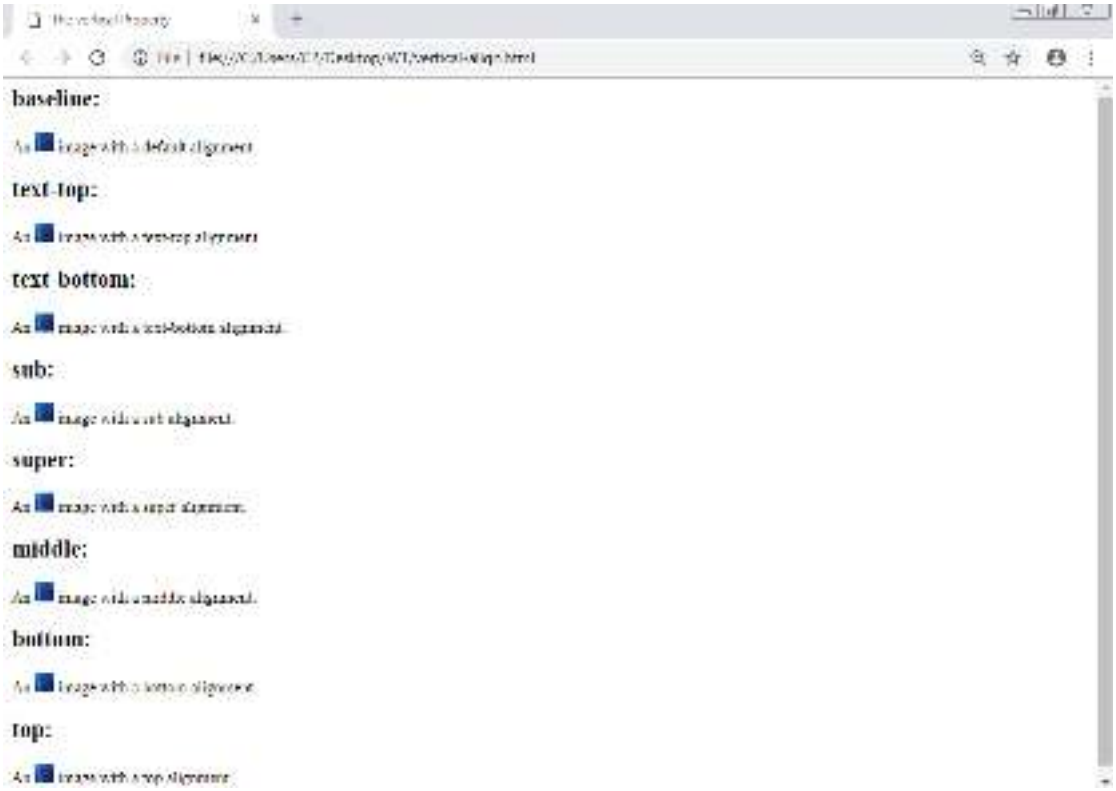
(viii) **text-bottom:** It aligns the bottom of the text and the bottom of the image at the bottom of the lowest text on the line.

Example

```
<html>
<head>
<style type = text/cs></style>
</head>
<body>
<title>The vertical Property</title>
<h2>baseline:</h2>
<p>An <img class= “a” src= “download.jpg” width=“20” height=“20” style = “vertical-align : baseline;”>
image with a default alignment.</p>
<h2>text-top:</h2>
<p>An <img class= “b” src= “download.jpg” width=“20” height=“20” style = “vertical-align : text-top;”> image with a
text-top alignment.</p>
<h2>text-bottom:</h2>
<p>An <img class= “c” src= “download.jpg” width=“20” height=“20” style = “vertical-align : text-bottom;” > image
with a text-bottom alignment.</p>
<h2>sub:</h2>
<p>An <img class= “d” src= “download.jpg” width=“20” height=“20” style = “vertical-align : sub;” > image with a sub
alignment.</p>
<h2>super:</h2>
<p>An <img class= “e” src= “download.jpg” width=“20” height=“20” style = “vertical-align : super;” > image with a
super alignment.</p>
<h2>middle:</h2>
<p>An <img class= “f” src= “download.jpg” width=“20” height=“20” style = “vertical-align : middle;” > image with a
middle alignment.</p>
<h2>bottom:</h2>
<p>An <img class= “g” src= “download.jpg” width=“20” height=“20” style = “vertical-align : bottom;” > image with a
bottom alignment.</p>
```

```
<h2>top:</h2>
<p>An <img class= “h” src= “download.jpg” width=“20” height=“20” “vertical_align : top;” > image with a top align-
ment.</p>
</body>
</html>
```

Output



2.2.3 Text Pseudo Classes

Q20. Explain in detail about different types of pseudo classes.

Answer :

Pseudo Classes and Pseudo Elements

Pseudo classes and pseudo elements apply the style to the elements that cannot be handled by traditional selectors. Pseudo classes and pseudo elements are used to provide the style to the first character of the first line of a paragraph.

Pseudo classes compare the elements by using the information other than the elements name or their attributes. Pseudo classes are preceded by a colon.

Pseudo elements deal with the subparts of an element. They are also preceded by a colon.

Pseudo classes and pseudo elements are represented as follows,

Selector: pseudo-element {declaration}

Selector: pseudo-class {declaration}

Types of Pseudo Classes

The different types of pseudo classes are as follows,

1. The :first-line Pseudo Class

The :first-line pseudo class applies the styles to the first line of the element. It is generally used for the <P> element. It has the following properties,

1. color
2. vertical-align
3. text-decoration
4. font

5. background
6. line-height
7. text-transform
8. word-spacing
9. letter-spacing
10. clear.

Example: p:first-line {color: Red; text-decoration: underline;}

2. The :first-letter pseudo Class

The :first letter pseudo class applies the styles to the first letter of the element. It is generally used for <P>,   elements.

Example: P: first-letter {font-size : 400% float : left ;}

3. The :before and :after Pseudo Elements

:before and :after pseudo elements are used for inserting some new content into an HTML document.

To insert any new thing before an existing element; :before pseudo element is used where as to insert any new thing after an existing element, the :after pseudo element is used.

The new content may be text, images, audio or video. These pseudo classes are not supported by the internet explorer before and : after pseudo classes use the 'content' property to add the content within the document. The following example shows the effect of : before pseudo class,

Example

```
<html>
<head>
<style type = "text/css" >
body >ol>li :before { content : "Examples:"; }
ol { list-style-type: none; }
</style>
</head>
<body>
<o>
<li> First example of :before class </li>
<li> Second example of :before class </li>
</ol>
</body>
</html>
```

Similarly :after pseudo class can be used as shown above but it adds the content after the element.

2.2.4 Selectors

Q21. What are selectors? Explain in detail about different types of selectors.

Answer :

Selectors

A selector provides a way to select the element (s) for applying the style rules. The elements can be selected in several ways by using the different types of selectors.

Types of Selectors

The following are eight different types of selectors in CSS,

1. Universal selector
2. The Type selector
3. The Class selector
4. The ID selector
5. The child selector
6. The descendent selector
7. The general sibling selector
8. The adjacent sibling selector
9. The attribute selector.

1. Universal Selector

The universal selector is represented as asterisk (*). It can be used to apply a rule to all the elements of a document.

Example: * { }.

2. The Type Selector

The type selector is used to apply the same rule to various elements. The type selector compares all of the elements present in the comma - delimited list. It reduces the size of stylesheet, saves the bandwidth and load on the server.

Example: h1 , h2 , p{ }

The above selector matches all of the h_1, h_2 and p elements of the document and applies the same rule to all of them.

3. The Class Selector

The class selector matches the rule with the element that has a class attribute and whose value is specified in the class selector.

Example: is the element in the document containing the "class" attribute with the value as " val ".

< p class = "val"> this is an example of class selector < /p >

The class selector can be used in the following two ways,

(a) If a user wants to apply the rule to all these elements that have a class attribute with the value " val ", then the class selector can be specified as follows,

.val{ }

(b) If a user wants to select all the elements that have a class attribute with the value " val " then , the class selector can be specified as follows,

p.val{ }

4. The ID Selector

It is used for unique style. It is same as the class selector but it uses the id attribute instead of the class attribute.

Example: Example of ID selector

The ID selector can be represented by using the pound (#) sign before the value of id attribute instead of using a period or full stop as used in class selector.

p # val { }

5. The Child Selector

The child selector is used to apply the style to an element that is the child of some other element. This selector includes less than symbol known as combinator to separate the direct child element and the parent element.

Example: td > b{ }

In the above example, <b> is the direct child of <td>. Hence, the style rule is applied to all such < b > elements that are direct children of all the <td> elements.

6. The Descendent Selector

The descendent selector is used to apply the style rule to an element that is a descendent of another element. It may not be a direct child at any level of nesting. It includes space symbol as the separator.

Example: table b { }

In the above example , the style rule is applied to all these <b> elements that are children of <td> and <th> elements. <b> element , can never be the direct child of <table> element , but at the level of nesting it is a descendent of <table> element i.e., <b> may be direct child of <td> or <th>. It is opposite to the child selector because , it selects all the children of <table> rather than just selecting the direct children.

7. The General Sibling Selector

The general sibling selector applies the style to an element that is sibling of another.

Example: h1~p{ }

A user can use this selector when both the sibling elements are required to be styled.

8. The Adjacent Sibling Selector

The adjacent sibling selector applies the style to an element that is the next sibling of another.

Example

Suppose a user want to provide a style to the first paragraph after every level of heading , then the adjacent sibling selector can be used as follows,

h_1 + p{ }

Using Child and Adjacent Sibling Selectors

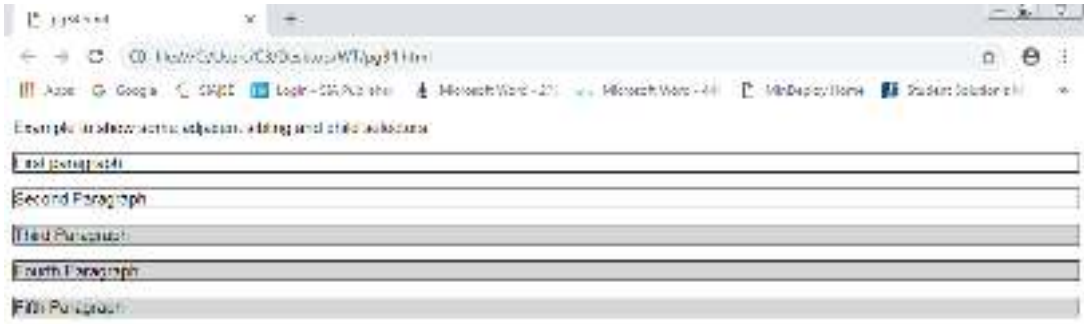
The child and adjacent sibling selectors are used to reduce the number of class attributes in an XHTML document. They can be used to control the presentation of pages. They make the mark up very clean. They provide a lot of flexibility in styling the document.

Example

exm.css

```
body {
color:#000000;
background-color:#ffffff;
font-size:15px;}
p {font-family:arial, verdana, sans-serif;}
div>p {border:1px solid #000000;}
p+p+p {background-color:#d6d6d6;}
<html>
<head>
<link rel= “stylesheet” type= “text/css” href= “exm.css”/>
</head>
<body>
<p>Example to show some adjacent sibling and child selectors</p>
<div>
<p>First paragraph</p>
<p>Second Paragraph</p>
<p>Third Paragraph</p>
<p>Fourth Paragraph</p>
<p>Fifth Paragraph</p>
</div>
</body>
</html>
```

Output



The first rule is applied to control the font of all the paragraph elements. The second rule is applied to provide a solid border of 1px to all the paragraph elements that are children of . The first paragraph is not the child of < div >. Hence, the second rule is not applicable to it. The third rule is applied to provide the color to last three paragraphs.

9. The Attribute Selector

Attribute selectors select the elements based on the availability of the value of one or more attributes. Attribute selectors are supported by Firefox and Opera but they are not supported by IE.

There are totally seven attribute selectors out of which four are provided by CSS2 and 3 are extended by CSS3.

(a) Existence Selector

The existence selector matches the elements that have the specified attribute.

Syntax

element [name of attribute]

(or)

[name of attribute]

Example

p [id] it selects all the paragraph elements in the document that have the attribute ' id '. If there is no element name in the selector, all of the elements are selected by default.

(b) Equality Selector

The equality selector selects the elements depending on their attribute value.

Syntax

element [name attribute = " value of attribute"]

(or)

[name of attribute = " value of attribute"]

Example

p [id = "note"] selects all the elements that have the attribute value "note".

(c) Space Selector

Space selector selects the elements that have same values.

Syntax

element [name of attribute ~ = " value"]

(or)

[name of attribute ~ = "value"]

Example

p [class ~ = "one"] selector matches with both of the following,

 ---

(d) Hyphen Selector

The hyphen selector matches an element with attribute value similar to the specified value or starting with the specified value followed by "-". The main purpose of this selector is to match the language sub code.

Syntax

element[attribute name | = "value"]

Example

p [long | = "tel"]

matches with,

 - -

 - -

(e) Prefix Selector

The prefix selector matches the elements in which the starting of attribute-value matches with the specified attribute value.

Syntax

element [name of attribute ∧ = " value of attribute"]

(or)

[name of attribute ∧ = "value of attribute"]

Example

p [id ∧ = "match"] matches

 matching -- as well as

 matchless ...

(f) Suffix Selector

The suffix selector matches the elements in which the ending part of attribute-value matches with the specified attribute value.

Syntax

element [name of attribute $ = "value-of-attribute"]

(or)

[name of attribute $ = "value-of-attribute"]

Example

p [href $ = ".com"] matches

 Google

(g) Substring Selector

The substring selector matches the element in which at least one occurrence of the specified attribute value is present.

Syntax

element [attribute-name * = "attribute-value"]

(or)

[attribute-name * = "attribute-value"]

Example

p [id * = "match"] selects all these paragraph elements that have the attribute-value as "match". The above selector matches <p id = "Bestmatch Bestmatch > substring .

2.2.5 Lengths

Q22. How lengths can be measured used in CSS?

Answer :

Lengths

In CSS, the lengths can be measured in three ways. They are,

1. Absolute Units

These units are fixed and the different length units in CSS properties are as follows,

(i) **pt:** It defines the length in points.

(ii) **pc:** It defines the length in picas.

(iii) **in:** It defines the length in inches.

(iv) **cm:** It defines the length in centimeters.

(v) **mm:** It defines the length in millimeters.

Example

An example to illustrate the absolute units in CSS length properties are as follows,

```
<html>
<head>
<style>
a {font-size: 4.5pc;}
b {font-size: 25pt;}
c {font-size: 1in;}
d {font-size: 2cm;}
e {font-size: 25mm;}
</style>
</head>
<body>
```

```
<title>Absolute Units</title>
<a>SIA</a><br>
<b>Publishers</b><br>
<c>and</c><br>
<d>Distributors</d><br>
<e>Pvt. Ltd</e>
</body>
</html>
```

Output



2. Relative Units

The three different types of relative units in CSS properties are as follows,

(i) **px:** px is known as pixel which is a small part of a digital image. This is used to define the size of the layout.

(ii) **em:** It is used to define the font-size of the element.

(iii) **ex:** It is used to define the font-height of the element.

Example

An example to illustrate the Absolute units in CSS Length properties is given below,

```
<html>
<head>
<style>
a {
font-size: 30px;
}
b {
font-size: 2ex;
}
c {
font-size: 16px;
line-height: 0.5em;
}
</style>
</head>
<body>
<title>Relative Units</title>
<a>SIA Publishers</a>
<b>and</b>
<c>Distributors Pvt Ltd.</c>
</body>
</html>
```

Output



3. Percentages

This unit is used provide a value in a relation to another value. By using the percentage value, the element occupies the width of the web browser.

Example

```
<html>
<head>
<style>
a {font-size: 80%;}
b {font-size: 100%;}
</style>
</head>
<body>
<title>Percentages</title>
<a>SIA Publishers and Distributors </a>
<b>Pvt Ltd.</b>
</body>
</html>
```

Output



2.2.6 Introducing the Box Model

Q23. Discuss in detail about box model.

Answer :

Box Model

This model helps in defining the layout of a web document. In HTML each element will be in the form of "box". Wherein the position of each box (or) element is determined based on the order (flow) of web page. This box can be of two types

(a) Block level box and

(b) Inline box.

(a) Block Level Box

A Block level box allows , paragraph, or heading to utilize the horizontal space available at the begining of the line.

(b) Inline Box

An Inline Box allows or image to utilize the space available before and after the inline element. A box model is responsible for managing the height and width of elements and also the horizontal and vertical space available beside the elements.

The below figure represents the architecture of the box model.

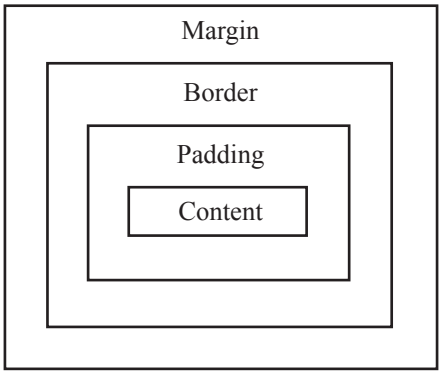


Figure: Diagramatical Representation of the Box Model

CSS model specifies the process of applying borders. padding, margin across the block level elements. The above area consists of the following.

(i) **Content Area:** It is an area which must include the content element like: video player, text, image etc. Dimensions of the content area are the content width and content height. The size of the content area is determined by evaluating the width, min width, max width, height, min height, and max height content properties.

(ii) **Padding Area:** It is an area that is surrounded by padding edge. The dimensions of the padding area are the padding width and the padding height. The size of the padding area is determined by evaluating the padding-top, padding-right, padding bottom, padding-left and shorthand padding properties. It inherits the background color designed by content area.

(iii) **Border Area:** It is an area that includes border edge. The dimensions of the border area are the border width and border height. The size of the Border area is determined by evaluating the border width and border properties.

(iv) **Margin Area:** It is the area that adds margin edge. It is an outmost area around the border area. The dimensions of the margin area are margin width and margin height. The size of the margin area is determined by evaluating the margin top, margin-right, margin bottom, margin left and shorthand margin properties.

Total width of an element is calculated as:

Element width = width + left padding + right padding + Left border + right border + left margin + right margin.

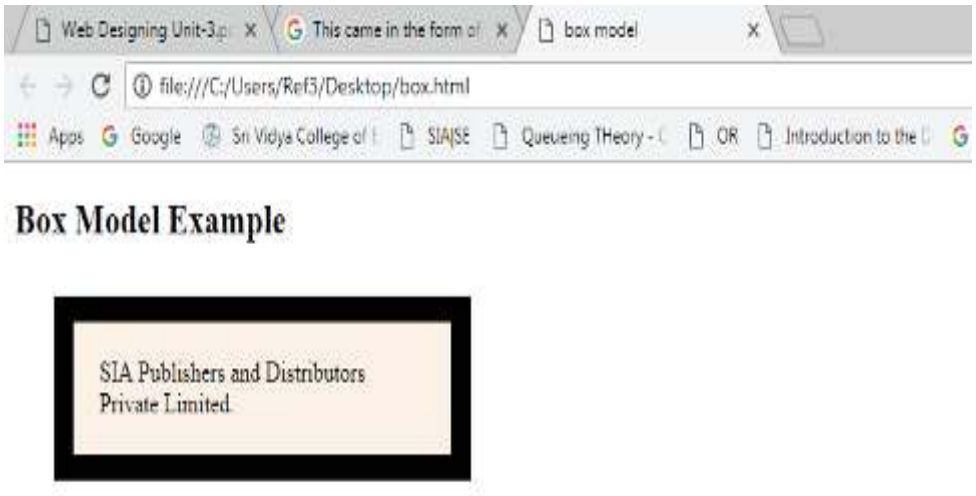
Total height of an element is calculated as:

Total element height = height + top padding + bottom padding + top border + bottom border + top margin + bottom margin.

Example

```
//CSS3  Box Model
    <!DOCTYPE html>
    <html lang= “en”>
    <head>
    <meta charset= “UTF-8”>
    <title> box model </title>
    <style>
    div {
       background-color: linen;
       width: 250px;
       border: 15px solid black;
       padding: 20px;
       margin: 30px;
    }
    </style>
    </head>
    <body>
    <h2>Box Model Example</h2>
    <div>SIA Publishers and Distributors Private Limited.</div>
    </body>
    </html>
```

Output



Q24. What are the different border properties? Explain.

Answer :

Border Properties

Border properties will set the borders of the box. The three different border properties are,

1. The border-color Property

This property is used to set the color of the border surrounding a box.

Syntax

a {border-color : colorname;}

The different colors in CSS are represented either with color names or hex codes, or RGB values. Some of the colors are as shown in the below table,

Color Name	Hex value	RGB value	RGB Percentage
yellow	#FFFF00	RGB (255, 255, 0)	rgb(100%,100%,0%)
black	#000000	RGB (0, 0, 0)	rgb(0%,0%,0%)
blue	#0000FF	RGB (0, 0, 255)	rgb(0%,0%,100%)
lime	#00FF00	RGB (0, 255, 0)	rgb(0%,100%,0%)

User can specify the colors for top, bottom, left and right of a border by using the properties such as border-top-column, border-bottom-color, border-left-color and border-right-color respectively.

2. The border-style Property

This property is used to set the style for border line.

Syntax

a {

border-style: value; }

The different border-style values in CSS are as follows,

- **none:** It displays the border with no line. If no value is specified then the value none is taken by default.
- **dotted:** It displays the border with a dotted line.
- **solid:** It displays the border with a solid line.
- **dashed:** It displays the border with dashed line.
- **hidden:** It is similar to the none but not in terms of the border-conflict resolution for table elements.
- **double:** It displays the border with two solid lines.
- **groove:** It displays the border that it carves into the page.
- **ridge:** It is completely opposite to the groove value.
- **inset:** It displays the border that looks like the border is embedded into the page.
- **outset:** It displays the border that looks like the border coming out from the canvas.

User can specify the style of the top, bottom, left and right of a border by using border-top-style, border-bottom-style, border-left-style and border-right-style properties respectively.

3. The border-width Property

This property is used to set the border width.

Syntax

a {

border-width: value; }

Where, the value may be thin or medium or thick.

User can specify the width of the top, bottom, left and right of a border by using the properties such as border-top-width, border-bottom-width, border-left-width and border-right-width respectively.

Expressing Border Properties Using Shorthand

This property is used to set all the properties of border in a single statement.

Syntax

a {

border: 5px, solid, green; }

User can specify the color of the top, bottom, left and right of a border by using border-top, border-bottom, border-left and border-right properties respectively.

Example

An example to illustrate all the border properties is as follows,

```
<html>
<head>
<style> (or) <style type = “text/ess”>
a {
border-style: solid;
border-width: thin;
border-color: red;
}
b {
border: 5px solid green;
}
</style>
</head>
<body>
<title>Border Properties Example</title>
<a>SIA Publishers and Distributors pvt Ltd.</a><br>
<br><b>SIA Publishers and Distributors pvt Ltd.</b>
</body>
</html>
```

Output



Q25. Explain in detail about padding property.

Answer :

Padding

The CSS padding property allows a user to set the spaces between the content and border of an element in HTML document. That is CSS property allows user to specify adjust padding for each side using the following properties.

- ❖ Padding-top
- ❖ padding-bottom
- ❖ Padding-left
- ❖ Padding-right.

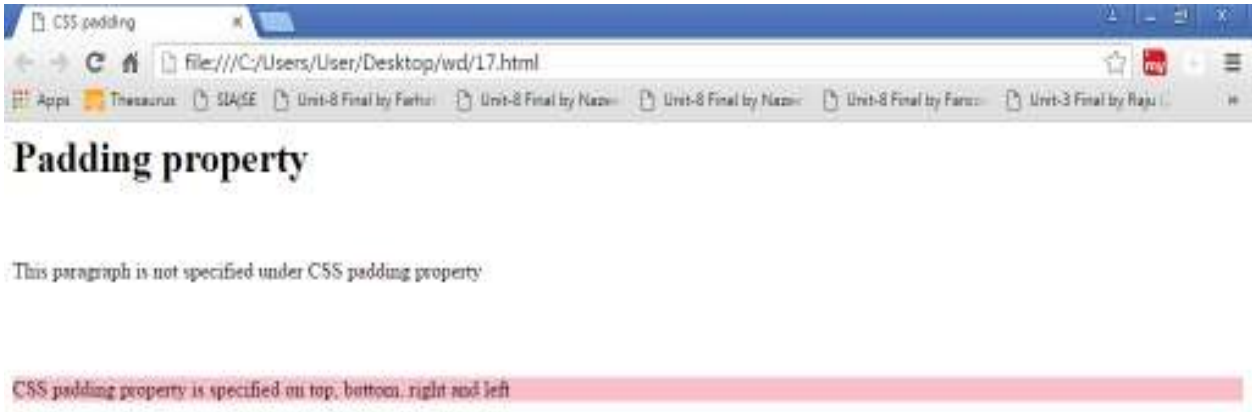
A padding property has following values,

1. **Length:** It defines padding in terms of pixels and centimeters.
2. **%:** It defines padding in the form of % of width of containing element.
3. **Inherit:** It defines padding value by inheriting from the parent element.

Program

```
<html>
<head>
<title> CSS padding </title>
<style>
p.e {
border:2px;solid: blue;
background-color:pink;
padding-top: 100 px;
padding-right: 50 px;
padding-bottom: 100 px;
padding-left: 75 px;
}
</style>
</head>
<body><h1> Padding property </h1>
<br/>
<P> This paragraph is not specified under CSS padding property </P>
<br/>
<br/>
<P class = “e”> CSS padding property is specified on top, bottom, right and left </P>
</body>
</html>
```

Output



Q26. Explain in detail about margin property.

Answer :

Margin

The CSS property allows the user to set the margin size of the HTML page. This property helps in adjusting white space outside the page border.

A margin can be specified on each side (i.e., top, bottom, right, left) using the following properties.

- ❖ Margin-top
- ❖ Margin-bottom
- ❖ Margin-left
- ❖ Margin-right.

A margin has following values.

(a) **Auto:** If a margin value is set to auto, then the browser automatically sets the margin length in terms of pixels, cm etc.

(b) **%:** It specifies that the margin should be % of width of containing element.

(c) **Inherit:** It defines that the value must be derived from its parent element.

Program

```
<html>
<head>
<style>
P
{
background-color : pink;
}
P.e {
border: 1px;
solid: yellow;
margin-top: 50 px;
margin-bottom: 50 px;
margin-right: 75 px;
margin-left: 80 px;
}
</style>
</head>
<body>
<h1> A paragraph without margin: </h1>
<br/>
<P class = “e”> This paragraph has top, bottom, left and right margin.
</P>
</body>
</html>
```

Output



Q27. Discuss in detail about the dimensions of a box.

Answer : Model Paper-III, Q10(a)

Dimensions of a Box

Every box has some dimensions. The properties to control the dimensions of a box are as follows,

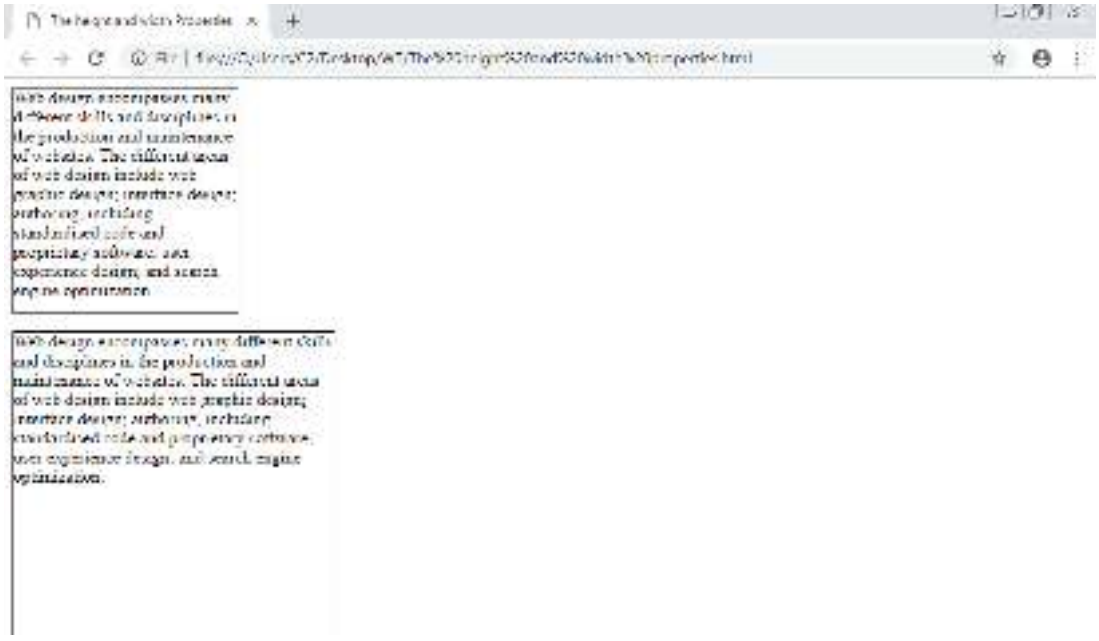
1. Height and Width Properties

The height and width properties are used to set the height and width of a box. They take the percentage, length or auto keywords as values.

Program

```
<html>
<head>
<style>
p.a {
height: 210px;
width: 210px;
 border: 1px solid black;
}
p.b {
height: 300px;
width: 300px;
border: 1px solid black;
}
</style>
</head>
<body>
<title>The height and width Properties</title>
<p class="a">Web design encompasses many different skills and disciplines in the production and maintenance of websites. The different areas of web design include web graphic design; interface design; authoring, including standardised code and proprietary software; user experience design and search engine optimization.</p>
<p class="b">Web design encompasses many different skills and disciplines in the production and maintenance of websites. The different areas of web design include web graphic design; interface design; authoring, including standardised code and proprietary software; user experience design and search engine optimization.</p>
</body>
</html>
```

Output



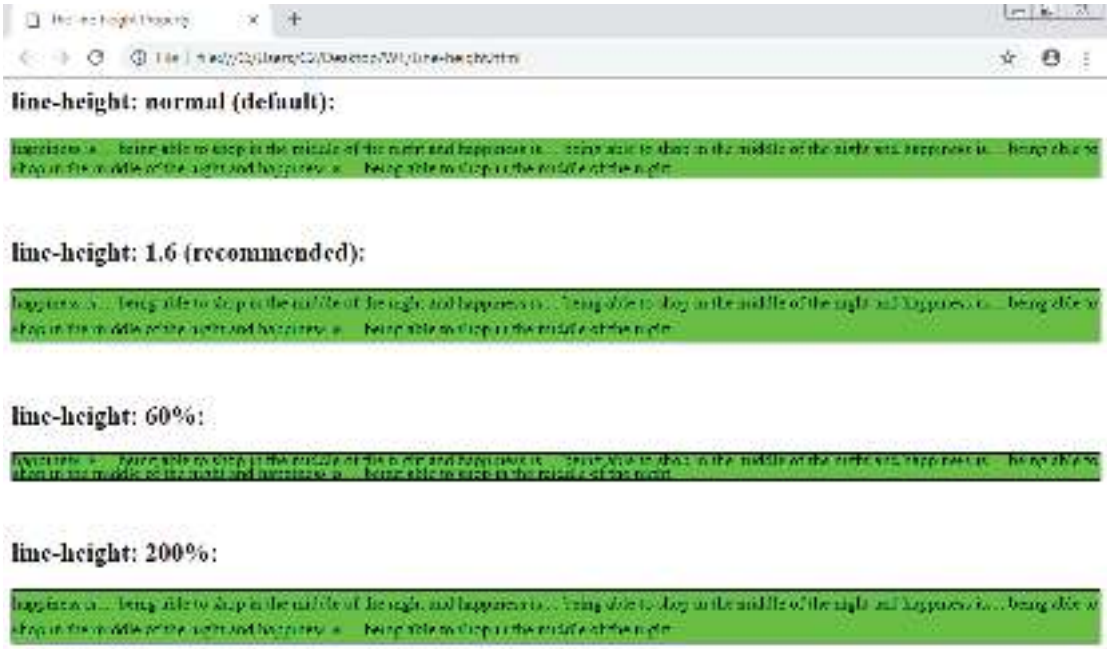
2. The line-height Property

This property is used to increase the space between the lines of text in a box. This property takes the length or a percentage as values.

Program

```
<html>
<head>
<style>
p {
background-color: lime;
border: 1px solid #000000;
}
p.a {
line-height: normal;
}
p.b {
line-height: 1.5;
}
p.c {
line-height: 80%;
}
p.d {
line-height: 150%;
}
</style>
</head>
<body>
<title>The line-height Property</title>
<h2>line-height: normal (default):</h2>
<p class="a">happiness is.... being able to shop in the middle of the night and happiness is.... being able to shop in the middle of the night and happiness is.... being able to shop in the middle of the night and happiness is.... being able to shop in the middle of the night</p><br>
<h2>line-height: 1.6 (recommended):</h2>
<p class="b">happiness is.... being able to shop in the middle of the night and happiness is.... being able to shop in the middle of the night and happiness is.... being able to shop in the middle of the night and happiness is.... being able to shop in the middle of the night</p><br>
<h2>line-height: 60%:</h2>
<p class="c">happiness is.... being able to shop in the middle of the night and happiness is.... being able to shop in the middle of the night and happiness is.... being able to shop in the middle of the night and happiness is.... being able to shop in the middle of the night</p><br>
<h2>line-height: 200%:</h2>
<p class="d">happiness is.... being able to shop in the middle of the night and happiness is.... being able to shop in the middle of the night and happiness is.... being able to shop in the middle of the night and happiness is.... being able to shop in the middle of the night</p>
</body>
</html>
```

Output



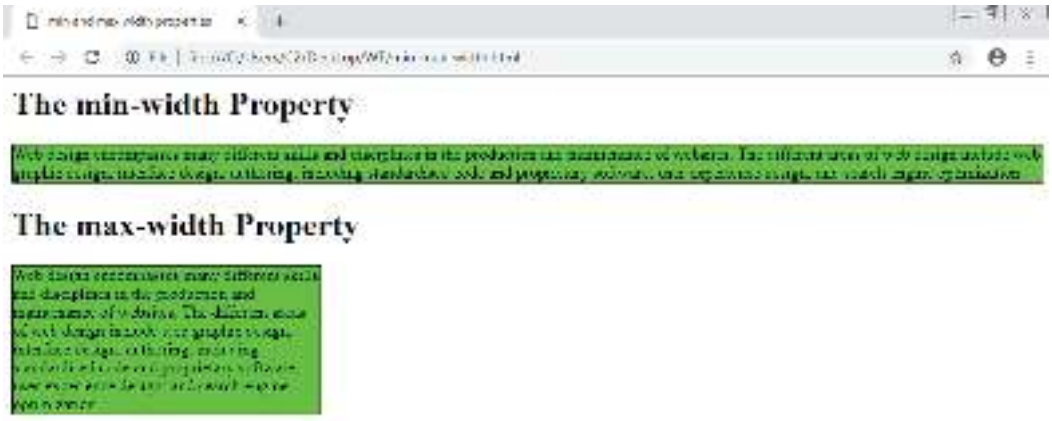
3. The max-width and min-width Properties

The max-width and min-width properties are used to set the maximum and minimum width for a box. These properties take the length and percentages as values. These properties will not accept negative values.

Program

```
<html>
<head>
<style>
p {
background-color: lime;
border: 1px solid #000000;
}
p.ex1 {
min-width: 100px;
}
p.ex2 {
max-width: 300px;
}
</style>
</head>
<body>
<title>min and max width properties</title>
<h1>The min-width Property</h1>
<p class="ex1">Web design encompasses many different skills and disciplines in the production and maintenance of websites. The different areas of web design include web graphic design; interface design; authoring, including standardised code and proprietary software; user experience design; and search engine optimization.</p>
<h1>The max-width Property</h1>
<p class="ex2">Web design encompasses many different skills and disciplines in the production and maintenance of websites. The different areas of web design include web graphic design; interface design; authoring, including standardised code and proprietary software; user experience design; and search engine optimization.</p>
</body>
</html>
```

Output



4. The min-height and max-height Properties

The min-height and max-height properties are used to set the minimum and maximum height for a box.

Program

```
<html>
<head>
<style>
p {
background-color: lime;
border: 1px solid #000000;
}
p.ex1 {
min-height: 100px;
}
p.ex2 {
max-height: 200px;
}
</style>
</head>
<body>
<title>min and max height properties</title>
<h1>The min-height Property</h1>
<p class="ex1">Web design encompasses many different skills and disciplines in the production and maintenance of websites. The different areas of web design include web graphic design; interface design; authoring, including standardised code and proprietary software; user experience design; and search engine optimization.</p>
<h1>The max-height Property</h1>
<p class="ex2">Web design encompasses many different skills and disciplines in the production and maintenance of websites. The different areas of web design include web graphic design; interface design; authoring, including standardised code and proprietary software; user experience design; and search engine optimization.</p>
</body>
</html>
```

Output



5. The overflow Property

This property is used when the content in the box exceeds the box dimensions (i.e., when the text overflows from the box). The different values in overflow property are,

(i) hidden: It hides the overflowing content.

(ii) scroll: It allows the user to see the text by scrolling.

Program

```
<html>
<head>
<style>
div.a {
background-color: yellow;
width: 200px;
height: 120px;
overflow: scroll;
}
div.b {
background-color: lime;
width: 200px;
height: 120px;
overflow: hidden;
}
</style>
</head>
<body>
<title>The overflow Property</title>
<h2>overflow: scroll;</h2>
<div class="a">Web design encompasses many different skills and disciplines in the production and maintenance of websites. The different areas of web design include web graphic design; interface design; authoring, including standardized code and proprietary software; user experience design; and search engine optimization.</div>
<h2>overflow: hidden;</h2>
<div class="b">Web design encompasses many different skills and disciplines in the production and maintenance of websites. The different areas of web design include web graphic design; interface design; authoring, including standardized code and proprietary software; user experience design; and search engine optimization.</div>
</body>
</html>
```

Output



2.3 MORE CASCADING STYLE SHEETS

2.3.1 Links

Q28. Explain how the links can be styled. Give an example.

Answer :

Styling the Links

The links can be styled by using the following pseudo classes presented in the table,

Pseudo-Class	Purpose
link	Provides the general style to links
active	Provides the styles for links that are currently being activated
visited	Provides the styles for links that have already been visited
hover	Provides the styles when someone is hovering the link

The following are the properties that can be used with these pseudo classes,

- **color:** provides color to the link
- **text-decoration:** underlines the link
- **background-color:** highlights the link when someone hovers the link.

Example

```
body {background-color: pink ;}
a {
    font-family: arial, Times new Roman;
    font-size : 15px; font-weight: bold; }
a: link {
        color: green;
        text-decoration; none; }
a: visited {
        color: blue;
        text-decoration : none; }
a: active {
        color: yellow;
        text-decoration: underline; }
a: link: hover {
        background-color: silver;
        text-decoration: underline; }
```

Q29. Explain in detail about background properties of CSS3 with an example.

Answer : Model Paper-I, Q10(b)

Background Properties

Background properties/module of CSS3 are used to in add color, images, attachments etc as background to an element. Besides this, it is also used for positioning the size of the background elements.

In CSS3, the color, images etc are specified in the background using Background property.

This property is a shorthand property that is used for defining each single background properties. The various background properties are as follows,

1. background – color
2. background – image
3. background – position
4. background – size
5. background – repeat
6. background – attachment
7. background – clip.

Syntax

background:color | image | position | size | repeat | attachment | clip

1. background – color Property

This property defines color 'of' background : behind the elements in a document.

Syntax

Selector {background – color : color | transparent;}

In the above syntax,

(a) color: It defines the color of the background.

(b) transparent: It specifies that no color must be added to background so that it should be transparent.

2. background – image Property

This property helps in adding multiple images as background to a text element.

Syntax

background – image : Url | none

In the above syntax,

(a) Url: It defines the exact location of the image.

(b) none: It displays empty background image.

3. background – position Property

This property specifies the initial position for background image.

Syntax

background – position: percentage | length | left | center | right.

In the above syntax,

(a) Bottom: It specifies that images should be positioned vertically

(b) Top: It specifies that images should be positioned horizontally

(c) Left: It specifies that images should be positioned at left corner

(d) Right: It specifies that images should be positioned at right corner.

(e) Center: It specifies that images should be specified both horizontal and vertical position.

(f) Length: It specifies the actual length of the pixels.

(g) Percentage: It specifies the percentage of the element size of the given document.

4. background – Size Property

This property defines the actual size of background image.

Syntax

background – size : length | percentage | auto | cover | contain.

In the above syntax,

(a) length: It specifies the width and height of the background image.

(b) percentage: It specifies the width and height of the background image as percentage.

(c) auto: It specifies that the background image should have default width and height values.

(d) cover: It specifies that images must be scaled in such a way that the background positioning area completely covered to the smallest size.

(e) contain: It specifies that images must be completely fixed within the display size.

5. background – repeat Property

This property defines the directions in which background images must be displayed.

Syntax

background – repeat : repeat | repeat-x | repeat-y | no-repeat.

In the above Syntax, repeat defines the repetitions of background image in horizontal and vertical directions, repeat-x defines that images must move horizontally, repeat-y defines that images must move vertically and no-repeat defines no repetition of background images.

6. background – attachment Property

This property is used for scrolling the background image or for display them at fixed position on the viewing area of the given document.

Syntax

background – attachment : scroll | fixed | local.

In the above syntax,

(a) fixed: It specifies that the background image must be fixed with respect to the viewport.

(b) scroll: It defines that, the background images must be scrolled within the viewport and translated with its containing element.

(c) local: It defines that, the background images must be scrolled with respect to the content elements.

7. background – Clip Property

It defines the area of region within which the background image or colon must be extended for an element.

background – Clip : border – box | padding – box | content – box.

In the above syntax,

(a) border – box: It defines that a background color | image can extend its position behind the edge if the is specified border.

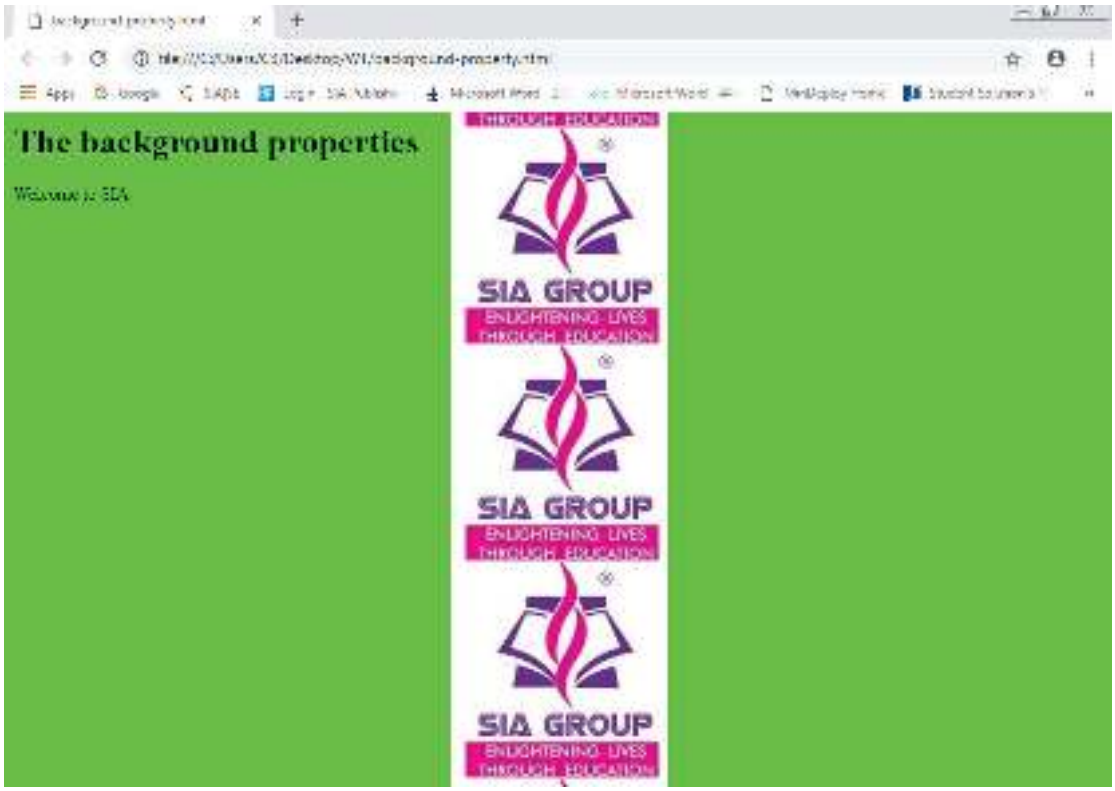
(b) padding – box: It defines that a background can extend its position over inner edge of the given padding border and does not allow any background below its border.

(c) content – box: It defines that a background can extend its position in content box. It does not allow any background below the padding area and border.

Example

```
<html>
<head>
<style>
body  {
background-image: url("download.jpg");
background-color: lime;
background-position: center;
background-size: auto;
background-repeat: repeat-y;
background-attachment: fixed;
background-clip: padding-box;
}
</style>
</head>
<body>
<h1>The background properties</h1>
<p>Welcome to SIA</p>
</body>
</html>
```

Output



2.3.2 Lists

Q30. What are the different properties associated with lists? Explain them in detail with an example for each.

Answer :

The different types of properties associated with lists are,

1. list-style-type
2. list-style-position
3. list-style-image
4. list-style
5. marker-offset.

1. list-style-type Property

The list-style-type property is used to control the shape or appearance of the bullet point or number (known as marker) The following are the standard styles for an unordered list,

(a) **none** : None

(b) **disk** (default) : A filled-in circle ex : • • •

(c) **circle** : An empty circle ex : ○ ○ ○

(d) **square** : A filled-in square ex : ■ ■ ■

The following are the values that can be used for the ordered lists,

(a) **decimal** : Number
Example: 1, 2, 3, 4, 5

(b) **decimal-leading-zero** : '0' before the number
Example: 01, 02, 03, 04, 05

(c) **lower-alpha** : Lowercase alpha numeric characters
Example: a , b , c , d , e

(d) **upper-alpha** : Uppercase alpha numeric characters
Example: A , B , C , D , E

(e) **lower-roman** : Lowercase roman numerals
Example: i, ii, iii, iv, v

(f) **upper-roman** : Uppercase roman numerals
Example: I , II , III , IV , V

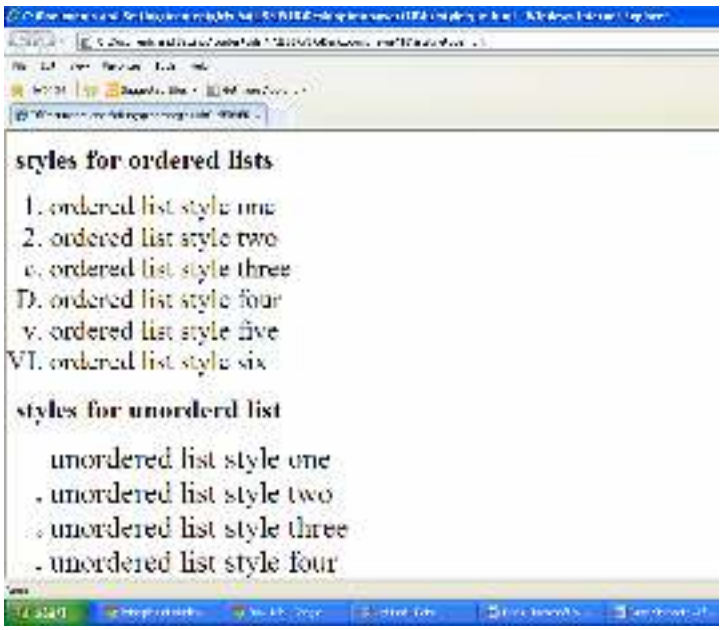
Program

lists styletype.html

```
<html>
<head>
<style type= "text/css">
li .a {list-style-type : decimal}
li .b {list-style-type : decimal-leading-zero}
li .c {list-style-type : lower-alpha}
li .d {list-style-type : upper-alpha}
li .e {list-style-type : lower-roman}
li .f {list-style-type : upper-roman}
li.h {list-style-type : none}
li.i {list-style-type : disc}
li.j {list-style-type : circle}
li.k {list-style-type : square}
</style>
</head>
<body>
<h1>styles for ordered lists</h1>
```

```
<ol>
<li class= “a”>ordered list style one</li>
<li class= “b”>ordered list style two</li>
<li class= “c”>ordered list style three</li>
<li class= “d”>ordered list style four</li>
<li class= “e”>ordered list style five</li>
<li class= “f ”>ordered list style six</li>
</ol>
<h1>styles for unordered list<h1>
<ul>
<li class= “h”>unordered list style one</li>
<li class= “i”>unordered list style two</li>
<li class= “j”>unordered list style three</li>
<li class= “k”>unordered list style four</li>
</ul>
</body>
</html>
```

Output



2. list-style-position property

The list-style property specifies the bullet points that have multiple lines of text by aligning the second line according to the beginning of the first line or. This property has 2 values, they are,

(a) inside : The second line is aligned underneath the start of the marker

(b) outside : The second line is aligned with the start of the first line

Example

```
<html>
<head>
<style type= “text/css”>
ul {list-style-position : inside ;}
ol {list-style-position : outside ;}
</style>
</head>
<body>
<ul>
```

```
<li> This bullet point specifies the effect of 'inside' value of the list-style-position property..</li>
</ul>
<ol>
<li> This specifies the effect of 'outside' value of the list-style-position property..</li>
</ol>
</body>
</html>
```

Output



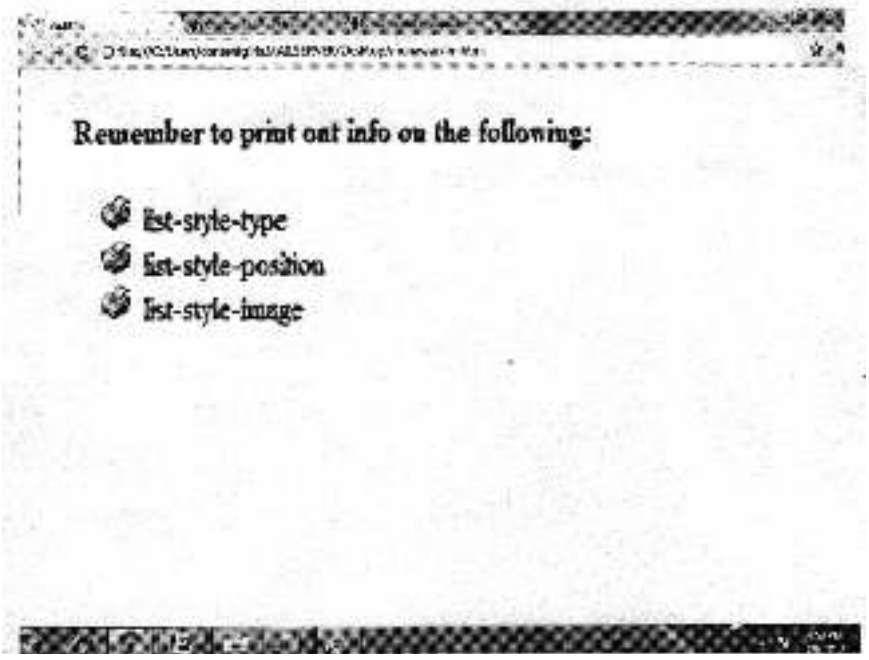
3. list-style-image Property

The list-style-image property specifies images as marker instead of a bullet points or numbers. It requires the URL of the image as its value.

Example

```
<html>
<head>
<style type= "text/css">
ul {list-style-image: url(/pix/printer_icon.gif)}
</style>
</head>
<body>
<h4> Remember to print out info on the following : </h4>
<ul class= "outside">
<li>list-style-type</li>
<li>list-style-position</li>
<li>list-style-image</li>
</ul>
</body>
</html>
```

Output



4. list-style Property

The list-style property can be used for representing the values of list-style-image , list-style-position and list-style-type at once.

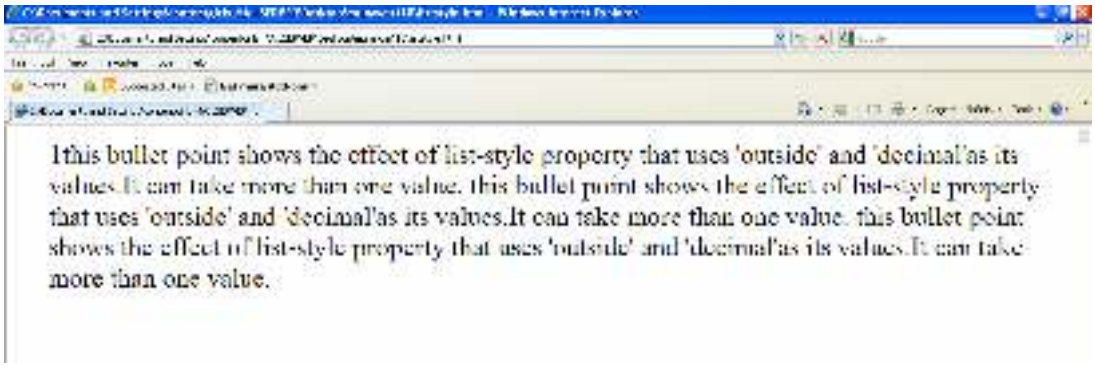
Example: ol {list-style : outside decimal ;}

Program

(liststyle . html)

```
<html>
<head>
<style type= “text /css”>
li {list-style : inside decimal ; }
</style>
</head>
<body>
<ol>
<li> this bullet point shows the effect of list-style property that uses ‘outside’ and ‘decimal’ as its values. It can take more than one value. This bullet point shows the effect of list-style property that uses ‘outside’ and ‘decimal’ as its values. It can take more than one value. This bullet point shows the effect of list-style property that uses ‘outside’ and ‘decimal’ as its values. It can take more than one value. </li>
</ol>
</body>
</html>
```

Output



5. marker-offset Property

The marker-offset property is used to specify the number of spaces between the marker and the text. The length is provided as value for it. It is not supported by IE2 and firefox2.

Example: li {marker-offset-3em ;}

2.3.3 Tables

Q31. What are the specific properties associated with tables? Explain them in detail with an example for each.

Answer :

The different <table> specific properties are as follows,

1. border-collapse
2. border-spacing
3. caption-side
4. empty-cells
5. table-layout.

1. border-collapse

The border-collapse property is used for the two purposes,

(a) It allows the border to control the appearance of the adjacent borders. The border is made to collapse with the borders of its adjacent cells.

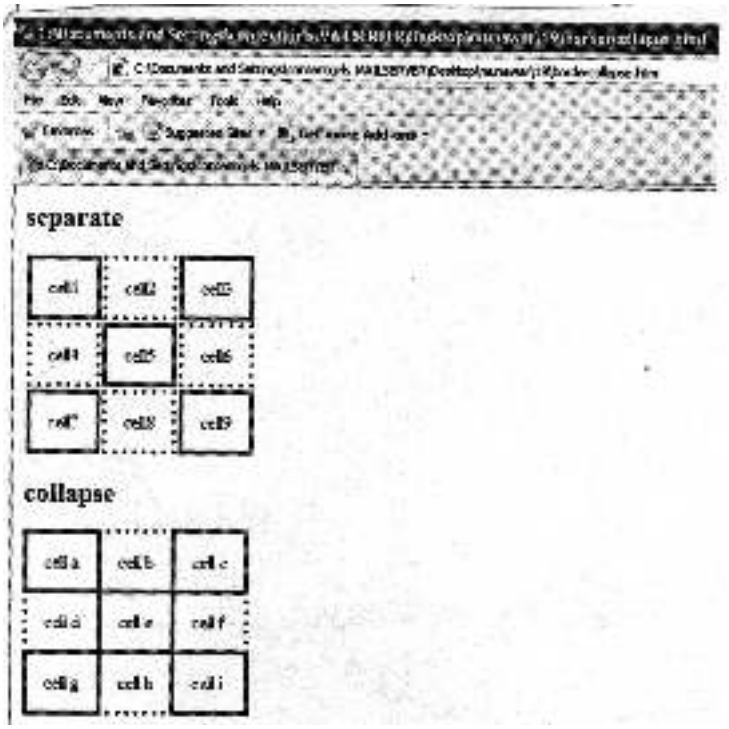
(b) It allows the cells border to maintain their own style. The border is separated from the borders of its adjacent cells.

The border-collapse property can take two values they are collapse and separate. The value collapse maintains the first purpose of the border-collapse property whereas the value 'separate' maintains the second purpose of the border-collapse property.

Example

```
<html>
<head>
<style type= "text/css">
table.table1 {border-collapse : separate ;}
table.table2 {border-collapse : collapse ;}
td.a {border-style : solid ; border-width : 4px ; border-color : green ; padding : 15px ;}
td. b {border-style-dotted ; border-width : 4px ; border-color : red ; padding : 15px ; )
</style>
</head>
<body>
<h2>separate</h2>
<table class= "table1">
<tr>
<td class= "a">cell1</td>
<td class= "b">cell2</td>
<td class= "a">cell3</td>
</tr>
<tr>
<td class= "b">cell4</td>
<td class= "a">cell5</td>
<td class= "b">cell6</td>
</tr>
<tr>
<td class= "a">cell7<td>
<td class= "b">cell8<td>
<td class= "a">cell9<td>
</tr>
</table>
<h2>collapse</h2>
<table class= "table2">
</tr>
<td class= "a">cell a</td>
<td class= "b">cell b</td>
<td class= "a">cell c</td>
</tr>
<tr>
<td class= "b">cell d</td>
<td class= "a">cell e</td>
<td class= "b">cell f</td>
</tr>
<tr>
<td class= "a">cell g</td>
<td class= "b">cell h</td>
<td class= "a">cell i</td>
</tr>
</table>
</body>
</html>
```

Output



2. border-spacing

The border-spacing property is used to specify the distance between any two cells in the table. The value of it can be one or more units of length. If it accepts one value then it can be applied for both the vertical and horizontal distance. If it accepts two values, the first value is specified for the horizontal distance and the second value is specified for the vertical distance. It is not supported by IE7.

Example

```
<html>
<head>
<style type= “text /css”>
table {width : 350px ; border-collapse : separate ;}
td {font-size : 17px ; text-align : center ; border-width : 4px ; border-color : green ; border-style : solid ; border-spacing
: 3pxpx ; 3px}
</style>
</head>
<body>
<table>
<tr>
<th>Fruits</th>
<th>color</th>
</tr>
<tr>
<td>Apple</td>
<td>green</td>
</tr>
<tr>
<td>Banana</td>
<td>yellow</td>
</tr>
<table>
</body>
</html>
```

Output



3. caption-side

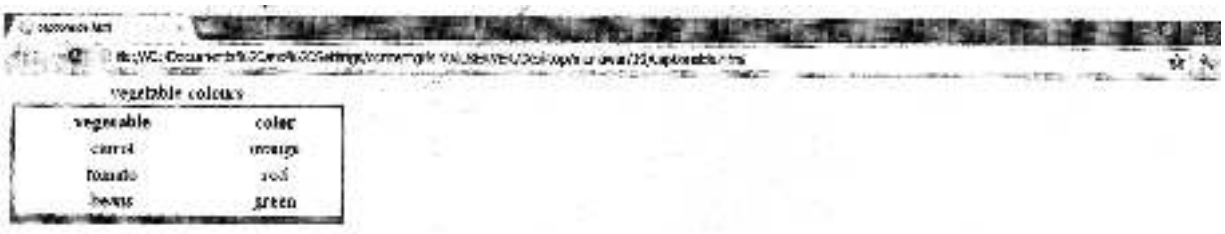
The caption-side property is used to specify the direction of the caption with respect to the table. The position of the caption can be specified by using four values they are top, right, bottom, left.

(i) **top :** The position of the caption will be at the top of the table.
(ii) **right :** The position of the caption will be at the right of the table.
(iii) **bottom :** The position of the caption will be at the bottom of the table.
(iv) **left :** The position of the caption will be at the left of the table.

Example

```
<html>
<head>
<style type= “text /css”>
caption {caption-side : top ; font-size : 20px ;}
table {border-style : solid ;}
th, td {font-size : 20px ; text-align : center ;}
table {width : 350px}
</style>
</head>
<body>
<table>
<caption>vegetable colours</caption>
<tr>
<th>vegetable</th>
<th>color</th>
</tr>
<tr>
<td>carrot</td>
<td>orange</td>
</tr>
<tr>
<td>tomato</td>
<td>red</td>
</tr>
<tr>
<td>beans</td>
<td>green</td>
</tr>
</table>
</body>
</html>
```

Output



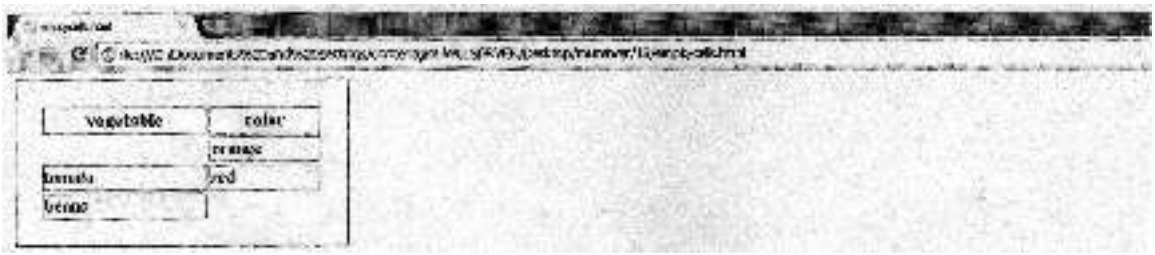
4. empty-cells

The empty-cell property is used to determine the border of an empty cell. If the empty-cell property has the value 'show' the border of an empty cell gets displayed. If it has value 'hide', the border of the empty cell cannot be displayed and if it has the value "inherit" (in case of nested tables) the empty cells border inherits the properties of its parent table.

Example

```
<html>
<head>
<style type= "text /css">
    table {width : 350px ; border-spacing : 3px ; border-collapse : separate ;
    empty-cells : hide ; padding : 25px ;}
    td , th {font-size : 20px ; border-style : solid ;}
</style>
</head>
<body>
<table border>
<tr>
<th>vegetable</th>
<th>color</th>
</tr>
<tr>
<td></td>
<td>orange</td>
</tr>
<tr>
<td>tomato</td>
<td>red</td>
</tr>
<tr>
<td>beans</td>
<td></td>
</tr>
</table>
</body>
</html>
```

Output



5. table-layout

The table-layout property allows the user to specify the table layout. It can take three values, namely fixed, auto and inherit. This property can only be used when the table is too large and requires long time to load.

(a) fixed

The value 'fixed' allows the browser to calculate the layout of the table by considering the fixed width specified for a cell in the column and by calculating the width of other cells in that column.

(b) auto

The value 'auto' allows the browser to calculate the layout of the table by considering each cell and calculating the size based on the settings for all cells. This value is used when the exact size of each column is not known.

(c) inherit

The value 'inherit' allows the browser to inherit the properties for a table from its parent table. It is used in case of nested tables.

Example

```
<html>
<head>
<style>
table {
border-collapse: collapse;
border:1px solid black;
}
th,td {
border: 1px solid black;
}
table.one {
table-layout: auto;
width: 180px;
}
table.two {
table-layout: fixed;
width: 180px;
}
</style>
</head>
<body>
<title>The table-layout property</title>
<h2>table-layout:auto Property</h2>
<table class="one">
<tr>
<th>Name</th>
<th>Date of joining</th>
<th>Salary</th>
</tr>
<tr>
<td>Nikhil</td>
<td>12-06-2017</td>
<td>15000</td>
</tr>
<tr>
<td>Vara</td>
<td>22-01-2013</td>
<td>25000</td>
</tr>
<tr>
<td>Veera</td>
<td>31-10-2018</td>
<td>10000</td>
</tr>
</table>
<h2>table-layout:fixed Property</h2>
<table class="two">
<tr>
<th>Name</th>
<th>Date of Joining</th>
<th>Salary</th>
</tr>
<tr>
<td>Nikhil</td>
<td>12-06-2017</td>
<td>15000</td>
</tr>
```

```
<tr>
<td>Vara</td>
<td>22-01-2013</td>
<td>25000</td>
</tr>
<tr>
<td>Veera</td>
<td>31-10-2018</td>
<td>10000</td>
</tr>
</table>
</body>
</html>
```

Output



2.3.4 Outlines

Q32. What are the different outline properties in CSS? Give an example.

Answer :

Outline Properties

The outline properties are used to highlight certain parts of the page without affecting the page content. The different outline- properties in CSS are as follows,

1. The outline-width property
2. The outline-style property
3. The outline-color property
4. The outline property

1. The outline-width Property

This property is used to set the width of the outline that is to be added to the box.

Example

```
a{outline-width: thick;}
```

2. The outline-style Property

This property is used to set the style of the outline that is to be added to the box. Dashed, dotted, solid are the different styles of the outline.

Example

```
a {outline-style: solid;}
    (or)
a {outline-style: dashed;}
    (or)
a {outline-style: dotted;}
```

3. The outline-color Property

This property is used to set the color of the outline that is to be added to the box. It accepts the color value either by color name, or RGB value, or hex color value.

Example

```
a { outline-color: blue; }
        (or)
a { outline-color: #0000FF ; }
        (or)
a { outline-color: rgb(0,0,255); }
```

4. The outline Property

This property is used to set the three properties at a time to the outline.

Example

```
a {outline: 5px dotted pink; }
```

A simple html program with all the outline properties is as follows,

outline-properties.html

```
<html>
<head>
<style>
p.a
{
border: 1px solid black;
outline-width: thick;
outline-style: solid;
outline-color: blue;
}
</style>
</head>
<body>
<title>Outline-Properties</title>
<p class="a">A solid outline in Blue color</p>
</body>
</html>
```

Output



2.3.5 The :focus and :activate Pseudo Classes

Q33. Discuss in brief about the :focus and :activate pseudo classes.

Answer :

The :focus Pseudo Class

The :focus pseudo class is used to add additional rules to the element that is focused.

Example

```
<html>
<head>
<style>
input:focus {
background-color: yellow;
}
</style>
```

```
</head>
<body>
<title>Focus Example</title>
<p>Enter the Login credentials :</p>
<form>
User name: <input type= “text” name= “User name”><br>
Password : <input type= “text” name= “password”><br>
<button type= “submit” form= “name form” value= “Login”>Login</button>
</form>
</body>
</html>
```

Output



The :activate Pseudo Class

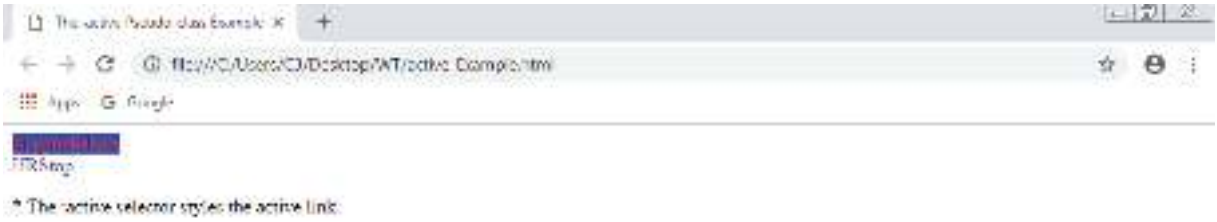
The :activate pseudo class is used to style a active link that is selected.

Example

active-Example.html

```
<html>
<head>
<style>
a:active {
background-color: blue;
}
</style>
</head>
<body>
<title>The: active Pseudo-class Example</title>
<a href= “http://www.siapublishers.com”>Sia publishers</a><br>
<a href= “https://siagroup.hrstoppro.com”>HRStop</a>
<p><b>*</b> The :active selector styles the active link.</p>
</body>
</html>
```

Output



2.3.6 Generated Content

Q34. Discuss in detail about content property in generated content.

Answer :

The Content Property

This property is used to specify the type of content that is to be added to the document. The various values that add different type of content to the document are as follows,

- **A string:** It is used to add a plain text consists alpha-numeric characters and it doesn't include quotes.
- **A Counter:** It is used to add a counter for numbering page elements.
- **A URL:** It is used to specify the URL of an image, text file or HTML file to be added to the document.
- **attr(x):** The value of the x element will be contained in this element.
- **open-quote:** It is used to add the suitable opening quote mark.
- **close-quote:** It is used to add the suitable closing quote mark.
- **no-open-quote:** This value doesn't use any opening quote marks.
- **no-close-quote:** This value doesn't use any closing quote marks.

Counters

Generally, <ol> element is used to create a number list. But, here the counter() function is used to create number list automatically by using a counter. When the browser comes across a new element then it automatically increments the number. This is mainly used when the user need to number the sections without formatting them as list.

Example

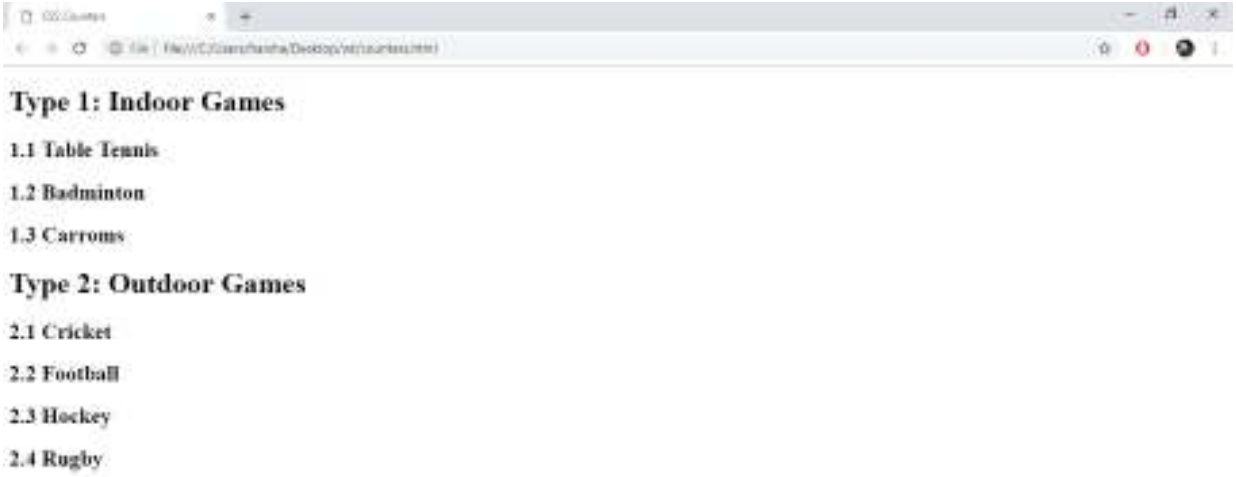
ex1.css

```
body {counter-reset: type; counter-reset: section; }
h1:before {content: "Type" counter(type) ":"; }
h1 {counter-increment: type; counter-reset: section;}
h2:before { content: counter(type) "." counter(section) " "; }
h2 {counter-increment: section; }
```

counters.html

```
<html>
<head>
<title>CSS Counters</title>
<link rel= "stylesheet" type= "text/css" href= "ex1.css" />
</head>
<body>
<h1> Indoor Games</h1>
<h2>Table Tennis</h2>
<h2>Badminton</h2>
<h2>Carroms</h2>
<h1> Outdoor Games </h1>
<h2>Cricket</h2>
<h2>Football</h2>
<h2>Hockey</h2>
<h2>Rugby</h2>
</body>
</html>
```

Output



Quotation Marks

open-quote and close-quote are the two values used in content property to specify the quote marks at the required position.

Example

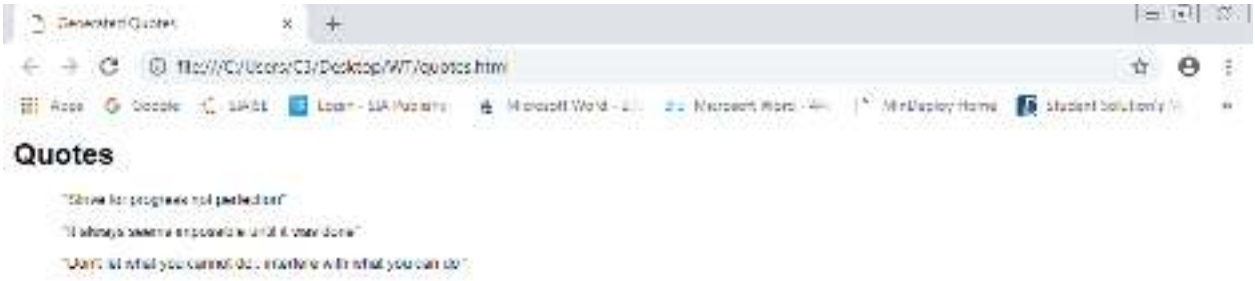
ex2.css

```
body {
color:#000000;
background-color:#ffffff;
font-family:arial, verdana, sans-serif;
font-size:12px;}
blockquote:before {content: open-quote;}
blockquote:after {content: close-quote;}
```

quotes.html

```
<html>
<head>
<title>Generated Quotes</title>
<link rel= “stylesheet” type= “text/css” href= “ex2.css” />
</head>
<body>
<h1>Quotes</h1>
<blockquote>Strive for progress not perfection</blockquote>
<blockquote>It always seems impossible until it was done</blockquote>
<blockquote>Don’t let what you cannot do.. interfere with what you can do </blockquote>
</body>
</html>
```

Output



2.3.7 Miscellaneous Properties

Q35. What are the different miscellaneous properties?

Answer :

The different miscellaneous properties in CSS are as follows,

1. The cursor property
2. The display property
3. The visibility property.

1. The cursor Property

This property is used to display the type of mouse pointer to be displayed when a user points over the element.

The various types of cursor types and their possible values are as follows,

(a) auto: This is the default value of the cursor. By default, browser sets its value.

Code

```
div
{
cursor: auto;
}
```

(b) **crosshair:** This indicates a crosshair or a plus symbol.

Code

```
div
{
cursor: crosshair;
}
```

(c) **default:** This indicates a normal arrow symbol.

Code

```
div
{
cursor: default;
}
```

(d) **pointer:** This indicates a pointing hand.

Code

```
div
{
cursor: pointer;
}
```

(e) **move:** This indicates a grasping hand using which user can move in a direction.

Code

```
div
{
cursor: move;
}
```

(f) **text:** This indicates a text which can be selected by a user.

Code

```
div
{
cursor: text;
}
```

(g) **wait:** This indicates that the specific program is busy.

Code

```
div
{
cursor: wait;
 }
```

(h) **help:** This indicates a question mark that is asking for help.

Code

```
div
{
cursor: help;
      }
```

(i) **url:** This indicates source of a cursor image file.

Code

```
div
{
cursor: url;
}
```

2. The display Property

This property is used to specify the display behavior (i.e., block, inline, none) of an element.

Syntax

```
class {
    display: block;
    }
    (or)
class {
    display: inline;
    }
    (or)
class {
    display: none;
    }
```

3. The visibility Property

This property is used to specify whether the element must be visible or invisible on the webpage. The different values of this property are as follows,

(a) **visible:** It makes the box and the elements visible to the user.

(b) **hidden:** It makes the box and the elements invisible to the user.

Example

```
<html>
<head>
<style>
p.one {
visibility: visible;
}
p.two {
visibility: hidden;
}
</style>
</head>
<body>
<title>visibility Property Example</title>
<p class="one">It is visible to the user</p>
<p class="two">It is invisible to the user</p>
</body>
</html>
```

Output



2.3.8 Additional Rules

Q36. Explain the various additional rules in CSS.

Answer :

Additional Rules

The additional rules in CSS are,

1. The @import rule
2. The !important rule
3. The @charset rule.

1. The @import Rule

This rule is used to import a style sheet into the current style sheet. This can be written in two ways,

@import “x.css”

(or)

@import url(“x.css”);

Example

```
@import “x.css”
p {
background-color: #00FF00;
font-family: times, verdana, Helvetica, sans-serif;}
h1{ font-size: 30pt; }
```

The above example, imports the x.css style sheet and executes.

2. The !important Rule

This rule is used to specify particular style sheet from multiple style sheets.

Example

```
<style>
p {
 color: green !important;
}
p {
color: yellow;
}
</style>
```

The above example provides the green color to the element because it is assigned with !important rule.

3. The @charset Rule

This property is used to indicate the character encoding that to be used in the style sheet. The @charset rule should be mentioned at the top of the style sheets without spaces.

Example

```
@charset “iso-8859-1”
```

2.3.9 Positioning and Layout with CSS

Q37. What are the different positioning schemes in CSS?

Answer :

There are three different positioning schemes in CSS. They are,

1. Normal positioning
2. Float positioning
3. Absolute positioning.

1. Normal Positioning

In CSS normal position is known as the default position. In this, the flow of block-level elements will be from top to bottom and the flow of inline elements will be from left to right.

Position Property

This property is used to set the position of the box. It takes different values and they are as follows,

(a) **static:** It is by default value just like the normal flow.

(b) **relative:** It sets the position of the element offset from the default position.

(c) **absolute:** It sets the position of the box using x and y coordinates from the top-left corner of the element.

(d) **fixed:** It sets the position of the box from top-left corner of the window and it cannot be changed further.

Example

ex7.css

```
body {
color:#000000;
background-color:#ffffff;
font-family:arial, verdana, sans-serif;
font-size:12px;}
p {
border-style:solid;
border-color:#000000;
border-width:2px;
padding:5px;
width:200px;
background-color:#FFFFFF;}
```

Normal-flow.html

```
<html>
<head>
<title>Normal Flow</title>
<link rel="stylesheet" type= "text/css" href= "ex7.css" />
</head>
<body>
<p>Paragraph <b>1</b>. The content that will be at the top of the page.</p>
<p>Paragraph <b>2</b>. The content that will be at the middle of the page.</p>
<p>Paragraph <b>3</b>. The content that will be at the bottom of the page.</p>
</body>
</html>
```

Output



2. Float Positioning

Float positioning is used to move block-level elements to the normal flow.

Example

ex.css

```
body {
color:#000000;
background-color:#ffffff;
font-family:arial, verdana, sans-serif;
font-size:12px;}
p {
border-style:solid;
border-color:#000000;
border-width:2px;
padding:5px;
width:200px;
background-color:#FFFFFF;
}
p.two {
position:relative;
left:40px;
top: -40px;
}
```

relative-positioning.html

```
<html>
<head>
<title>Relative Positioning</title>
<link rel= “stylesheet” type= “text/css” href= “ex.css” />
</head>
<body>
<p>Here is paragraph <b>one</b>. This will be at the top of the page.</p>
<p class= “two”>Here is paragraph <b>two</b>. This will be in the middle of the page.</p>
<p>Here is paragraph <b>three</b>. This will be at the bottom of the page.</p></body>
</html>
```

Output



3. Absolute Positioning

Absolute positioning scheme is used to fix the position of an element. To do this, set the position property of the element to absolute.

Example

ex9.css

```
body {
color:#000000;
background-color:#ffffff;
```

```
font-family:arial, verdana, sans-serif;
font-size:12px;}
div.page {
position:absolute;
left:50px;
top: 100px;
padding:20px;
border-style:solid; border-width:2px; border-color:#000000;}
```

absolute-positioning.html

```
<html>
<head>
<title>Absolute Positioning</title>
<link rel= “stylesheet” type= “text/css” href= “ex9.css” />
</head>
<body>
<div class= “page”><p>Here is paragraph <b>1</b>. This will be at the top of the page.</p>
<p class= “two”>Here is paragraph <b>2</b>. This will be in the middle of the page.</p>
<p>Here is paragraph <b>3</b>. This will be at the bottom of the page.</p>
</div>
</body>
</html>
```

Output





2.4 PAGE LAYOUT

2.4.1 Understanding the Site's Audience

Q38. Write about web designing principles.

Answer :

Web Designing Principles

1. Understanding the Site Aims

The site aims and ensures whether user has defined the object for the site clearly. A site may have several purposes, by asking some questions such as product or service that the user is dealing with, what is special about it etc., will clarify us what user is trying to achieve.

A site might be designed to push a product or service or to teach new skills to create a community etc. Each of these have their own set of questions, but it is important to know what the site is trying to achieve and what is required to meet those goals.

2. Whom You Expect to Visit Our Site

While creating a web site, every the page is created for the target audience also other than clients. So, through the design process, it might be choosing the position of elements or adding a feature a technology etc., user should keep in mind what the target audience want. And it is important to understand the target audience first.

It is important for a site related to company to provide information that a customer needs and is interested in and not the information for investors or any other. So, we need to be clear with the answers for the questions such as who will be visiting our site and what for, what might be their underlying motivation etc.

3. Defining Our Sites Content

After knowing the goals of our site, the aimed visitors and how often the contents is to be changed. The programmer need to decide the actual content of the site. The content should be what the visitors expect from our site, not just what we want them to see. It should include a detailed description about the product and services, time taken to complete a service and who will perform it etc.

User need to observe other sites which address the similar topic and look what they do and don't do, so that user can present in different way and stand better. We should add all the important things from our logo and branding to copyright notice, privacy policy etc.

4. Grouping and Categorization

All the ideas covered by user now be grouped together. If the site is about advertising several products and services then. They all can be grouped and placed together and can even be divided into subgroups. The information about the company, its history etc., can be placed in about us section. In this way all the related information can be grouped and categorized.

On the whole there should not be more than six or seven sections and these form primary or global navigation items of our site. There might be subsection with several pages of their own and more than seven subsections in each category. These form secondary or category navigation.

These categories and subcategories are like table of contents and form the basis of navigation of our site. These should take part in main menu. The customers should be able to easily navigate and understand our products services or subject.

5. Creating a Site Map

Once you have got idea about which sections and pages are needed to add to your site, then you can start up drawing a site map as you imagined. It should be like a family tree or folder in windows explorer. If at all the categories have subcategories or more than one page, they should be shown as the subcategories or sub pages of the first page. The site should start-up with the home page with all the required main categories at the top of the page. Example if 'department' is one of the main category then, each of the department in this category will be the subsections and each of them can be displayed in separate pages. These subsections can also have subsections in addition such as,

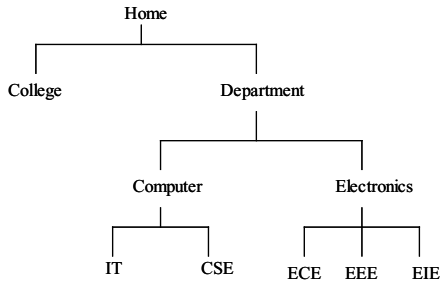


Figure: Site Map

Once you have drawn the site map, you will know about the number of pages on your site, the information to be displayed on each page and the links to be appeared on each page. Finally, you need to observe that, does your site contain all the information for which you expect the users to visit your site and how the user's should navigate through the site for the information they required.

6. Identification of Key Elements for Each Page

The last step that has to be performed before starting to design a page is to identify what are the key elements that should appear on each page. The elements such as branding or logos, primary navigation, category navigation, heading, main content, search box and space for self promotions advertising other companies, products or services will be included in these elements.

These elements will be reflecting the aim of the size. A site must be less cluttered for it to be easily navigation. For this, everything should not be included in each page we need to have good design and simple navigation and avoid putting everything on each page for the users to find what they want on the page easily.

2.4.2 Page Size

Q39. How the size of the page can be decided before designing?

Answer :

Before designing the pages, the designer need to understand what all is needed to be done to design a site and what information should be contained on each page. The first step of it would be to decide the size of the page to be used for the site. After deciding the page size, the page may not look alike on all the computers because different visitors will have different size monitors with different resolutions. So, while deciding the page size the designer need to consider the aspects such as different size monitors with different resolutions and people may also prefer to browse the small screen which shows even toolbars and other applications as a part of it.

While designing and building our site, the designer need to use sensible dimensions for a page otherwise he would end up with rebuilding the whole site once again when the page does not fit on all monitor screen's. The pixels that make up a picture are the subject matter of screen resolution. For example, if a screen has 1280 × 1024 resolution, it means that the screen is 1280 pixels width and 1024 pixels height.

The below table shows the percentage of visitors www.xyz.com who have different screen resolutions.

Resolution	Jan.-2008	Jan.-2007	Jan.-2006	Jan.-2005	Jan.-2004
640 × 480	0%	0%	0%	0%	1%
800 × 600	8%	13%	21%	28%	37%
1024 × 768	48%	53%	58%	54%	49%
1152 × 864	3%	3%	3%	3%	3%
1280 × 1024	28%	22%	12%	10%	6%

Table: Statistics of Screen Resolution

In the above table, one can observe that, in Jan 2008, 8% of users had 800 × 600 resolution and almost 90% of users had 1024×768 or higher resolution. So, designer need to decide the width of the page by considering all the users with all the screen resolutions. The design should be usable on 21-inch monitor at 1280 × 1024 resolution as well as on 15-inch monitor at 800 × 600 resolution. The designer should consider different toolbars which appear in the window because most of the users will be having menu or toolbars as a part of their windows horizontally. Users must not scroll horizontally rather they should scroll vertically. And all the main parts of the page should appear within top 500 pixels of page so that the user can easily understand what is the page about without scrolling.

Deciding the Width of a Page

Generally, the designer does not want the page to be scrolled horizontally that is from left to right to view the content. Therefore, set the width of a page to 1024 pixels that fits to the screen. The designer should consider different toolbars which appear in a window because most of the users will be having menu or toolbars as a part of their windows horizontally. Therefore, the width of a page for the information will be between 960 to 980 pixels.

Deciding the Height of a Page

The designers expect that the user should scroll vertically rather than to scroll horizontally to view the information that they required. Note that, after loading a page the user should know about the type of information to be displayed in a page without scrolling down. The designer should consider different toolbars that appear in a window because most of the users will have menu or a taskbar. Therefore, on an average the height of a page for content will be in between 570-600 pixels.

Q40. Explain fixed-width design.

Answer :

The web pages whose dimensions such as pixels, ems and centimeters are fixed to some value are called fixed-width design . The designers have the complete control on the appearance of page because they are one who decide and know the size of the page. They cannot be stretched or shrinked or resized by the users. The proportions of the page remain same but the design may appear in different size on different resolution.

When the width of the page is fixed it remains same, how much ever large the user's browser window might be. If the user's window is smaller than the layout, then horizontal scroll bars appear on the screen and if the user's window is larger than our layout then there will be gaps at left and right sides of the page. The below figure shows a fixed width layout.

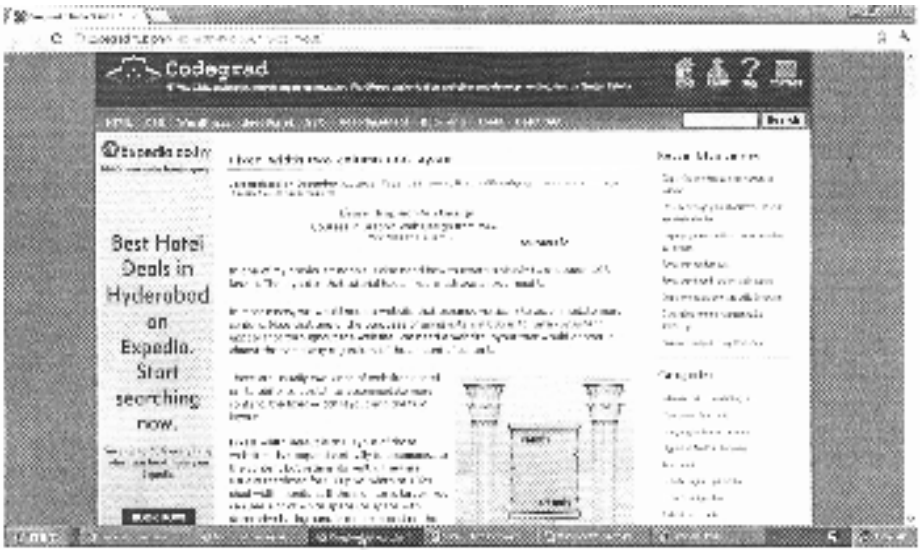


Figure: Fixed-width Layout

The html code for the above web page will be,

```
<body>
    <div  id = "fixed-width layout">
        <!--content of the page -->
    </div>
</body>
```

Here, the element hold the page by carrying the id attribute whose value will be the page. The CSS for the above page is as follows,

```
div#page
{
    width : 800px;
    margin-left : auto;
    margin-right : auto;
    background-color : red;
    border : 1px solid #666666;
    padding : 20px;
    font-size : 12px;
    text-align : left;
}
```

The width attribute in the above example is most often assigned the value in pixels. Here, it is assigned 800 pixels to make the page 800 pixels wide no matter the browser may be more or less than 800 pixels. The margin-left and margin-right are assigned the values auto which indicate the page is centered in the browser. When dimensions of the page is fixed to between 700 and 1000 pixels then the page can be used only on desktop computers or on laptop. The users with small-screen devices such as PDAs or mobile phones or set-top boxes cannot access because their resolutions are less than the computer screens.

Q41. Explain liquid design.

Answer :

The web page whose dimensions are not fixed is called liquid design. These pages are stretchable and skewable upto the browser window. If browser size is increased the page size also increases and if the browser size is decreased the page size also decreases with it. The below figure shows the liquid design,

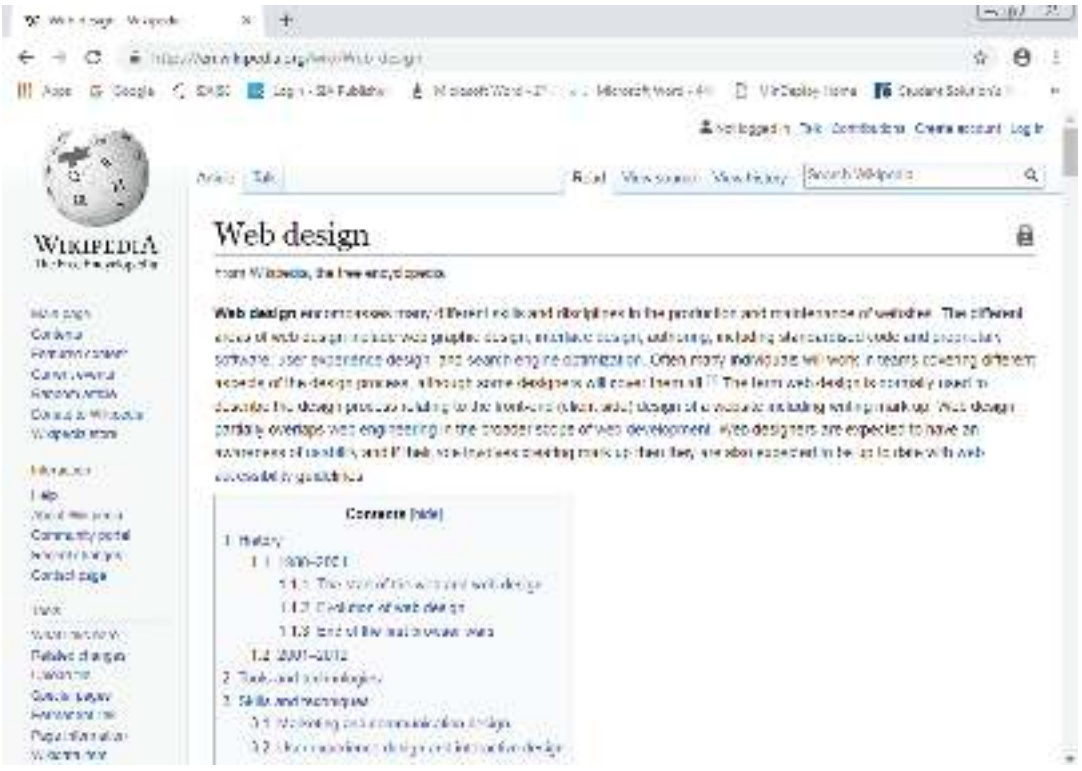


Figure: Liquid Design

The html code for the above web page would be like this,

```
<body>
    <div id = "liquid design">
        <!-- content of the page -->
    </div>
</body>
```

The CSS for the above page would be,

```
div. page
{
    width : 95%
    background-color : red;
    border : 1px solid #222222;
    padding : 20px;
    font-size : 10px;
}
```

Here, the width property is set to value 95%, that means the page occupies up to 95% of the window leaving the gap of 5% of the window around it. The page size increases and decreases along with the window size retaining the border around it.

Q42. Differentiate between fixed-width design and liquid design.

Answer :

	Fixed-width Design		Liquid Design
1.	The design of webpage whose dimensions are fixed is known as fixed-width design.	1.	The design of webpage whose dimensions are not fixed is known as liquid design.
2.	In this, the designers can control over the view of a page.	2.	In this, the designers cannot control over the view of a page.
3.	It can not be stretched or shrinked or resized by the user.	3.	It can be stretched and skewed to the browser window.
4.	The size of the page remains same even if the window size has changed.	4.	The size of page varies with the change in browser's window size.
5.	It is used in the sites that consists of more text.	5.	It is used in sites that features many lists and online tools or applications.
6.	**Example** http://new.bbc.co.uk	6.	**Example** www.maps.google.com www.amazon.com

Q43. Give the advantages and disadvantages of fixed width design and liquid design.

Answer :

Advantages of Fixed-width Design

1. The line length of the text can be controlled without bothering about the size of user's window.
2. The page size remains the same.
3. The designer has full control on the position and appearance of items on the page.
4. The width and position of elements can be controlled exactly by pixel values.

Disadvantages of Fixed-width Design

1. The text may not fit in the space allotted to it, if its size is set to large.
2. The web page might look smaller if browsed at higher resolution than it is designed for and page elements will not be clear enough to read.
3. Page must be in the middle of the window leaving gaps around it sometimes.
4. The container elements which are used to control the page layout might remain at the end of the code. By this the page becomes cluttered and can easily break.
5. This design is applicable only on the devices whose size and screen resolution is same as for desktop computers.

Advantages of Liquid Design

1. The page expands and fills the browser window without leaving any gaps around it.
2. The users are allowed to set font size lager than intended because page layout is stretchable.
3. The page can even fit in a small size window, there is no need for scrolling it.

Disadvantages of Liquid Design

1. The width of the sections of the page need to be controlled, otherwise the design looks different or might display gaps around elements or elements may squash together.
2. If the window is very wide, then the lines of text become long making it hard to read. And if the window is too narrow, then each line might have a word or two.

2.4.3 Designing Pages

Q44. Discuss in brief about the composition and grids.

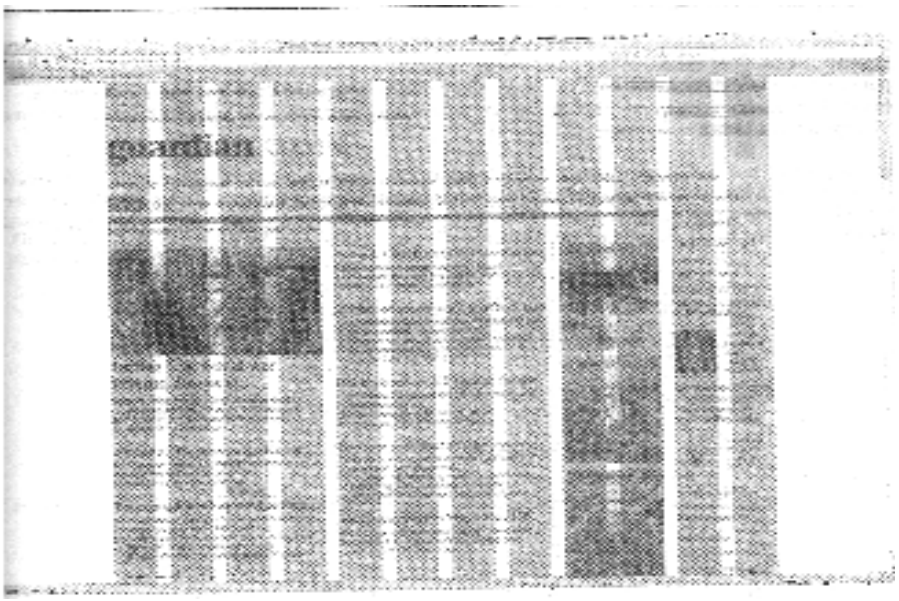
Answer : Model Paper-II, Q10(b)

Composition and Grids

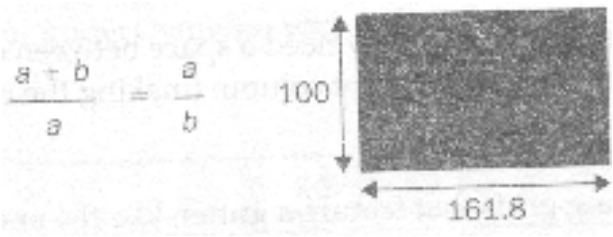
Composition in a visual art will organize the visual elements. A grid is used to arrange or organize these visual elements on a web page to be displayed. The famous UK news paper website is as shown in below figure,



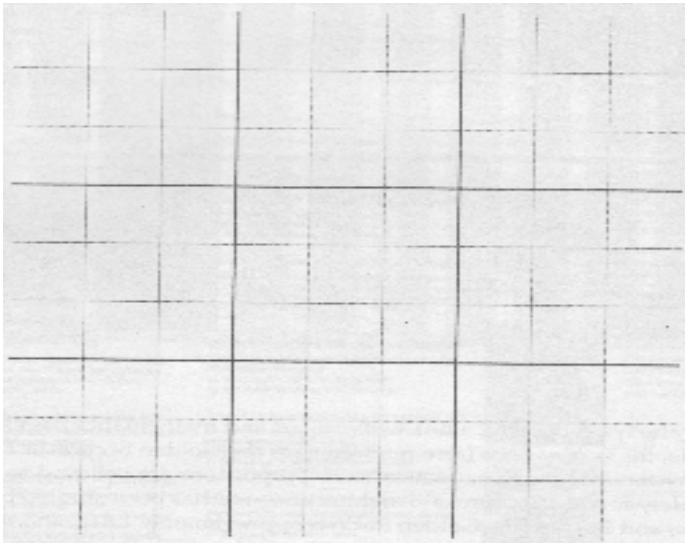
The following figure with the set of super imposed lines indicates how the site is designed using the grid system.



Since the main objective is to create a good design, there exists a link between mathematics and art for many centuries. For instance, many artists make use of Golden Ratio because of renaissance. The concept of golden ratio was studied by many mathematicians (Euclid and Pythagoras) because of its pleasing nature and architecture. The approximate value of golden ratio is 1.618 that states a+b is to a as a is to b. The designers creates some complex grids based upon the golden ratio (i.e., as shown in the below figure).

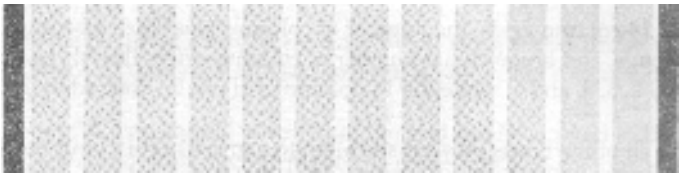


Drawing a grid on the basis of golden ratio becomes complicated. Therefore, use some other alternative method such as rule of thirds to draw grids i.e., divide the page into 3*3 equal parts and figure out the intersecting points. Suppose the width of a web page is 960 pixels then it is divided into three columns each of 320 pixels. And also vertical lines should be drawn for each 320 pixels width. This will create grid boxes which will be further divided into thirds.



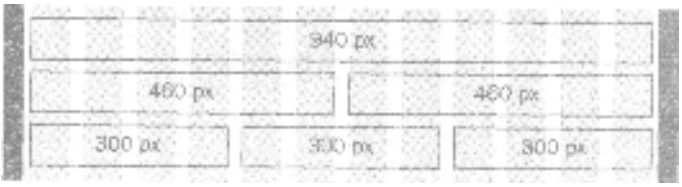
There should be some space between the columns in order to avoid the jumping of text or image from one column to another column. The space is called as "Gutter".

Consider a page of width 960 pixels. Make the page as a 12 column grid which is as shown in the below figure,

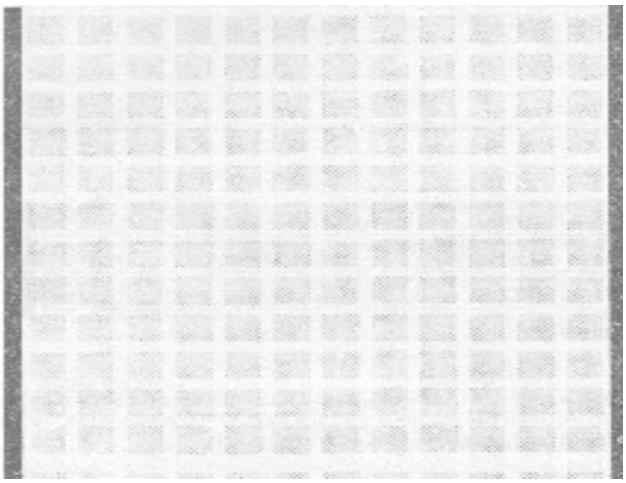


In the above figure, each column is of width 80 pixels and every column is separated by 20 pixels.

A grid is used to create two columns of equal size in layout or single column layout that is as shown in the below figure,



Designers might even make use of same units of measurement for vertical spaces to produce vertical lines. But this is not possible, so vertical measurements are used to divide by the width of single column. A 12 column grid in the form of vertical lines set with same distance is shown as follows,

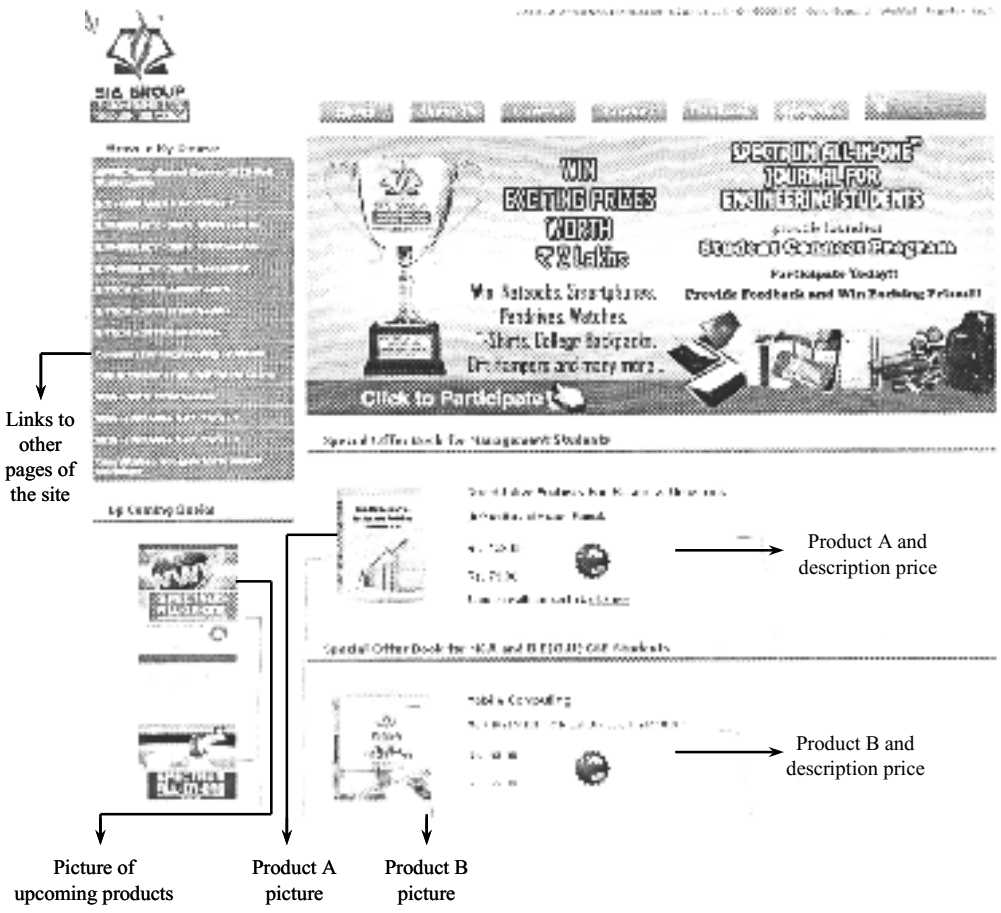


Q45. Explain the process of sketching the placement of elements.

Answer :

The designer must decide where the elements must be placed on page after knowing the number of pages, the pages to be linked, main elements of the page and type of layout to be used. This should be done before building the page. Leaving all other things such as colors, fonts, images or other design issues, designer need to sketch out where the elements should be placed and how much space they occupy.

Rather than focusing on visual presentation it is required to focus on whether every element needed by the user and the data with which the user can interact is included in the page. This process is called wireframing. The below figure shows the example of wireframe for a website.



Figure

After creating a wireframe model, designer need to check whether the end users are easily able to find for what they visited the site. With this he/she get to know where the links navigate from the site and get a good idea of functionality of the model. This is important to know because the users when visit the designed site prototype will focus on visual elements rather than on the proposed function. Thus, designer can ensure that the user or client is focusing on structure and function of the content rather than on its appearance by using skeletal model.

At this stage designer need to decide which elements will go on the top of the page. The mostly used functions should take higher priority as per the general rule. So, while designing the site designer need to think from visitors point of view to make the site successful.

Visitors mostly scan the pages to know whether they are useful to them or not. So, designer need to provide the elements to focus on by making the heading and subheading clear and concise. The items given below should be visible to users as soon as the page loads.

- ❖ Branding
- ❖ Global and subsection navigations
- ❖ Heading of the page
- ❖ Enough description about the page
- ❖ Promotions or advertising and
- ❖ Option to search.

The items given below need not be visible when the page loads,

- ❖ Details about the remaining page
- ❖ Advertising
- ❖ Unwanted links to related information or other sites
- ❖ Footer navigation such as copy right, terms and condition, privacy policies etc.

Q46. What are the default and common page elements used in designing?

Answer :

Default Page Elements

While designing a website or page for companies, remember that easily identifiable style will make the brand stronger. So, by using default page elements such as logo and colors in the site designing, the users can easily identify brand of the site. The designer should ask the company for the digital copy of the logo instead of scanning it from brochures. The electronic versions of the ad's or leaflets can also be used. The color scheme of the site can feature the company colors. To make the site look professional some of the materials such as photographs products or work done for any of the customers along with any text can also be used if the company is able to provide.

Common Page Elements

There should be some degree of consistency maintained across all pages of a site. This helps the users to identify the site from its look. So, elements which appear on every page starting from branding and primary navigation need to be checked. If these are placed in the same place on every page, the users can learn how to use the site more quickly and easily.

If at all our site dedicates a page to each product or service that we offer, or article or story then each of them should follow the consistent design. Their heading, logo, description, header, footer and primary navigation should be placed in the same place across all the pages.

Q47. Which of the page elements are grouped together?

Answer :

A web page contains several related elements. All the related elements are grouped together. Example, elements of a login form or details of a product are grouped together. The below methods help to group the elements and make them clear that which elements are related to each other.

1. **Location**

   Similar element must be placed in same location or nearer to each other.

2. **Color**

   The foreground and background color of the related elements can be selected as same so that they can be identified easily.

3. **Styles**

   The interrelated items can be applied with same styles. Such as using similar buttons for navigation items.

4. **Borders and Padding**

   The groups can be separated by creating a gap between them to identify the groups of related elements easily.

5. **Headings**

   By using headings, the user can understand the type of content available under it.

Q48. Briefly explain how a home page must be designed.

Answer :

While designing a site, the front page should be in such as that it impresses because it is the page displayed on the first visit of site. The main purpose of the site should be shown in the front page itself such as by using a one-sentence tagline or a logo.

All the tasks for which that user's will most likely visit the site need to be emphasized. And also the information for which the users visit the site should be made possible to find as soon as possible. The links names, images and headings should begin with important keywords so that they make the user understand their purpose and what are they about.

For example, a website of a company should cover the details of its product and about company's marketing department. If the users doesn't find the information they need then the site will not have expected level of visitors to their site. Not only this, the site must also address the needs of the visitors.

Q49. How a content page should display its content?

Answer :

Most of the sites such as news sites and e-commerce sites use the content pages for displaying their articles, product details etc. A content page should display its content in an easily understandable way for the users. The information should be placed or displayed in the content pages in the following ways.

- ❖ The information provided must be consistent for each item when there are several products or services.
- ❖ Equivalent information of the similar products should be placed in all the pages. For this it is enough to design only one article page or product page and other pages will follow that template.
- ❖ When lot of information is available to be displayed in the same page, then present it in a clean manner so that users to focus on the content. It would be better to maintain gaps between different elements, make the text to be split into appropriate subheadings and short paragraphs because wide lines of text are difficult to be read.
- ❖ Images that are used should be relevant for the product, service or topic in the question. It would be better if they are placed at left or right side of the page with text around them. There should be enough space between different images and text around them.
- ❖ The page need to be action oriented if they are displaying product information and also allow the customer to locate or select the item, to find required information about a product and also buy lit.
- ❖ Content pages should avoid using unrelated text and images which make users distracted from knowing the actual purpose of the page.

Q50. What influence does images have on design?

Answer :

The images that are incorporated in sites have a strong influence on the visitors. Use of good images, logos and graphics in the sites attract most of the visitors towards it. If the front page has a poor photo then the users does to visit that site how much ever good the content might be.

The photographs that are shot especially for that company which represent the product or company or life-style of company are featured in the site mostly. The quality of the images depend upon the budget for the site. If clients already have the photos then they are used or else if the hired clients budget is large then photographers are being hired in order to take appropriate shots. Most of the multinational companies and popular brands are hiring photographers to take impressive photos and their sites contain impressive graphics.

There are number of stock photo sites from which the appropriate images can be obtained but they need to be paid.

Q51. What are the different tactics used to make the site easier to scan?

Answer :

The different tactics used to make the site easier to scan are as follows,

1. Use Subheadings

In some cases, it is difficult for the user to scan the lengthy text. So, divided the text into descriptive subheadings. Note that, while splitting the text the meaning should not be broken.

2. Text should be Concise in Some Specific Headings

Provide the heading in such a way that user should understand the text provided by it.

3. Use Bold and Other Styles

Highlight the important keywords so that it is helpful for the users. This avoid users from searching for the key terms, using which they can decide whether the page is useful to them or not.

4. Stimulate Users with Visuals

Provide the visuals such as images or videos to provide better understanding of the concept.

5. Make Links Descriptive

Provide the description for the text links then only user will get to know what will be the content when the link is opened.

2.4.4 Coding Your Design

Q52. What are the different properties that are used to position the items on the page. Discuss in detail about IE Box Model Bug.

Answer :

Properties that used for Position the Items on the Page

The different properties that are used to position the items on the page are as follows,

1. The XHTML Property

It is used to group the items of the XHTML pages to chunks or blocks.

2. The CSS margin Property

It is used to separate the blocks.

3. The CSS width Property

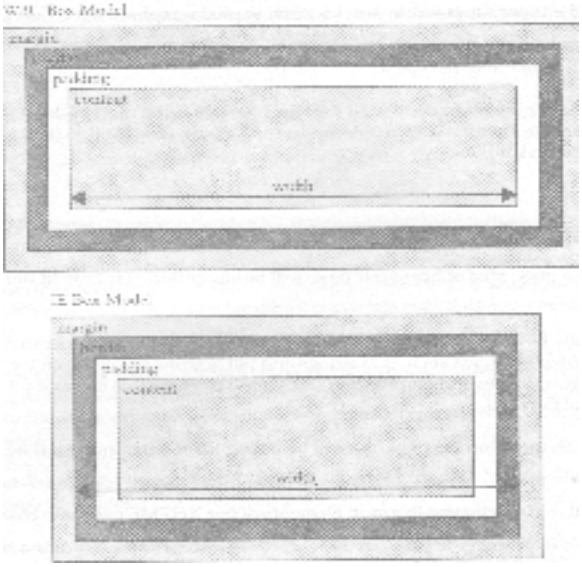
It is used to set the width of the box.

4. The CSS float Property

It is used to align the elements to the left or right of the page.

The IE Box Model Bug

The programmer must use DOCTYPE in the XHTML pages to represent them correctly in internet explorer. If the DOCTYPE is not declared then the internet explorer will apply the different versions of the box model to the remaining browsers. With this the pages will look bigger in IE than in other browsers, this is shown in below figure,



The following are the two different declarations that make the IE version 6 and later versions to render the page correctly.

- **Transactional XHTML DOCTYPE**

  < !DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitionalt//EN"

  "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd" >

- **Strict XHTML DOCTYPE**

  < !DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitionalt//EN"

  "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd" >

Q53. Explain the process of working with grid.

Answer :

A grid can be added to the background of the page by using CSS background-image property when the page is designed using the fixed width design. This will help programmer to confirm the items position. For this, a 1-pixel height image is required to be repeated vertically on the grid.

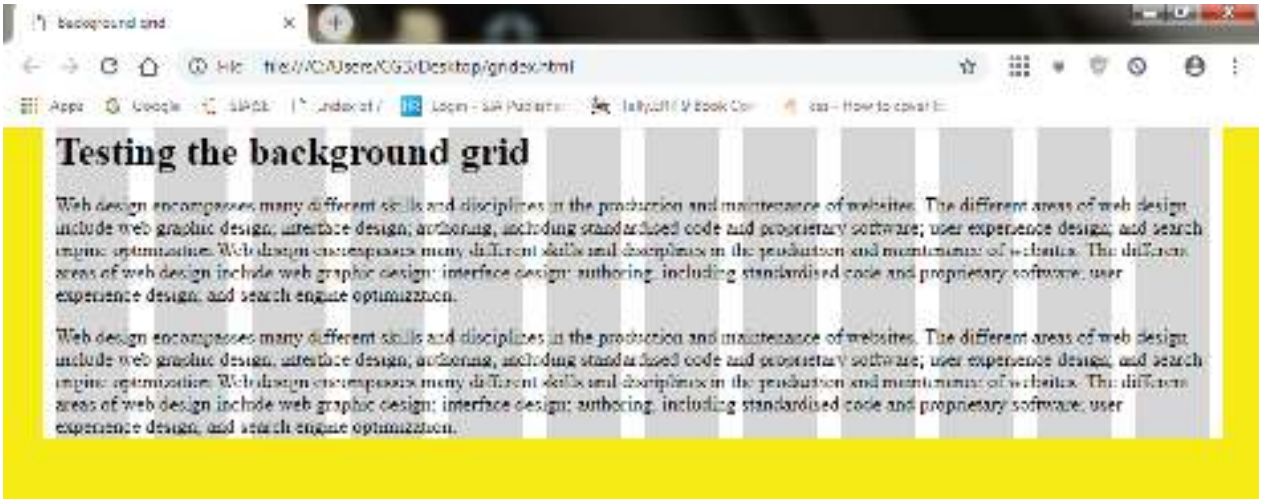
Example

```
<?xml version= "1.0" ?>
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitionalt//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns= "http://www.w3.org/1999/xhtml" lang= "en" xml:lang= "en">
<head>
<title>Background grid</title>
<link rel= "stylesheet" type= "text/css" href= "s6.css" />
</head>
<body>
<div id= "frame">
<div id= "page">
<h1>Testing the background grid</h1>
<p>Web design encompasses many different skills and disciplines in the production and maintenance of websites. The different areas of web design include web graphic design; interface design; authoring, including standardised code and proprietary software; user experience design; and search engine optimization. Web design encompasses many different skills and disciplines in the production and maintenance of websites. The different areas of web design include web graphic design; interface design; authoring, including standardised code and proprietary software; user experience design; and search engine optimization.</p>
<p>Web design encompasses many different skills and disciplines in the production and maintenance of websites. The different areas of web design include web graphic design; interface design; authoring, including standardised code and proprietary software; user experience design; and search engine optimization. Web design encompasses many different skills and disciplines in the production and maintenance of websites. The different areas of web design include web graphic design; interface design; authoring, including standardised code and proprietary software; user experience design; and search engine optimization. </p>
</div>
</div>
</body>
</html>
```

s6.css

```
body {
background-color:yellow;
margin:0px;
padding:0px;}
h1 {
margin:0px;}
#frame {
width:960px;
margin-left:auto;
margin-right:auto;
background-image:url("image.jpg");
background-repeat:repeat-y;}
#page {
width:940px;
margin-left:auto;
margin-right:auto;}
```

Output



In the above example, two items are used. First one is the id attribute with the frame value (width of the grid) and the second one is the id attribute with page value.

Q54. Explain the multi-column layouts with examples.

Answer :

A layout that contains multiple layouts is called multi-column layout. Examples of it are two column layout, four column layout etc.

Two-column Layout

A two-column layout contains two columns with multiple rows in a page. It contains heading that spans for both the columns. It contains a navigation bar below the header and footer at the bottom of the page.

Example

ex4.css

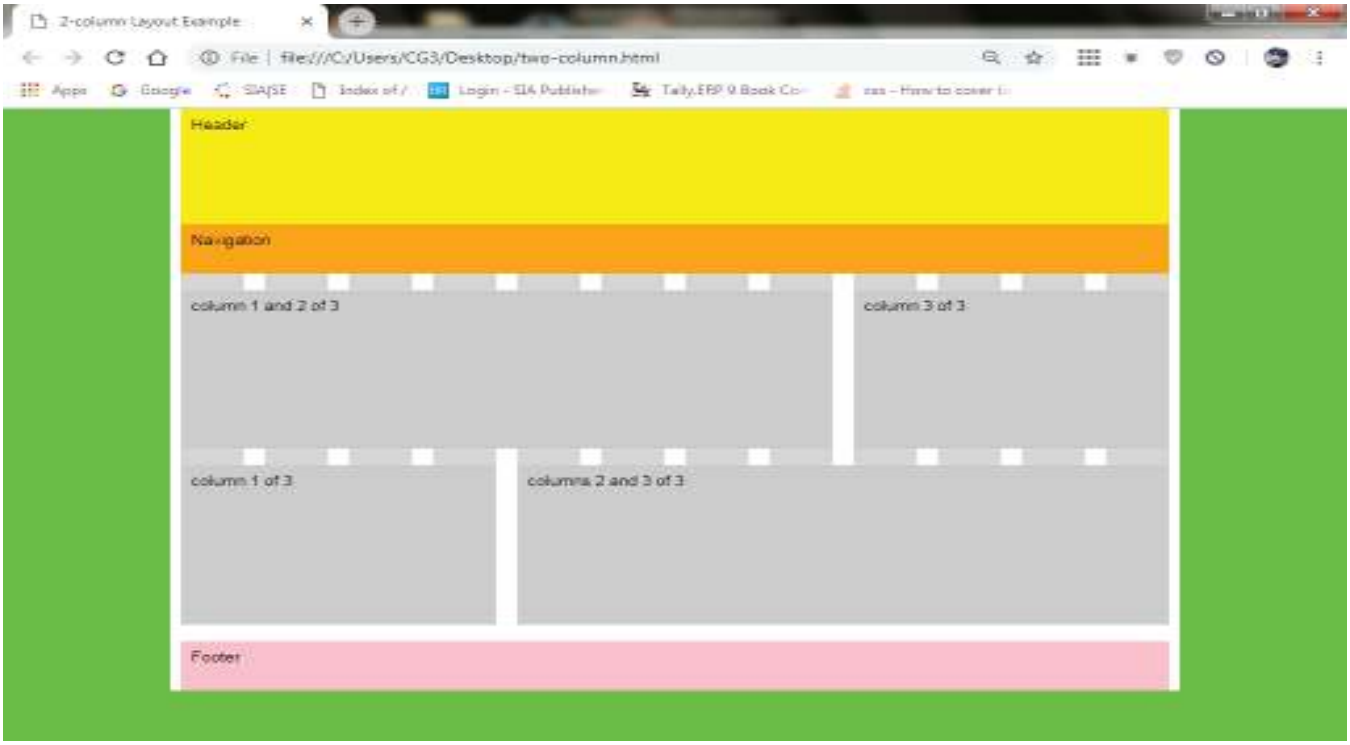
```
body {
margin:0px;
background-color:lime;
font-family:arial, verdana, sans-serif;
text-align:center;}
#frame{
margin-left:auto;
margin-right:auto;
text-align:left;
width:960px;
background-image:url("image.jpg");
background-repeat:repeat-y;}
#page {
margin:0px 10px 10px 10px;}
#header {
background-color:yellow;
padding:10px;
height:120px;}
#navigation {
background-color:orange;
padding:10px;
height:40px;}
#footer {
background-color:pink;
```

```
padding:10px;
height:40px;
clear:both;
border-top:20px solid #ffffff;}
/* 3 columns */
.column1of3, .column2of3, .column3of3 {
float:left;
width:280px;
background-color:#cccccc;
padding:10px;
margin-top:20px;
height:173px;}
.column1of3, .column2of3, .columns1and2of3 {
margin-right:20px;}
.columns1and2of3, .columns2and3of3 {
float:left;
width:600px;
background-color:#cccccc;
padding:10px;
margin-top:20px;
height:173px;}
```

two-column.html:

```
<html>
<head>
<title>2-column Layout Example</title>
<link rel= “stylesheet” type= “text/css” href= “ex4.css” />
</head>
<body>
<div id= “frame”>
<div id= “page”>
<div id= “header”>Header</div>
<div id= “navigation”>Navigation</div>
<div class= “columns1and2of3”>column 1 and 2 of 3</div>
<div class= “column3of3”>column 3 of 3</div>
<div class= “column1of3”>column 1 of 3</div>
<div class= “columns2and3of3”>columns 2 and 3 of 3</div>
<div id= “footer”>Footer</div>
</div>
 </div>
 </body>
</html>
```

Output



Four-column Layout

A four-column layout contains four columns with multiple rows in each of them. It contains heading that might span for all the four columns. It contains a navigation bar below the header and footer at the bottom of the page.

Example

ex5.css

```
body {
margin:0px;
background-color:lime;
font-family:arial, verdana, sans-serif;}
#frame {
margin-left:auto;
margin-right:auto;
width:960px;
background-image:url("image.jpg");
background-repeat:repeat-y;}
#page {
margin:0px 10px 10px 10px;}
#header {
background-color:yellow;
padding:10px;
height:120px;}
#navigation {
background-color:orange;
padding:10px;
height:40px;}
#footer {
background-color:pink;
padding:10px;
height:40px;
clear:both;
border-top:20px solid #ffffff;}
/* 4 columns */
```

```
.column1of4, .column2of4, .column3of4, .column4of4 {
float:left;
width:200px;
background-color:#cccccc;
padding:10px;
margin-top:20px;
height:175px;}
.columns1and2of4 {
float:left;
width:440px;
background-color:#cccccc;
padding:10px;
margin-top:20px;
height:175px;}
.columns2and3and4of4 {
float:left;
width:680px;
background-color:#cccccc;
padding:10px;
margin-top:20px;
height:175px;}
.column1of4, .column2of4, .column3of4, .columns1and2of4 {
margin-right:20px;}
```

four-column.html

```
<html>
<head>
<title>4-column Layout Example</title>
<link rel= “stylesheet” type= “text/css” href= “ex5.css” />
</head>
<body>
<div id=“frame”>
<div id= “page”>
<div id= “header”>Header</div>
<div id= “navigation”>Navigation</div>
<div class= “column1of4”>column 1 of 4</div>
<div class= “column2of4”>column 2 of 4</div>
<div class= “column3of4”>column 3 of 4</div>
<div class= “column4of4”>column 4 of 4</div>
<div class= “columns1and2of4”>columns 1 and 2 of 4</div>
<div class= “column3of4”>column 3 of 4</div>
<div class= “column4of4”>column 4 of 4</div>
<div class= “column1of4”>column 1 of 4</div>
<div class= “columns2and3and4of4”>columns 2 and 3 and 4 of 4</div>
<div id= “footer”>Footer</div>
</div>
</div>
</body>
</html>
```

Output



2.4.5 Developing for Mobile Devices

Q55. What are the important aspects about the usage of web on mobile?

Answer :

Most of the users think that the use of internet on mobile phones is not much relevant to them or their company. But, by considering the following observations user may change their decisions.

- ❖ The success rate of certain high profile mobile devices such as iPhone and Blackberry.
- ❖ The statistics that represent number of users who are using the web or internet on mobile devices.

The British government has conducted a survey in 2009. It says that the 11% of the UK web users are using web through mobiles. This is equal to the 1/5th of the web users over the world.

Understanding the Medium and How it is Used

The main aspects to use the web on mobiles are as follows,

1. Context: What is the mobile user doing?

People use their mobile phone while moving from one place to another when they are in hurry. And they can be used with one hand. Using mobile, the user can get the same result as the personal computer. One more advantage is mobile phones are portable devices. Mobile phones are also used to perform some specific things such as finding nearby places, weather report, latest news regarding sports, politics and etc.

2. Relationship: How do users treat their mobile phones?

The relationship between the user and the mobile phones is personal. Most of the users have their own mobile phones rather than having a personal computer (Desktop computer). Desktop computer is shared between the friends and family where the mobile phones are treated as personal. A user can carry mobiles phones everywhere 24 by 7.

3. Capabilities: How do features of mobiles compare with desktop PCs?

The comparison between the features of mobiles and desktop PCs are as follows,

Feature	Mobile Phones	Desktop PCs
Location Specific Services	It is provided with the GPS or triangulation to specify the user location.	The Desktop PCs don't have any GPS system.
Built-in Camera	They are provided with Built-in camera.	These are not provided with Built-in cameras. User can connect the web cameras whenever required.
Built-in Payment System	The mobile phones are provided with the payment system where a user can have debit/credit card details.	These PCs also have the payment system but most of the users prefer mobile phone payment systems.
Voice Data	Mobile phones are integrated with voice data. They allow phone calls.	Desktop PCs allows phone calls through skype.
Other Features	Mobile phones are provided with the services like SMS, MMS and etc.	PCs doesn't provided with the SMS & MMS services.

Limitations of Mobile Phones Over Desktop Computers

Some of the limitations of mobile phones over desktop computers are as follows,

- **Small Screens**
  The screens of mobile phones are very small compared to desktop computers.
- **Connection may be Isolated**
  When a user is moving from one place to another, the connection may be isolated sometimes because of the weak signals. And some rural areas don't have a proper connection.
- **Slower Connection Speeds**
  The speed of mobile networks is slow compared to the other broadband services.
- **Less Control over the Appearance**
  Previously, most of the phones are introduced with the default fonts, font size, and images to be displayed on the mobile screen. Therefore, it is not users choice.
- **Input Devices are Harder to Use**
  Even though the mobile users are highly skilled in using mobile, they cannot use the keyboard and mouse with the same accuracy and speed.

Q56. Explain how to approach the mobile development.

Answer :

The development of mobiles will depend on the mobile strategies and the resources available to the manufacturer company. The different approaches for the mobile development are mainly categorized into three types. They are,

1. Do not Develop a Mobile Site

The simple and cheapest way for the mobile development is 'do nothing'. Most of the mobile phones provide access to the content of the site. Some of the famous mobile phones like Blackberry provides a resized website to fit to the screen and it permits the user to zoom into the content of the pages.

Drawback

This approach provides the content of the site which does not show the main aim of the implementation to the users or visitors.

2. Provide an alternative View of your Site

The middle ground is used to re-use the information that is available. But, provide the information in a simple, easy and user understandable way. The two common approaches in middle ground are as follows,

(i) Add a secondary style sheet to the same pages. Therefore, the mobile devices can make use of different set of CSS rules to control the flow of the pages that appear to the users. This type of approach helps the user to use a page with different dimensions, font styles, font size and layout.

Example

<link href= "ex1.css" rel= "stylesheet" type= "text/css" media= "handled">

This approach cannot be applicable to all the devices because all the mobile phones are not familiar to use this version of the style sheet. The mobile phones that are familiar are required to download the same file i.e., created for the desktop computers which may take long time to download the file.

(ii) Use various set of templates when the visitors visit the site through a mobile phone. Here, some of the famous companies use the same content with different designs rather than using a latest style sheet.

Drawback

These approaches provide the content of the site which doesn't show the main aim of the implementation to the users or visitors.

3. Create a Separate Mobile that is Optimized with its own Content

If the user has all the resources for the development of a site that is useful to the mobile users then it clearly explains the aim of the site to the visitors.

Example

http://m.facebook.com/ and http://m.youtube.com/.

These two are mainly designed for the mobiles users only.

Q57. What are the different key based on which the creation of site for mobile devices is decided?

Answer :

Domain Name

The most frequently asked query about the mobile site development is whether a top-level domain is required for the site or not. This is because the top-level domain called .mobi is designed especially only for the mobile devices. In recent times, most of the companies acquired different addresses like mobile.domainname.com, m.domainname.com, and domain-name.com/mobile.

Example

m.facebook.com, or youtube.com/mobile etc.

Device Detection

Device detection is used to find whether a user is on personal computer or on mobile phone and provides the suitable site to the user.

The two URLs http://wurfl.sourceforge.net or www.andymoore.info are used to make a good start to the development.

Provide a link of the complete website to the user only when mobile protection is provided to them and direct users to the mobile version of the site. If the user is satisfied with this link then you can target the same site to the user in future.

Provide the link of mobile version of the site the main site so that, users may get to know about the mobile site and have chances of using.

Languages for Creating Mobile Web Sites

Most of the mobile phones are able to display the basic CSS and XHTML. But some of them doesnot display the accurate results as expected. Some devices display the site styles and font in different ways. Technically, most of mobiles support two subsets of XHTML i.e., XHTML basic and XHTML-mobile profile. The DOCTYPE declaration for the XHTML-profile is as follows,

<!DOCTYPE html PUBLIC "-//WAPFORUM//DTD XHTML Mobile 1.0//EN"

"http://www.wapforum.org/DTD/xhtml-mobile10.dtd" >

Design Tips

The main aim of the mobile browsers is to provide the content in such a way that the user can understand easily and complete their tasks quickly. Therefore, the design of the site should be good. The different tips to be followed in the design of a mobile site are as follows,

(a) Content

Some of the tips to be followed while providing the content in a page are as follows,

1. Provide a title to the page using which user should easily understand the type of information the page consists of.
2. Use simple semantic markup like paragraphs in <p> elements, headings in <h> elements and etc.
3. Use simple, clear and user understandable language.
4. User should be able to download the images in short time.

(b) Color

Some of the mobiles are not provided with good screen resolution and contrasts, so follow the below tips at the time of designing to provide good screen resolution and contrasts.

1. Try not to use background colors.
2. Try to provide enough contrast in the design.

(c) Navigation

Navigation is also one of the important concept in designing a page.

1. Designers should provide the main navigation link in the main page only. If not provide the minimum number of options.
2. Use the access keys for accessing links.
3. Designers should provide the home link and the back link at the end of each page for easy navigation. Ensure that the links are not repeated.

(d) Links

Rules to be followed while providing the links in the page are as follows,

1. Provide the links in blue color and remaining text in black color. Try to ignore the other colors to avoid confusion.
2. Some of the famous mobile phones like blackberry and iPhone automatically identify the mobile numbers so try to provide the numbers in <a> tag.
3. Mention the active link clearly because it is difficult to find the active link in mobile phones.

(e) Forms

In mobile devices filling a form is difficult. Therefore, ignore the forms as much as possible.

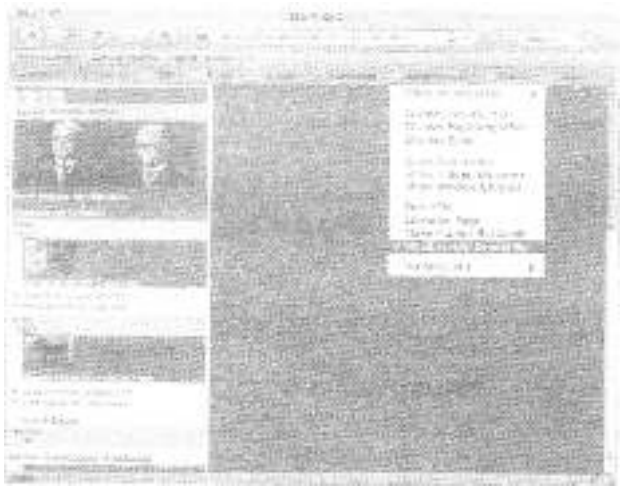
1. Use the links instead of using form controls.
2. Do not hide the password with dots(.) and asterisks(*) because in mobile devices the appearance of password does not make any problem.
3. Try to ignore the logins to make an efficient site.

Testing

Testing of a mobile site on all the different mobiles is not much easy because many types of phones with different versions available. The tester will choose some mobiles and test the mobile site in those mobile phones. But, it is possible to compare the view of site in your mobile and other mobiles. Some of the emulators that replicate about the abilities of the phone are as follows,

- iPhoney (www.marketcirce.com/iphoney).
- Nokia Mobile Browser Simulator (www.forum.nokia.com/)
- Blackberry (http://na.blackberry.com/eng/developers/resources/simulators.jsp)
- Opera Mini Simulator (www.operamini.com/)
- dotMobi emulator- Simulates Sony Ericsson K750 and Nokia N70 (http://mtld.mobi/emulator.php)
- Openwave phone simulator (http://developer.openwave.com/dvl/tools_and_sdk/phone_simulator).

Firefox web developer toolbar is another toolbar that permits users to test pages on small screen (https://addons.mozilla.org/en-US/firefox/addon/60). This type of feature is available from the miscellaneous menu, under the small screen rendering option as shown in the below figure,



2.5 DESIGN ISSUES

2.5.1 Typography

Q58. What is Typography? Discuss some of the issues while designing a page.

Answer :

Typography

Typography refers to an art and technique used in type design that contains decisions related to type faces, point size, line length, letter spacing, etc. That is it needs a lot of efforts to make the page easy to read, navigate and to provide the messages clearly. Some of the design issues while designing a page are as follows,

Line Length-A Measure

A measure is referred as the number of characters available in a single line of text. It counts the letters, punctuations and spaces also. Generally, the line length will be 45 to 75 characters and for a small column the length would be 30 to 50 characters.

Every line should end properly and the readers should have to understand the starting of the next line before the user lost the reading flow. If the length of the line is long then it is very difficult to go back and re-read.

Font Sizes

Font sizes allow the designer to design a page that looks like more balanced. Select the font size in pixel which follows scale of size such as,

```
body {font-size: 12px;}
h1 {font-size: 24px ;}
h2 {font-size: 18px ;}
h3 {font-size: 16px ;}
```

But specifying the font sizes in pixels is not allowed in internet explorer. Therefore, select the font size in em or percentages for a similar effect.

All the browsers take the default font size as 16 pixels. Thus, create the font sizes using this default feature.

```
body {  font-size: 75%; }      // 16*0.75=12
h1 {font-size: 200%; }         // 12*2=24
h2 {font-size: 150%; }         // 12*1.5=18
h3 {font-size: 133%; }         // 12*1.33=16
```

The font sizes can also be specified in em.

```
body {font-size: 100%; }
h1 {font-size: 0.75em; }       // 16*0.75=12
h2 {font-size: 1.5em; }        // 16*1.5=24
h3 {font-size: 1.3px; }        // 16*1.33=18
```

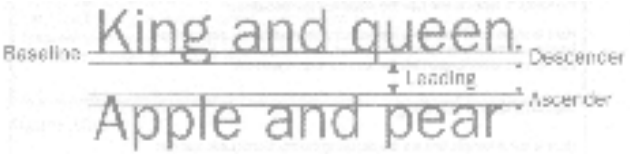
Choosing Fonts

Choose the fonts that are familiar to the users and install on their systems. Because, if the user don't have the font installed on their system then it is waste of providing font in that style. Some of the fonts that are used by many users are as follows,

- **Serif:** Times New Roman, Times, Georgia
- **Monospace:** Courier, Courier New
- **Sans-serif:** Helvetica, Verdana, Arial

Line Height - Leading

Leading is referred as the vertical space between the lines of text. The text that lies in the baseline and the height of the letters like o, u and v are collectively known as the x height. The parts of letters that will rise above the x height are known as ascenders and the parts of the letters that will drop below the x height are known as descenders. Leading (line height) is measured from the bottom line of the descenders to the top line of the ascenders that is shown clearly in the below figure,



CSS Resets

When the designer controls the designs to the pixel, the output may be changed from one browser to another browser. This can be achieved by providing a set of rules at the time of designing the style sheet (set the padding and margins of the elements to zero). The full version of different set of rules for this is Eric Meyer's CSS Reset and one can see that from http://meyerweb.com/eric/tools/css/reset. For example, a few lines that specifies a long list of elements and a set of handful of their properties is as follows,

html, body, div, span, applet, object, iframe, h1, h2, h3, h4, h5, h6, p, blockquote, pre, a, abbr, acronym, address, big, cite, code, del, dfn, em, font, img, ins, kbd, q, s, samp, small, strike, strong, sub, sup, tt, var, b, u, i, center, dl, dt, dd, ol, ul, li, fieldset, form, label, legend, table, caption, tbody, tfoot, thead, tr, th, td{

```
margin: 0;
padding: 0;
border: 0;
outline: 0;
font-size: 100%;
vertical-align: baseline;
background: transparent; }
body {line-height: 1; }
```

Include the above mentioned lines at the top of the style sheet or import them using @import or @include commands.

Line Height - Baseline Grids

The baseline grids are used to align the bottom of the main text to the grid. These are used by many typographers called as baseline grids.

This involves in the following steps,

- ❖ Reset the size of the padding and margins to the items by using some CSS Reset rules.
- ❖ Decide the line height of the body text i.e., the gap between the lines.

Here, the line height is 18 pixels and the distance between the text is 12 pixels and to make sure the text lines up I created a GIF with 1 pixel width by 18 pixels height which is as shown in the below figure,



Setting Headings on the Grid

To set the headings on the grid set the line-height property in CSS to an integer that can be divided equally into the height of the grid which is as shown in the below figure,



Practically, the height above the heading should be more than the height below the heading. This can be solved by using the following lines,

- ❖ Initially, divide the baseline grid size by the font size.

  For <h1> 18 ÷ 24 = 0.75

  For <h2> 18 ÷ 18 = 1

  For <h3> 18 ÷ 14 = 1.286

- ❖ Set the margin-top, margin-bottom and line-height to the resulting values.

```
p {
font-size:0.75em;          /* 16 x 0.75 = 12 */
line-height:1.5em;}        /* 12 x 1.5 = 18 */
h1 {
font-size:1.5em;           /* 16 x 1.5 = 24 */
line-height:0.75em;        /* 24 x 0.75 = 18 */
margin-top:1.2em;
margin-bottom:0.3em; }
h2 {
font-size:1.125em;         /* 16 x 1.125 = 18 */
line-height:1em;           /* 18 x 1 = 18 */
margin-top:1.7em;
margin-bottom:0.3em; }
h3 {
font-size:0.875em;         /* 16 x 0.875 = 14 */
line-height:1.286em;       /* 14 x 1.286 = 18 */
margin-top:2.172em;
margin-bottom:0.4em; }
```

Now, check the measurements in the browser and adjust them when required.

Aligning Other Items to the Grid

The two different ways to keep the text on a grid are as follows,

1. Set the line height as the body of the text.
2. Set the line height to 3/4th of the baseline grid height.

With the second approach add a margin at the top of the item to confirm that the very first line is on the grid.

The important points to be remember while aligning other items to the grid are as follows,

- The units of the padding and the border should be same then only the text appears on the grid even if the text size is changed.
- Padding and borders on boxes can throw items off-grid; if this is required then set the margin around the item within the box to confirm that the text appears on the grid.

Space between Letters - Tracking

Tracking is referred to as the space between the letters. In CSS, the letter-spacing property is used to control the space between the items on the page. The letter spacing is required when dealing with the headings that is in uppercase letters because reading the headings is little bit difficult than reading a sentence.

Example

```
h1.header_spacing {
letter-spacing: 0.1em;
}
```

White Space

White space or a negative space is referred as the space between the items in the page. It is used to specify the gaps between the items in a page. The two different white spaces are,

- **Micro White Space:** It is the space defined between the elements like headings, text and lists.
- **Macro White Space:** It is the space defined between the lengthy elements in a page.



The above figure explains the space between the large elements (Macro white space).

Consider the below figure that shows a page with some items that are separated with different spaces.



- On the left part, no element has margin and padding.
- In the middle part, the elements are with normal default margins with even spacing.
- On the right part, the headings are closer to the text.

Chunking

Chunking is the process of collecting independent blocks of information and grouping them together to produce a lengthy content. Consider the below figure that explains the use of white spaces that groups the related items.



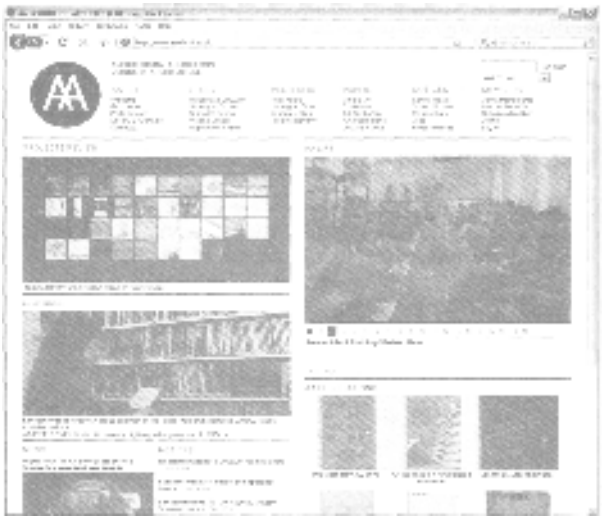
Another way to chunk the information is using a border on a box, or providing a box a background color. Consider the below example that uses a background color to group the information together.



Contrast

The contrast of a page should be good between the items and the background to provide better presentation to the users even when they have bad quality monitors or light focus on their screens.

Even if the contrast is high then the user may suffer. Therefore provide enough contrast. For instance consider the below figure,



Notice that the black color text on the white background color is not much easier to the user to read but the white color text on the black color background is good enough. In this examples, the contrast can be reduced by providing a dark gray color text on a white background rather than using black on white. In the same way, light gray on a black background color is easy to read.

In case if you are using an image as background then choose the low contrast image to read the page easily and clearly.

2.5.2 Navigation

Q59. What is the need of navigation? List the methods of navigation.

Answer :

Although site is planned in a perfect manner, the users will use the site in their own way. For example, a site contains various pages that can be visited from its home page, but the visitors might visit from some other page that is linked to other site links. The users visit according their needs. So, site must be designed in such a way that visitors find the required information easily and quickly. There are three methods of navigation using which the users navigate the site.

1. Users can navigate by using the menu that is provided.
2. Users can navigate by browsing through the links that are provided in between of text or any other parts of page rather than the menu.
3. Users can also navigate by searching for relevant items of information.

Q60. How menus helps the users to navigate your site? Explain.

Answer :

A keypart of any website that features more than one page is called a menu. A menu represents various sections of a site. It allows the users to get where they want to go on site. A site can contain more than one menu. It can have a primary navigation in one menu and secondary navigation in a submenu or in a separate menu. The guiding rules for the menus are as follows,

1. Menus Must Focus on What the Users Want to Achieves

For all the sites especially commercial sites, the main priority of the menus must be to satisfy the users satisfy the following,

- ❖ Provide the required information for users in the size.
- ❖ Provide this information in clear and concise manner.
- ❖ Priorities the item for which the site is mostly visited.

2. Menus must be Separated from the Content

A menu must be easily identifiable for the users to navigate the site. This can be achieved by the following techniques,

- ❖ By using different font size for the menu compared to main content
- ❖ By adding extra space around menu
- ❖ By placing the menu is a box with colored background or separating a menu from the main content with a time.
- ❖ By using images for the menus. But large images slow down the loading of a site.

3. Icons Used as Menus must be Understandable and Add a Link also

Many web sites use images called as icons for links as menus. For example, a magnifying glass indicates search features. Such type of images must be understandable if not the users may not use them. Mostly the icons such as a house for home page, envelope for email address, a question for help files are used. It would be better if a link is provided in words with the icons that are less common.

4. Menus must be Quickly and Easily Readable

The visitors usually do not read the pages while browsing, rather they scan them. So, the menus should be distinct and clear from the main part of the page so that users can easily scan and register the navigation.

The images used on a menu must be large enough and the text must contrast with its background with enough size. Their links also must be short and concise. For example, a link "Home" is easily readable and understandable.

5. Menu Items must be Logically Grouped

When many pages are available then use submenus. The menu items must be carefully grouped so that visitors can easily understand where to look for the required information with searching through several categories. The submenus must be easily distinguishable from the main menu and should be clear which items belong to which section. The submenus mostly appear in different color background, smaller font, an idented position, or an alternate color from the main menu.

6. Menu Items must be Easy to be Selected

A menu item should be in such a way that a user can easily select it. A menu item which is small or has less space between menu items or moving seems harder to select. And the menu items which are large are easier to select.

Make sure while creating the menus that they work on all the main browsers and avoid complex code in menus. The experimental types of menus that move and slide between the items as the user moves the mouse over an item need fine control of a pointing device such menus look great on experimental site.

7. Menus must Load Quickly

Every visitor to the site may not have a internet connection. So, menus must load quickly an any connection. Although speed of connection improves, a menu should load within a couple of seconds. If it takes longer than 8 seconds to load then users think that page is not loading or browser is frozen. The users might reload the page or go to another page.

Loading speed is very important for the menus that contain flash or graphics. The loading time doubles when the user want to change the image or appear highlighted when mouse is rolled over an image.

8. Menus must be Consistent Across a Site

A site with many pages requires a menu with many navigation items. When submenus are introduced the navigation becomes difficult and also varies from page to page. But it is important to maintain the primary navigation consistent across all the pages.

The submenus in each section must be in the same position on every page and also look similarly to help the users to know the exact place to navigate.

Q61. How a user can navigate your site using links? Explain.

Answer :

The web pages not only provide menus for the users to navigate the site, they also contain hyper links in the text which is in the body of the document. The users can also navigate using these links. These links are not the part of main menu. Two different types of links are as follows,

1. Text links
2. Image links.

1. Text Links

The text links will be represented in blue color with a underline by default. It is suggested by some of the experts that, it would be easy for the users to identify the links if they are left to their default appearance. Using a different color from the main menu for the links makes it easy to tell which text makes up the link.

In addition to this one can also change the appearance of the link as the user hovers the link, by changing their color, gaining a background color or the links getting underlined. Changing the shade of the links which the users have already visited reminds them where they have been earlier.

2. Image Links

Images can used as links in menus, advertisements, photo, graphical icons etc. Two attributes need to be used when an image is used as link. One attribute is 'alt' which tells the description of the image and another is the title of the link. In case if not using CSS, user need to add the border attribute for the images that are used as links.

Q62. How a user can navigate your site using search feature? Explain.

Answer :

A search feature allows the user to find the information that they need from our site by looking a keyword relating to the information. Hence, user can also navigate the site using the search feature. This features decreases the difficulty of the users in finding what they came for to the site and saves them from learning the navigation scheme.

Search Feature makes the Site more Usable

Search feature is important as our site grows. This feature can be implemented in many ways. User can add simple features in some ways and some methods need advanced programming experience.

In large commercial sites whose content is stored in the database, user can use programming commands such as queries to ask for the required information. User can even use an indexing application to index the site using which the search feature becomes available.

The sites which doesn't use databases or indexing tools can simply added a searched feature by creating a search box when the users enter their queries in the search box, they will be sent to the server of the company which offers the search service and then the server will return the answers to the user. Google is one of the best known company which offers this service for free.

Adding Google Search to our Site

Google is most widely used search engine on the internet at present. It offers a very powerful and flexible service and user can use this service to provide the search feature on the site. User have to do this by registering for it while writing the code for the site. The instructions, setup for it are very simple on the site and service is also free.

2.5.3 Tables

Q63. Explain in detail about tables.

Answer :

The different techniques that allows the user to read the tables quickly are as follows,

1. Padding
2. Headings
3. Shading rows.

1. Padding

The padding is used to define the space between each item and the border.

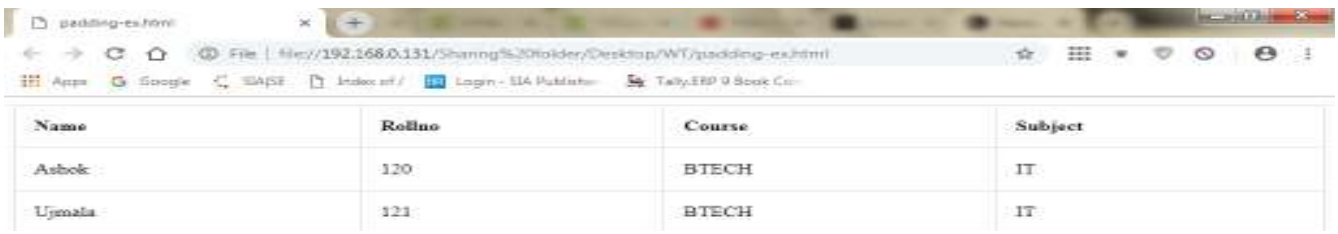
Example

```
th, td{
padding: 10px;
}
```

Program

```
<html>
<head>
<style>
table, td, th {
border: 1px solid #ddd;
text-align: left;
}
table {
border-collapse: collapse;
width: 100%;
}
</style>
</head>
<body>
<table>
<tr>
<th> Name </th>
<th> Rollno </th>
<th> Course </th>
<th> Subject</th>
</tr>
<tr class = “even”>
<td> Ashok </td>
<td> 120 </td>
<td> BTECH </td>
<td> IT </td>
</tr>
<tr class = “odd”>
<td> Ujmala </td>
<td> 121 </td>
<td> BTECH </td>
<td> IT </td>
</tr>
</table>
</body>
</html>
```

Output



Name	Rollno	Course	Subject
Ashok	120	BTECH	IT
Ujmala	121	BTECH	IT

2. Headings

Generally, all the heading of a table should be in bold, also make the heading in bold and uppercase and the upper line of the heading should be a dark line that separates the content from the headings.

Example

```
th{
border_bottom: 2px solid #FFFFFF;
text-transfom: uppercase;
background-color: yellow;
text-align: left;
}
```

Program

```
<html>
<head>
<style>
table, td, th {
border-bottom:2px solid red;
text-transform:uppercase;
background-color:yellow;
text-align:left;}
table {
border-collapse: collapse;
width: 100%;
}
th, td {
padding: 15px;
}
</style>
</head>
<body>
<table>
<tr>
<th> Name </th>
<th> Rollno </th>
<th> Course </th>
<th> Subject</th>
</tr>
<tr class = “even”>
<td> Ashok </td>
<td> 120 </td>
<td> BTECH </td>
<td> IT </td>
```

```
</tr>
<tr class = “odd”>
<td> Ujmala </td>
<td> 121 </td>
<td> BTECH </td>
<td> IT </td>
</tr>
</table>
</body>
</html>
```

Output



Shading Alternate Rows

This helps the users to follow along the lines. In this, every alternate row will be shaded with some light color than heading color. Note that the, background color should be in contrast with the text so that user can read the table data easily.

Example

```
th, td{
padding: 10px;
margin: 0px; }
th{
border_bottom: 2px solid #FFFFFF;
text-transfom: uppercase;
background-color: yellow;
text-align: left;
.even {background-color: #0F0F0F;}
.number {text-align: right; }
}
```

Program

```
<html>
<head>
<title> Font-family Example</title>
<link rel = “stylesheet” type = “text/css” href = “s1.css”/>
</head>
<body>
<table>
<tr>
<th> Name </th>
<th> Rollno </th>
<th> Course </th>
<th> Subject</th>
```

```
</tr>
<tr class = "even">
<td> Ashok </td>
<td> 120 </td>
<td> BTECH </td>
<td> IT </td>
</tr>
<tr class = "odd">
<td> Ujmala </td>
<td> 121 </td>
<td> BTECH </td>
<td> IT </td>
</tr>
</table>
</body>
</html>
```

Output



2.5.4 Forms

Q64. What is a form? Discuss the steps to be followed before designing the form.

Answer :

Form

Form is a webpage which contains various elements like text fields, checkboxes, buttons etc. Online forms can be used for different purposes.

Example

Online registration, order entry, subscription etc.

Steps to be followed before Designing the Form

Before designing the form the designer should follow the below steps,

1. List the Required Information

List all the requirements that you want to provide in the form. For a registration form the following information is required,

(a) Login information

(b) User name

(c) User address

(d) Contact details.

The above listing include the exact information as,

(a) Login Information

It includes user id and password

(b) User Name

It includes First name and last name

(c) User Address

It includes colony, city, state, pin code

(d) Contact Details

It includes mobile number, e-mail id, address

2. Group the Related Information

After listing the required information check for the related information and group them accordingly. In the above example, the related information can be grouped like the following,

- ❖ Name and E-mail id
- ❖ Login information
- ❖ Other details

Q65. Explain the process of designing a form.

Answer :

Design the form when you know all the form controls that are to be added to the page by grouping them together and labeling them accordingly.

Selecting the Right Type of Form Control

After estimating the approximate length and layout of the form. Decide the different form controls to be added to the page.

The user can enter the text using three types of form controls. They are,

- ❖ Using <input> element whose type attribute has text as its value (when the text is one line).
- ❖ Using <textarea> element (when the text is more than one line).
- ❖ Using <input> element whose type attribute has password as its value (when the text is sensitive).

The user can choose one of the following controls when he/she wants to choose one option from all,

- ❖ Radio buttons when only one option is required to be selected from multiple.
- ❖ Check boxes or multiple checkbox when multiple options are required to be selected from multiple.

Note that while declaring the form controls, the names of the controls should be related to the information provided. The different control names to describe the type of form controls use the following,

1. txtName for text areas and text boxes
2. chkName for checkboxes
3. selName for select boxes
4. radName for radio buttons.

1. Textboxes

Textbox control adds textboxes onto the form present on editor window. A textbox that allows the user to add text in the area present in textbox. A text box control is added to order form. That is, a text box control basically accepts input from users. If the text is small then you can use one line textbox or else use multi-line text area. For example, users are asked for their favorite subject and why.

2. Radio Buttons and Check Boxes

- ❖ Radio button control adds radio button onto a form. A radio button can be used for selecting single options from a set of options.
- ❖ Check box control adds check boxes to a form. A check box allows a user to select options by checking or unchecking it.
- ❖ Dropdown list control adds list of data as drop down list. A drop down list is used for making a single selection from the list.

Grouping Controls

Grouping of various controls can be done by using the <fieldset> element or by splitting the form into several pages.

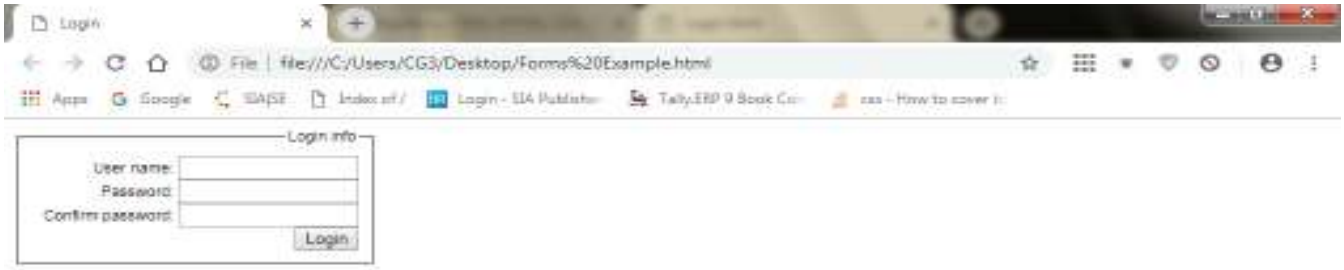
Using <fieldset> Elements

It is used to group all the related elements in a form.

Example

```
<head>
<title>Login</title>
<link rel= “stylesheet” type= “text/css” href= “s1.css” />
</head>
<body>
<form name= “Example” action= “ ” method= “post”>
<fieldset>
<legend>Login info</legend>
User name: <input type= “text” size= “15” name= “txtUserId” /><br />
Password: <input type= “password” size= “15” name= “txtPassword” /><br />
Confirm password: <input type= “password” size= “15” name= “txtPasswordConfirm” /><br />
<input type= “submit” value= “Login” />
</fieldset>
</form>
</body>
</html>
```

Output



Splitting a Form into Separate Pages

When the length of the form is more then split the page into different pages to make the user comfortable and avoid users from scrolling. For example, consider the below example that is divided into four forms for an efficient use.

But splitting the page into several pages leads to some new complexities i.e., the program needs to remember all the things the user entered in between the steps.

Q66. Write in detail about layout of forms.

Answer : Model Paper-III, Q10(b)

Layout of Forms

The layout of a form will attract the user to use the pages. Some of the issues concerned with the layout of forms are as follows,

Labeling Controls

The labeling of the form controls should be clear and user should quickly understand what actually it is. The two different types of labels are as given below,

- **Explicit labels:** These labels use <label> element.
- **Implicit labels:** These labels use normal text and markup text to the control.

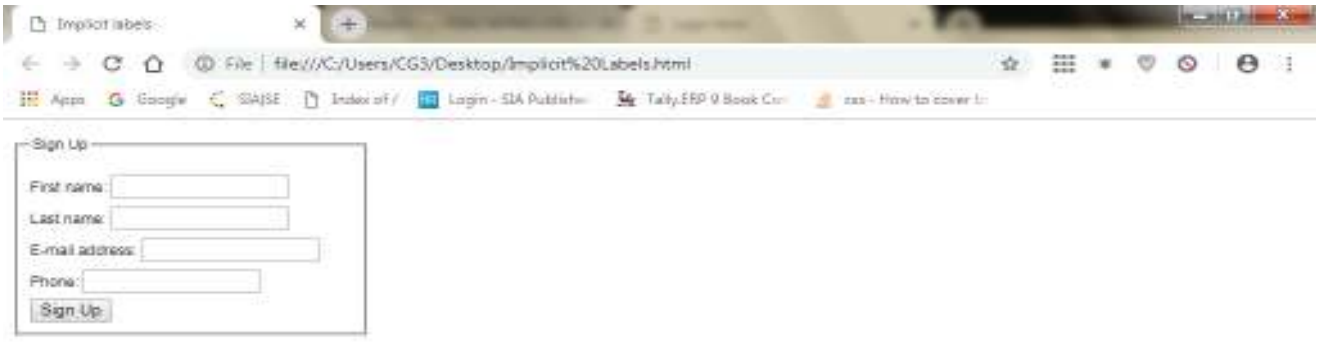
Generally, for checkboxes and radio buttons the label should appear at right to them. For textboxes, the label should appear to the left of the text box or above the text box. And for buttons the label should be on the button itself.

Example

The simple example to illustrate the implicit labels is as follows,

```
<html>
<head>
<title>Implicit labels</title>
<link rel= “stylesheet” type= “text/css” href= “s2.css” />
</head>
<body> <form name= “Example” action= “ ”  method= “post”>
<fieldset>
<legend>Sign Up</legend>
First name: <input type= “text” name= “txtFirstName” size= “15” /><br />
Last name: <input type= “text” name= “txtLastName” size= “15” /><br />
E-mail address: <input type= “text” name= “txtEmail” size= “15” /><br />
Phone: <input type= “text” name= “txtPhone” size= “15” /><br />
<input type= “submit” value= “Sign Up” />
 </fieldset>
</form>
</body>
</html>
```

Output



Example

The simple example to illustrate the explicit labels is as follows,

```
<?xml version= “1.0” ?> <!DOCTYPE html PUBLIC “-//W3C//DTD XHTML 1.0 Transitionalt//EN”
“http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd”>
<html xmlns= “http://www.w3.org/1999/xhtml” lang= “en” xml:lang= “en”>
<head>
<title>Implicit labels</title>
<link rel= “stylesheet” type= “text/css” href= “s3.css” />
</head>
<body>
<form name= “Example” action= “ ” method= “post”>
<fieldset>
<legend>Sign Up</legend>
<label for= “firstName”>First name: </label>
<div class= “control”>
<input type= “text” name= “txtFirstName” size= “20” id= “firstName” />
</div>
<label for= “lastName”>Last name: </label>
<div class= “control”>
<input type= “text” name= “txtLastName” size= “20” id= “lastName” />
```

```
</div>
<label for= “email”>E-mail address: </label>
<div class= “control”>
<input type= “text” name= “txtEmail” size= “20” id= “email” />
</div>
<label for= “phone”>Phone: </label>
<div class= “control”>
<input type= “text” name= “txtPhone” size= “20” id= “phone” />
</div>
<input type= “submit” value= “Sign Up” />
</fieldset>
</form> </body>
</html>
```

Output



Keep Relevant Information Next to or above Form Controls

Keeping relevant information next to or above the form controls is one more requirement for a good design. For example, consider the below figure that consists of two separate text boxes with no indications. This type of declaration makes the user feel difficult. Therefore keep the require information near to the form controls.

Required information is referred to as the information that is to be entered without fail. To provide such type of fields in the form follow the below rules,

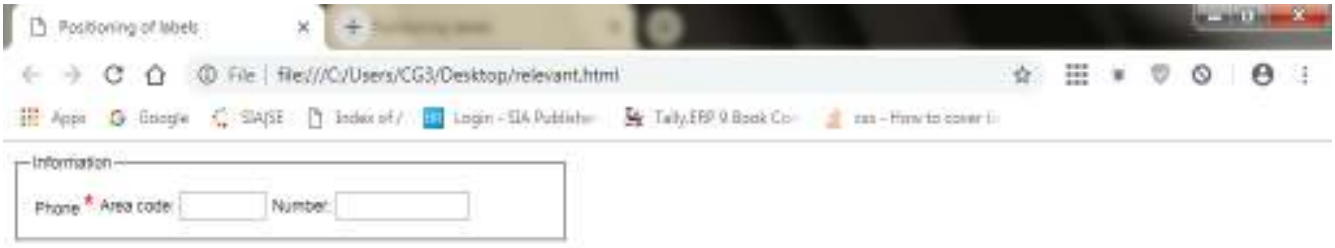
- ❖ Use an asterisk (*) symbol to indicate it as mandatory or required field.
- ❖ Use the “required” keyword next to the field.

Example

```
<?xml version= “1.0” ?> <!DOCTYPE html PUBLIC “-//W3C//DTD XHTML 1.0 Transitionalt//EN”
“http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd”>
<html xmlns= “http://www.w3.org/1999/xhtml” lang= “en” xml:lang= “en”>
<head>
<title>Positioning of labels</title>
<link rel= “stylesheet” type= “text/css” href= “s4.css” />
</head>
<body>
<form name= “Example” action= “ ” method= “post”>
<fieldset>
<legend>Information</legend>
<table>
<tr>
<td class= “label”>Phone <span class= “required”>*</span></td>
```

```
<td>Area code:&amp;nbsp;<input type= “text” name= “txtTelAreaCode” size= “5” />
Number:&amp;nbsp;<input type= “text” name= “txtTelNo” size= “10” /></td>
</tr>
</table>
</fieldset>
</form>
</body>
</html>
```

Output



Careful Placement of Buttons

Provide the button at the required and proper place because misplacement of buttons leads the user to a wrong page. For example, in online shopping websites the “add to cart” button will be exactly positioned at the product if it is misplaced then it leads to the selection of other product.

Using the title Attribute on Form Controls

The title attribute is used to provide additional information to the users when the cursor moving over the form controls.

Example

```
<?xml version= “1.0” ?>
<!DOCTYPE html PUBLIC “-//W3C//DTD XHTML 1.0 Transitionalt//EN”
“http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd”>
<html xmlns= “http://www.w3.org/1999/xhtml” lang= “en” xml:lang= “en”>
<head>
<title>Using the title attribute</title>
<link rel= “stylesheet” type= “text/css” href= “s5.css” />
</head>
<body>
<form name= “frmExample” action= “ ” method= “post”>
<fieldset>
<legend>Enter the OTP</legend>
OTP:&amp;nbsp;<input type= “text” name= “txtOTP” title= “Enter the otp sent to your registered mobile number to complete
the transaction” />
</fieldset>
</form>
</body>
</html>
```

Output



Color in html is known to be one of the important features to convey the information but depending too much on the colors to pass the information is not good for the design. It must be confirmed that there is sufficient contrast between the colors and make the difference clearly.

Using CSS with the Form Elements

Usage of CSS to control the elements of a form (like textbox, text area, buttons etc.,) makes the page more popular.

Example

```
<?xml version= “1.0” encoding= “iso-8859-1”?>
<!DOCTYPE html PUBLIC “-//W3C//DTD XHTML 1.0 Transitional//EN”
“http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd”>
<html xmlns= “http://www.w3.org/1999/xhtml”>
<head>
<title>Search form</title>
<meta http-equiv= “Content-Type” content= “text/html; charset=iso-8859-1” />
<style type= “text/css”>
input.search {
width:335px;
height:37px;
padding:10px 10px 10px 55px;
background-image:url(s.jpg);
border:none;
background-color:#cccccc;
font-size:30px;} #search_container input: focus {
outline: 0;}
</style>
</head>
<body>
<input type= “text” name= “search” class= “search” />
</body>
</html>
```

Output



Suppose, basic form is the main part of a page then the basic form contains many form controls. Login page, theasures.com pages are the examples of large form inputs.

INTERNAL ASSESSMENT

OBJECTIVE TYPE

I. Multiple Choice

1. CSS stands for _____. [ ]

   (a) Common Style Sheet (b) Cascading Synchronized Sheet

   (c) Cascading Style Sheet (d) None of the above

2. _____ property is used to specify the size of the font. [ ]

   (a) The font_size (b) The size

   (c) The font-size (d) The fontsize

3. _____ property places the text in the browser's window according to the specified value. [ ]

   (a) text-align (b) browser-align

   (c) align (d) All the above

4. _____ value preserves the white spaces and does not break the line until
 element is used. [ ]

   (a) Post (b) Pre

   (c) Nowrap (d) Normal

5. The _____ pseudo class applies the styles to the first line of an element. [ ]

   (a) :first-line (b) :f_line

   (c) first_line (d) None of the above

6. _____ selector applies the style to an element that is the next sibling of another. [ ]

   (a) Sibling (b) Attribute

   (c) Next sibling (d) Adjacent sibling

7. Which of the following is the way to represent colors in CSS? [ ]

   (a) hexa decimal code (b) RGB value

   (c) Color name (d) All the above

8. Which of the following methods help to group the elements? [ ]

   (a) Styles (b) Location

   (c) Color (d) All the above

9. The web page whose dimensions are not fixed is called as _____. [ ]

   (a) Fixed design (b) Fixed-width design

   (c) Liquid design (d) Layout design

10. _____ rule is used to import a style sheet into the present style sheet. [ ]

    (a) @important (b) !import

    (c) @import (d) import

II. Fill in the Blanks

1. _____ is a document written in any styling language.
2. _____ attribute indicates the URL of the style sheet.
3. The _____ is used to control the amount of space between each word.
4. The _____ selector can be used to apply a rule to all the elements of a document.
5. _____ in a visual art is defined as the organization of visual elements.
6. The web page whose dimensions are not fixed is called as _____.
7. _____ is a webpage that contains various form controls.
8. _____ is used to define the space between each item and the border.
9. _____ is the process of collecting independent units of information and grouping them together to form a large unit.
10. _____ is the vertical space between the lines of text.

KEY

I. Multiple Choice

1.	(c)	2.	(c)	3.	(a)	4.	(b)	5.	(a)
6.	(d)	7.	(d)	8.	(d)	9.	(c)	10.	(c)

II. Fill in the Blanks

1. Style sheet
2. href
3. White-space property
4. Universal
5. Composition
6. Liquid design
7. Form
8. Padding
9. Chunking
10. Leading

III. Very Short Questions and Answers

Q1. What is a style sheet?

Answer :

Style sheet is a document written in any styling language. It consists of style information which can be prepared using the specific formatted style rules.

Q2. What is the purpose of content property?

Answer :

Content property is used to specify the type of content that should be added to the document. The various values that add different type of content to the document are as follows,

- A String
- A Counter
- A URL
- attr(x)
- open-quote
- close-quote
- no-open-quote
- no-close-quote.

Q3. List the different relative units in CSS.

Answer :

The different relative units in CSS are as follows,

1. px
2. em
3. ex.

Q4. Define a measure in Typography.

Answer :

Typography refers to an art and technique used in type design which comprises of decisions relating to type faces, point size, line length, letter spacing, etc.

Q5. What is a form?

Answer :

Form is a webpage which contains elements like text fields, checkboxes, button etc. These web pages look like normal white sheet of paper consisting of significant spaces information to be entered by users.

UNIT 3

LEARNING AND WORKING WITH JAVASCRIPT, PUTTING YOUR SITE ON THE WEB

Marketed by:
SIA GROUP

LEARNING OBJECTIVES

After studying this unit, a student will have thorough knowledge about the following key concepts,

- Script tag , Document Object Model
- Document object, Forms collection and Images collection with its properties and methods
- Variables, Operators and its types and Functions
- Control Statements, Looping and Events
- Built-in Objects with its properties and methods
- Form Validation and Regular Expressions
- Select Box Options, Radio Buttons and check boxes,
- Form Enhancements and Java script Libraries
- Meta Tags, http-equiv and content attributes
- Pre-publishing Tests, Pre-release Tests, Link Checking and Checking Color Depths.
- Meta tags and its attributes
- Testing the site
- Different things to get the site on web
- Different pointers to make more people aware of the site
- Understanding the visitors.

INTRODUCTION

JavaScript is a programming language that supports to object oriented methodologies. It facilitates the inclusion of certain executable data along with it. JavaScript can also contain dynamic data which can interact with the users, dynamically create HTML context and can control the actions of browser. It provides various types of operators, conditional statements, iterative statements, arrays, objects etc. JavaScript provides different types of objects such as math object, string object, date object, boolean and number object and also some event handling techniques.

JavaScript can interact with web pages in various ways. Implementation of Javascript is described in this unit.The javascripts can be created using basic scripts and Pre-written javascript libraries. The developed site is represented to users and is put on the web. Tests are performed on it to validate the documents, links, and to make sure that the site works in other browsers.

PART-A
SHORT QUESTIONS WITH SOLUTIONS

Q1. Write the structure of java script program.

Answer :

The structure of javascript code begin with the tag <script language = "javascript">. This tag validates the javascript and minimizes the occurrence of errors. It ends with </script>.

```
<script language = "javascript">
{
javascript code;
}
</script>
```

Q2. Write a short note on DOM.

Answer :

Document object model or DOM is an interface using which given programs/scripts can dynamically alter (arrange, format etc.) the contents of a given web document. Here, one has to remember that DOM remains a language and platform independent interface. Also, the given web document can further be processed (if required) and the result of which can be induced in it again.

Q3. What is a variable in java script? How to declare a variable?

Answer : Model Paper-I, Q6

Variable

A variable is a data item that stores the values defined by the user. It can be manipulated at any time. In JavaScript, the variables are same as the other programming language variables. If a value is assigned to a variable then this is called assignment.

Declaring a Variable

Variables declared with 'var' keyword.

```
var variable = value;
var a = 10;
```

Q4. Define functions in javascripts.

Answer : Model Paper-II, Q6

A function is a self contained block of code that performs specific and well defined tasks. It simplifies complex task by dividing the program into modules.

A function definition is the complete description of a particular operation. Actually a function definition tells what a function does and how it performs. A function definition contains a function body (a block of statements) in addition to function name, arguments list and return type.

Q5. Write about 'string' object.

Answer :

String function forms one of the important consequences in JavaScript. In general terms, strings refer to series of characters (such as alphabets, special characters, numerics or combination of them) that are closed within double quotes. Like math, even strings are treated as object which supports numerous methods.

Length is the main property used for string object. It is used to specify the count or number of a characters given in a string.

Q6. Write a short note on Date Objects.

Answer :

Date forms one of the important specifications in almost every programming discipline. Hence, Java scripts provides a date class for this purpose. This class includes a set of methods to access the date.

Globally, date/time is recognized in two ways i.e., UTC and local time. Universal time as the name suggests is a standard that is followed all over the world. On the other hand, local time is the time of the system where the script is currently residing.

Date object can be created in the following way,

var currentdate = new date ( );

On the execution, the date object simply captures the date of local system (in which it is executing) at that instant and returns the value (which is nothing but the time in milliseconds) to current date variable. The returned value is the time started from 0 clock, January 1st 1970. The date constructor in the above declaration is empty (i.e., without any parameters). Different values can be obtained by passing parameters to this constructor.

Q7. What is an array in java script?

Answer : Model Paper-II, Q6

An array contains a set of similar items. But in javascript an array can contain different items. Arrays here are similar to structure variables. But they can be accessed randomly by making use of index and the array name. Thus, arrays in javascript are also called as associative arrays. This feature makes javascript arrays distinct than arrays of other languages.

An array can be declared as follows,

var arr;

Elements can be allocated to this array by using the below statement,

arr = new Array(10);

An array is not initialized while it is allocated.

Q8. List any two mathematical functions with syntax and examples.

Answer :

(i) **Function** : min( )

Description : Displays minimum of two or list of numerical values entered into it.

Usage : document.writeln(min(2, 5)); //output 2

(ii) **Function** : sqrt( )

Description : It displays the square root of the value entered into it.

Usage : document.writeln(sqrt(4)); //output 2

Q9. Write a short note on meta tags.

Answer : Model Paper-I, Q7

The meta tag provides the information related to the page indexing for the search engines. The meta information is simply defined as "information about information".

Example

<meta name = "description" content = "SIA publishers and distributors pvt Ltd"/>

Generally, a meta tag can take four universal attributes (dir, lang, xml:lang and title) and four other attributes that are specific to the <meta> element. They are as follows,

1. schema
2. name
3. content
4. http-equiv.

Here, the name and content attributes, http-equiv and content attributes are used together.

Q10. What are the two different stages of testing?

Answer :

The two different stages in the process of testing are given below,

1. Pre-publishing Tests

The tests that are performed on the system before exposing them to the users are known as pre-publishing tests.

2. Pre-release Tests

The tests that are performed on a server before publishing are known as pre-release tests.

Q11. List the different points used to make people aware of the site.

Answer : Model Paper-II, Q7

The different pointers to make more people aware of the site are as follows,

1. Always mention the URL
2. Forums, communities, and social Networks
3. Check for trade sites and directories
4. Reciprocal links
5. Look at your competition
6. Make it easy for people to link to you
7. Pay-per-click advertising
8. Display advertising
9. Create an E-mail news letter.

Q12. Write a short note on domain name.

Answer : Model Paper-III, Q7

The domain name is an address that is provided to the users of the market for communicating through online/internet.

Example

siapublishers.com, hrstoppro.com

There are many domain regestration companies for registering a domain name for the site. So, choose a company and register your domain name. Most of the companies provide suffix as.com where as some other companies provides the suffix based on the country. For example, .co.in for India, .co. for UK, and .ru for Russia, so on. .me.uk is the suffix given to the personal sites in UK, .info is provided to the information-based sites, .org is provided to the organizations.

PART-B

ESSAY QUESTIONS WITH SOLUTIONS

3.1 LEARNING JAVASCRIPT

3.1.1 How to Add Script to Your Pages

Q13. How a script is added to the webpage.

Answer : Model Paper-I, Q11(a)

The web pages contain the XHTML code as well as javascript code. Javascripts can be added to the webpage in two ways. One way is to add the code within <script> and </script> tags. The purpose of the <script> tag is to notify the browser that the contents inside the <script> tag are to be considered as the script and they should be interpreted by the respective interpreter. Consider the below example,

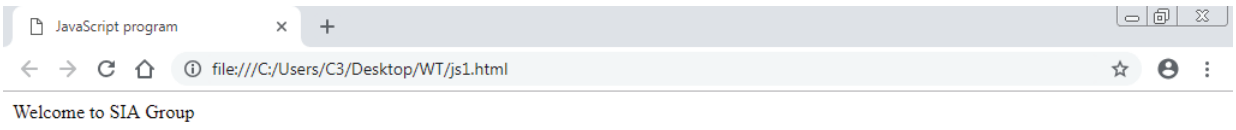
```
<script language = "JavaScript">
Javascript statements
</script>
```

In the above code snippet, the language attribute is optional. Multiple scripts can be inserted in an XHTML document. All together form a single program along with the order in which they are declared. The interpreter is responsible for interpreting the program along with statements from top to bottom.

Example

```
<html>
<head>
<title>JavaScript program </title>
</head>
<body>
<script language = "JAVASCRIPT">
document.writeln("Welcome to SIA Group");
</script>
</body>
</html>
```

Output



The second way to add the javascript into XHTML is to identify the code available in the external source file. This file should be made to refer from within the XHTML. This file should be saved with extension .js. It is the XHTML program which refers this external file. The <script> tag contains an attribute src that refers the external javascript file name.

```
<script language = "JavaScript" src = "source.js">
</script>
```

The source file must be located in the directory which contains the XHTML file. Even this method allows multiple javascript files to be referred. A script tag for every file must be used in case of multiple files. This method is useful when the javascript code is lengthy.

Example

external.html

```
<html>
<head>
<title>JavaScript program </title>
</head>
<body>
<script src="js2.js"></script>
</body>
</html>
```

js2.js

```
document.writeln("Welcome to SIA Group");
```

Output



A single xhtml document can have any number of scripts and can be used in three different sections as follows,

(i) <head>

The script element is used in the <head> section of the page and is called whenever an event is triggered.

(ii) <body>

The script element is used in the <body> section of the page. Whenever a script is run a page is loaded.

(iii) External File

When an link is provided in <head> section then the script is treated as if it is in this <head> section. It occurrs whenever an event is triggered. When it is provided in <body> section then the script is treated as it it is in the <body> section and executed when the page is load.

Q14. Write about comments and <no script> element in Javascript.

Answer :

Comments

Comments in Javascript can be used in the following two ways,

(i) A comment is provided in a one line by using two forward slash(//) after the statement in the same line shown as below,

```
<script type = "text/javascript">
document.write("javascript Program")          //This text is displayed on screen
</script>
```

(ii) A comment can be provided in multiple lines by using the opening characters /* and closing characters */ shown as below,

```
/* this is a javascript program.
It contains multiline comments*/
```

<noscript> Element

This tag displays an alternative content with in a document for the javascript users who are disabled. It can contain any XHTML content provided by the author.

3.1.2 The Document Object Model

Q15. Explain about Document Object Model.

Answer :

Document Object Model

Document object model or DOM is an interface using which given programs/scripts can dynamically alter (arrange, format etc.) the contents of a given web document. Here, one has to remember that DOM remains a language and platform independent interface. Also the given web document can further be processed (if required) and the result of which can be induced in it again.

Level 0 HTML Document Object Model

The following figure illustrates the level 0 HTML document object model which contains different child objects that specifies the collection of similar tags present in a document.

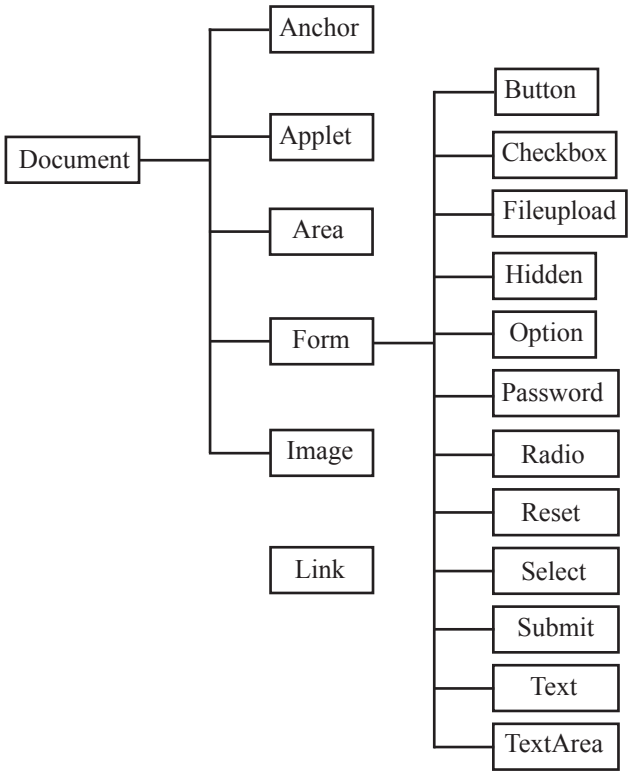


Figure: Level 0 HTML Document Object Model

(i) Anchor Collection

It specifies the anchor elements used in a document for linking.

(ii) Applet Collection

It specifies the applets present in a document.

(iii) Area Collection

It specifies the image maps which use an <area>element present in the document.

(iv) Forms Collection

It specifies the <form> tags present in the document.

(v) Image Collection

It specifies the images present in the document.

(vi) Link Collection

It specifies the hyperlinks present in the document.

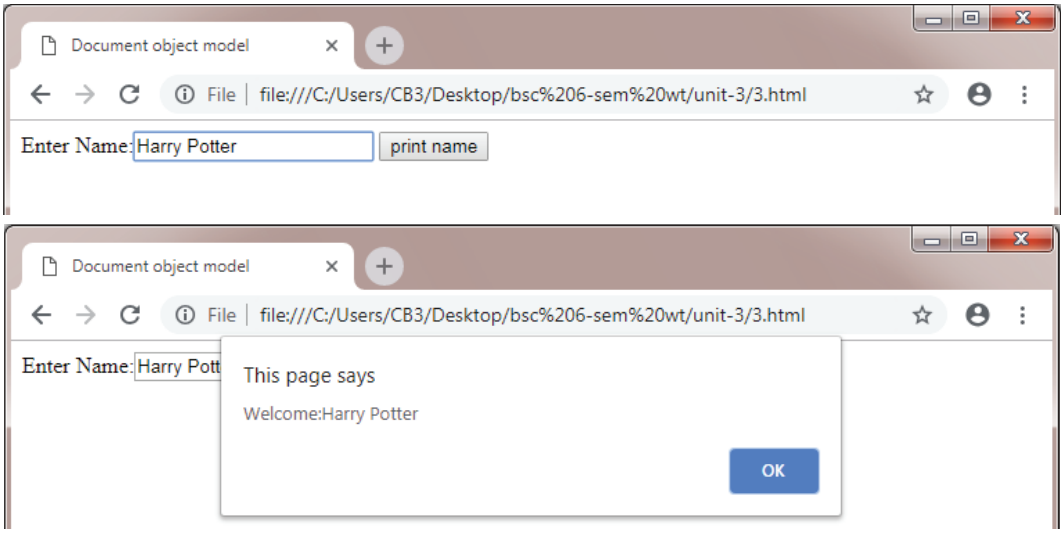
Forms contains the child objects which are used to specify various types of form controls that are shown on the form such as Button, Checkbox, File upload, Hidden, Option, Password, Radio, Reset, Select, Submit, Text, TextArea.

Example

Consider a document is accessed by using a DOM that contains one form and two links.

```
<html>
<head>
<title>Document object model</title>
</head>
<body>
<script type="text/javascript">
function display()
{
var name=document.form1.name.value;
alert("Welcome:"+name);
}
</script>
<form name="form1">
Enter Name:<input type="text" name="name"/>
<input type="button" onclick="display()" value="print name"/>
</form>
</body>
</html>
```

Output



DOM provides a script to access the content of the form (forms collection) and the two links (links collection).

Accessing Values Using Dot Notation

The object can be accessed by a dot notation.

Example

document.links[0].href

In the above example, the first link in the document is accessed by the links i.e., a child object of the document object.

Here,

(i) The document specifies that document object is accessed by a user.

(ii) Links present in the links collection.

(iii) [0] specifies the first link of the document.

(iv) href specifies to retrieve the property for the given link.

Q16. Write about document object with its properties and methods.

Answer :

Document Object

The word "Document" in terms of programming aspects refers to the page which will be displayed as soon as the browser window loads. It is a web page which can be created and displayed on browser.

Properties of Document Object

The following are the different properties of a document object.

1. **alinkcolor**

The color that has to be used to display the links in the document is defined by alinkcolor property.It can be used for both read and write.

2. **bgcolor**

The bgcolor property indicates the background color of the document. It can be used for both read and write.

3. **fgcolor**

The fgcolor property indicates the foreground color (i.e., the text color in the document). It can be used for both read and write.

4. **lastModified**

The lastModified property indicates the date on which the document was modified recently.

5. **linkcolor**

The linkcolor property indicates the color that has to be used for hyperlinks while displaying them in the document.

6. **referrer**

The referrer property indicates the URL of the document that refers the current document.

7. title

This property defines the title of the document as it is defined between the <title></title> tags.

8. vlinkcolor

The vlinkcolor property indicates the color that has to be used for the links that are already visited by the current user while displaying them in the document.

The above properties can be accessed by using dot notation.

Example

Title of the document can be accessed by using the following syntax,

document.title

Methods of Document Object

The following are the methods of document object.

(i) write(String)

It is used for adding text or elements to the document.

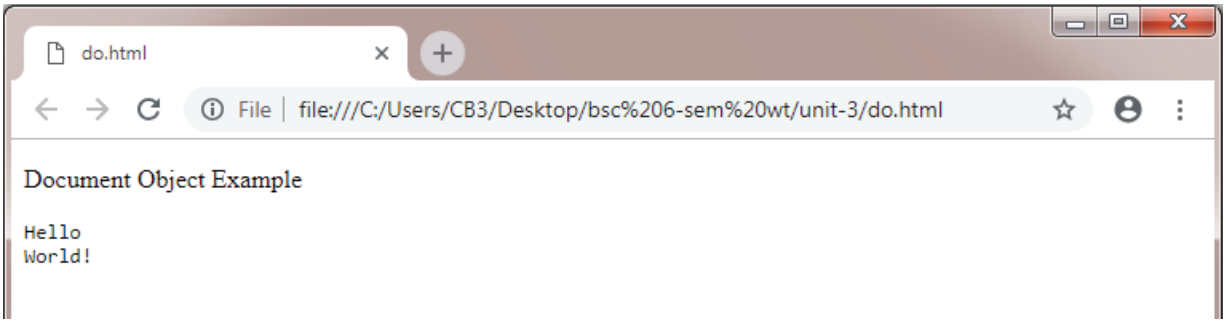
(ii) writeln(String)

It is used for adding a new line at the ending of the output.

Example

```
<!DOCTYPE html>
<html>
<body>
<p>Document Object Example</p>
<pre>
<script>
document.writeln("Hello");
document.writeln("World!");
</script>
</pre>
</body>
</html>
```

Output



Q17. Explain about forms collection and form elements with their properties and methods.

Answer :

Forms Collection

A form object is an array that contains elements of a form. The elements of this array object of the form can be accessed by using elements property of that object. These elements contain only the form fields. The individual elements are accessed using ordinal number or the name of the element as an index.

Properties of Form Objects

The following are the properties of form objects,

(i) action

It indicates the action attribute present in the <form> element. It can be used for both read and write.

(ii) length

It specifies the number of form controls present in the form. It is used only to read.

(iii) method

It indicates the value of an method attribute as get or post in the <form> element. It is used for both read and write.

(iv) name

It specifies the name attribute in the <form> element. It is used only to read.

(v) target

It specifies the target attribute in the <form> element. It is used for both read and write.

Methods of Form Objects

The following are the methods form objects,

(a) submit( )

A form is submitted using submit( ) method. When the submit( ) method of the form is clicked, then the handler of the submit event is called. All the form elements are checked inside the handler of submit event whether they contain correct value. A form is submitted by calling submit( ) only when it passes this phase successfully. The URL that is specified by the action attribute is loaded or current page is loaded if no action attribute is specified.

All the desired tasks need to be performed before applying submit method. There must be a reference to the form object by using HTML DOM or other methods to invoke the submit( ) method on it.

Example

```
document.forms[0]. submit( );
```

(b) reset( )

The reset( )method is used to set the form fields to their default values. User need to have a reference to the form element before invoking this method.

Form Elements

The following are the different things used with the elements in a form.

(i) textfields

This is used to read the user entered data or write the new data.

(ii) checkboxes and radiobuttons

This is used for testing whether checked or unchecked.

(iii) buttons

This is disabled whenever an user chooses any option.

(iv) select Boxes

This is used for selecting an option from the list of pre-define options.

Properties of Form Elements

The following are the properties of the form elements used in the form.

(i) checked

Checked property returns true when a form element is checked and it returns false when form element is not checked. It is used for both read and write. This property is used by checkboxes and radiobuttons.

(ii) disabled

Disabled property returns true when it is disabled. It is used for both read and write. This property is used by all form elements except hidden elements.

(iii) form

Form property is used to return a part of reference. It is used only to read. This property is used by all form elements.

(iv) length

Length property is used to return the count or the number of options present in the <select> element. This is used only to read. This can be used by select boxes.

(v) name

Name property is used to access the value of the name attribute. It is used only to read. This can be used by all the form elements.

(vi) Selectedindex

Selectedindex property is used to return the index number of the selected item in the form. It is used for both read and write. This can be used by select boxes.

(vii) type

Type property is used to return the type of a form control. It is used only to read. This can be used by all the form elements.

(viii) value

The value property is used to access the value attribute given to the element or the content of a text input. It is used for both read and write. This can be used by all the form elements.

Methods of Form Elements

The following are the methods of form elements.

(i) blur( )

This is used by all the form elements except hidden elements. It forwards the focus present on the currently active element to the next element in the tabbing order.

(ii) click( )

This is used by all the form elements except text elements. It simulates the mouse click on particular element.

(iii) focus( )

This is used by all the form elements except hidden elements. It provides focus on the element.

(iv) select( )

This is used by text elements except hidden elements. It is used to select the text of the element.

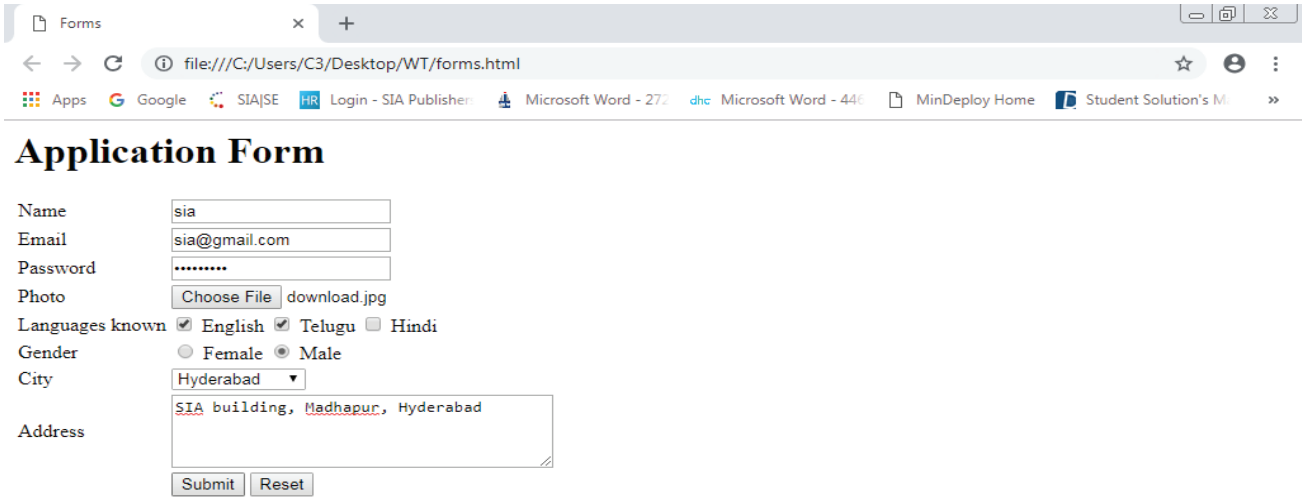
An example to illustrate different form elements is as given below,

Example

```
<html>
<title>Forms</title>
<body>
<h1>Application Form</h1>
<form>
    <table>
      <tr>
        <td>
          <label for= “name”>Name</label>
        </td>
        <td>
          <input type= “text” id= “name” name= “username”>
        </td>
      </tr>
      <tr>
        <td>
          <label for= “email”>Email</label>
        </td>
        <td>
          <input type= “text” id= “email” name= “usermail”>
        </td>
      </tr>
      <tr>
        <td>
          <label for= “pass”>Password</label>
        </td>
        <td>
          <input type= “password” id= “pass”>
        </td>
      </tr>
      <tr>
        <td>
          <label for= “photo”>Photo</label>
        </td>
        <td>
          <input type= “file” id= “photo”>
        </td>
      </tr>
      <tr>
        <td>
          <label>Languages known</label>
        </td>
        <td>
          <label>
            <input type= “checkbox”> English</label>
          <label>
            <input type= “checkbox”> Telugu</label>
      <label>
```

```
            <input type= “checkbox”> Hindi</label>
        </td>
      </tr>
      <tr>
        <td>
          <label>Gender</label>
        </td>
        <td>
          <label>
            <input type= “radio” value= “f” name= “gender”>
          Female</label>
          <label>
            <input value= “m” type= “radio” name= “gender”>
          Male</label>
        </td>
      </tr>
      <tr>
        <td>
          <label for= “city”>City</label>
        </td>
        <td>
          <select id= “city” name= “city”>
          <option>--Select City--</option>
              <option>Hyderabad</option>
              <option>Warangal</option>
              <option>Bangalore</option>
              <option>Chennai</option>
          </select>
        </td>
      </tr>
      <tr>
        <td>
          <label>Address</label>
        </td>
        <td>
          <textarea rows= “4” cols= “40”></textarea>
        </td>
      </tr>
      <tr>
        <td></td>
        <td>
          <input type= “submit” value= “Submit”>
          <input type= “reset”>
        </td>
      </tr>
    </table>
  </form>
    </body>
</html>
```

Output



Q18. Discuss about images collection with their properties.

Answer :

Images Collection

Images collection specifies the references for images. These are also referenced by name or using index number provided in the collection. The src attribute for the second image can be denoted by the following syntax.

document.images[1].src

Properties of Image Object

The following are the properties of an image object.

(i) border

The border property of the <img> element is used to specify the width of the border in terms of pixels. It can be used for both read and write.

(ii) complete

The complete property is used to specify whether the given image is loaded successfully. It is used only for reading.

(iii) height

The height property of an <img> element is used to specify the height of the image as pixels. It is used for both read and write.

(iv) hspace

The hspace property of an <img> element is used to specify the space above and below the image to separate it from the surrounding elements. It is used for both read and write.

(v) lowsrc

The lowsrc property of <img> element is used to specify the lower resolution version of the image. It is used for both read and write.

(vi) name

The name property of an <img> element is used to specify the name of an image. It is used for both read and write.

(vii) src

The src property of an <img> element is used to specify the location of file. It can be used for both read and write.

(viii) vspace

The vspace property of an <img> element is used to specify the space between the left and right of the image that is separated from the surrounding elements. It can be used for both read and write.

(ix) width

The width property of an <img> element is used to specify the width of an image interms of pixels. It can be used for both read and write.

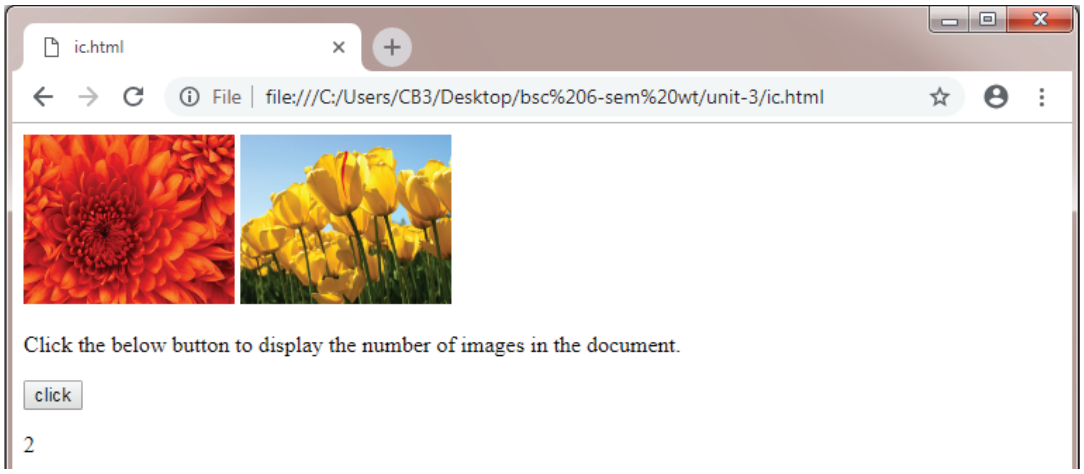
Methods

Image object doesnot have any methods in the level 0 DOM.

Example

```
<!DOCTYPE html>
<html>
<body>
<img src="D:\image.jpg" alt="flower" width="150" height="120">
<img src="D:\image1.jpg" alt="flower" width="150" height="120">
<p>Click the below button to display the number of images in the document.</p>
<button onclick="cal()">click</button>
<p id="demo"></p>
<script>
function cal()
{
var y = document.images.length;
document.getElementById("demo").innerHTML = y;
}
</script>
</body>
</html>
```

Output



Q19. What are the different types of objects in Javascript.

Answer :

The three different types of objects in javascript are as follows,

(i) W3C DOM Objects

Different browsers contain different objects that are provided to have control over a document.

(ii) Built-in Objects

Javascript contains many objects that are in built such as date, math, browser etc., these objects have their own usage and functionality. For example, date object is used for providing date and time functions.

(iii) Custom Objects

When an advanced javascript is started then Javascript objects are created based on their functionality for example, validation object is used for validating forms.

3.1.3 Variables, Operators, Functions

Q20. Explain about variables in Javascript.

Answer :

Variable

A variable is a data item that stores the user defined values. It can be manipulated at any time. In JavaScript, the variables are same as the other programming language variables. If a value is given to 'a' variable then this is called assignment.

Variable is declared with "var" keyword.

Syntax

var variable = value1, value2, ... valuen;

Example

```
var a, b, c;
a = 10
b = 20 + a
c = b * a
```

Here, a and b are two variables.

Names of Variables

A name given to a variable is called variable name. In JavaScript, there are some rules for naming variables,

1. A variable name should begin either with an underscore (_) or a letter.
2. Spaces cannot be used in variable names.
3. Variable Names in JavaScript are case-sensitive. The meaning of 'a' is different from 'A' and 'aA' is different from 'AA', etc.
4. Unlike programming languages, reserved keywords in JavaScript cannot be used as variable names.

Example

College_name, a, A, sum, _ sum.

Declaration of a Variable in Java Script

A variable is created by preceding it with the keyword 'var' despite of any data will be stored in that variable.

Syntax

var variable_name;

Example

var x;

Assigning a Value to a Variable

A value can be assigned to a variable by using an assignment operator. The value can be changed even though the value for a variable is already assigned.

Example

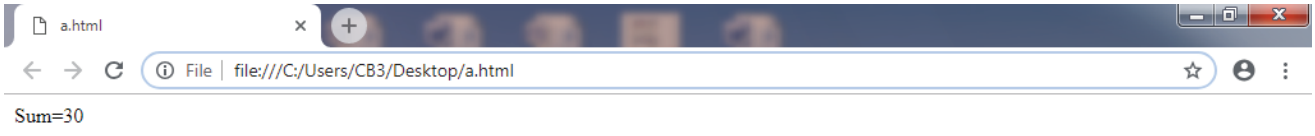
```
Var a = 10
    a = 25
```

Now, the value for variable a is changed to 25.

Example

```
<html>
<body>
<script language="javascript">
var a=10,b=20,c=a+b;
document.write("Sum=",c);
</script>
</body>
</html>
```

Output



Lifetime of a Variable

Generally, JavaScript supports outward scoping. This means that, the code present in a curly brackets can have access to the code present outside it. But, the code outside curly brackets cannot have access to the code present inside the curly brackets.

Basically, scope rules are defined for variables and functions which specifies the portion of the program where a variable or a function can be accessed.

Scope Rules for Variables

The scope rules for variables are of two types,

(i) Local scope

(ii) Global scope.

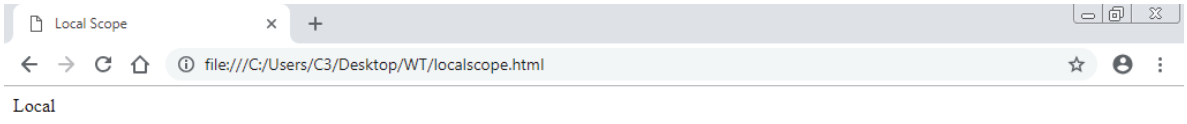
(i) Local Scope

If the variable is declared inside a function then the variable is said to have a local scope. In this case, a variable can be accessed only within the function itself.

Example

```
<html>
<head>
<title>Local Scope</title>
</head>
<body>
<script language="JAVASCRIPT">
var x = "Global";
function printValue()
{
    var x = "Local";
    document.writeln(x);
}
printValue();
</script>
</body>
</html>
```

Output



(ii) Global Scope

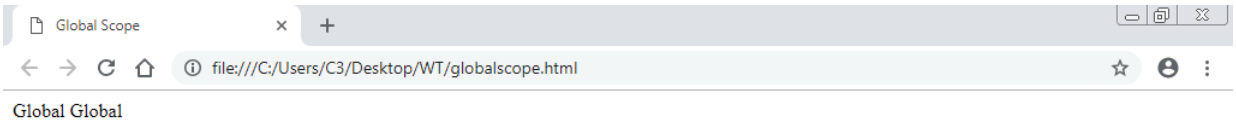
If the variable is declared out of function then the variable is said to have global scope. In this case, the variable can be accessed any where in the program.

Example

```
<html>
<head>
<title>Global Scope</title>
</head>
<body>
<script language="JAVASCRIPT">
var x = "Global";
function chkScope()
```

```
{
var y = "Local";
y = x;
document.writeln(y);
}
chkScope();
document.writeln(x);
</script>
</body>
</html>
```

Output



JavaScript does not support the blocked scope. This is the scope wherein every set of braces will have their own scope.

Q21. What is an operator? What are the different types of operators?

Answer : Model Paper-II, Q11(a)

Operator

An operator performs mathematical or logical operations. They are used to manipulate data and variables called operands. It tells the computer about the operation to be performed on the operands.

Types of Operators

The various types of operators used in javascript are as follows,

1. Arithmetic Operators

Arithmetic operators are the mathematical operators that are used to perform basic arithmetic operations. Various arithmetic operators are as follows,

(i) Addition (+)
(ii) Subtraction (–)
(iii) Multiplication (*)
(iv) Division (/)
(v) Modulus (%)
(vi) Increment (++)
(vii) Decrement (– –).

(i) Addition (+) Operator

It is a addition/plus operator which performs numeric addition. It performs addition operation when used on integers and it performs concatenation when used on strings.

Examples

(a) var sum = 6 + 4; //result will be 10
(b) var value = "Hello" + "World"; //result will be Hello World

(ii) Subtraction(–) Operator

It is a subtraction/minus operator that is used to perform numerical subtraction.

Example

result = 5 – 2; //result will be 3.

(iii) Multiplication(*) Operator

It is a multiplication operator that is used to perform numerical multiplication.

Example

result = 6 * 2; //result will be 12

(iv) Division(/) Operator

It is a division operator which is used to perform numerical division.

Example

result = 10/2; //result will be 5

(v) Modulus(%) Operator

It is a modulus operator which returns remainder as the result.

Example

result = 5 % 2; //result will be 1

(vi) Increment(++) Operator

It is an increment operator which increments the value by 1. There are two types of increment operators namely pre increment and post increment. The pre increment operator returns the final result after incrementing. The post increment operator returns the value before incrementing the value.

Example

(a) Post Increment

x = 2;

y = x + +; //y value will be 2

(b) Pre Increment

x = 2

y = + + x //y value will be 3

(vii) Decrement(– –) Operator

It is a decrement operator that decrements the value by 1. There are two types of decrement operators, namely pre decrement and post decrement. The pre decrement operator returns final result after decrementing the value. The post decrement operator returns the initial value before decrementing it.

Example

(a) Post Decrement

x = 2;

y = x – –; //y will be 2

(b) Pre Decrement

x = 2

y = – – *x*; //y will be 1

Example

```
<html>
<head>
<title>Arithmetic operators</title>
</head>
<body>
<script language = "JAVASCRIPT">
var a=5;
var b=4;
var linebreak = "<br/>";
document.write("a+b=" + (a+b));
document.write(linebreak);
document.write("a - b ="+ (a-b));
document.write(linebreak);
document.write("a*b ="+ (a*b));
document.write(linebreak);
document.write("a / b ="+ (a/b));
document.write(linebreak);
document.write("a % b ="+ (a%b));
document.write(linebreak);
document.write("++a =" + (++a));
document.write(linebreak);
document.write("a++ =" + (a++));
document.write(linebreak);
document.write("--b ="+ (--b));
document.write(linebreak);
document.write("b-- ="+ (b--));
document.write(linebreak);
</script>
</body>
</html>
```

Output



a+b=9
a - b =1
a*b =20
a / b =1.25
a % b =1
++a =6
a++ =6
--b =3
b-- =3

2. Assignment Operators

Assignment operators are used to assign values to variables. The basic assignment operator is '=' which assigns the value at the right side to the operand at its left side. The below tables shows various assignment operators.

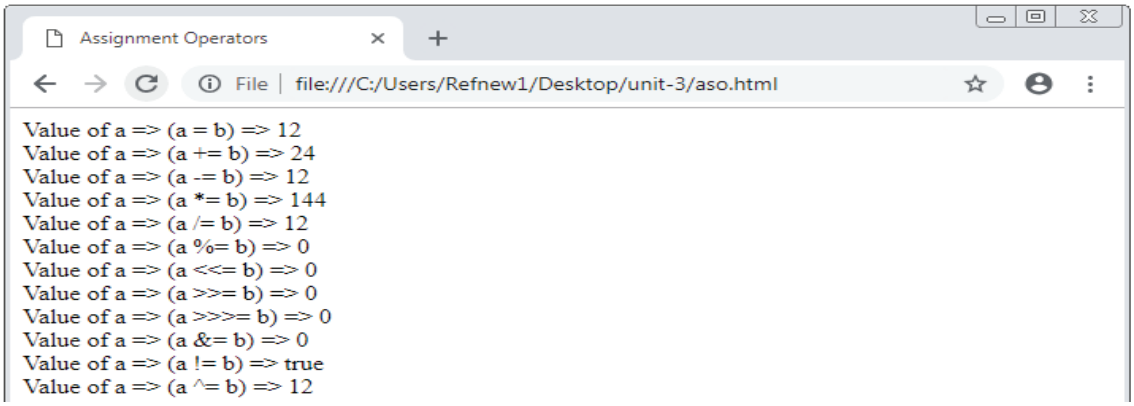
	Operator	Example
(i)	+ =	x + = y (or) x = x + y
(ii)	– =	a – = b (or) a = a – b
(iii)	* =	a* = b (or) a = a * b
(iv)	/ =	a / = b (or) a = a / b
(v)	% =	a % = b (or) a = a % b
(vi)	<< =	a << = b (or)a = a << b
(vii)	>> =	a >> = b (or) a = a >> b
(viii)	>>> =	a >>> = b (or) a = a >>> b
(ix)	& =	a & = b (or) a = a & b
(x)	| =	a | = n (or) a = a | b
(xi)	^ =	a ^ = b (or) a = a ^ b

Example

```
<html>
<body>
<title>Assignment Operators</title>
<script type="text/javascript">
var a = 15;
var b = 12;
var linebreak = "<br />";
document.write("Value of a => (a = b) =>");
result = (a = b);
document.write(result);
document.write(linebreak);
document.write("Value of a => (a += b) =>");
result = (a += b);
document.write(result);
document.write(linebreak);
document.write("Value of a => (a -= b) =>");
result = (a -= b);
document.write(result);
document.write(linebreak);
document.write("Value of a => (a *= b) =>");
result = (a *= b);
document.write(result);
document.write(linebreak);
document.write("Value of a => (a /= b) => ");
```

```
result = (a /= b);
document.write(result);
document.write(linebreak);
document.write("Value of a => (a %= b) =>");
result = (a %= b);
document.write(result);
document.write(linebreak);
document.write("Value of a => (a <<= b) =>");
result = (a <<= b);
document.write(result);
document.write(linebreak);
document.write("Value of a => (a >>= b) =>");
result = (a >>= b);
document.write(result);
document.write(linebreak);
document.write("Value of a => (a >>>= b) =>");
result = (a >>>= b);
document.write(result);
document.write(linebreak);
document.write("Value of a => (a &= b) =>");
result = (a &= b);
document.write(result);
document.write(linebreak);
document.write("Value of a => (a != b) =>");
result = (a != b);
document.write(result);
document.write(linebreak);
document.write("Value of a => (a ^= b) =>");
result = (a ^= b);
document.write(result);
document.write(linebreak);
</script>
</body>
</html>
```

Output



3. Comparison Operators

These operators are used to compare numeric and string operands. Some of the comparison operators are as follows,

- **>:** It returns true if the operand at left side is greater than the operand at right side.
- **<:** It returns true if the operand at left side is less than the operand at right side.
- **> =:** It returns true if the operand at left side is greater than or equal to the operand at the right side.
- **< =:** It returns true if the operand at left side is less than or equal to the operand at the right side.

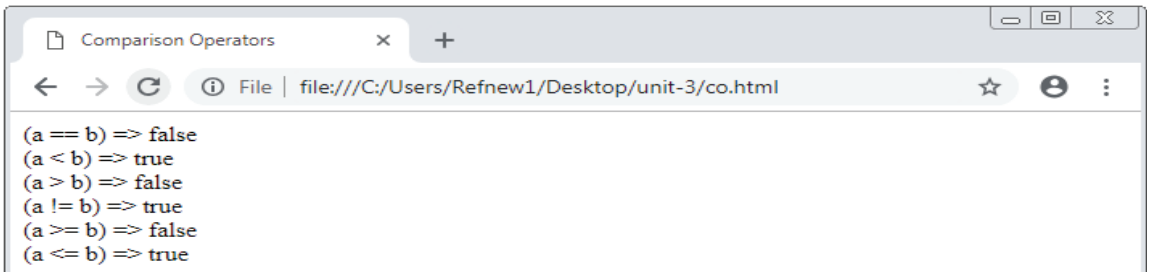
= = (equal) and ! = (not equal) Operator

The equal to (= =) operator is used to compare two operands. It returns true if they are equal. The not equal (! =) operator is used to compare the operands for the opposite condition. It returns true if the two operands are not equal. But they work differently when used with different data types. The boolean, numeric and string types are tested by value whereas arrays, functions and objects are tested by reference. If the operands differ in their type, they are converted to same type.

Example

```
<html>
<body>
<title>Comparison Operators</title>
<script type= “text/javascript”>
var a =7;
var b =25;
var linebreak = “<br />”;
document.write(“(a == b) =>”);
result = (a == b);
document.write(result);
document.write(linebreak);
document.write(“(a < b) =>”);
result = (a < b);
 document.write(result);
document.write(linebreak);
document.write(“(a > b) =>”);
result = (a > b);
document.write(result);
document.write(linebreak);
document.write(“(a != b) =>”);
result = (a != b);
document.write(result);
document.write(linebreak);
document.write(“(a >= b) =>”);
result = (a >= b);
document.write(result);
document.write(linebreak);
document.write(“(a <= b) =>”);
result = (a <= b);
document.write(result);
document.write(linebreak);
</script>
</body>
</html>
```

Output



4. Logical or Boolean Operators

The operators which return the boolean values true or false are called as logical operators or boolean operators. Various logical operators are &&, | | and !.

(a) && Operator

&& operator is used for checking whether the two conditions are true.

Example

(x > y && a = = 1)

The above expression when evaluated returns true if the two conditions are satisfied.

(b) \\ Operator

\\ operator is used for checking if any one of the condition is true among the given expression.

Example

(x > 5 \\ a = 1)

The above expression when evaluated returns true if any one of the condition is satisfied.

(c) ! Operator

! operator is used for checking the given condition is true or not.

Example

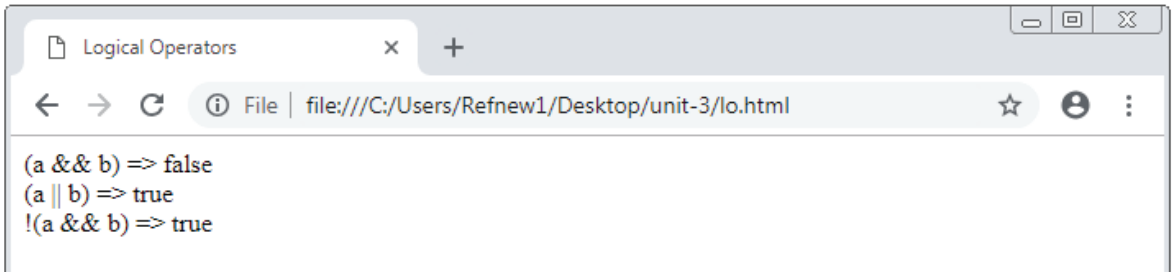
!(x > y)

The above expression when evaluated returns true if the values given to it satisfy the condition.

Example

```
<html>
<body>
<title>Logical Operators</title>
<script type="text/javascript">
var a = true;
var b = false;
var linebreak = "<br />";
document.write("(a && b) =>");
result = (a && b);
document.write(result);
document.write(linebreak);
document.write("(a || b) =>");
result = (a || b);
document.write(result);
document.write(linebreak);
document.write("!(a && b) =>");
result = (!(a && b));
document.write(result);
document.write(linebreak);
</script>
</body>
</html>
```

Output



5. String Operators

"+" operator is used for adding the text to the strings.

Example

word1 = "Hello"

word2 = "World"

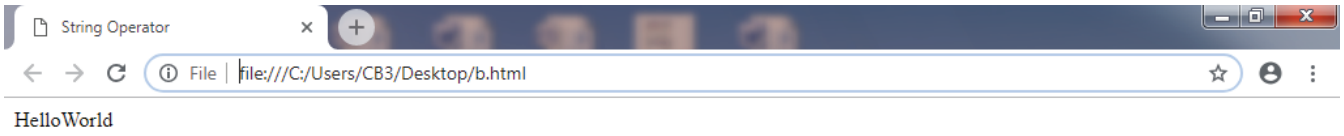
word = word1 + word2

Concatenation is defined as the addition operation on two strings

Program

```
<html>
<body>
<title>String Operator</title>
<p id="demo"></p>
<script>
var a = "Hello";
var b = "World";
c=a+b;
document.write(c);
</script>
</body>
</html>
```

Output



Q22. Explain about functions used in Javascript.

Answer :

Functions

A function is a self contained block of code that performs specific and well defined tasks. It simplifies complex task by dividing the program into modules. They are as follows,

(i) Function Declaration

Function declaration declares a function that consists of a function type (or return type), function name parameter (or argument) list and a terminating semicolon. Function declaration is also called function prototype. A function must be declared before it is invoked.

Syntax

Function function_name (parameters)

(ii) Function Definition

A function definition is the complete description of a particular operation. Actually a function definition tells the operation to be performed. A function definition contains a function body (a block of statements) in addition to function name, arguments list and return type.

Syntax

```
Function function_name(parameters)
{
    statements;
    ———
    ———
    ———
}
```

(iii) Function Call

A function can be called by using function name and a list of actual parameters enclosed in paranthesis. When a function call is encountered, the control is transferred to the respective function, it is executed and the return value is returned to the function which called it.

Syntax

function_name(parameters)

Example

```
<HTML>
<HEAD>
<TITLE>Functions</TITLE>
<SCRIPT>
function hello( )
{
document.write("Hello");
}
hello();          //function calling
</SCRIPT>
</HEAD>
<BODY>
</BODY>
</HTML>
```

Output



Return Statement

Return statement is used to return a value or a result. This statement is used to specify the value that is returned when a function is called.

Example

Consider the below example that calculates the area of a rectangle by using a function,

Program

```
<html>
<head>
<script type="text/JavaScript">
function Area_Rectangle(length, width)
{
area =  length * width;
return area;
}
</script>
<title>Area of a Rectangle</title>
</head>
<body>
<form name="frmArea">
Enter the length and width of your rectangle to calculate the area:<br />
Length: <input type="text" name="txtLength" size="4" /><br />
Width: <input type="text" name="txtWidth" size="4" /><br />
<input type="button" value= "Calculate Area" onclick= "alert(Area_Rectangle(document.frmArea.txtLength.value, document.frmArea.txtWidth.value))" />
</form>
</body>
</html>
```

Output



3.1.4 Control Statements, Looping, Events

Q23. What are the different types of conditional or control statements.

Answer :

Conditional Statements

Conditional statements can be defined as the statements get executed based on a specific condition. If the condition is true, block of statements are executed. Otherwise, control flow comes out of the block.

1. if

'if' is the most basic control statement which supports execution of certain statements after satisfying a condition contained in it.

Syntax

```
if (condition)
statement;
```

Example

```
<html>
<head>
<title>Example of If Statement</title>
```

```
</head>
<body>
<script language = "JAVASCRIPT">
var x=2;
if(x>0)
{
document.write(x + "is a positive number");
}
</script>
</body>
</html>
```

Output



2. if-else

'if-else' is an extension to 'if' control statement. The only difference between them is the 'else' statement contain certain block of statement(s) which get's executed only if its superior 'if' condition becomes false.

Syntax

```
if (condition)
statement;
else
statement;
```

Example

```
<html>
<head>
<title>Example of If-else Statement</title>
</head>
<body>
<script language = "JAVASCRIPT">
var x=-2;
if(x>0)
{
document.write(x + "is a positive number");
}
else
{
document.write(x + "is a negative number");
}
</script>
</body>
</html>
```

Output



3. Switch

A 'switch' statement is a multi-way decision making statement.

Syntax

```
switch(expression)
{
    case constant1: statement-1;
                    break;
    case constant2: statement-2;
                    break;
    case constantn: statement-n;
                    break;
    default: default-statement;
                    break;
}
statement next;
```

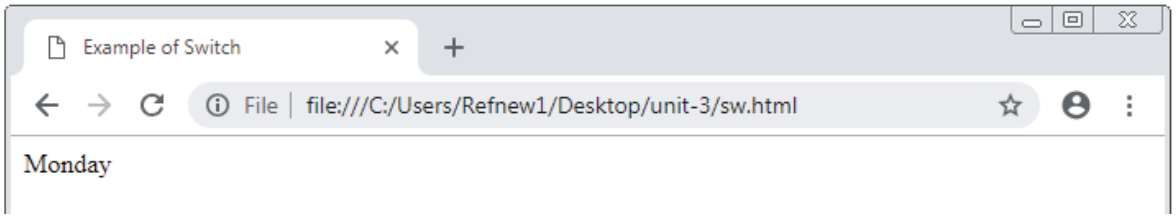
Here, the expression is a valid expression and constant is the result of the expression. When the constant value is a character, it has to be enclosed within single quotes.

The value of the expression is evaluated and it is compared with constant case values. When match is found, the corresponding statement block associated with the case is executed.

Example

```
<html>
<title>Example of Switch</title>
<body>
<script>
switch(new Date( ).getDay( ))
{
case 0:
day = "Sunday";
break;
case 1:
day = "Monday";
break;
case 2:
day = "Tuesday";
break;
case 3:
day = "Wednesday";
break;
case 4:
day = "Thursday";
break;
case 5:
day = "Friday";
break;
case 6:
day = "Saturday";
break;
} document.write(day);
</script>
</body>
</html>
```

Output



Q24. What are the different types of loops?

Answer :

Looping

Looping is a process of executing a statement until the given condition is true. It is necessary to terminate the loop as soon as the required task is completed. Hence, a condition is used to control the loop before or after the execution of every block of code. Various types of loops used in JavaScript are illustrated as follows,

1. for loop
2. while loop
3. do-while loop.

1. for Loop

A set of instructions can be executed repeatedly for a specific number of times by using for loop. The general syntax of it is as follows,

Syntax

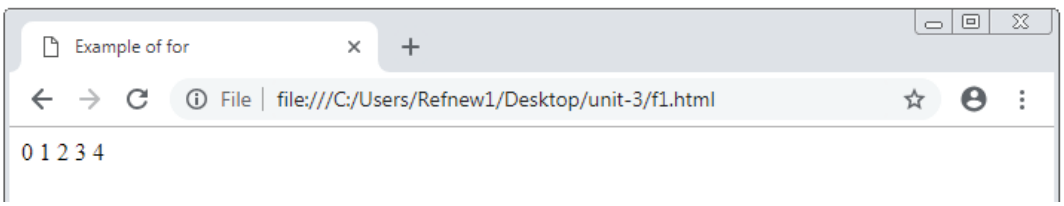
```
for([initial exp]; [condition]; [update exp])
{
    statements;
}
```

The for loop contains three parts namely the initial expression, a conditional expression and the update expression. The initial expression will be executed only once while entering the loop. The body of the loop will be executed if and only if the condition is true. The 'update exp' will be executed at the end of the loop to update the variable. This procedure will be continued until the condition satisfies.

Example

```
<html>
<title>Example of for</title>
<body>
<script>
for(i=0;i<5;i++)
{document.writeln(i+ ' ');}
</script>
</body>
</html>
```

Output



2. while Loop

This loop is used to execute a set of statements until the condition remains true. The general syntax of it is as follows,

Syntax

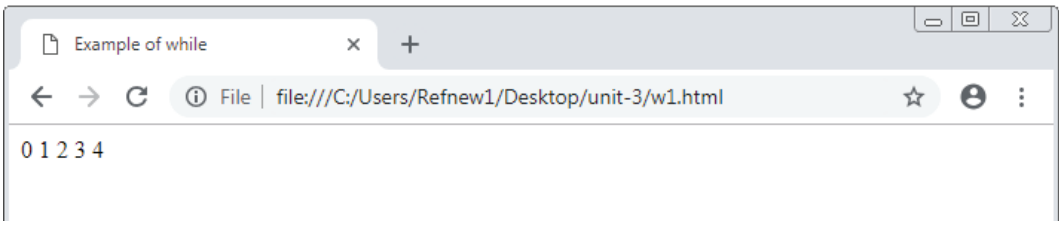
```
while(condition)
{
    Statements;
}
```

The condition will be executed at the beginning of the loop and in every iteration. The body of the loop will be executed only if the condition is true. If not, the control of execution goes to the end of while loop. The variable will be updated at the end of the loop.

Example

```
<html>
<title>Example of while</title>
<body>
<script>
i=0;
while(i<5)
{
document.writeln(i+' ');
i++;
}
</script>
</body>
</html>
```

Output



3. do-while Loop

do-while loop is same as that of while loop but the condition for it is specified at the end of the loop. The general syntax of it is as follows,

Syntax

```
do
{
    Statements;
}
while(condition);
```

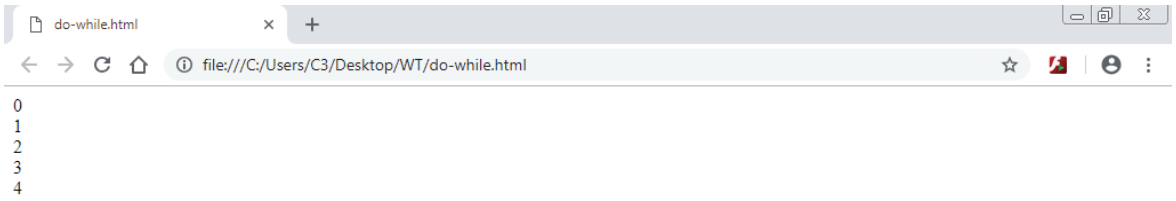
The body of the loop gets executed atleast once even if the specified condition is false.

Example

```
<html>
  <body>
    <script type="text/javascript">
```

```
        var i = 0;
        do
        {
          document.write(+ i + "<br />");
          i++;
        }
        while (i < 5);
    </script>
  </body>
</html>
```

Output



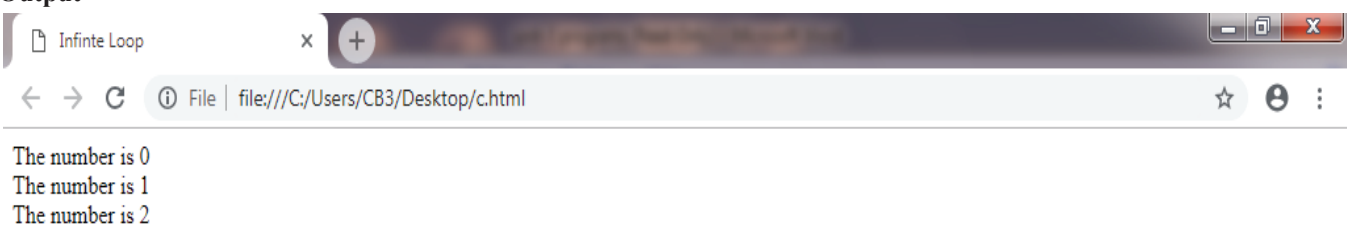
Infinite Loops and Break Statement

An expression that returns true will end with an infinite loop. This will lead to hangup of the program and may even crash the computer, webbrowsers are trying to find infinite loops and stop that loop. A break statement is used to skip the infinite loop. This is used to prevent a loop from executing multiple times.

Example

```
<html>
<body>
<title> Infinte Loop</title>
<script>
var text = "";
var i = 0;
while (i < 5) {
document.write("The number is " + i + "<br/>");
i++;
if (i == 3)
{
break;
}
}
</script>
</body>
</html>
```

Output



In the above program, the execution terminates when a break statement is encountered.

Q25. Explain about events in Javascript with its event handlers.

Answer :

Event

Responses provided by the browser as a response to user's interactions are referred as events. For example, playing audio clip as soon as the page is loaded, generating informative text when mouse pointer is moved through a certain region of the web page, submtting user entered data to the server when submit button is clicked etc., are the types of events.

Event Handler

Once the event is generated, there is often requirement of code to process these events. Such code is known as event handler. In general, event handlers are of two types.

(a) Interactive Event Handlers

The event handlers which simply rely on the user's activity to be invoked are interactive event handlers.

Example: onClick, onBlur, etc.

(b) Non-Interactive Event Handlers

The event handlers which do not rely on the user's to be invoked are non-interactive event handlers.

Example: onPageLoad.

Types of Events

The two types of events are used to trigger scripts are as follows,

1. Window Events

Window events are the events that occur as soon as something happens to the window. They occur when a page loads or windows load or when the focus moves to and from a window or frame.

2. User Event

User events are the events that occur when user interacts with page elements by using mouse or keyboard or other pointing device.

Commonly Used Event Handlers

Event	Description
onload	It occurs when the page loads (this can be used to load images, documents, frameset).
onunload	It occurs when the user quits a document or frameset i.e., a page is unloaded.
onclick	It occurs when a link, client side image map area, or a form element (like button) is clicked.
ondbclick	It occurs when an element is double clicked.
onmousedown	It occurs when the user presses a mouse button
onmouseup	It occurs when the user releases a mouse button
onmouseover	It occurs when the mouse is moved over a link or an area of client side image map.
onmousemove	It occurs when the user moves the mouse.
onmouseout	It occurs when the mouse is moved out of a link or an area of client side image map.
onkeypress	It occurs when the key is pressed and released over the element.
onkeydown	It occurs when the user presses a key.
onkeyup	It occur when the user releases the key.
onfocus	This event will be called when an element like a document, window, frameset, or form element gets the current input focus.
onsubmit	It occurs when the user presses a submit button.
onreset	It occurs when the user clicks the reset button.
onselect	It occurs when the text is selected in a text field or a text area.
onblur	It invokes as soon as any text, data or image turns blur (or object losses its focus).
onchange	It invokes whenever data in any of the HTML control (textarea, textfield etc.) gets changed.

Example

```
<a href = " "
onmouseover = "document.images.link.src = 'images/click-img1.fig';"
onmouseout = "document.images.link.src = 'images/click-img2.gif';">
<img src = "images/click-img2.gif" width = "150" height = "100" border = "0" name = "links">
</a>
```

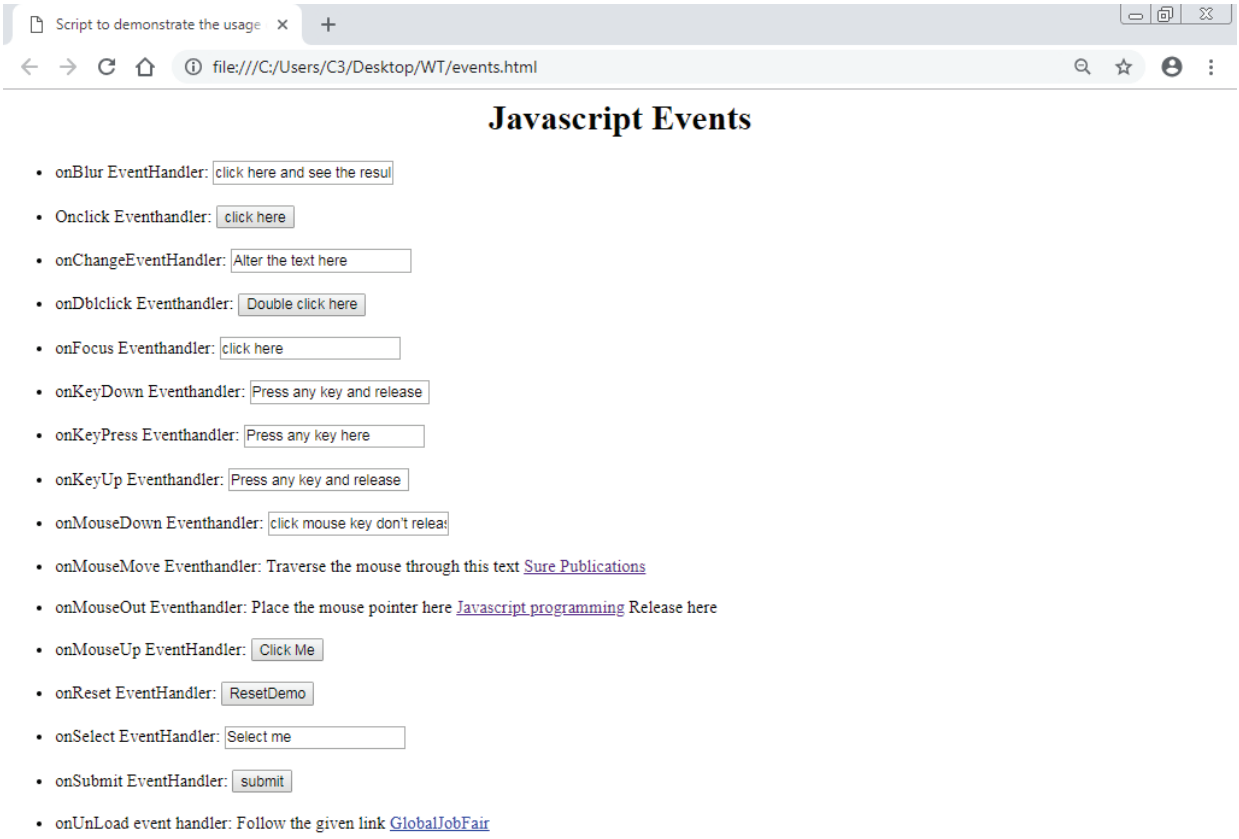
Program

Events

```
<HTML>
<HEAD>
<TITLE>Script to demonstrate the usage of frequently used events</TITLE>
</HEAD>
<BODY onLoad='alert("Examples of javascript events")' >
<H1 align="center">Javascript Events</H1>
<ul>
<li>onBlur EventHandler:
<input type=text value="click here and see the result"
size="20" onBlur= 'alert("NOT CLEAR!")'>
<BR>
<BR>
<li>Onclick Eventhandler:
<input type=button value="click here" onClick= 'alert("Clicked")'>
<BR>
<BR>
<li>onChangeEventHandler:
<input type=text value="Alter the text here" size=20 onChange='alert("Changed")')>
<BR>
<BR>
<li>onDblclick Eventhandler:
<input type=button value="Double click here" onDblClick='alert("Double clicked")'>
<BR>
<BR>
<li>onFocus Eventhandler:
<input type=text value="click here" onFocus='alert("Focus gained")'>
<BR>
<BR>
<li>onKeyDown Eventhandler:
<input type=text value="Press any key and release" size=20 onKeyDown='alert ("Key Down!")'>
<BR>
<BR>
<li>onKeyPress Eventhandler:
<input type=text value="Press any key here" size=20 onKeyPress='alert ("key was pressed")'>
<BR>
<BR>
<li>onKeyUp Eventhandler:
```

```
<input type=text value="Press any key and release" onKeyUp='alert("Key up")'>
<BR>
<BR>
<li>onMouseDown Eventhandler:
<input type=text value="click mouse key don't release"size=20
onMouseDown='alert("Mouse Down")'>
<BR>
<BR>
<li>onMouseMove Eventhandler:
Traverse the mouse through this text
<A href=" "onMouseMove='alert("mouse traversed")'>Sure Publications</A>
<BR>
<BR>
<li>onMouseOut Eventhandler:
Place the mouse pointer here
<A href=" "onMouseOut='alert("mouse out")'>Javascript programming</A> Release here
<BR>
<BR>
<li>onMouseUp EventHandler:
<input type="Button" value="Click Me" onMouseUp='alert("MouseUp")'>
<BR>
<BR>
<li>onReset EventHandler:
<input type="RESET" value="ResetDemo">
<BR>
<BR>
<li>onSelect EventHandler:
<input type="text" value="Select me" size=20 onSelect='alert("TextSelected")'>
<BR>
<BR>
<li>onSubmit EventHandler:
<input type="submit" value="submit">
<BR>
<BR>
<li>onUnLoad event handler:
Follow the given link <a href="https://globaljobfair.com">GlobalJobFair</a>
</A>
<BR>
<BR>
</ul>
</BODY>
</HTML>
```

Output



3.1.5 Built-in Objects

Q26. What are the different types of Built-in objects in Javascript?

Answer :

The following are the different types of Built-in objects in javascript.

1. String Object

For answer refer Unit-III, Page No. 205, Q.No. 27.

2. Date Object

For answer refer Unit-III, Page No. 208, Q.No. 28.

3. Math Object

For answer refer Unit-III, Page No. 211, Q.No. 29.

4. Window Object

For answer refer Unit-III, Page No. 215, Q.No. 31.

5. Array Object

For answer refer Unit-III, Page No. 213, Q.No. 30.

Q27. Discuss about string object with its properties and methods.

Answer :

String Object

String function forms one of the important consequences in JavaScript. In general terms, strings refer to set of characters (such as alphabets, special characters, numerals, combination of them) enclosed within double quotes. Like math, even strings are treated as object which supports numerous methods.

Property of String Object

Length

It is the property used for string object. It is used to specify the count or number of a characters given in a string.

Methods

The following are the different methods or string objects.

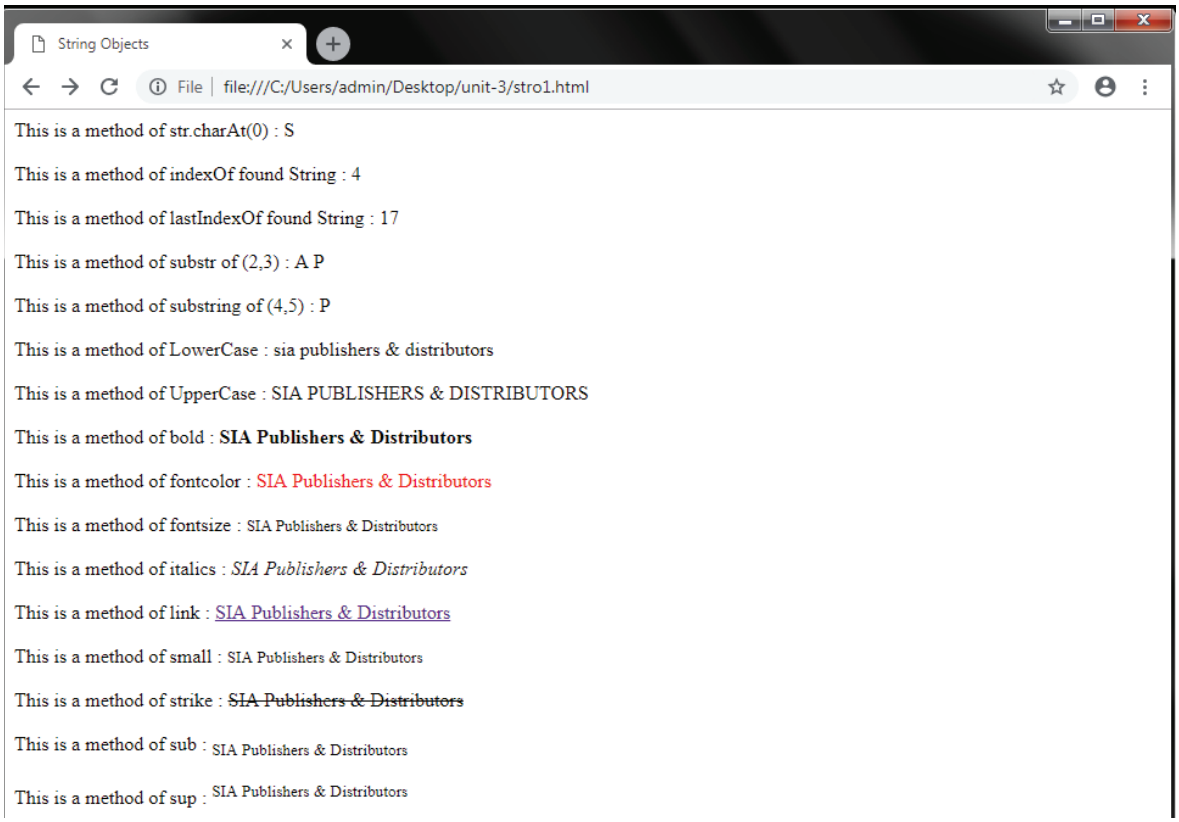
	Method	Description
1.	anchor(name)	It creates an anchor element with the name attribute.
2.	big( )	It is used to display the text as given in a <big> element.
3.	bold( )	It is used to display the text as given in a <bold> element.
4.	charAt(index)	It returns a character whose index value is supplied as an argument to the method.
5.	fixed	It is used to display the text as given in a <tt> element.
6.	fontcolor(color)	It is used to display the text as given in a <font> element along with the value of color attribute.
7.	fontsize(fontsize)	It is used to display the text given as in a <font> element along with the value of size attribute.
8.	indexof(search value, [FromIndex])	It usually takes character(s) as argument and returns the numeric value, which is nothing but the position of that character(s) in the given string.
9.	italics( )	It is used to display the text given as in a <i> element.
10.	lastIndexOf(search value, [fromIndex])	It again takes character(s) as an argument and returns the numeric value, which is nothing but the position of that character (s) in the given string. The difference between lastIndexOf( ) and index Of( ) is that the prior start searching string (for the given character(s)) from the end of the string and post from the beginning. But both of the methods returns numeric values starting from the beginning of the string.
11.	link(targetURL)	It is used to create a link in the given document.
12.	small( )	It is used to display the text as given in the <small> element.
13.	strike( )	It is used to display the text as given in the <strike> element.
14.	sub( )	It is used to display the text as given in the <sub> element.
15.	substr(start, [length])	It takes two arguments “index” and “length” respectively. The index argument specifies the initiating character and the length argument specifies the length of characters to be displayed.
16.	substring(start position, end position)	It takes two arguments “index” and “end” respectively. Here, index argument specifies the initiating character and the end specifies the last character to be displayed.
17.	sup( )	It is used to display the text given as in the <sup> element.
18.	toLowerCase( )	It takes a string of all uppercase characters and convert them into their equivalent lowercase strings. If the string contains few non-characters, then they remain unchanged.
19.	toUpperCase( )	It takes a string of all lowercase characters and converts them into their equivalent upper case string. If the string contains few non-characters, then they remain unchanged.

Example

```
<html>
<head>
<title>String Objects</title>
</head>
<body>
<script type=”text/javascript”>
var str = new String( “SIA Publishers & Distributors” );
var URL = “http://www.siapublishers.com/”;
```

```
var index = str.indexOf( “Publishers” );
document.writeln(“This is a method of str.charAt(0) :” + str.charAt(0));
document.write(“<br /><br />This is a method of indexOf found String :” + index );
var index = str.lastIndexOf( “Distributors” );
document.write(“<br /><br /> This is a method of lastIndexOf found String :” + index );
document.write(“<br /><br />This is a method of substr of (2,3) :  “  + str.substr(2,3));
document.write(“<br /><br />This is a method of substring of (4,5) :  “  + str.substring(4,5));
document.write(“<br /><br /> This is a method of LowerCase : “+str.toLowerCase( ));
document.write(“<br /><br /> This is a method of UpperCase : “ +str.toUpperCase( ));
document.write(“<br /><br /> This is a method of bold : “ +str.bold());
document.write(“<br /><br /> This is a method of fontcolor : “ +str.fontcolor( “Red” ));
document.write(“<br /><br /> This is a method of fontsize : “ +str.fontsize( 2 ));
document.write(“<br /><br /> This is a method of italics : “ +str.italics( 4));
document.write(“<br /><br /> This is a method of link : “ +str.link( URL ));
document.write(“<br /><br />This is a method of small : “ +str.small());
document.write(“<br /><br /> This is a method of strike : “ +str.strike());
document.write(“<br /><br />This is a method of sub : “ +str.sub());
document.write(“<br /><br /> This is a method of sup : “ +str.sup());
</script>
</body>
</html>
```

Output



Q28. Explain about date object with its properties and methods.

Answer :

Date Object

Date is one of the important specifications in almost every programming discipline. Hence, JavaScript has reserved a separate class in this aspect. This date class includes variety of methods to access the date.

Globally date/time is recognized in two ways i.e., UTC(Universal Time Coordinated) and local time. Universal time as the name suggests is a standard followed through out the world. On the other hand, local time is the time of the system where the script is currently residing.

Methods of Date Object

The following are the methods of date object.

Method		Description
1.	date( )	It is used to return a date object.
2.	getDate( )	It is used to return the date of a date object i.e., from 1 to 31
3.	getDay( )	It is used to return the day of a Date object. i.e., from 0 to 6 where 0 = Sunday, 1 = Monday.
4.	getMonth( )	It is used to return the month of a Date object i.e. From 0 to 11 where 0 = january.
5.	getFullYear( )	It is used to return the year of a date object i.e., four digits.
6.	getYear( )	It is used to return the year of a Date object by using only two digits i.e., from 0 to 99.
7.	getHours( )	It is used to return the hours of a Date object. i.e., from 0 to 23.
8.	getMinutes( )	It is used to return the minutes of a Date object. i.e., from 0 to 59.
9.	getSeconds( )	It is used to return the seconds of a Date object. i.e., from 0 to 59.
10.	getTime( )	It is used to return the number of milliseconds from midnight 1/1/1970.
11.	getTimezoneOffset( )	It is used to return the time difference between the user's computer and GMT.
12.	parse( )	It is used to return the string date value that contains the number of Milliseconds from january 01 1970 00 : 00 : 00.
13.	setDate( )	It is used to set the date of the month in the Date object i.e., from 1 to 31.
14.	setFullYear( )	It is used to set the year in the Date object. i.e., four digits.
15.	setHours( )	It is used to set the hours in the Date object i.e., from 0 to 23.
16.	setMinutes( )	It is used to set the minutes in the Date object. i.e., from 0 to 59.
17.	setMonth( )	It is used to set the months in the Date object. i.e., from 0 to 11 where 0 = january.
18.	setSeconds( )	It is used to set the seconds in the Date object. i.e., from 0 to 59.
19.	setTime( )	It is used to set the milliseconds after 1/1/1970.
20.	setYear( )	It is used to set the year in the Date object. i.e., from 00 to 99.
21.	toGMTString( )	It is used for converting the Date object to string and set to GMT time zone.
22.	toLocaleString( )	It is used for converting the Date object to a string and set to the current time zone of the user.
23.	toString( )	It is used for converting the Date object to the string.

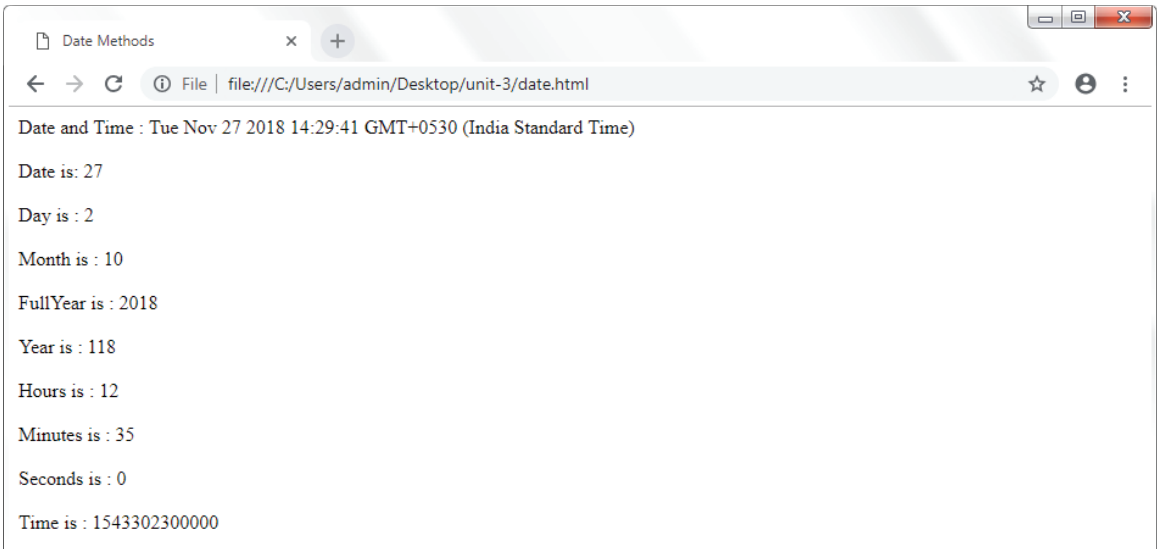
Example

(i) Getting the Values

```
<html>
<head>
<title>Date Methods</title>
</head>
<body>
```

```
<script type="text/javascript">
var dt =  Date();
document.write("Date and Time :" + dt );
var dt = new Date( "Nov 27, 2018 12:35:00" );
document.write("<br/><br/>Date is:" + dt.getDate() );
document.write("<br/><br/>Day is :" + dt.getDay() );
document.write("<br/><br/>Month is :" + dt.getMonth() );
document.write("<br/><br/>FullYear is :" + dt.getFullYear() );
document.write("<br/><br/>Year is :" + dt.getYear() );
document.write("<br/><br/>Hours is :" + dt.getHours() );
document.write("<br/><br/>Minutes is :" + dt.getMinutes() );
document.write("<br/><br/>Seconds is :" + dt.getSeconds() )
document.write("<br/><br/>Time is :" + dt.getTime() );
document.write("<br/><br/>TimezoneOffset is :" + tz );
</script>
</body>
</html>
```

Output

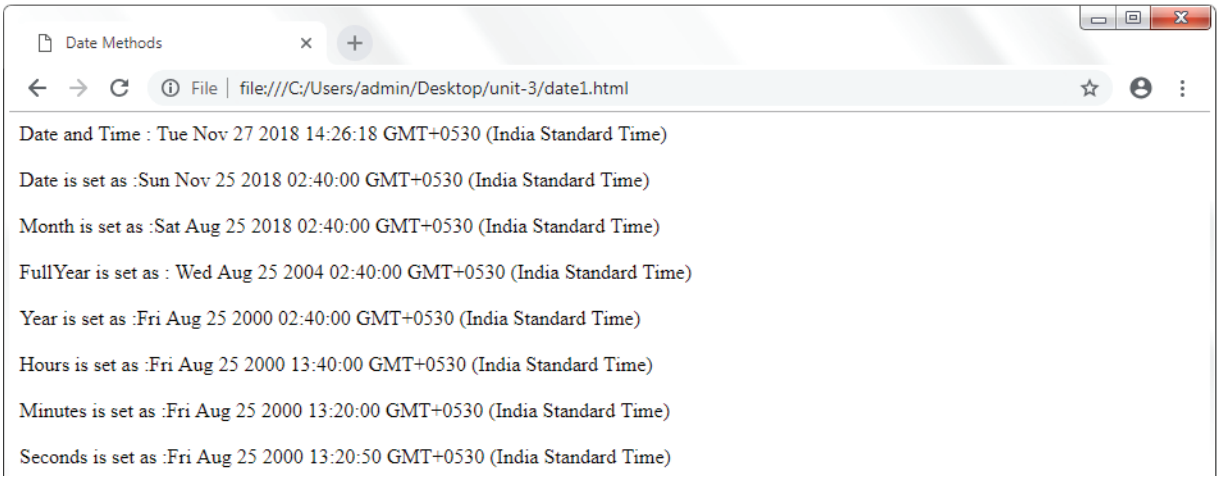


(ii) Setting the Values

```
<html>
<head>
<title>Date Methods</title>
</head>
<body>
<script type="text/javascript">
var dt =  Date();
document.write("Date and Time :" + dt );
var dt = new Date( "Nov 27, 2018 02:40:00" );
document.write("<br/>");
```

```
document.write("<br/>");
dt.setDate(25);
document.write("Date is set as :"+dt );
document.write("<br/>");
document.write("<br/>");
dt.setMonth(07) ;
document.write("Month is set as :"+dt );
document.write("<br/>");
document.write("<br/>");
dt.setFullYear(2004);
document.write("FullYear is set as :"+dt );
document.write("<br/>");
document.write("<br/>");
dt.setYear(2000 );
document.write("Year is set as :"+dt );
document.write("<br/>");
document.write("<br/>");
dt.setHours(13);
document.write("Hours is set as :"+dt );
document.write("<br/>");
document.write("<br/>");
dt.setMinutes(20) ;
document.write("Minutes is set as :"+dt );
document.write("<br/>");
document.write("<br/>");
dt.setSeconds(50) ;
document.write("Seconds is set as :"+dt );
document.write("<br/>");
</script>
</body>
</html>
```

Output



Q29. Write about Math object with its properties and methods.

Answer :

Math Object

Javascript has introduced various mathematical functions in order to perform various mathematical operations. These mathematical functions or methods are entities of "Math" object.

Properties of Math Object

The following are the properties of math object.

	Property	Description
1.	E	It is used to return the value of base of a natural logarithm.
2.	LN2	It is used to return the value of a natural logarithm of base 2.
3.	LN10	It is used to return the value of a natural logarithm of base 10.
4.	LOG2E	It is used to return the value of a base-2 logarithm of E.
5.	LOG10E	It is used to return the value of a base-10 logarithm of E.
6.	PI	It is used to return the value of Pi.
7.	SQRT1-2	It is used to return the value of 1 divided by the square root of 2.
8.	SQRT2	It is used to return the value of square root of 2.

Methods of Math Object

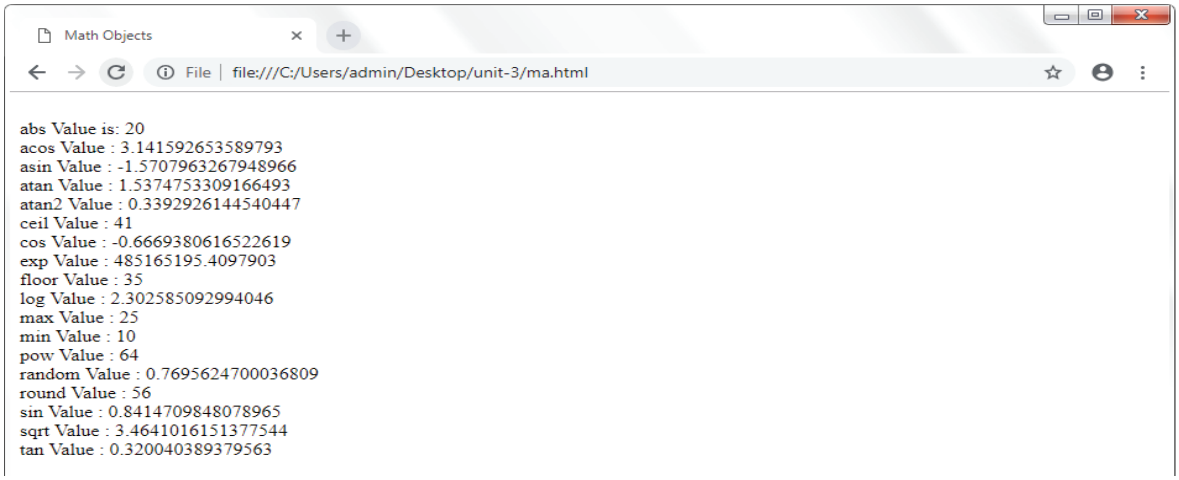
The following are the methods or math object.

	Method	Description
1.	abs(x)	It displays the absolute value of the numeric entity entered into it.
2.	acos(x)	It displays the arc cosine value of x.
3.	asin(x)	It displays the arcsine value of x.
4.	atan(x)	It displays the arctangent value of x.
5.	atan2(y, x)	It displays the tangent angle from the x-axis to a point.
6.	ceil(x)	It displays the rounded value of the integer entered into it. The values displayed as output always remains greater than the value supplied.
7.	cos(x)	It displays trigonometric cosine value for the numeric value passed to it.
8.	exp(x)	It displays exponential(i.e., $e^{numeric}$) value of the numeric value passed to it.
9.	floor(x)	It usually rounds the numeric(supplied to it) to the largest integer.
10.	log(x)	It displays logarithmic($\log_e$) equivalent value for the numeric value supplied.
11.	max(x, y)	It displays maximum of two numerical values entered into it.
12.	min(x, y)	It displays minimum of two numerical values entered into it.
13.	pow(x, y)	When two numerals n1, n2 are supplied to it, then n1 forms the power of n2 and the result displayed as an output.
14.	random( )	It displays the random number between 0 and 1.
15.	round(x)	It rounds the value entered to its nearest integer.
16.	sin(x)	It displays trigonometric sine value for the numeric value passed to it.
17.	sqrt(x)	It displays the square root of the value entered into it.
18.	tan(x)	It displays trigonometric tangent values for the numeric value passed to it.

Example

```
<html>
<head>
<title>Math Objects</title>
</head>
<body>
<script type="text/javascript">
var value = Math.abs(20);
document.write("<br /> abs Value is:" + value );
var value = Math.acos(-1);
document.write("<br />acos Value :" + value );
var value = Math.asin(-1);
document.write("<br />asin  Value :" + value );
value = Math.atan(30);
document.write("<br />atan Value :" + value );
var value = Math.atan2(30,85);
document.write("<br />atan2 Value :" + value );
var value = Math.ceil(40.15);
document.write("<br />ceil  Value :" + value );
var value = Math.cos(40);
document.write("<br />cos Value :" + value );
var value = Math.exp(20);
document.write("<br />exp Value :" + value );
var value = Math.floor(35.9);
document.write("<br />floor Value :" + value );
var value = Math.log(10);
document.write("<br />log Value :" + value );
var value = Math.max(25, 7);
document.write("<br />max  Value :" + value );
var value = Math.min(12,10);
document.write("<br />min  Value :" + value );
var value = Math.pow(4,3);
document.write("<br />pow Value :" + value );
var value = Math.random( );
document.write("<br />random Value :" + value );
var value = Math.round( 55.7 );
document.write("<br />round Value :" + value );
var value = Math.sin( 1 );
document.write("<br />sin  Value :" + value );
var value = Math.sqrt( 12 );
document.write("<br />sqrt Value :" + value );
var value = Math.tan( 60 );
document.write("<br />tan Value :" + value );
</script>
</body>
</html>
```

Output



Q30. Write about Arrays in Javascript.

Answer :

Array

An array contains a set of similar items. But, in javascript an array contains different items. Arrays are similar to structure variables and they can be accessed randomly by making use of index and the array name. Thus, arrays in javascript are also called as associative arrays. This feature makes javascript arrays distinct than arrays of other languages.

Creating and Accessing the Array Elements

An array can be declared as follows,

var arr;

Elements can be allocated to this array by using the below statement,

arr = new Array(10);

An array is not initialized while it is allocated.

Arrays in javascript are considered as objects. They can also be created by making use of constructors. Constructors in javascript are of three types. They are as follows,

- Array( );
- Array(number of elements);
- Array(List of elements separated by commas);

In the above statements, the first constructor will not accept any arguments. It creates an empty array. Elements can be added to it when required. Consider the below statement,

var arr = new Array( );

It creates an array with length zero. Consider the below statement,

arr[3] = 5; //length 4

It adds an element at the location arr[3]. In addition to this, it also creates arr[0], arr[1] and arr[2]. But the values of them are not defined but they can be assigned later on. If the value that is assigned is greater than or equal to the length of the array, then the size of the array can be increased as per the requirement. This method is useful when the number of elements are not known while array declaration.

The length property of an array object is used to store the number of array elements i.e., size of an array. The size of an array can be changed modified by assigning a new value to the length property which is greater than the previous value of current length.

Example

```
Week = new Array("Sunday", "Monday", "Tuesday", "Wednesday");
document.write(week.length);
```

tostring( ) Method

tostring( ) method is used for converting the array into a string.

Example

```
document.write('value is' + week.tostring( ));
```

Methods

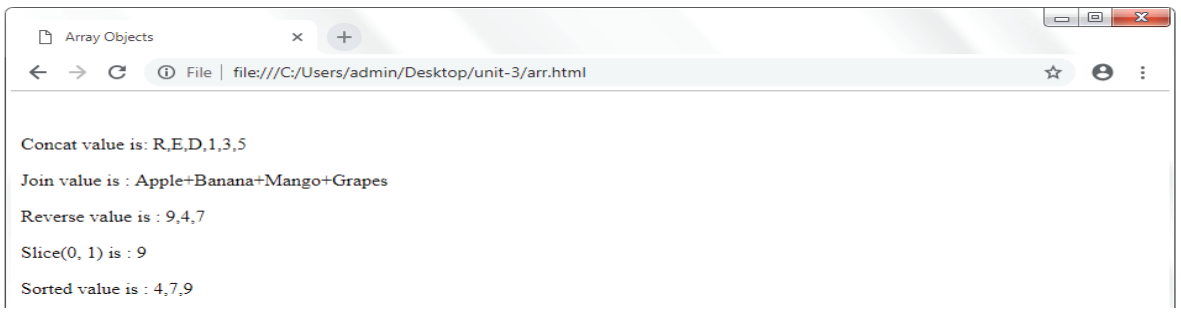
The following are the methods of an array.

	Method	Description
1.	concat( )	This method concatenates or merges arrays and returns a new array.
2.	join(separator)	This method joins the elements in an array into one string by using a default comma as a separator. It is also possible to specify any separator between the array elements.
3.	reverse( )	It is used to return the array elements in an reverse order.
4.	slice( )	It is used to return a selected part of elements of an array.
5.	sort( )	It is used to return the sorted array elements in an alphabetical or numerical order.

Example

```
<html>
<head>
<title>Array Objects</title>
</head>
<body>
<script type="text/javascript">
var alphabets = ["R", "E", "D"];
var numbers = [1, 3, 5];
var result = alphabets.concat(numbers);
document.write("<br /> <br /> Concat value is:" +result  );
var Fruits= ["Apple", "Banana", "Mango", "Grapes"];
var str = Fruits.join("+");
document.write("<br /><br /> Join value is :" + str );
var num = [7, 4, 9].reverse();
document.write("<br /> <br /> Reverse value is :" + num );
document.write("<br /><br /> Slice(0, 1)  is :" +num.slice( 0, 1) );
var sorted = num.sort();
document.write("<br /><br />  Sorted value is :" + sorted );
</script>
</body>
</html>
```

Output



Q31. Discuss about window object with its properties and methods.

Answer :

Window Object

Each and every browser window or a frame contains a window object that is created by an instance of a <body> or <frameset> element.

Example

The status property is used to change the text displayed on the browser's status bar.

```
<script type = "text/javascript">
function statusBarTexts( )
{
window.status = "can you see the window";
}
</script>
```

The above code can be called in the <body> element as shown below,

```
<body onload = "statusBarText( )">
```

Properties of Window Object

The following are the properties of a window object.

Property		Description
1.	closed	A boolean value is used to indicate as true if a window is closed.
2.	defaultstatus	It is used to display the default message on the browser's window status bar.
3.	document	It is used to show a document object which is saved in that window.
4.	frames	An array consists of named child frames in the current window.
5.	history	A history object that has the information of visited URL's in the same window.
6.	location	The location object specifies the URL of the current window.
7.	name	The name is used to specify the name of the window.
8.	status	A status is a temporary message that is set at any time to display in the status bar.
9.	statusbar	A statusbar specifies whether the status bar can be visible or not. It can be known by a boolean value (true or false).
10.	toolbar	The toolbar specifies whether the toolbar can be visible or not. Its can be known by a boolean value (true or false).
11.	top	Top specifies the reference for the topmost window when different windows are opened on the desktop.
12.	window	Window specifies the currently opened window or a frame.

Methods of an Window Object

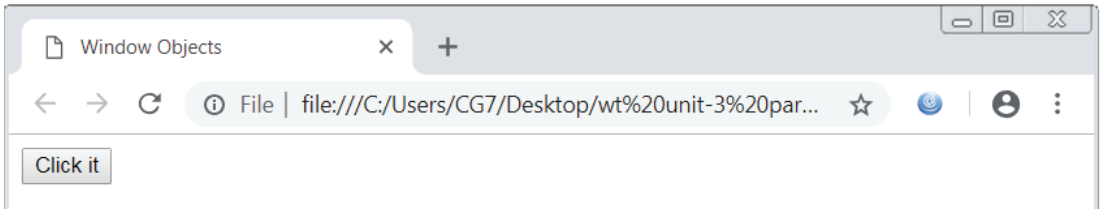
The following are the methods of an window object.

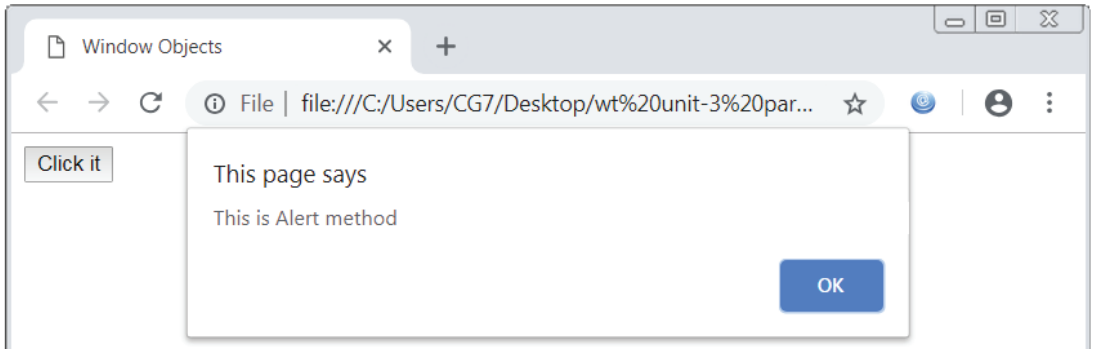
	Method	Description
1.	alert( )	It is used for displaying an alertbox with message and ok button.
2.	back( )	It has the same effect as the browser's back button.
3.	blur( )	It is used for removing and focus or making it to loose its focus from the current window.
4.	close( )	It is used to close the current window.
5.	confirm( )	It is used to confirm whether the user want to perform any action or not by choosing any one option ok or cancel shown in the dialog box.
6.	focus( )	It is used to provide focus to a particular window and make it as a top.
7.	forward( )	It is similar to the clicking of the browser's forward button.
8.	home( )	It is used to allow the user to reach their browser's designated homepage.
9.	moveBy(horizantal pixels, vertical pixels)	It is used to move the window by a given number of pixels with its current co-ordinates.
10.	moveTo(Xposition, Yposition)	It is used to move the top left side of the window to the given x-y coordinates.
11.	openlURL$_1$name [$_1$features])	It is used for opening a new browser window.
12.	print( )	It is used to print the content of the current window.
13.	prompt( )	It is used for creating a dialog box to enter an input by the user.
14.	stop( )	It is used same as clicking the stop button in the browser.

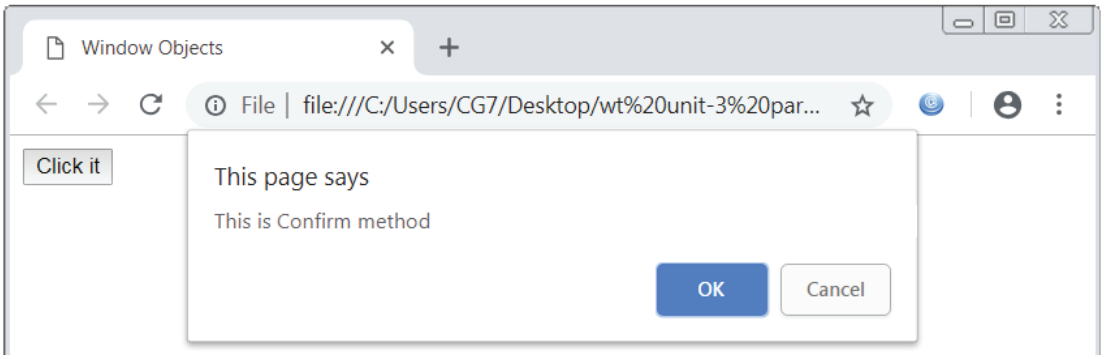
Example

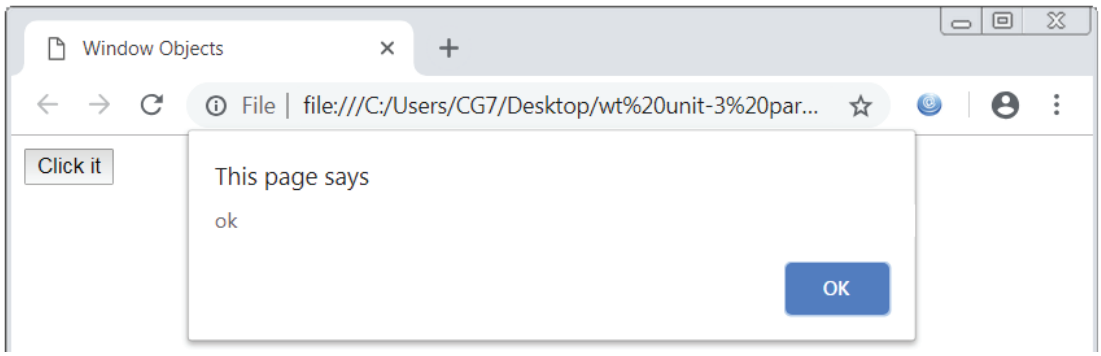
```
<html>
<body>
<title>Window Objects</title>
<script type= "text/javascript">
function msg()
{
alert("This is Alert method");
var a= confirm("This is Confirm method");
if(a==true){
alert("ok");
}
else{
alert("cancel");
}
var b= prompt("This is Prompt method");
alert("I am "+b);
open("http://www.google.com");
setTimeout(
function(){
alert("Google page is closed")
},2000);
var myWindow = window.open(" ", " ", "width=200, height=100");
myWindow.document.write("<p>A new window!</p>");
myWindow.blur();
}
</script>
<input type="button" value="Click it" onclick="msg()"/>
</body>
</html>
```

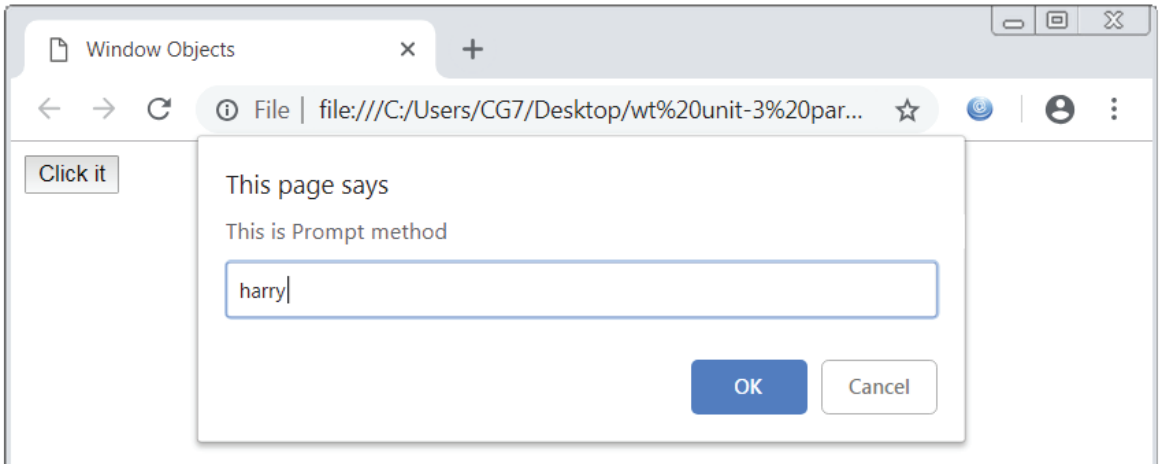
Output

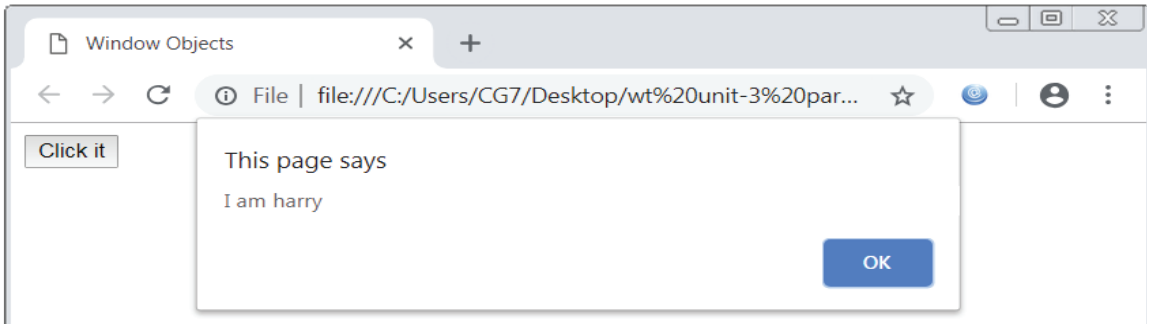

Window Objects
File | file:///C:/Users/CG7/Desktop/wt%20unit-3%20par...
Click it

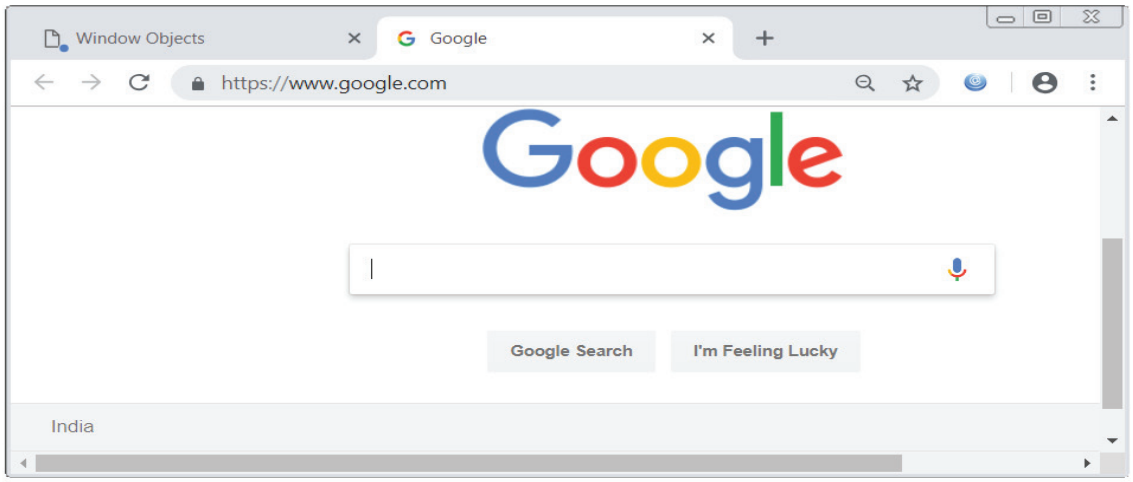

Window Objects
File | file:///C:/Users/CG7/Desktop/wt%20unit-3%20par...
Click it
This page says
This is Alert method
OK

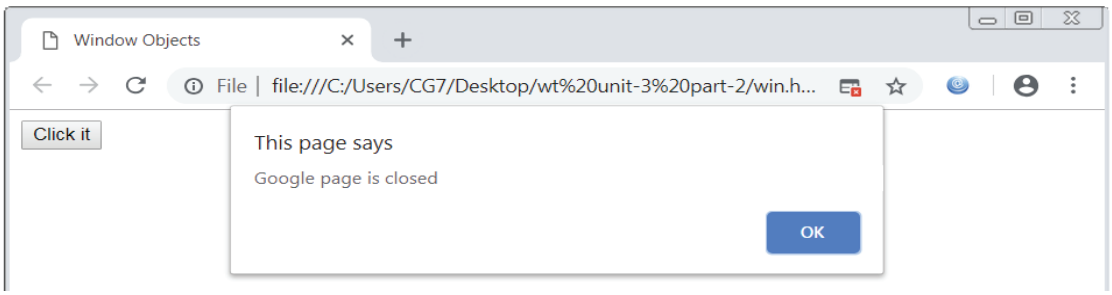

Window Objects
File | file:///C:/Users/CG7/Desktop/wt%20unit-3%20par...
Click it
This page says
This is Confirm method
OK
Cancel


Window Objects
File | file:///C:/Users/CG7/Desktop/wt%20unit-3%20par...
Click it
This page says
ok
OK


Window Objects
File | file:///C:/Users/CG7/Desktop/wt%20unit-3%20par...
Click it
This page says
This is Prompt method
harry
OK
Cancel







3.2 WORKING WITH JAVASCRIPT

3.2.1 Practical Tips for Writing Scripts, Form Validation

Q32. What are the practical tips for writing Java Scripts?

Answer :

The following are the tips used for writing java scripts.

(i) There are many sites available on the internet which provide javascirpt code for the programmers. The examples of such sites are www.HotScripts.com, www.JavaScriptKit.com, www.webreference.com/programming/javascript/, http://JavaScript.Internet.com. The programmer need to check these sites on internet before writing the script.

(ii) The programmer can write usable code in addition to reusing the scripts and folders of others. Consider an example, user creates many sites which has similar form that permits the users to contacts with the site owners. A form contains different fields which are required to be filled/entered. A script can be created only once by a user and is used on any contact form for any number of times.

(iii) External java script files are used for writing the script in one page and using it in more than one page. Example a newsletter sign up from is created on every page of the site and the text entered by a user in the address field is checked by using a script. If this script saved in any external javascript file then the script is not included in each page.

(iv) A special script folder is created to save the external scripts. This is used for improving the organization of site and directory structure.

Q33. Write about form validation.

Answer : Model Paper-III, Q11(a)

Form Validation

Form validation is the process of verifying or validating the data that is entered by the user. That is form validation is done inorder to check whether the user has entered data in all the mandatory fields and whether the user entered valid data in valid fields etc. JavaScript is used to perform form validation.

Basic Validation

It checks whether all the mandatory fields have been filled. It uses a loop for this purpose. This loop goes through all the fields in the form for checking the data.

Date Format Validation

It checks the correctness of the data entered in the form. This requires necessary logic in the code of the form.

Validation can be performed on both client-side and server-side. Java script allows the data to be validated on client side itself so as to enable faster validation on server-side. This feature of JavaScript is attracting most of the developers.

Client-side Validation

The client side validation is generally performed on forms that gather information from the users. The JavaScript programs written for this purpose connect event handlers to the form elements. This enhances the user interface and restricts the invalid data to be sent to the server.

Server-side Validation

Web servers are responsible for performing server-side validation. It is basically done after the user input is sent to the server.

Server-side javascript performs execution of program/script on webserver. Server-side scripts are usually written using JSP. These languages are later embedded in HTML documents on servers file system. It (server-side script) deals with creation of interactive websites. A user can directly connect to the databases using server-side statements.

Data validation is carried out with the help of certain function and attributes.

A form can be checked in different ways. Wherever a user clicks on submit button of the form then it is triggered by the onsubmit event handler present in the <form> element. It is used for calling the validation function that is present in some other script or in the head of the current document. This function returns true when a form is sent and it returns false when an error occurred while sending a form. This has to be informed to the user. The values are assigned in the validation functions to return whether script contains errors or not. Initially, true value is assigned and the script is verified with the user entered data. If an error is found then it returns false value.

Example

Validation_Example.html

```
<html>
<head>
<title>Form validation Example</title>
<script type=”text/JavaScript”>
function validate(form)
{
var returnValue = true;
var username = form.txtUserName.value;
var password1 = form.txtPassword.value;
var password2 = form.txtPassword2.value;
if(username.length < 6)
{
returnValue = false;
alert(“Your user name should be at least \n 6 characters”);
form1.txtUserName.focus();
}
if (password1.length < 8)
```

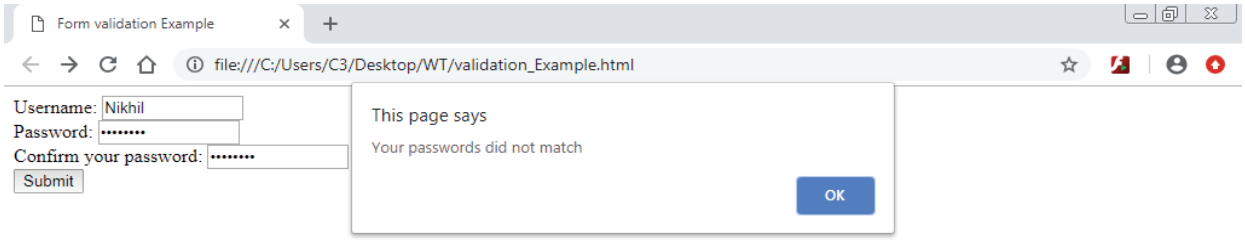
```
{
returnValue = false;
alert("Your password should be at least \n 8 characters");
form1.txtPassword.value = " ";
form1.txtPassword2.value = " ";
form1.txtPassword.focus();
}
if (password1 != password2)
{
returnValue = false;
alert("Your passwords did not match");
form1.txtPassword.value = "";
form1.txtPassword2.value = "";
form1.txtPassword.focus();
}
return returnValue;
}
</script>
</head>
<body>
<form name= "form1" method= "post" action= "thanku.html"  onsubmit= "return validate(this);">
<div>
<label for= "txtUsername">Username:</label>
<input type= "text" name= "txtUserName" id= "txtUserName" size= "12" />
</div>
<div>
<label for="txtPassword">Password:  </td></label>
<input type="password" name="txtPassword" id="txtPassword" size="12" />
</div>
<div>
<label for= "txtPassword2">Confirm your password:</label>
<input type= "password" name= "txtPassword2" id= "txtPassword2" size= "12" />
</div>
<div>
<input type="submit" value="Submit"  />
</div>
</form>
</body>
</html>
```

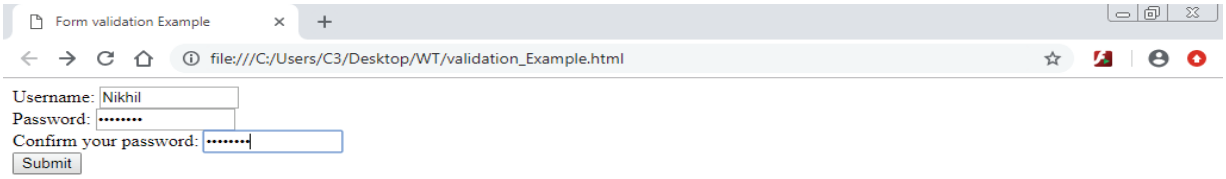
thanku.html

```
<html>
<head></head>
<body>
<p> Thank you for your submission </p>
</body>
</html>
```

Output

When the user enters two different passwords then the output will be as shown below,





When the user enters the correct details then the output will be as follows,



Q34. How the text fields in a form are checked?

Answer :

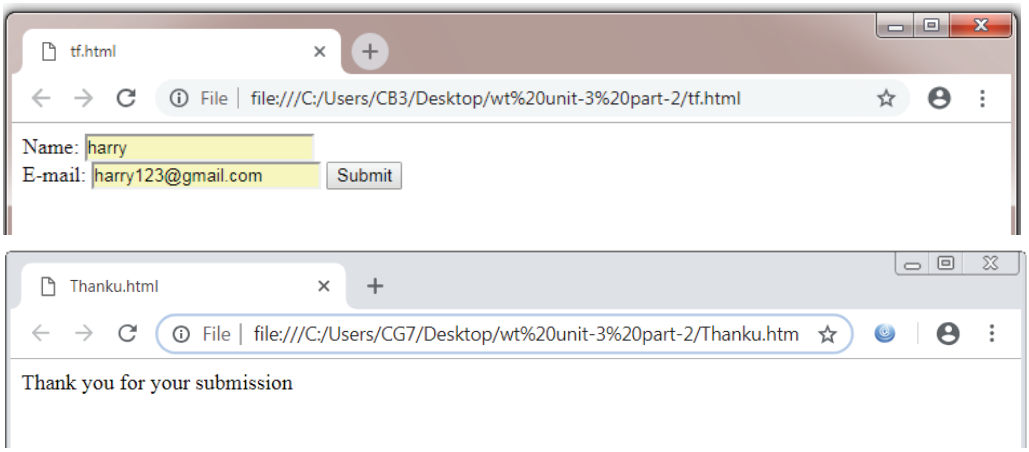
Consider the below example,

```
<!DOCTYPE html>
<html>
<head>
<script>
function validateF() {
      var n = document.forms[“Form1”][“Name”].value;
      var e = document.getElementById(“email”).value.indexOf(“@”);
if (n == “”) {
alert(“Name field must be filled “);
 return false;
}
if (e == -1) {
alert(“E-mail id is invalid”);
return false;
}
}
</script>
</head>
<body>
<form name=“Form1” action=“Thanku.html” onsubmit=“return validateF()” method=“post”>
Name: <input type=“text” name=“Name”><br>
E-mail: <input type=“text” id=“email” size=“20” name=“mail”>
<input type=“submit” value=“Submit”>
</form>
</body>
</html>
```

Thanku.html

```
<html>
<body>
<p> Thank you for your submission </p>
</body>
</html>
```

Output



The <form> tag contains onsubmit attribute whenever as user clicks on submit button then the value specified to it is run/executed. The validate function is called to return true or false.

When the above code is executed then the following things are checked.

(i) Username should be of minimum length

(ii) Password should be of minimum length

(iii) The two passwords should match.

Validate( ) function can be implemented in two ways,

(a) By creating a single function for checking form

(b) By creating re-usable functions.

(a) Single Function Approach

A single function is created to check a form by using validate( ) function. The validate function is called and notified about the current form by passing the parameter value.

```
function validate<form>
{
}
```

This keyword is used in the <form> element to call a function to indicate a form.

Variable contains the values of the user entered in the text controls. Value property is used to store the value entered by the user.

Example

For answer refer Unit-III, Page No. 219, Q.No. 33, Topic: Example.

(b) Reusable Functions Approach

Reusable functions are also used for checking a form. JavaScript files contain functions which can be reused.

validate( ) function is used to call two more functions present in the javascript library.

(i) Validate( ) Confirm Password( )

It checks for the passwords field whether they are valid.

(ii) Validate Maximum Length( )

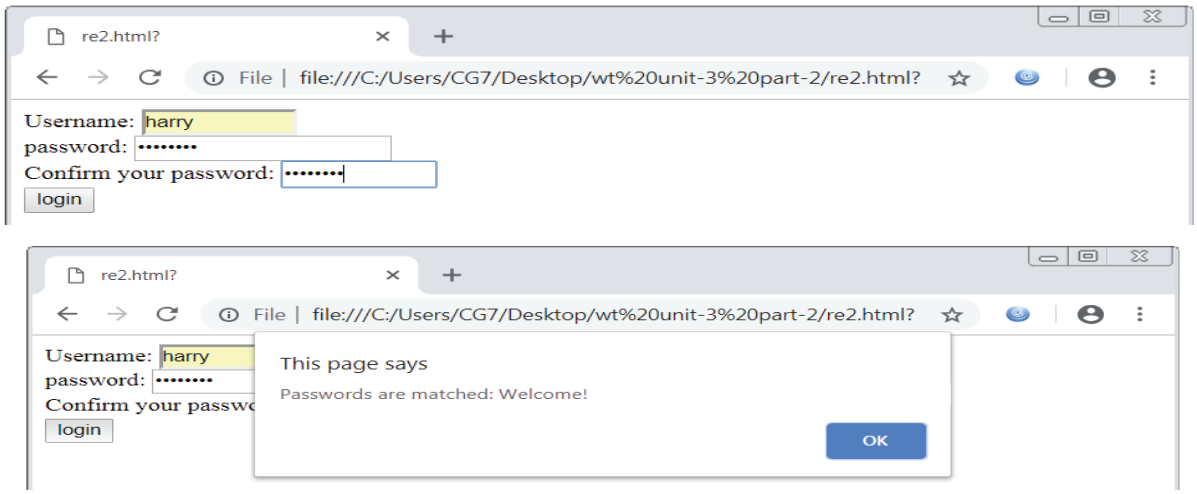
It checks for the minimum number of characters entered by the user in the field.

Example

```
<!DOCTYPE html>
<html>
  <head>
     <script>
       function checkPassword(form)
        {
          password1 = form.password1.value;
          password2 = form.password2.value;
          if (password1 == ‘’)
             alert (“Please enter the Password”);
          else if (password2 == ‘’)
             alert (“Please re-enter to confirm password”);
          else if (password1 != password2) {
             alert (“\n Passwords did not match!”)
             return false;
          }
          else{
             alert(“Passwords are matched: Welcome!”)
             return true;
          }
        }
     </script>
  </head>
  <body>
<form onSubmit = “return checkPassword(this)”>
<label for = “Username”>Username:</label>
<input type = “text” id = “Username” size = “10”/>
<span class = “msg”></span>
</div>
<div>
<label for = “password1”>password:</label>
<input type = “password” id = “password1” size “10”/>
<span class = “msg”></span>
</div>
<div>
<label for = “password2”>Confirm your password:</label>
<input type = “password” id = “password2” size = “10”/>
<span class = “msg”></span>
</div>
<div>
<input type = “submit” value = “login”/>
</div>
<div>
</form>
</body>
</html>
```

The above function is used for checking whether the user has entered minimum number of characters.

Output



replace( ) method

The replace( ) method in javascript is used for replacing certain characters in the document with the specified characters.

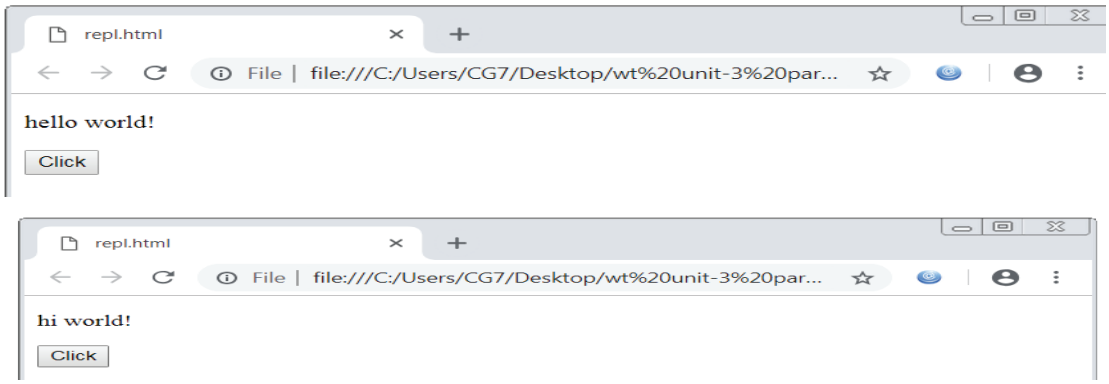
Syntax

string.replace(Old String, new String);

Example

```
<!DOCTYPE html>
<html>
<body>
<p id=”demo”>hello world!</p>
<button onclick=”rep()”>Click</button>
<script>
function rep() {
   var str = document.getElementById(“demo”).innerHTML;
   var str1 = str.replace(“hello”, “hi”);
   document.getElementById(“demo”).innerHTML = str1;
}
</script>
</body>
</html>
```

Output



Q35. Explain about regular expressions.

Answer :

Regular Expression

A regular expression is a string used in the process of pattern matching. In JavaScirpts, regular expressions are objects supporting numerous predefined methods. Following are two examples of declaring regular expressions.

Method 1

```
var sample = "world";
var data = "Welcome to the world of stars";
data.match(sample);
```

Predefined method is used in order to search the string "data" using the regular expression "sample".

Method 2

```
var sample = new RegExp ("world");
var data = "welcome to the world of stars";
sample.exec(data);
```

Predefined method is used in order to search the string "data" using the regular expression "sample".

Method1 refers to static way of declaring the regular expression. Whereas method 2 is dynamic

The above two methods contain set of characters that are defined as regular expression. In order to implement the pattern matching concept efficiently. The regular expression should contain certain symbolic templates. The usage of these symbolic templates are governed by strict grammatical rules.

Following are the regular expression grammar applied while using the symbolic templates in javascripts.

\n

It is used for linefeed.

\r

It is used for carriage return.

\t

It is used for tab.

\v

It is used for vertical tab

\f

It is used for form-feed

\d

It is used for specifying a digit i.e, 0 to 9.

\D

It used for specifying the non-digit [i.e., (^ 0 – 9)].

\w

It is used to specify a word or a alphanumeric character i.e., [a – z or A – Z or 0 –9]

\W

It is used to specify a non-word character i.e., (^ a – z or A – Z or 0 – 9]

\s

It is used for a white-space character i.e.,[\t \v \n \r \f]

\S

It is used for a non-white space character i.e., [^\t \v \n \r \f]

The carriage return or linefeeds can be replaced with the
 tag as shown below,

```
replace (/ \r | \n | \r \n | \n \r | /g), <'br/>';
```

Example

```
<!DOCTYPE html>
<html>
<body>
<title>Regular Expression</title>
<p>Click the below Button</p>
<button onclick="func()">Click</button>
<script>
function func() {
var str = "hello world\n";
var pattern1 = /\n/;
var result1 = str.search(pattern1);
document.write("This is the result of \\n :" +result1);
var str = "good morning\r";
var pattern2 = /\r/;
var result2 = str.search(pattern2);
document.write("<br> This is the result of \\r : " +result2);
var str = "good \t afternoon";
var pattern3 = /\t/;
var result3 = str.search(pattern3);
document.write("<br> This is the result of \\t : " +result3);
var str = "good evening\v";
var pattern4 = /\v/;
var result4 = str.search(pattern4);
document.write("<br> This is the result of \\v : " +result4);
var str = "good night\f";
var pattern5 = /\f/;
var result5 = str.search(pattern5);
document.write("<br> This is the result of \\f : " +result5);
var str = "monday 03-12";
var pattern6 = /\d/g;
var result6 = str.match(pattern6);
document.write("<br> This is the result of \\d :" +result6);
var str = "This is Alexa";
var pattern7 = /\D/g;
var result7 = str.match(pattern7);
document.write("<br> This is the result of \\D :" +result7);
var str = "Hello siri!";
var pattern8 = /\w/g;
var result8 = str.match(pattern8);
document.write("<br> This is the result of \\w :" +result8);
var str = "06/12 & 07/12";
var pattern9 = /\W/g;
var result9 = str.match(pattern9);
document.write("<br> This is the result of \\W :" +result9);
```

```
var str = "Wel come to the party";
var pattern10 = /\s/g;
var result10 = str.match(pattern10);
document.write("<br> This is the result of \\s :" +result10);
var str = "Harry is good";
var pattern11 = /\S/g;
var result11 = str.match(pattern11);
document.write("<br> This is the result of \\S :" +result11);
}
</script>
</body>
</html>
```

Output



Q36. How the characters are tested using test( )?

Answer :

The characters in the regular expression are verified using test method.

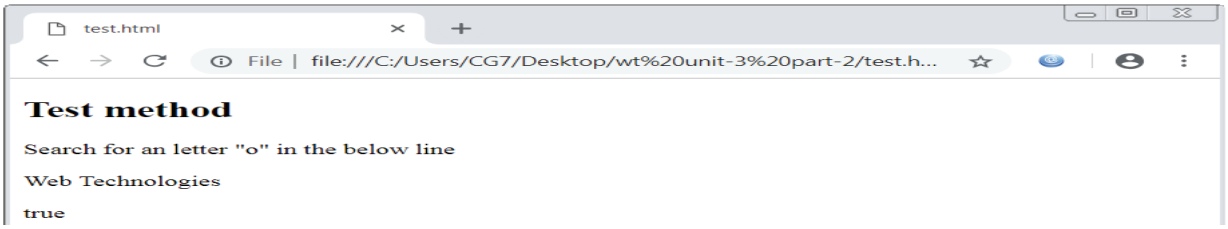
test( ) Method

It takes string as it's argument and searches for a match in it by considering the sample characters string. Upon success it return true, else false.

Example

```
<!DOCTYPE html>
<html>
<body>
<h2>Test method</h2>
<p>Search for an letter "o" in the below line </p>
<p id="demo">Web Technologies</p>
<p id= "demo1"></p>
<script>
str= document.getElementById("demo").innerHTML;
document.getElementById("demo1").innerHTML = /o/.test(str);
</script>
</body>
</html>
```

Output



Q37. Discuss about the following,

(a) Select box options

(b) Radio buttons.

Answer :

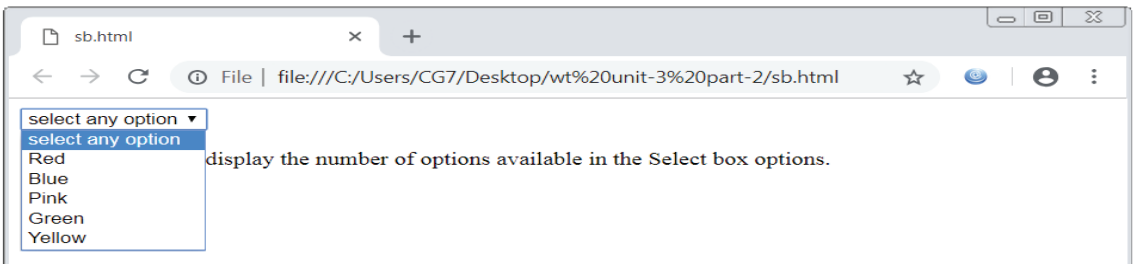
(a) Select Box Options

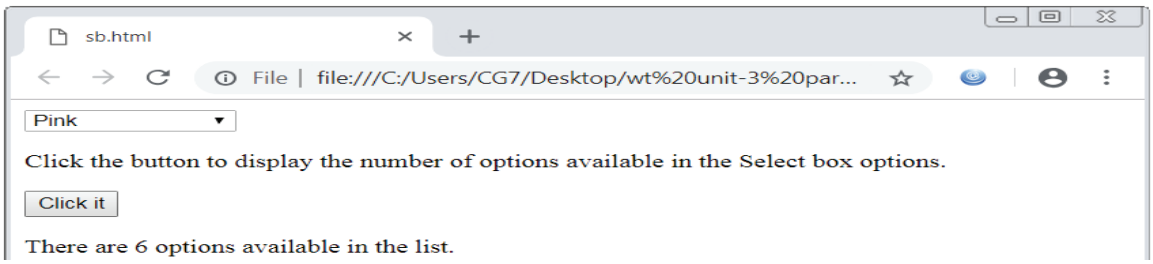
Select Box has the select object that contains selected Index property that specifies the option selected by user. Its value starts form 0.

Example

```
<!DOCTYPE html>
<html>
<body>
<form>
<select id= “Colors”>
<option> select any option</option>
<option>Red</option>
<option>Blue</option>
<option>Pink</option>
<option>Green</option>
<option>Yellow</option>
</select>
</form>
<p>Click the button to display the number of options available in the Select box options.</p>
<button onclick= “func1()”>Click it</button>
<p id=”demo”></p>
<script>
function func1() {
var x = document.getElementById(“Colors”).options.length;
document.getElementById(“demo”).innerHTML = “There are “ + x + “ options available in the list.”;
}
</script>
</body>
</html>
```

Output





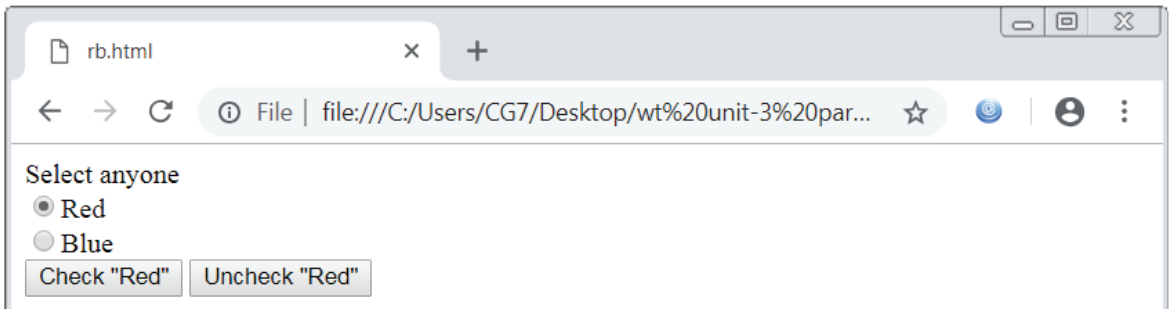
(b) Radio Buttons

A form contains a set of radio buttons from which only one can be selected. An array is used to represent the set of radio buttons in javascript to specify which option is selected.

Example

```
<!DOCTYPE html>
<html>
<body>
<form>
Select anyone<br>
<input type="radio" name="colors" id="Red">Red<br>
<input type="radio" name="colors" id="Blue">Blue
</form>
<button onclick="check()">Check "Red"</button>
<button onclick="uncheck()">Uncheck "Red"</button>
<script>
function check() {
document.getElementById("Red").checked = true;
}
function uncheck() {
document.getElementById("Red").checked = false;
}
</script>
</body>
</html>
```

Output



Q38. Write about check boxes.

Answer :

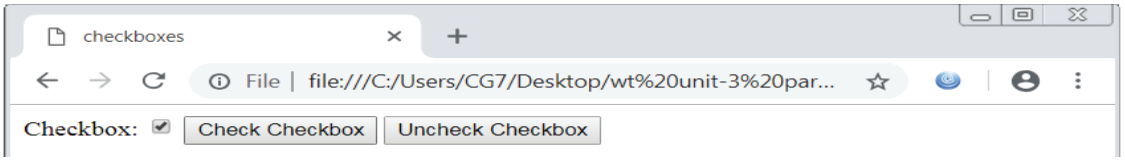
Check Boxes

Check boxes provide a set of choices to the user from which only one item can be selected. An array is used to represent the group of check boxes that share the same name.

Example

```
<!DOCTYPE html>
<html>
<body>
<title>checkboxes</title>
Checkbox: <input type="checkbox" id="Check">
<button onclick="check()">Check Checkbox</button>
<button onclick="uncheck()">Uncheck Checkbox</button>
<script>
function check() {
document.getElementById("Check").checked = true;
}
function uncheck() {
document.getElementById("Check").checked = false;
}
</script>
</body>
</html>
```

Output



Q39. How the form submission is prevented?

Answer :

Preventing a Form Submission Until a Checkbox has been Selected

The function is used to check whether the checkbox is cheked or not. If it returns true to then the form gets submitted. When the function returns false then the user will be prompted to check the box.

Example

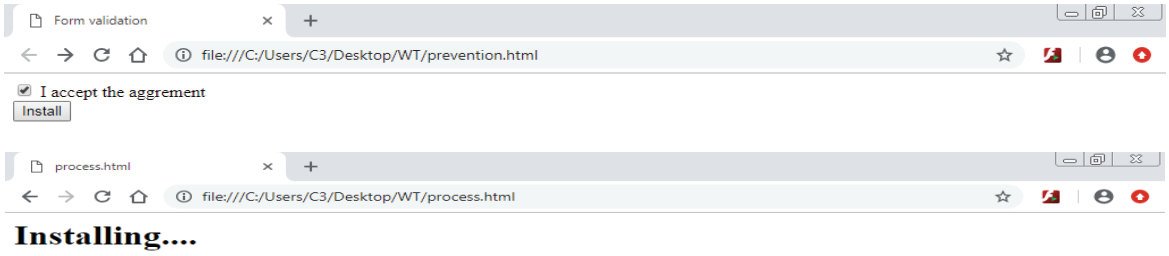
prevention.html

```
<html>
<head>
<title>Form validation</title>
</head>
<body onload="document.myform.submit_btn.disabled=true">
<form name= "myform" action= "process.html" method= "post">
<input type= "checkbox" value= "0" name= "chk_Agree" id= "chk_Agree"
onclick= "document.myform.submit_btn.disabled=false" />
I accept the aggrement
<br/><input type= "submit" name= "submit_btn" value= "Install" /><br />
</form>
</body>
</html>
```

process.html

```
<html>
<head></head>
<h1>Installing....</h1>
</html>
```

Output



3.2.2 Form Enhancements

Q40. Write short notes on following,

(i) Focusing on first form item

(ii) Auto tabbing between fields.

Answer : **Model Paper-I, Q11(b)**

(i) Focusing on First Form Item

Several users might visit the web page and interact with it through a form. So, rather than allowing them to move the mouse and click on the form elements like text boxes etc., programmers can provide focus to the form elements as soon as the page loads. Mostly the form with login box or search option will be provided with focus option.

In order to provide focus to the first text input on a form, add onload event handler to the <body> element of the document. This will choose the form control to be highlighted and provides focus to it by using the control's focus( ) method. The cursor will be flashing in this particular form control ready for the user to enter some text into it when the page loads. The onload event fires as soon as the page gets loaded.

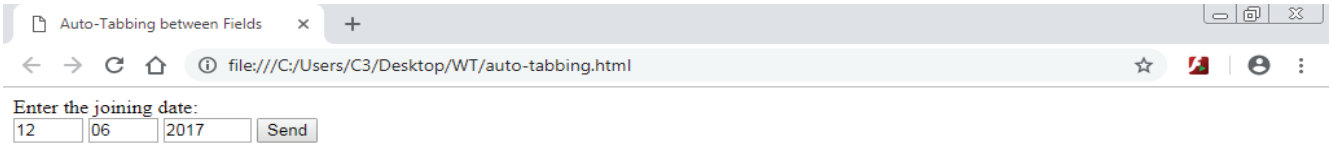


(ii) Auto Tabbing between Item

The focus( ) method can be used to focus a particular control. Consider the example of adjusting data in the system. The focus can be moved among the three boxes for the user to enter day, month and year.

```
<form name = "Form1">
Enter the joining date : <br/>
<input name = "txtm" id = "txtm" size = "3" maxlength = "2" onkeyup = "if(This.value.length> = 2)
this.form.to.focus( );"/>
<input.name = "to" id = "to" size = "3" maxlength = "2" onkeyup = "if(this.value.length> = 2)
this.form.ty.focus( );"/>
<input name = "ty" id = "ty" size = "5" maxlength = "4" onkeyup = "if(this.value.length> = 4)
this.form.submit.focus( );"/>
<input type = "submit" name = "submit" value = "send"/>
</form>
```

In the above code onkeyup event handler is used to check the equality between length of text that is entered by the user and the required number of characters for that field. The focus will be moved to the next box when user enters the required number of characters. The this keyword will represent the current form control and the value property represents the value entered for a control. The length property indicates the length of value in the current form.



Q41. Discuss about disabling a text input.

Answer :

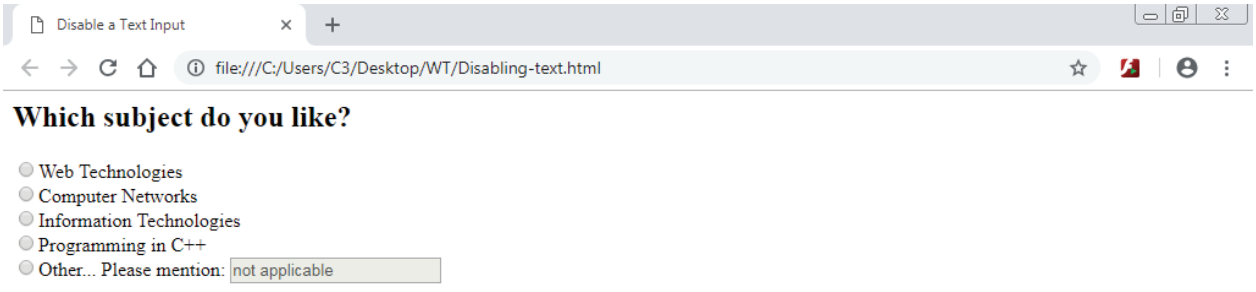
The programmers might sometimes need to disable the text input until a given condition is satisfied. For example the text "agree to terms and condition" option will be disabled in a form until the checkbox beside it is selected. Consider the below code.

```
<body onload = "document.frmRef.txt.disabled = true";
document.frmRef.txt.value = "not applicable">
<h2> How do you know about us? </h2> <form name = "frmRef">
<input type = "radio" name="Hear" value = "1" onclick = "handleOther(this.value);"/> Web Technologies <br/>
<input type = "radio" name = "Hear" value = "2" onclick = "handleOther (this.value);"/> Computer Networks <br/>
<input type = "radio" name = "Hear" value = "3" onclick = "handleOther(this.value);"/> Information Technologies <br/>
<input type = "radio" name = "Hear" value = "4" onclick = "handleOther(this.value);"/> Programming in C++ <br/>
<input type = "radio" name = "Hear" value = "5" onclick  "handleother(this.value);"/> Other .... Please mention <br/>
</form>
</body>
```

The above form will ask the user about how they know about the site. Radio buttons are used to provide options for the user to select. Initially, the text is disabled by using the onload( ) event handler of <body> element.

When the user selects an option from set of radio buttons, then the text beside the selected radio button gets enabled. For each selection the onclick event calls the "handleOther( )" function to which the form control value is passed as parameter. This function contains a simple "if else" statement that checks whether the value of selected form control and text "other" are equal or not. If the selected radio button has the value "other" then the "disabled" property of the textbox is set to false. The text box will be disabled for all other options and its value will be set to "not applicable".

```
function handleOther (sRadio)
{
if(sRadio = = "other")
{
document.frmRef.txtOther.disabled = false;
document.frmRef.txtOther.value = ' ';
}
else
{
document.frmRef.txtOther.disabled = true;
document.frmRef.txtOther.value = 'not applicable';
}
}
```



Q42. Explain in brief about the following,

(i) Case conversion

(ii) Trimming spaces from beginning and end of fields.

Answer :

(i) Case Conversion

In some circumstances, the case of user input need to be changed from lowercase to uppercase or vice versa since the Java script is case-sensitive. There are two methods provided by Javascript's string object for conversion of text case.

- toLowerCase( ) : It changes the text from uppercase to lowercase
- toUpperCase( ) : It changes the text from lowercase to uppercase.

```
<form>
<input type = "text" name = "case" size = "20" onblur = "this.value = this. value.toLowerCase( );"/>
</form.>
```

(ii) Trimming Spaces from Beginning and End of Fields

The programmer might sometimes need to remove the spaces from the starting and ending of the form field. For this purpose the substring( ) method of the string object is used.

substring (beg, end)

The above method returns the part of string that is specified by start and end points. The default value for end is end of string.

The initial positions of the first character must be '0' since they are zero based. To remove the spaces from the beginning of the string, the substring( ) is used twice. The first call retrieves the first character entered by the user. The second call will remove the space if the user input is 'space'.

First call of substring( )

```
while (this.value.substring(0, 1) = = ' ')
```

Second call of substring( )

```
this.value = this.value.substring(1, this.value.length);
```

This statement will select from second character to end of string in control value. This will be the new form value because the first space is removed. Removing of first space is done in the onblur event handler.

```
<form>
<input type = "text" name = "frame" size = "100" value = "Enter some text, let the first character be a space"
onblur = "while(this.value.substring (this.value.length-1, this.value.length) = = ' ')
this.value = this.value substring (0, This.value.length);"/>
<br/>
</form>
```

The above process is similar to trim any of the trailing spaces but it is reversed.

The first call of substring( ) will collect the last character of string and removes it at all it is a blank.

```
<form>
<input type = "text" name = "+Name" size = "100" value = "Enter any text, let the first character be space." onblur =
"while(this.value.substring (this.value.length.1, this.value.length = = ' ') )
this.value = this.value.substring (0, this.value.length-1);"/> <br/>
</form>
```

An alternative method for the above code is to use regular expression.

```
<form>
<input type = "text" name = "remove space" size = "100" value = "Enter some text, let the first character be space."
onblur = "this.value = this.value.replace (/^\  \s+/, '')
replace(/\ s + $/,'');"
/> <br/>
</form>
```

The above code removes both the trailing and leading space.

Q43. Write in short about the following,

(i) Selecting all the content of a text area

(ii) Check and uncheck all checkboxes.

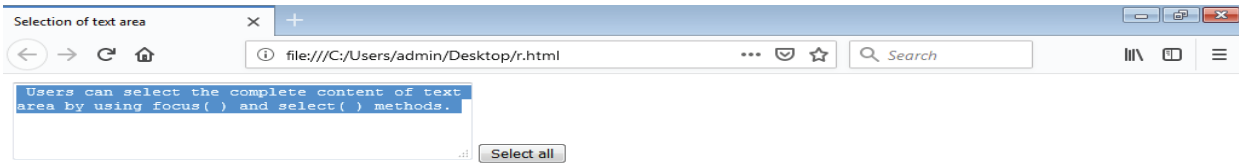
Answer :

(i) Selecting All the Content of a Text Area

Users are allowed to select the complete text of the text area by using focus( ) and select( ) methods. The selectAll( ) function accepts two parameters.

```
<html>
<head> <title> Selection of text area
<script language="JavaScript">
function selectAll(str) {
str.focus();
str.select();
}
</script>
</head>
<body>
<form name="form1" >
<textarea name="myText" rows="4" cols="10"> Users can select the complete content of text area by using focus( ) and select( ) methods. </textarea>
<input type= "Button" name= "btnSelectAll" value= "Select all"
onclick= "selectAll(document.form1.myText);" />
</form>
</body>
</head>
</html>
```

Output



The button through which user can select all will have a onclick event handler that calls SelectAll( ) function. It provides the from control focus through focus( ) method and selects the control through select( ) method.

(ii) Check and Uncheck All Checkboxes

When a set of checkboxes are used the user must be able to select all the checkboxes or reset all of them at a time. This is possible by using check( ) and uncheck( ) methods.

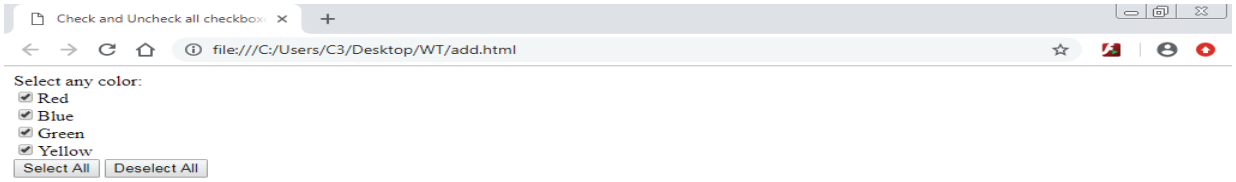
```
function check(field)
{
for(var i = 0; i > field.length; i++)
{
field[i].checked = true;
}
}
function uncheck(field)
{
for(var i = 0; i < field.length;i++)
```

```
{
field [i].checked = false;
}
}
```

The above functions work only when there are multiple checkboxes in the group and two buttons in the form. These buttons will call the check or uncheck function by passing in the array of checkbox elements that are under same group.

```
<form name = "frm" action = " "> select any colors <br/>
<input type = "checkbox" name = "color" value = "1"/> Red <br/>
<input type = "checkbox" name = "color" value = "2"/> Blue <br/>
<input type = "checkbox" name = "color" value = "3"/> Green <br/>
<input type = "checkbox" name = "color" value = "4"/> Yellow <br/>
<input type = "button" value = "Select All" onclick = "check(document.frm.color);"/>
<input type = "button" value = "Deselect All" onclick = "uncheck(document.frm.color);"/>
</form>
```

Output



The above code can be combined into a function that will be called through a button.

```
function checkUncheckAll (field)
{
var frm = field.form, i = 0;
for (i = 0; i < frm.length;i++)
{
if(frm [i].type = = 'checkbox' & & frm[i].name ! = 'checkall')
{
frm [i].checked = field.checked;
}
}
}
```

3.2.3 Javascript Libraries

Q44. Define javascript libraries.

Answer :

The javascript files which consist of functions that can be used in various pages are referred to as javascript libraries. Some of the different types of such scripts are as follows,

1. **Animated Effects and Drag-and-Droplists**

   For answer refer Unit-III, Page No. 236, Q.No. 46(i).

2. **Lightboxes and Modal Windows**

   For answer refer Unit-III, Page No. 237, Q.No. 46(ii) and Page No. 238, Q.No. 47(i).

3. **Sortable Tables**

   For answer refer Unit-III, Page No. 238, Q.No. 47(ii).

4. **Calender Controls**

   For answer refer Unit-III, Page No. 239, Q.No. 48(i).

5. **Auto-completing Text Inputs**

   For answer refer Unit-III, Page No. 239, Q.No. 48(ii).

Q45. Explain about the animated effects using scriptaculous.

Answer :

Javascript allows several types of animated effects by using scriptaculous. It allows users to perform tasks such as animation, adding drag-and-drop functionality, building editing tools and creating autocompleting text inputs. In addition to these, it even allows to create DOM fragments. It contains huge number of functions to create the animations. The real life objects or concepts are represented by using the javascript objects. For scriptaculous library the object called effect is used to handle the animated effects. Various methods are used to create different types of effects. For example, the shake( ) of effect object method can be used to create shaking effect, fade( ) method can be used to create fading effect. These methods work only when prototype. js and scriptaculous .js libraries are placed in scriptaculous.

<script src = "scripts/scriptaculous/prototype.js" type = "text/javascript"> </script>

<script src = "scripts/scriptaculous/scriptaculous.js" type = "text/javascript"> </script>

Consider a page that contains three effects namely shake( ), shrink( ) and Fade( ). When one of the box is clicked the respective effect will be triggered.

 Effect.shake( )

 shake the box

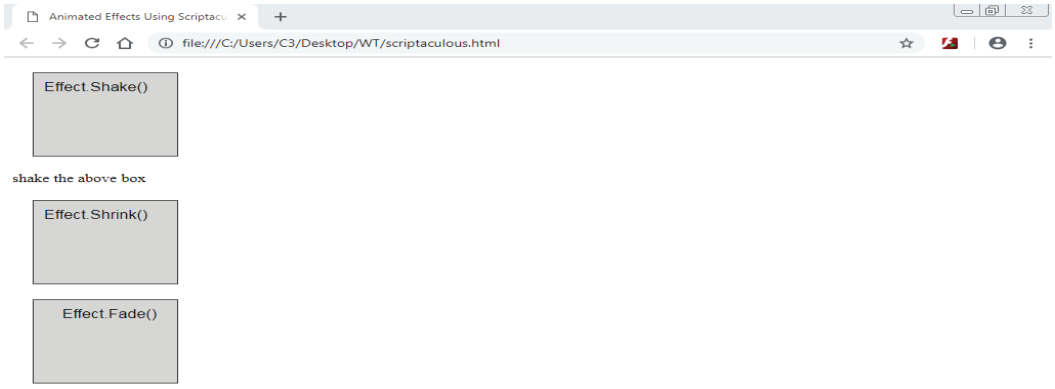
 effect.shrink

<div d = "demo" onclick = "effect.Fade(this);

window.settimeout('Effect.Appear(\'effect-fude\')',3000);">

Effect.Fade( )

The above code results the following output,



Q46. Write in short about the following,

(i) Drag and drop sortable lists using scriptaculous

(ii) Creating a lightbox.

Answer :

(i) Drag and Drop Sortable Lists Using Scriptaculous

Consider an example of reordering the lists by dragging and dropping the elements. Creation of a simple unordered list is as follows,

<script src = "scripts/prototype.js" type = "text/javascript"> </script>

<script src = "scripts/scriptaculous.js" type = "text/javascript"> </script>

<stype type = "text/css">

li { porder; 1Px solid # 000000;

Padding: 10Px;

margin-top: 10px;

list-style-type:none;

width:150 Px;}

```
</style>
</head>
<body>
<ul id = "items">
<li id = "item1"> <Item1 </li>
<li id = "item2"> Item2 </li>
<li id = "item3"> Item3 </li>
</ul>
```

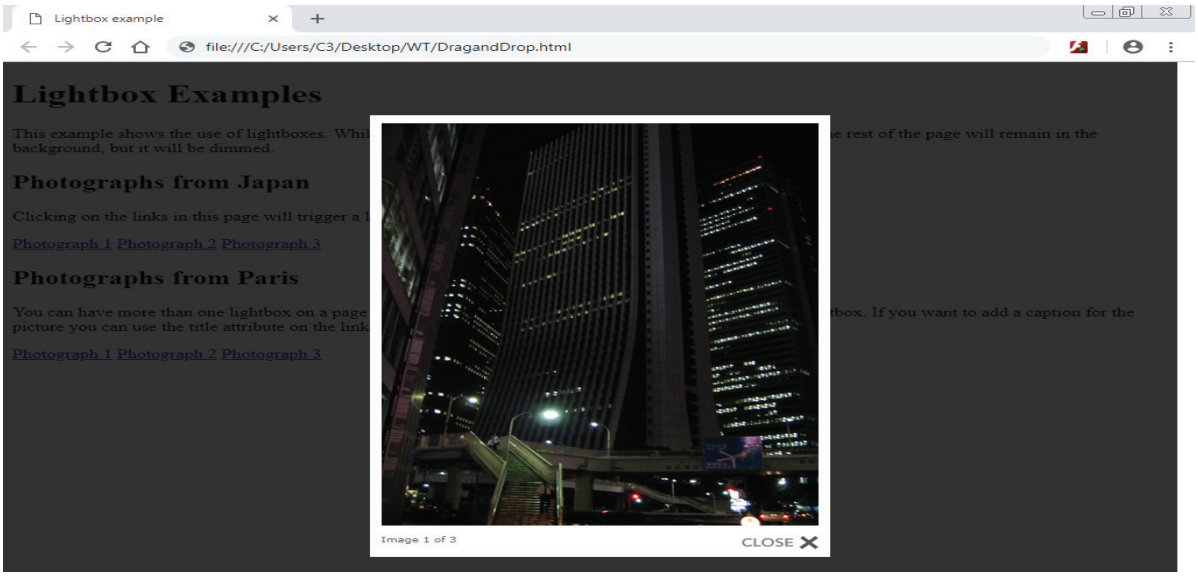
The above list can be made sortable by adding the <script> element

```
<script type = "text/javascript" language = "javascript">
sortable.create("items", {dropOnEmpty:true, constraint: false});
</script>
```

The list is made sortabel by using the sortable object and create( ) method of scriptaculous library and this method accepts few parameter, one is the value of id attribute of unordered list and the other is option that indicates the working of the list.

(ii) Creating a Lightbox

A lightbox is an image that will open in foreground of browser even without reloading the page. It will be shown at the middle of the page.



The above light box can be created by adding the below three scripts.

```
<script type = "text/javascript" src = "scripts/scriptaculous/prototype.js>
</script>
<script type = "text/javascript" src = "scripts/scriptaculous.js? load = effects, builder"> </script>
<script type = "text/javascript" src = "scripts/lightbox.js"> </script>
```

Code for creating the lines to the images in lightbox is as follows,

```
<a href = "images/img1.jpg" rel = "light box[Japan]">
Image1</a>
<a href = "images/img2.jpg" rel = "light box[Japan]">
Image2</a>
<a href = "images/img3.jpg" rel = "light box[Japan]">
Image3</a>
```

Users can even add multiple lightboxes on the same page but with different names.

Q47. Explain in brief about the following,

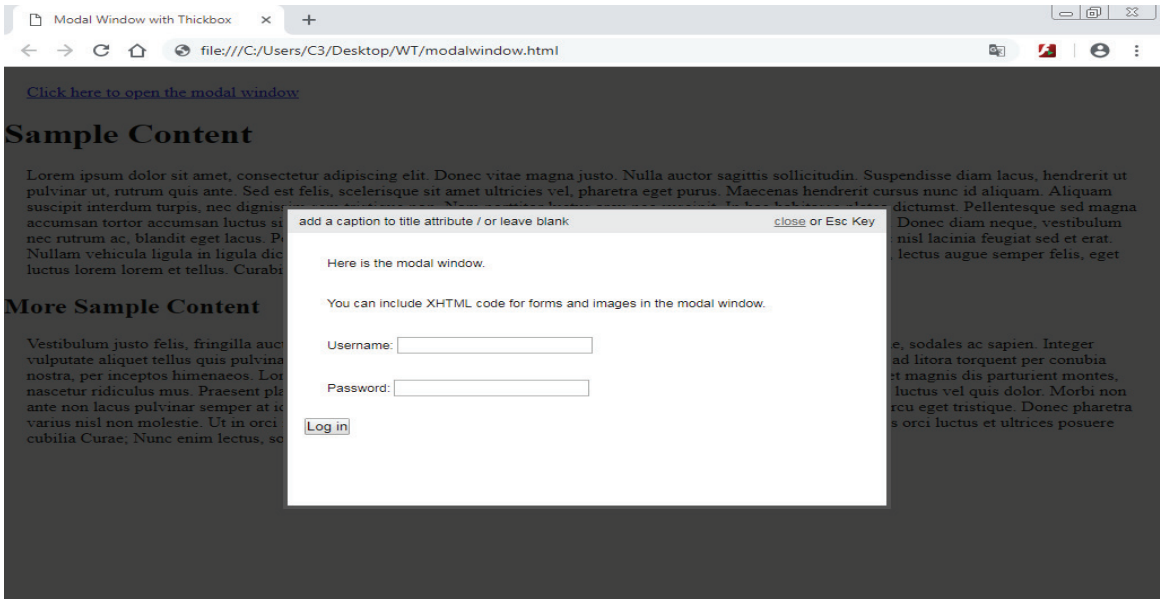
(i) Creating a Modal Window

(ii) Sortable tables with Mochikit.

Answer :

(i) Creating a Modal Window

A model window is a child window with which the user need to interact to return to main window.



The above model window makes use of script "Thickbox" along with Jquery library.

<script type = "text/javascript" src = "scripts/jquery/jquery –1.3.2.js">

</script>

<script type = "text/javascript" src = "scripts/thickbox js">

</script>

<link rel = "stylesheet" href = "scripts/thickbox.css" type = "text/css" media = "screen"/>

Code to add the content to modal window is,

 <P>

Here is the modal window

</P>

This code can be disabled on the page by using style sheet rule. Other than this, a link can also be added to the modal window by using <a> element. It should start with class attribute with value thickbox.

(ii) Sortable Table with Mochikit

Users are allowed to create sortable table to reorder the table contents just by clicking the column heading. It is useful to handle large tables where several users sort the data in several ways. The below table contains set of students details. The up arrow beside "Id" heading represents that tables contents are sorted according to the id.

This concept makes use of a javascript library called Mochikit. The code to create sortable table is as follows,

<script type = "text/javascript" src = "scripts/Mochikit/Mochikit.js"> </script>

<script type = "text/javascript" src = "scripts/mochikit/examples/sorttab/sorttab.js">

</script>

The headers must be differentiated from coloumns by using CSS styles. Inorder to sort the table the below three things are needed.

- The <table> element needs id attribute with value sortable table.
- The <th> element must contains an attribute mochi for every column.
- The <td> elements first row must be mochi:content.

Q48. Discuss about the following,

(i) Creating calendar with YUI

(ii) Auto completing text inputs with YUI.

Answer :

(i) Creating Calendar with YUI

YUI stands for Yahoo User Interface library. It is a large library that contains various types of functionalities in several scripts. It is split into four sections, described as follows,

1. The heart of YUI contains three scripts namely Yahoo Global Object, DOM collection and event utility. Sometimes all three are needed and sometimes only one or two of them are required.
2. The library utilities that offer the functionality that are useful in various tasks like libraries to create animated effects, drag-and-drop functionality and image loading.
3. The library consists of scripts for creating various types of user interface controls like calendars, color pickers, text editors and image carousels. The 'cheat sheets" help to create the user interface components quickly.
4. A set of CSS stype sheets are used with UI controls to represent as desired.

Consider an example of YUI to add a calender to the web page. This needs three javascript files from YUI to be added and a calender is script file.

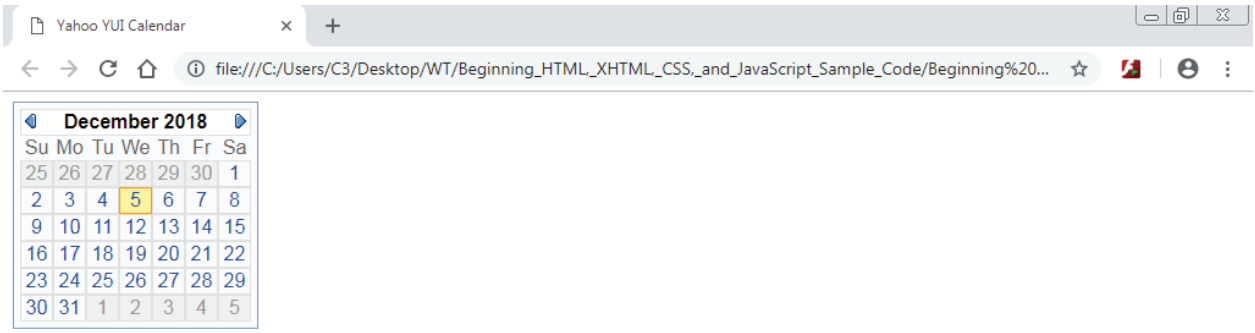
```
<script type = "text/javascript" src = "scripts/yui/yahoo/yahoo.js"> </script>
<script type = "text/javascript" src = "scripts/yui/event/events.js"> </script>
<script type = "text/javascript" src = "scripts/yui/dom/dom.js"> </script>
<script type = "text/javascript" src = "scripts/yui/build/calendar"> </script>
```

A css file must above be added with YUI download.

```
<link type = "text/css" rel = "stylesheet" href = "scripts/YUI/sam/calendar.css">
```

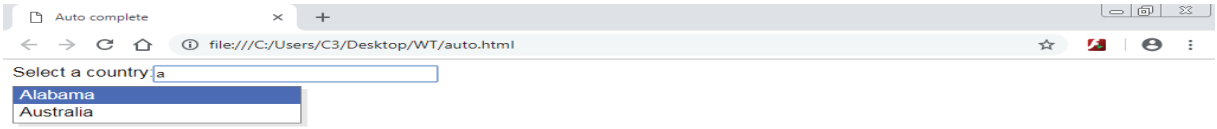
Finally, this can also be included in the script that calls the YUI library and files the element with calender

```
<script>
Yahoo.namespace("ex.calender");
Yahoo.ex.calender.init = func( )
{
Yahoo.ex.calender.c1 = new Yahoo.widget.calender("c1" "c1container");
Yahoo.ex.calender.c1.render( );
}
Yahoo.util.event.onDomReddy(Yahoo.ex.Calendar.init);
</script>
```



(ii) Auto Completing Text Inputs with YUI

To illustrate the auto completion of text input with YUI consider the example of creating text input where users are provided with help such as options for the users to type.



Initially, add three JavaScript files:

<script type = "text/javascript" src = "scripts/yui/yahoo/yahoo.js"> </script>

<script type = "text/javascript" src = "scripts/yui/event/event.js"> </script>

<script type = "text/javascript" src = "scripts/yui/dom/dom.js"> </script>

Now, add the animation and data source library utilities.

<script type = "text/javascript"

src = "scripts/yui/animation/animation.js"> </script>

<script type = "text/javascript"

src = "scripts/yui/datasource/ds.js"> </script>

Finally, add the autocomplete.js JavaScript file:

<script type = "text/javascript"

src = "scripts/yui/automcomplete/ac.js"> </script>

Add the text input and a to provide suggestions about the data to be entered

select a country :

<input id = "Input" type = "text">

Now a JavaScript array is created with the possibilities user this to enter

<script type = "text/javascript">

example.arr = [

"Alebama"

"Australia"

// other states go here

];

</script>

Finally add the JavaScript the page to tie the text input form control to the array and call the Auto-Complete function

<script type = "text/javascript">

yahoo.example.ACJSArray = new func( ){

//Instantiate first JS array data source

this.OACDS = new yahoo.widget.DS_JSArray(yahoo.example.statesArray);

// Instantiate first AutoComplete

this.oAutoComp = new yahoo.widget.AutoComplete('Input', 'Container', this.oACDS);

this.oAutoComp.prehighlightClassName = "yui-ac-prehighlight";

this.oAutoComp.typeAhead = true;

this.oAutoComp.useShadow = true;

this.oAutoComp.minQueryLength = 0;

var sInputValue = yahoo.util.Dom.get('statesinput').value;

if(sInputValue.length = = 0) {

var oSelf = this;

setTimeout(function( ){oSelf.sedQuery(sInputValue);}, 0);

}

});

};

</script>

3.3 PUTTING YOUR SITE ON THE WEB

3.3.1 Meta Tags

Q49. Explain in detail about the meta tags.

Answer :

Meta Tags

The meta tag is used to provide the information related to the page indexing for the search engines. The meta information is simply defined as "information about information".

Example

<meta name = "description" content = "SIA publishers and distributors pvt Ltd"/>

Generally, a meta tag can take four universal attributes (dir, lang, xml:lang and title, and four other attributes that are specific to the <meta> element. They are as follows,

1. schema
2. name
3. content
4. http-equiv.

Here, the name and content attributes, http-equiv and content attributes are used together.

(i) The Name and Content Attributes

This attribute type is used to specify the properties of a complete page.

Example

<meta name = "description" content = "SIA publishers and distributors pvt Ltd.">

Here, the value of name is "description", and the content provides the company information.

The value of the name attribute can be anything because it doesnot have publishing standards. But, there exists some predefined values for the name attribute and they are as follows,

❖ **description**

It is used to provide the description of a web page. The description value should not exceed 200 characters. Already some search engines like Google displays only the first 100 characters. Thus, the main information must be provided in the first 100 characters.

Example

Consider a <meta> tag with value as description displaying information about SIA company.

<meta name = "description" content = "SIA Publishers, a leading Educational publishers trusted by millions of students across twin states of Telangana & Andhrapradesh"/>

But, when it is sent to a search engine, it shows the following,

"SIA Publishers, a leading Educational publishers trusted by millions of students across twin..."

❖ **keywords**

It is used to specify a list of comma separated keywords that are used in the site indexes. For example, tutorialpoint website consists of coding ground with various languages that might use keywords such as,

<meta name = "keywords" content = "C, C++, java, C#, ASP.NET, ADO.NET, python, R"/>

Note that the text in the content should not exceed 200 characters, and the keywords that are not related to the site can not be used.

A user can also use the lang keyword along with the keywords and description to specify the language.

Example

If the language used is americal English. Then,

<meta name = 'keywords', content = "go, went, gone" lang = "en-us"/>

If the language used is french. Then,

<meta name = "keywords", content = "aller, all*é*, parti" lang = "fr"/>

❖ **robots**

It is used to specify indexing for search engine. Using this value you can say to a search engine that not to index the given page or from following links on a page.

Example

<meta name = "robots" content = "noindex,nofollow"/>

Here, the meta tag informs the search engine not to do index to the present page and not to follow any of the links in the page.

Here, the content attribute has some different values. They are,

- **all:** It informs the search engine to index all the pages.
- **none:** It informs the search engine not to index any page.
- **index:** It informs the search engine to index the current page.
- **noindex:** It informs the search engine not to index the current page.
- **follow:** It informs the search engine to follow the links from the current page.
- **nofollow:** It informs the search engine not to follow any link in the current page.

(ii) Scheme Attribute

The scheme attribute is used to specify the format or scheme for a property value. Consider the example of working with dates. The date format is different in different countries. In India, mostly date is written in dd-mm-yyyy format and in united states, the date can be written in mm-dd-yyyy format. In such cases. Use the following,

/*India*/ <meta scheme = "India" name = "date" content = "12-11-2018"/>

/*US*/ <meta scheme = "US" name = "date" content = "11-12-2018"/>

Note that, this attribute is not supported in HTML5.

(iii) http-equiv And Content Attributes

For answer refer Unit-III, Page No. 242, Q.No. 50.

Q50. Explain about the http-equiv and content attributes.

Answer :

The combination of http-equiv and content attributes are used to specify the header values. When a browser requests a page, the headers are sent along with the request, where the header consists of information like browser type, OS, date, format, screen resolution and some other information about the configuration of a user. And, the server responds to the request and sends the requested page along with the header that consists of information like date and time of the page sent to the client, date and time of the page altered previously and the type of the web server.

The header can also consists of information about page expiration, and a period of time to refresh the page etc. The values that are accepted by http-equiv attribute are as follows,

expires

It is used to specify the date and time of page expiration.

Example

<meta http-equiv = "expires" content-"Tue, 13 April 2018 10:59:59 GMT"/>

It means that the page will be expired on Tuesday, 13th April 2018 at 10:59:59 A.M.

pragma

It is used to prevent a browser from caching a page altogether.

Example

<meta http-equiv = "pragma" content = "no-cache"/>

refresh

It is used to refresh a page after a given period of time.

Example

<meta http-equiv = "refresh" content = "10; www.cricbuzz.com"/>

author

It is used to provide the author name of the document.

Example

<meta http-equiv = "author" content = "SIA Team of Experts"/>

Content-Type

It is used to set the character encoding.

Example

<meta http-equiv = "Content-Type" content = "ISO-8859-1"/>

3.3.2 Testing Your Site

Q51. What are the two different states of testing? Explain about validating HTML, XHTML and CSS.

Answer : **Model Paper-II, Q11(b)**

The two different stages in the process of testing are given below,

1. Pre-publishing Tests

The tests that are performed on your system before showing it to anyone else are known as pre-publishing tests.

2. Pre-release Tests

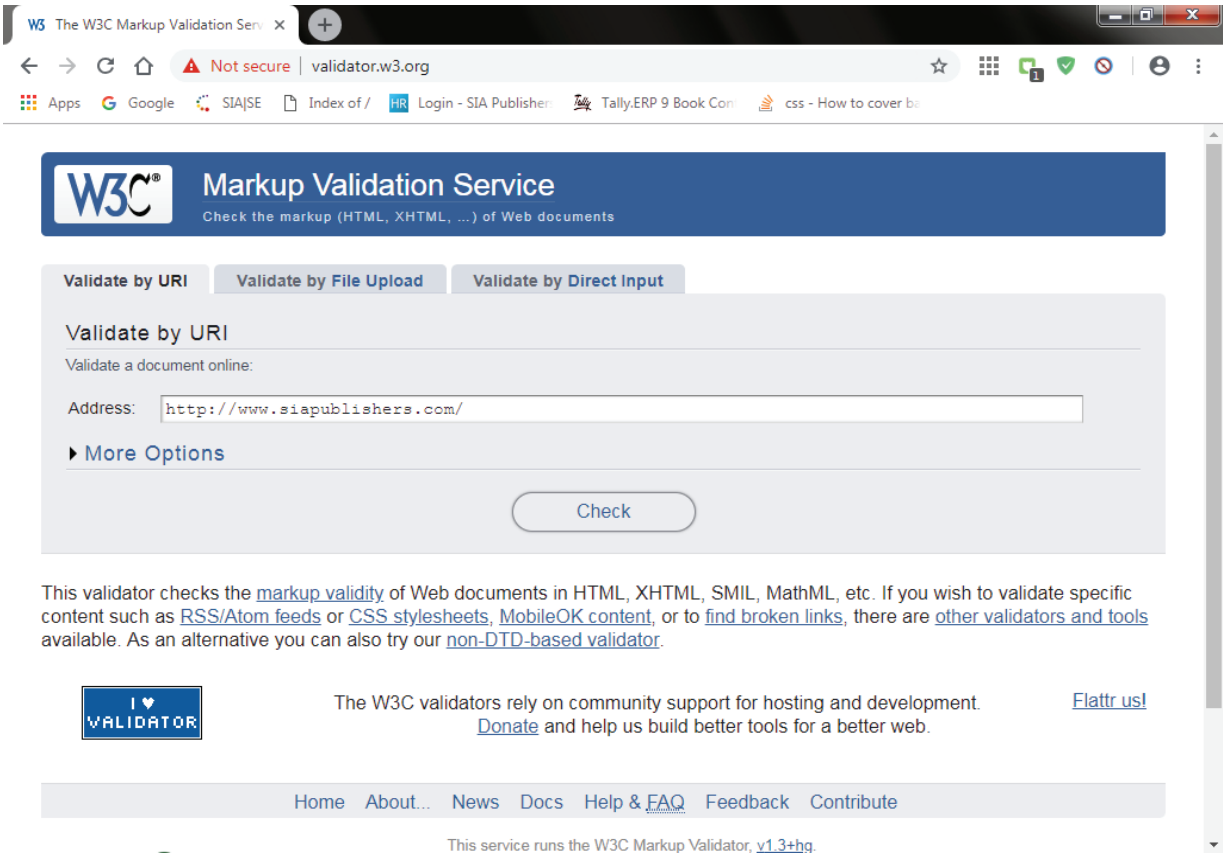
The tests that are performed on a server before publishing are known as pre-release tests.

Validating HTML, XHTML and CSS

A validator is used to validate the opening and closing tags, the various attributes of the respective tags etc. Suppose if a simple tag like </td> is missed in the code, it doesnot make any effect on the present browser but, it might not provide same result on other systems when used.

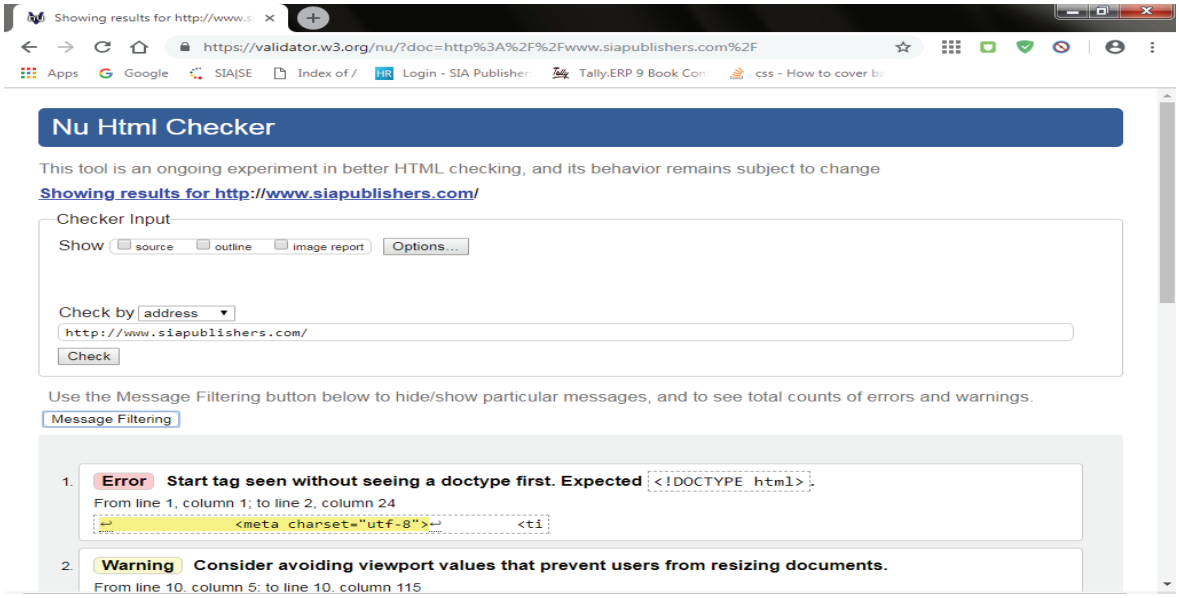
Every version of HTML and XHTML has its own set of rules that directs the version of the language called as schema or DTD. Here, the validator is used to check whether the page follows the rules of schema or DTD. The validator provides tags that are missing in the page or any missing part of the markup.

An online validation tool at http://validator.w3.org/ is provided by W3C. It allows user to enter the URL of the page. The markup validation service screen will be as shown in the below figure,



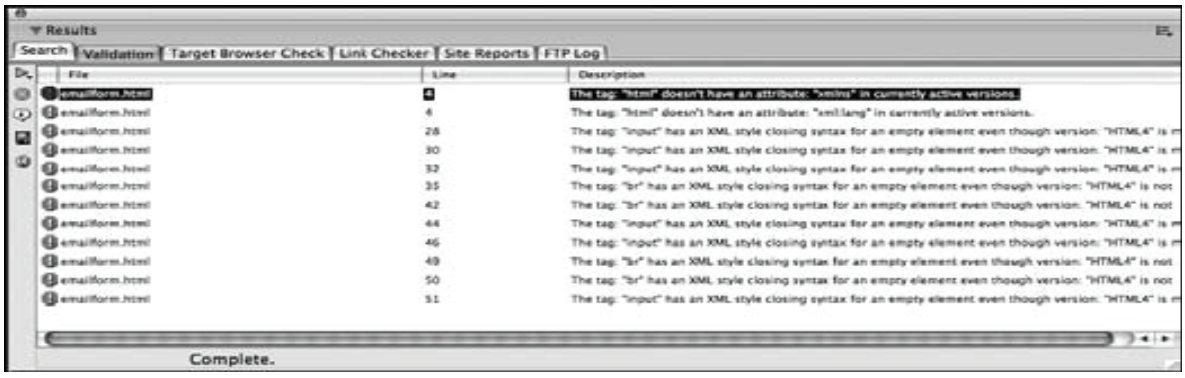
When the URL is entered, it will check for the errors in the page.

Example

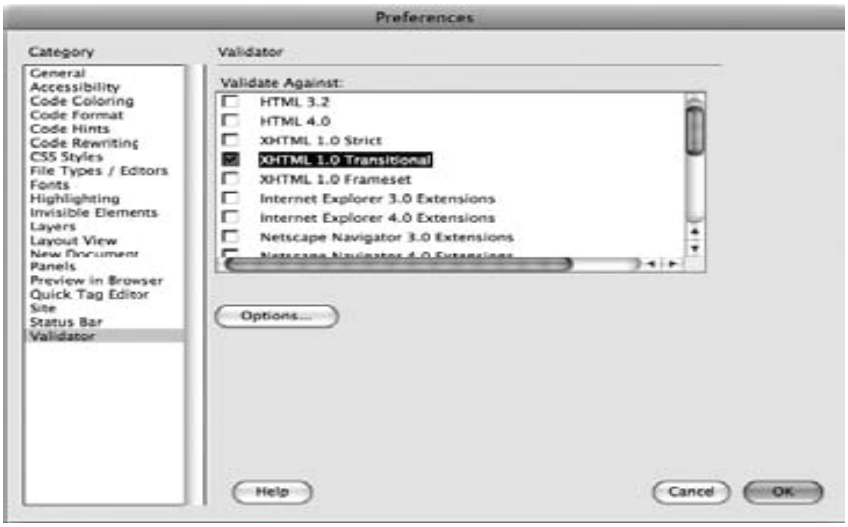


One disadvantage of W3C validation tool is that it validates the documents one after the other.

There are some other authoring tools that are used to validate the complete site. For example, Dreamwaver consists of many new features to perform validation for XHTML pages in its Dreamweaver Mx version. A user can validate a page by simply clicking on "shift + F6". It generates the errors in the page as shown in the below diagram,



To make the Dreamweaver work properly, right-click on the run panel and select the setting dialog box in windows or option-clic on mac operating system. The dialog box displays a wide range of document standards. Note that, all the checkboxes should be unchecked except the one which user want to check. For example, if a user is trying to validate a XHTML 1.0 Transactional document, then the respective box should be checked. This is shown in the below figure,



Q52. Discuss in detail about different stages in testing a site.

Answer :

The different stages in testing a site are as follows,

1. Validating HTML, XHTML and CSS
2. Link checking
3. Checking color depths
4. Accessibility checkers
5. Checking in different versions of browsers
6. Pilot testing
7. Proofreading.

1. Validating HTML, XHTML and CSS

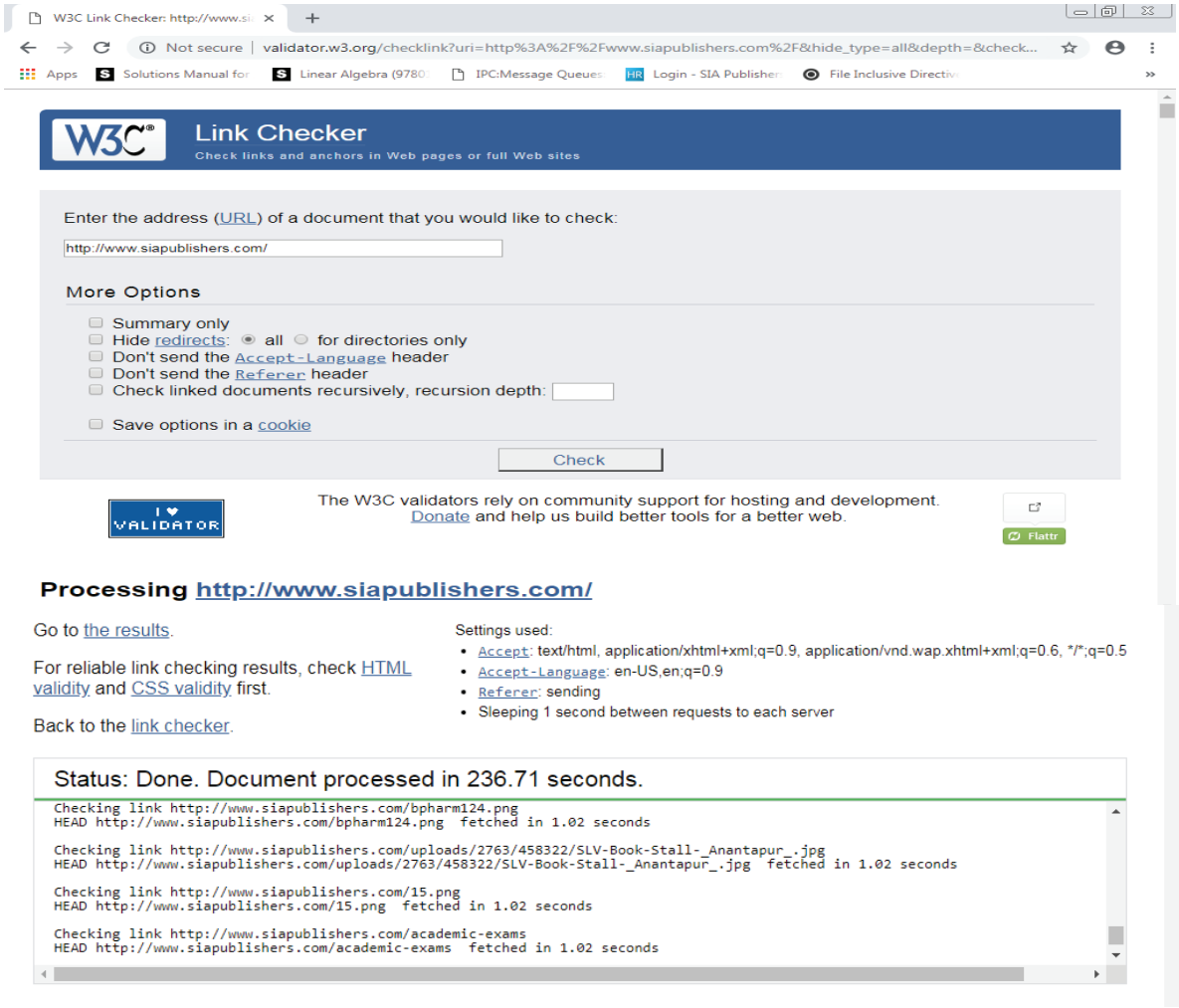
For answer refer Unit-III, Page No. 243, Q.No. 51, Topic: Validating HTML, XHTML and CSS.

2. Link Checking

The link of a web page need to be checked before and after it is published on web. There are many tools to check the links of the pages. Some of the tools charge some money and the other are free. The following are the different free link checking tools.

1. HTMLHELP's link valet at "www.htmlhelp.com/tools/valet/
2. The W3C's link checker at "http://validator.w3.org/checklink/

The following figure shows the result for checking the link at "http://validator.w3.org/checklink/",

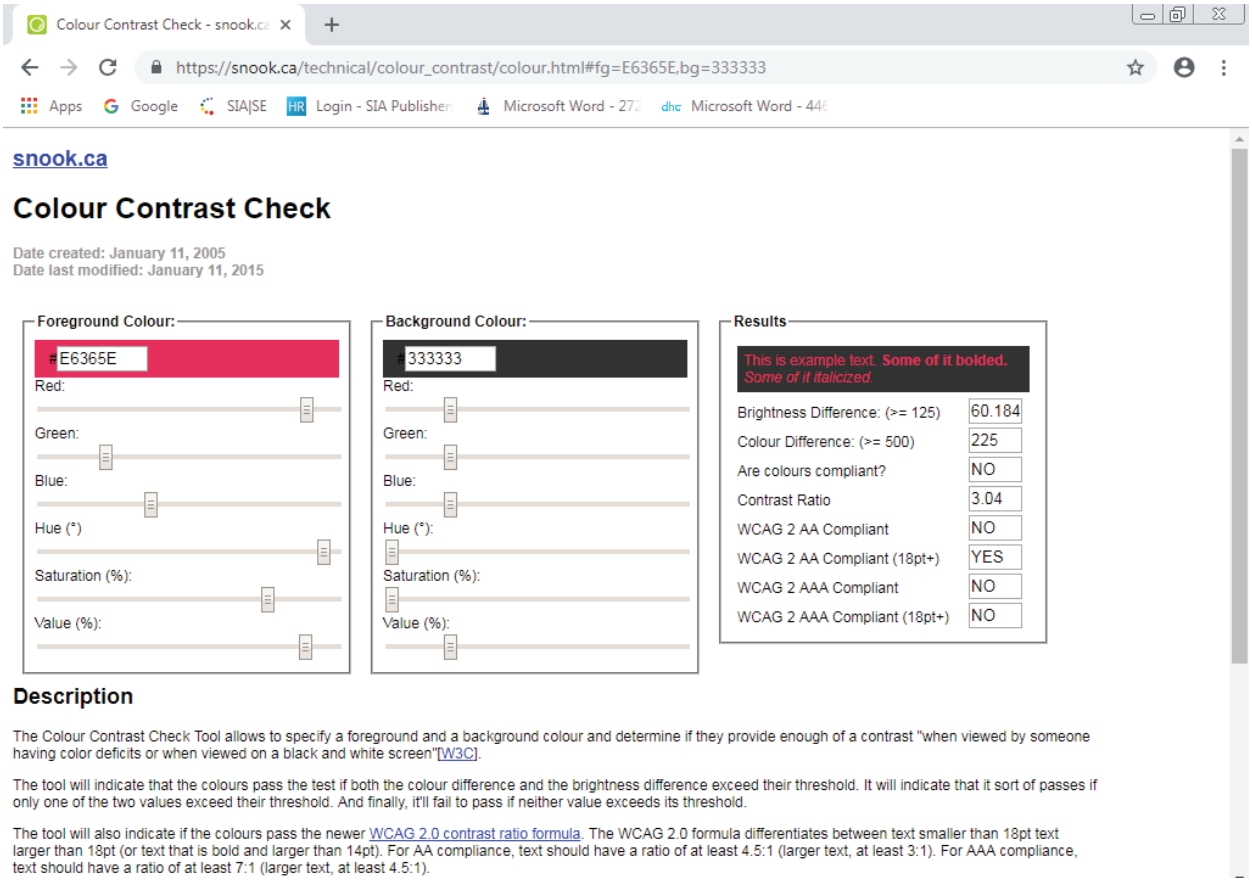


Dreamweaver has its own tools for checking links. This can be done by simpling clicking on ctrl + shift + F9.

3. Checking Color Depths

Check the color contrast to confirm whether it is clearly visible to people with colorblindness. This can be done by some color checking tools. Some of them are as follows,

- http://graybit.com/main.php
- www.vischeck.com/vischeck/vischeckURL.php
- www.snook.ca/technical/colour_contrast/colour.html.



4. Accessibility Checkers

Accessibility checkers are used to check whether the site has all the accessibility requirements. This also helps the site to run efficiently on a wide range of devices that access the web. Some of the different online Accessibility checkers are as follows,

- www.section508.info/
- www.webaim.org/resources/wave

For instance, the above tools check whether an 'alt' attribute is used as an alternative but it doesnt inform about the text that is to be displayed an behalf of the alternate image.

5. Checking in Different Versions of Browsers

It is very important to check the site in different versions of browsers. This is because, some of the browsers support the elements used in coding and some don't. Therefore, check the site in as many browsers as possible. Atleast, check in the latest versions of browsers because the versions of the browser in a system cannot be installed simultaneously.

Most importantly, the site need to be checked in IE6, chrome, opera and safari browsers.

There are some tools that permit user to take screenshots about the presentation of web site in the different browsers. Some of them are as follows,

- www. browsercom.com/
- http://browserlab.adobe.com/
- http://browsershots.org/.

7. Proof Reading

Proofreading is important to make the site content with out accurate errors. This is because, the errors in the site makes the client to feel bad about the site and the developers. Therefore, proof reading of the webpage avoids the errors and attracts the viewers towards the site.

6. Pilot Testing

Pilot testing is defined as validation of different modules of the site under the real-time circumstances. This test should be done before publishing the site. In this testing, the end user will ask to perform the different tasks on the site. For example, consider the site you have developed is an online book store. Some of the tasks that end users should try are given below,

- ❖ Find a story book that would be helpful to a 7-years kid.
- ❖ Find the cost of a specific book.
- ❖ Find the process of visiting the store when it is opened.
- ❖ Find whether a student can get all the requirements in the store or not.

Observe the operations performed by end users and help them in finding their requirement. And also observe what was the time taken to complete the given tasks and navigation from one page to another.

There are two types of observers while performing the above tasks. One observer watches operations performed by the user and the other asks the user to say something while performing the tasks called as "talking-aloud protocol". At that time, they provide some positive or negative information about the working of the site. These are helpful in making the site effective to the users.

The example of transcript that an observer end up with is as follows,

1. "I am supposed to find a book for Nithisha"
2. "Looking at....home, store..."
3. "Clicking on store"
4. "List of novels appears on the left, in which some of them not heard yet"
5. "Pictures on the right showing the "novels, storybooks..."
6. "Click on the image for novels..."
7. "Shows authors, so click the required...."
8. "Looking at the images..."
9. "This one is looking good; click on the image"
10. "Doesn't do anything... even it shows extra information about it"
11. "Click on "two-states"
12. "There, that one looks good"

Provide cameras on users shoulders to know the operations performed by them and their results including the audio.

3.3.3 Taking the Leap to Live

Q53. What are different things required to get the site on web.

Answer :

The different things required to get the site on the web are as follows,

1. The Domain name
2. The hosting space
3. Transfer the site to a new server using an FTP program.

1. The Domain Name

The domain name is an address that is provided to the users for communicating through online/internet.

Example

siapublishers.com, hrstoppro.com

There are many domain regestration companies to register a domain name for the site. So, choose a company and register the domain name. Most of the companies provide suffix as.com where as some other companies provide the suffix based on the country. For example, .co.in for India, .co. for UK, and .ru for Russia, so on. .me.uk is the suffix given to the personal sites in UK, .info is provided to the information-based sites, .org is provided to the organizations.

The different things that need to be checked and confirmed before choosing a domain name are as follows,

1. Check the availability of your domain name.
2. Check whether the name of the site is similar to any other sites or not.
3. Ensure that the name of the site should be simple to remember. For example, for a company sia publishers and distributors private Ltd, choose www.siapublishers.com rather than choosing www.siapublishersanddistributorsprivateLtd.com.
4. Register a name in such a way that it can also be used in mails. For example, vc@siapublishers.com.

Some domain registration companies provide hosting also. But, choose an other company for hosting then only the registration company to point the domain to the provided hosting company server.

Remember that, a domain name can also be registered before building the site. But, should not be content in it until and unless the site is built completely.

2. Hosting

Users can create their own web pages through ISP's. Inorder to create this, the ISP allots a specific amount of disk space to each user. So that they can create set of web pages. The users using Internet should be allowed to access these web pages just like any other web pages on internet. The additional hardware required for this apart from disk space is web server with in the premises of ISP. Therefore, this server plays as host to the subscribes's web pages and also if any request arises, the server issue them to any user i.e., web client or browser.

3. Transfer Your Site to a New Server Using an FTP Program

For answer refer Unit-III, Page No. 249, Q.No. 55.

Q54. What are the various key considerations for choosing a host?

Answer :

The various key considerations for choosing a host are as follows,

1. Bandwidth

Bandwidth is defined as the amount of data that is permitted to the users on the web site. Suppose the average size of the site is 100 KB with all the images and documents, and allow 100 visitors to visit the site per month, let each access be 10 pages, then the required bandwidth is $100 \times 100 \times 10 = 100000$ kb(or 100 MB).

Never mention that the website is highly popular because the host companies may charge more. Suppose, if the site exceeds the size of bandwith then the company will charge some additional amount.

2. Country

Choose the server in a country in which the site has most visitors. Then the data can be transfered quickly and can be visible to the users in milliseconds. For example, the visitor in India will view the data quickly from the host server in India.

3. Backups

Backup is defined as an other copy of the site that is stored when any problem occurs in the computer. The host companies stores the copy of the site in case of server break down and some other backups of data when the site breaks suddenly.

Suppose if the site updates frequently request the hosting company to check for the frequent backups.

4. Disk Space

Disk space is defined as the size of memory to store the files on a computer from which the site will run. Generally, the disk space will be in Mega Bytes (MB) or Giga Bytes (GB). It depicts the size of the site. Ensure that the diskspace should be more than the size of the files that run the site.

5. Data Centers

A data center is defined as a large group of servers that hosts various sites. Most of the companies tell that they have a multimillion data centre which doesn't makes the site a big operation. This is because maximum of the companies hire the data from the huge data centres.

6. Uptime

Uptime is defined as the percentage of time that the site is available to the user. The uptime for most of the companies is 99 out of 100 minutes. But, this also shorts 1 minute that means 87.6 hours for a year (approximately 4 days).

7. E-mail Accounts

Most of the hosting companies offer E-mail service with the web hosting. Here, consider the following two factors,

- ❖ The size of mailbox is provided.
- ❖ The number of mailboxes that are given.

Here, some hosting companies provide unlimited size of mailbox and limited number of mailboxes, some provide unlimited number of mailboxes and with limited size of mailbox. In such cases, it is difficult to provide the data to as many mails as user needs. Finally, some of the hosting companies allow the user to use the maximum of mailboxes and sizes but here, the size will depend on the storage device.

8. Statistics Packages

When a user requests for a file from the system then the server will record some information about the user like IP address, OS, browser and stores them in a file called as log file. The statistic packages are used to check the log file to find the useful information from such as like the operations performed by the user on the site? Keywords used by them to redirect the page and so on.

9. Shared Versus Dedicated Hosting

A web server that has many sites in it is known as a shared host because, only a few people visit the smaller site.

Some web sites have many visitors all the time. Some times the users may use the data more than the provided band width. It means the site needs to pay some additional amount to the hosting companies. Thus, it makes the site more popular. In such cases, the host own website with the server that is known as decidated server. This means the complete server will be sticked to the site only.

Q55. Explain about putting your site on a server using FTP.

Answer :

Putting the site on a server using FTP is an easy and efficient way. FTP stands for "File Transfer Protocol" that is used to send the binary files across the internet. This method has a transfer rate greater than the HTTP so that user can send the files faster than HTTP.

Maximum of the host providers need user to use the FTP to send the files to a server across the internet that means user requires a FTP program to put the site on a web server.

Generally, FTP consists of two windows, one displays the folders or files in the computer and the other displays the folders in the server. Some of the different FTP programs to send the files into the server are shown in the below table,

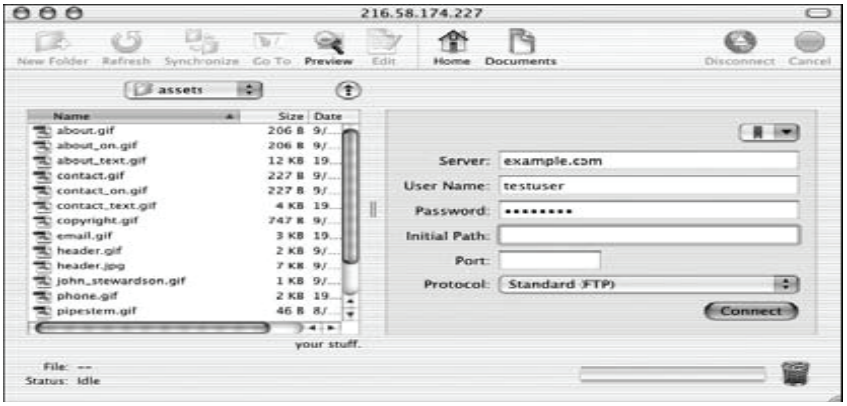
Product Name	Operating System	Path/URl
Cute FTP	windows and MAC OS X	www.cuteftp.com/
FireFTP	windows and MAC OS X	http://fireftp.mozdev.org/
FTPX	windows	www.ftpx.com/
Transmit	Mac	www.panic.com/transmit/
Fetch	Mac	www.fetchsoftworks.com/

The above mentioned programs follow same principle with small variations in the code.

Once the registration is done with the FTP, it sends the details to FTP the site to its servers. It includes the following details:

- ❖ An FTP Address (like ftp.example.com)
- ❖ An FTP Username (The same name as the domain username)
- ❖ An FTP Password (The same password as the domain password).

The process of entering the details into the FTP program called Transmit on a Mac is as shown in the below figure,



Some of the hosting companies support SFTP or secure FTP which is the variant of FTP. SFTP is known to be highly secured and standard because the details like username and password will be encrypted when they sent to the server.

Q56. Discuss in brief about the importance of directory structures and URLs.

Answer :

Importance of Directory Structures and URLs

Most of the people develop their very first website on their desktop or computer using relative URLs for all the images, styles sheets, links and scripts. That means, all the links provided in the coding of a site will work on the web server without modifications.

When a site is opened, it shows the URL in the browser window as like given below,

C:\User\webdesigning\example\Ex1.html

But when the site can switch to new site or URL then the URL will appears as follows,

http://www.ex.com/Ex1.html

Generally, in a site the relative URLs for images, styles, links and scripts will be declared as,

But, when the site is switched to a new site or URL onto the web server, it doesn't load the source image file. Therefore, provide the image URL something like the following,

Thus, the page loads the image until and unless the example directory is deleted for the computer.

3.3.4 Telling the World About Your Site

Q57. What are the different pointers used to make more people aware of the site?

Answer : **Model Paper-III, Q11(b)**

The different pointers used to make more people aware of the site are as follows,

1. Always mention the URL
2. Content is king
3. Forums, communities, and social Networks
4. Check for trade sites and directories
5. Reciprocal links
6. Look at your competition
7. Make it easy for people to link to you
8. Pay-per-click advertising
9. Display advertising
10. Create an E-mail news letter.

1. Always Mention the URL

Mention the URL of the site where ever possible. Provide the URL in the brouchers, business cards, letterheads, adverts and posters then the people will get to know about the site.

2. Content is King

Content is the king of the page. If the content provided is effective, useful, entertaining and informative then the people will attract to the site easily. Try to provide the information in an easy and unique way rather than providing regularly.

Provide the updated information and update the existing information periodically by mentioning, "last update on particular time and date". Then, the user will get to know the information in the page is latest.

Publish the benefited from a Press Release, in PR news services like PR web(www.PRweb.com). Thus, the other bloggers or journalist will choose the content and mentions it.

3. Forums, Communities and Social Networks

Different hobbies and industries have their own forums and online communities. Group of people that interested in the same topic can be found on the social media like Facebook, Twitter etc.

Suppose, if user is a member of online community which is related to the site then they can start your profile in it with its URL and photographs, and providing the URL of the site in the social media profiles. Then the site will become popular among the people.

Interact with the other people by reacting and replying to their posts rather than only posting the site information and publishing the website.

4. Check for Trade Sites and Directories

Now a days, most of the trades and hobbies have their own site which provides the list of all other sites. Check for the site which is relevant to the information of the site and request the relative sites.

For the listing of related sites, some trades may charge some money. In such situations, know the average number of visitors visits the site and the monthly traffic before subscribing to their sites. Try to take a trail period to check the monthly traffic and the number of visitors per month.

5. Reciprocal Links

Most of the smaller websites provide reciprocal links that means, their site will link to present site for linking itself to them. In such cases, do not provide others links on the front page because it leads to populate their site than present site. With this, the site will get better traffic from the sites that are related to the information in the site.

6. Look at Your Competition

Find the different competitors of the site and check who is linking to them. And, request them to link to the site as well.

7. Make it Easy for People to Link to You

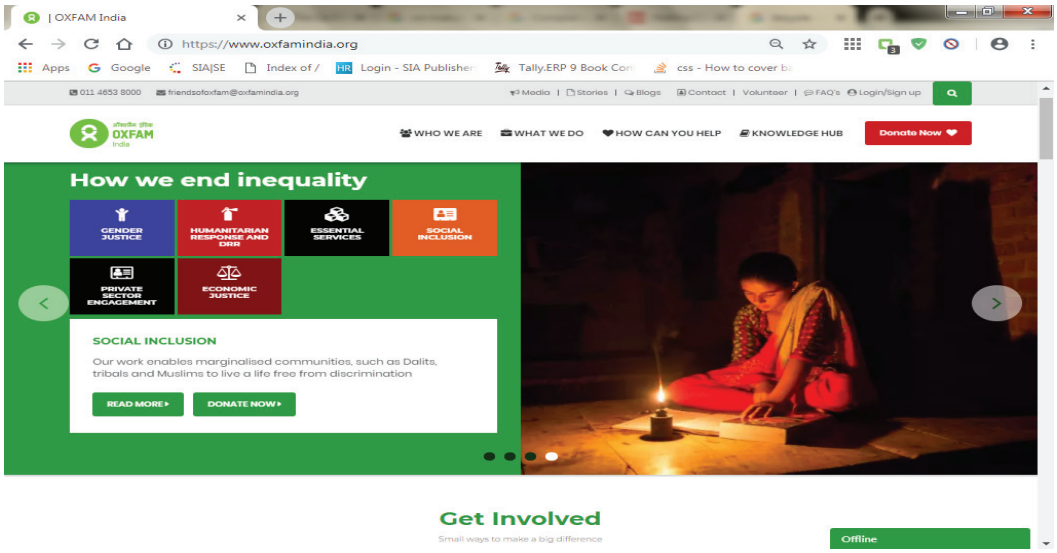
Most of the webpage provides links to bookmark a page on services like Email, Twitter and facebook, etc. This is shown in the below figure,



A simple and easier way to link these options is using the website known as www.addthis.com/ that offeres some simple tools to create a link to share the link of the site with others.

Create a page with banners or button that helps the visitors to link to the site easily. Some websites like Oxfam include the ads that are with attractive images and effective content.

Example



8. Pay-per-click Advertising

Pay-per-click is a type of advertising techniques in which marketer pay a specified amount for listing the web site on all search engine results, which will appear by entering various keywords.

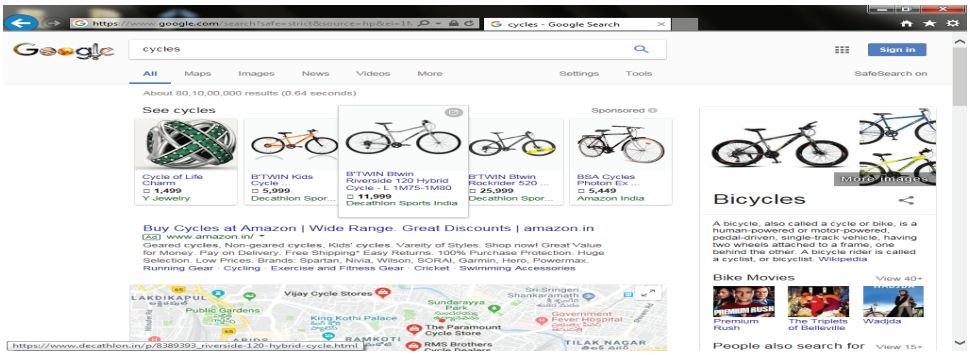
Two main factors decides the order of the advertisements to be appeared on the page. They are,

- ❖ The amount of money paid for every click of an ad.
- ❖ The count of users that clicks on the ad.

This is known as a best and cost-effective way to produce the traffic to the site. Some other like microsoft and Yahoo provides similar services called adcenter and Yahoo! search marketing respectively.

Example

An example for pay-per-click advertising is given below,



9. Display Advertising

Display advertising is an online advertising technique that forces the user to pay some amount for advertising about the site on some other sites. It includes banner ads.

The companies charge for adverting the site in two ways. They are,

1. Cost-per-thousand impressions, named as CPM.
2. Cost to occupy some space on a particular site for some months or years.

10. Create an E-mail Newsletter

E-mail newsletter is a feature that is required when the content of the page is changing frequently. In this, when the content is changed then an email will be sent to the subscriber saying the content is updated.

The following are the different tools to create and manage the e-mail lists,

- ❖ www.MailChimp.com
- ❖ www.CampaignMonitor.com.

These tools are used to send mail to the visitors and also helps in creating the subscription forms that are used to signup for the users.

3.3.5 Understanding Your Visitors

Q58. What are the two different ways to learn about the visitors to the site?

Answer :

The two different ways to learn about the visitors to the site are as follows,

1. Log file analysis
2. Analytics software.

1. Log File Analysis

When a user requested for a file from the system then the server will record certain information about them like IP address, operating system, browser type and stores them in a file called as log file. Most of the companies offers some tools to analyze the visitors information.

2. Analytics Software

Analytics software uses a script for checking the log file and collected information about the visitors that is similar to other. It also offers the tools to analyze the collected information.

Google ofters a free service tool called as Google Analytics (www.google.com/analytics). This provides a detail analysis about the visitors like how they entered the site, how many times they have visited, how many files they have downloaded (referred as hits). This uses the different static packages and collects the information which is useful to the site, and analyzes it.

The example of a report from the Google Analystics is as shown in the below figure,



Q59. What are the ten different things to check in google analytics report for a site?

Answer :

The ten different things to check in google analytics report for a site are as follows,

1. Pageviews

It displays the number of pages visited by the visitors on the site.

2. Views

It displays the number of times that visitors visited the site in a period of time.

3. Unique Visits

It displays the number of visitors visited the site in a period of time. Here, each visitor is counted once.

4. Referres

Referrers are defined as the different sites that sends the visitors to visit the page. These increases the traffic for the site. User can also view the referres and work together to maintain the same traffic for the site.

5. Search Terms

Search terms are terms that a user enters in the search engine like google before redirecting to the site. This is used to know about the different terms they used for visiting the site.

6. User's Browser, Operating System and Screen Resolution

It displays the type of browser, operating system and screen resolution from which user visited the site.

7. Top Content

It displays the highly populated sections and pages of the site.

8. Top Entry Pages

It displays the pages that are visited firstly and mostly in the site.

9. Top Exit pages

It displays the pages from which the users left.

10. Bounce Rate

It displays the count of the visitors who left from the same page that they have entered.

Q60. Explain the concept of measuring everything.

Answer :

The locations of the site visitors can be tracked by using Google Analytics. In addition to this, it also helps user to measure the success of marketing activity and that permits user to find the Return on Investment (ROI). And using this user can also find the time to initiate the marketing activity.

Example

Consider that user can spend a $200 in a month on Google Adwords and it consumed a day to set everything, writing ads, and checking how the ads are running. Here, it may cost something but no one will do all these things with out pay.

Confidence is necessary to measure each and every awareness and marketing activities that user performs. Google Analytis also permits user to provide an additional information to the URL. Thus, one can measure the effectiveness of the links. The additional information provided by Google Analytics is as follows,

Term	Description	Query string property
Campaign term	It provides different keywords for the campaign	utm_term
Campaign name	It provides a readable term using which the source can be determined	utm_campaign
Campaign source	It provides the referring site	utm_source
Campaign medium	It provides the medium like a blog, e-mail, PPC, display	utm_medium
Campaign content	It is used to differentiate multiple links from the same campaigns	utm_content

Google Analytics URL Builder is a handy tool that allows user to add these links to the URL

Example

http://www.example.com/?utm_source = referrer.org&utm_medium = display&utm_term=cheese&utm_content= cheddar&utm_campaign = taste%2Bthe%2Ocheese

INTERNAL ASSESSMENT

OBJECTIVE TYPE

I. Multiple Choice

1. The structure of a javascript code begins with the tag _____. [ ]

   (a) <scripting javascript> (b) <script type = "language/javascript">

   (c) <script language = "type/javascript"> (d) <script "Code = javascript">

2. _____ is a programming language with direct support to the object oriented methodologies. [ ]

   (a) JavaScript (b) Java

   (c) C++ (d) None of the above

3. _____ function is used to convert the all lowercase letters to the equivalent upper case letters/characters. [ ]

   (a) toUpper() (b) toUpperCase()

   (c) Upper() (d) None of the above

4. Which of the following is an object based array function _____. [ ]

   (a) push( ) (b) reverse( )

   (c) pop( ) (d) All the above

5. Which of the following is an object provided by javascript? [ ]

   (a) Match object (b) Date object

   (c) Both (a) and (b) (d) None of the above

6. Which of the following method is uesd to manipulate an array? [ ]

   (a) reverse( ) (b) slice( )

   (c) sort( ) (d) All the above

7. _____ function is used to display the minimum of two number given to it. [ ]

   (a) Min( ) (b) Minimum( )

   (c) Both (a) and (b) (d) None of the above

8. Which of the following is a universal attribute of meta tag. [ ]

   (a) dir (b) title

   (c) lang (d) All the above

9. _____ is defined as a large group of servers that hosts various sites. [ ]

   (a) Data center (b) Server

   (c) Host (d) None of the above

10. _____ displays the number of times that visitors visited the site in a period of time. [ ]

    (a) Unique visits (b) Visits

    (c) Views (d) Page views

II. Fill in the Blanks

1. _____ performs execution of program/script on webserver.
2. JavaScript is _____ side scripting language.
3. Arrays in javascript are also called as _____.
4. _____ is a data item that stores the user defined values.
5. _____ function is used to combine two strings.
6. _____ is a self contained block of code that performs specific and well defined tasks.
7. _____ function displays the rounded value of the integer.
8. _____ is used to provide the information regarding the page indexing to the search engines.
9. _____ is defined as verifying different modules of the site under the real-time circumstances.
10. _____ displays the number of pages visited by the visitors on the site.

KEY

I. Multiple Choice

1. (c) 2. (a) 3. (b) 4. (d) 5. (c)
6. (d) 7. (a) 8. (d) 9. (a) 10. (c)

II. Fill in the Blanks

1. Server-side javascript
2. Client
3. Associative arrys
4. Variable
5. concat( )
6. Function
7. ceil( )
8. Meta tag
9. Pilot testing
10. Page views

III. Very Short Questions and Answers

Q1. What is javascript?

Answer :

Java script is a programming language that support object oriented methodologies. It facilitates inclusion of certain executable data along with it.

Q2. What are the uses of javascript?

Answer :

The uses of javascript are as follows,

(i) Java script is multi-functional i.e., it can be used as a client side scripting as well as server side.

(ii) It is used to provide an easy way of accessing document objects as well as it can manipulate almost all of these objects.

Q3. Define variable in JavaScirpt.

Answer :

A variable is a data item that stores the user defined values. It can be manipulated at any time. In JavaScript, the variables are same as the other programming language variables.

Q4. Define bandwidth.

Answer :

Bandwidth is defined as the amount of data that is permitted to the users to use on the site.

Q5. Define uptime.

Answer :

Uptime is defined as the percentage of time that the site is available to the user.

Unit-wise Important Questions

Unit-1

SHORT QUESTIONS

Q1. Write about different formatting tags.

Answer : Important Question

For answer refer Unit-I, Page No. 2, Q.No. 1.

Q2. Write about various controls of forms.

Answer : Important Question

For answer refer Unit-I, Page No. 2, Q.No. 2.

Q3. Write about types of lists.

Answer : Important Question

For answer refer Unit-I, Page No. 3, Q.No. 3.

Q4. Write about the tags used for creating table.

Answer : Important Question

For answer refer Unit-I, Page No. 3, Q.No. 4.

Q5. What is a frame in HTML? Give an example.

Answer : Important Question

For answer refer Unit-I, Page No. 3, Q.No. 5.

Q6. Explain <img> tag in HTML.

Answer : Important Question

For answer refer Unit-I, Page No. 4, Q.No. 6.

Q7. What are head, title, body tags in HTML?

Answer : Important Question

For answer refer Unit-I, Page No. 4, Q.No. 7.

Q8. What is hyperlink? Explain the anchor tag with syntax.

Answer : Important Question

For answer refer Unit-I, Page No. 4, Q.No. 9.

Q9. Define with syntax <FONT> tag.

Answer : Important Question

For answer refer Unit-I, Page No. 4, Q.No. 10.

ESSAY QUESTIONS

Q10. Explain in detail about HTML.

Answer : Important Question

For answer refer Unit-I, Page No. 5, Q.No. 11.

Q11. Discuss about presentational elements.

Answer : Important Question

For answer refer Unit-I, Page No. 15, Q.No. 20.

Q12. Explain about images and supported attributes in HTML.

Answer : Important Question

For answer refer Unit-I, Page No. 30, Q.No. 31.

Q13. Explain how basic table is created in xhtml.

Answer : Important Question

For answer refer Unit-I, Page No. 46, Q.No. 48.

Q14. What are different types of Form controls in xhtml?

Answer : Important Question

For answer refer Unit-I, Page No. 56, Q.No. 55.

Q15. Discuss about frameset.

Answer : Important Question

For answer refer Unit-I, Page No. 69, Q.No. 64.

Unit-2

SHORT QUESTIONS

Q1. What is CSS?

Answer : Important Question

For answer refer Unit-II, Page No. 80, Q.No. 1.

Q2. Write about style rule.

Answer : Important Question

For answer refer Unit-II, Page No. 80, Q.No. 2.

Q3. List the properties that influence the appearance of the font.

Answer : Important Question

For answer refer Unit-II, Page No. 80, Q.No. 4.

Q4. What is a selector? List the types of selectors in CSS.

Answer : Important Question

For answer refer Unit-II, Page No. 81, Q.No. 5.

Q5. Write about vertical-align property.

Answer : Important Question

For answer refer Unit-II, Page No. 81, Q.No. 6.

Q6. What are the different properties that are used to position the items on the page.

Answer : Important Question

For answer refer Unit-II, Page No. 81, Q.No. 7.

Q7. Discuss in brief about tracking.

Answer : Important Question

For answer refer Unit-II, Page No. 81, Q.No. 8.

Q8. Write a short note on,

(i) list-style-position property

(ii) list-style-image property.

Answer : Important Question

For answer refer Unit-II, Page No. 81, Q.No. 9.

Q9. List the advantages of fixed-width design and liquid design.

Answer : Important Question

For answer refer Unit-II, Page No. 81, Q.No. 10.

ESSAY QUESTIONS

Q10. How style sheets are specified into HTML document?

Answer : Important Question

For answer refer Unit-II, Page No. 82, Q.No. 12.

Q11. Explain how the fonts can be controlled. List and explain each of the properties that directly affect the font and its appearance.

Answer : Important Question

For answer refer Unit-II, Page No. 89, Q.No. 18.

Q12. Discuss in detail about the dimensions of a box.

Answer : Important Question

For answer refer Unit-II, Page No. 111, Q.No. 27.

Q13. Explain in detail about background properties of CSS3 with an example.

Answer : Important Question

For answer refer Unit-II, Page No. 117, Q.No. 29.

Q14. Discuss in brief about the composition and grids.

Answer : Important Question

For answer refer Unit-II, Page No. 142, Q.No. 44.

Q15. Write in detail about layout of forms.

Answer : Important Question

For answer refer Unit-II, Page No. 165, Q.No. 66.

Unit-3

SHORT QUESTIONS

Q1. What is a variable in java script? How to declare a variable.

Answer : Important Question

For answer refer Unit-III, Page No. 174, Q.No. 3.

Q2. Define functions in javascripts.

Answer : Important Question

For answer refer Unit-III, Page No. 1744, Q.No. 4.

Q3. What is an array in java script?

Answer : Important Question

For answer refer Unit-III, Page No. 175, Q.No. 7.

Q4. Write a short note on meta tags.

Answer : Important Question

For answer refer Unit-III, Page No. 175, Q.No. 9.

Q5. List the different points used to make people aware of the site.

Answer : Important Question

For answer refer Unit-III, Page No. 175, Q.No. 11.

Q6. Write a short note on domain name.

Answer : Important Question

For answer refer Unit-III, Page No. 175, Q.No. 12.

ESSAY QUESTIONS

Q7. How a script is added to the webpage.

Answer : Important Question

For answer refer Unit-III, Page No. 176, Q.No. 13.

Q8. What is an operator? What are the different types of operators?

Answer : Important Question

For answer refer Unit-III, Page No. 188, Q.No. 21.

Q9. Write about form validation.

Answer : Important Question

For answer refer Unit-III, Page No. 219, Q.No. 33.

Q10. Write short notes on following,

(i) Focusing on first form item

(ii) Auto tabbing between fields.

Answer : Important Question

For answer refer Unit-III, Page No. 231, Q.No. 40.

Q11. Write in short about the following,

(i) Drag and drop sortable lists using scriptaculous

(ii) Creating a lightbox.

Answer : Important Question

For answer refer Unit-III, Page No. 236, Q.No. 46.

Q12. Discuss about the following,

(i) Creating calendar with YUI

(ii) Auto completing text inputs with YUI.

Answer : Important Question

For answer refer Unit-III, Page No. 239, Q.No. 48.

Q13. What are the two different states of testing? Explain about validating HTML, XHTML and CSS.

Answer : Important Question

For answer refer Unit-III, Page No. 243, Q.No. 51.

Q14. What are the different pointers used to make more people aware of the site?

Answer : Important Question

For answer refer Unit-III, Page No. 250, Q.No. 57.

Q15. Explain the concept of measuring everything.

Answer : Important Question

For answer refer Unit-III, Page No. 253, Q.No. 60.

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LAB PRACTICAL WITH SOLUTIONS

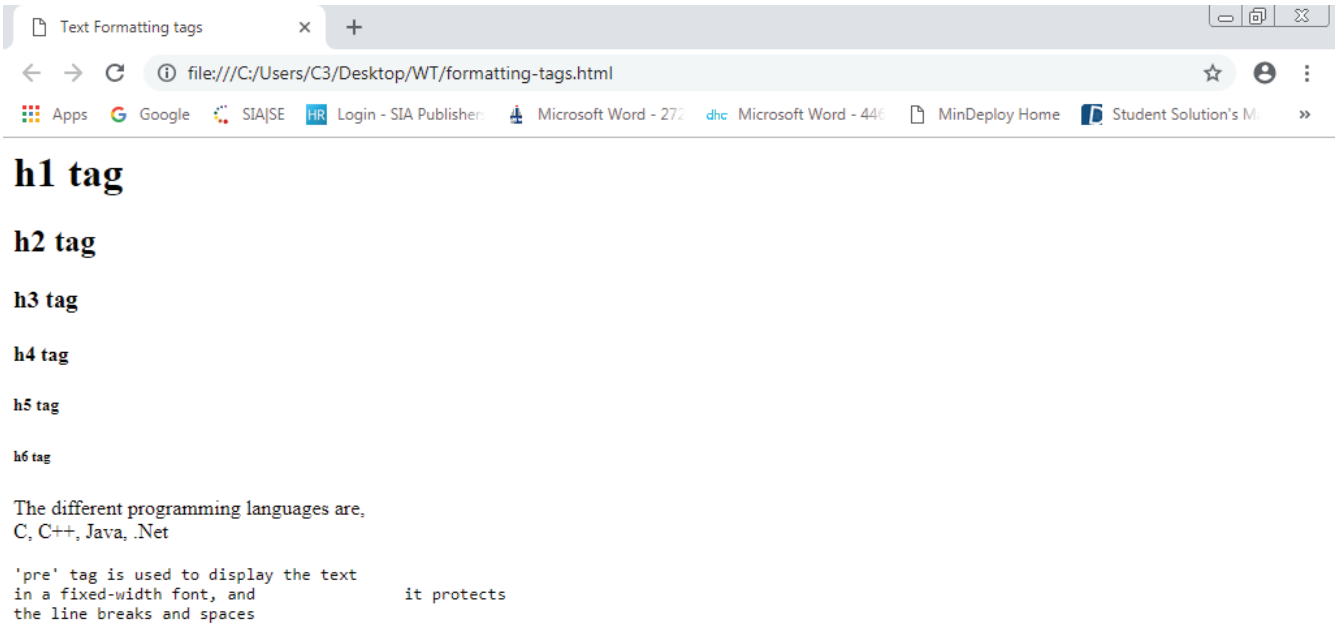
**Q1. (a) Write a HTML program using basic text formatting tags, <hn>, <p>,
, <pre>.**

Answer :

Program

```
<html>
<title>Text Formatting tags</title>
<body>
<h1>h1 tag</h1>
<h2>h2 tag</h2>
<h3>h3 tag</h3>
<h4>h4 tag</h4>
<h5>h5 tag</h5>
<h6>h6 tag</h6>
<p>The different programming languages are,<br>C, C++, Java, .Net <p>
<pre>
'pre' tag is used to display the text
in a fixed-width font, and        it protects
the line breaks and spaces
</pre>
</body>
</html>
```

Output



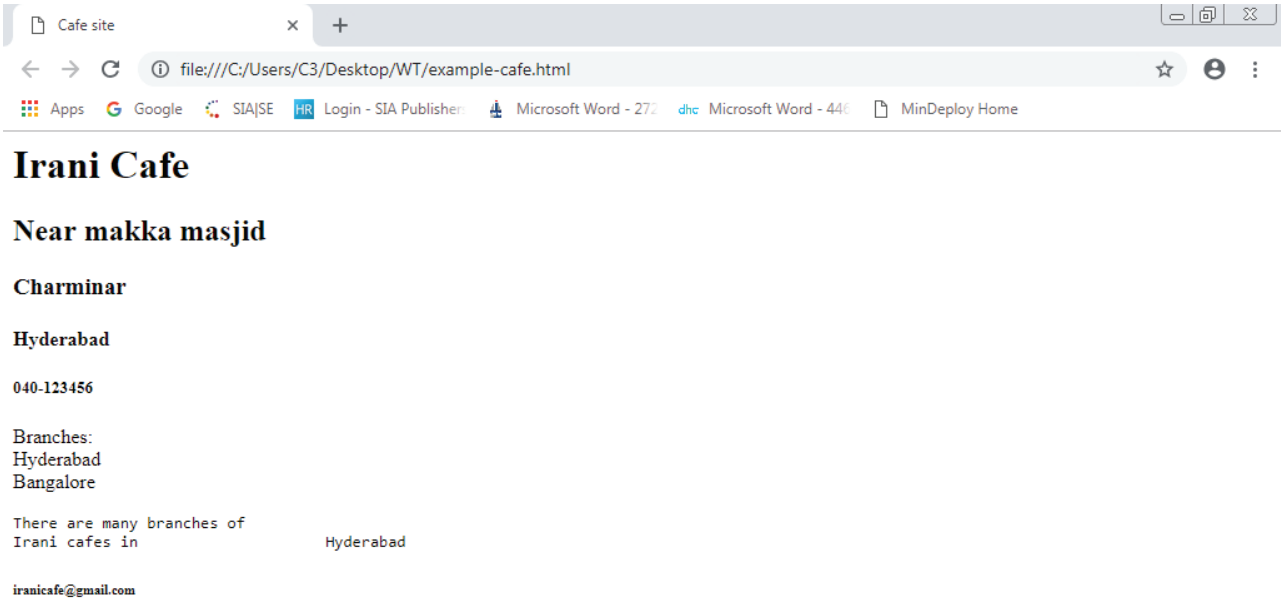
(b) Write a HTML page for Example Cafe using above text formatting tags.

Answer :

Program

```
<html>
<title>Cafe site</title>
<body>
<h1>Irani Cafe</h1>
<h2>Near makka masjid </h2>
<h3>Charminar</h3>
<h4>Hyderabad</h4>
<h5>040-123456</h5>
<p>Branches:<br>Hyderabad<br>Bangalore</p>
<pre>
There are many branches of
Irani cafes in                  Hyderabad
</pre>
<h6>iranicafe@gmail.com</h6>
</body>
</html>
```

Output



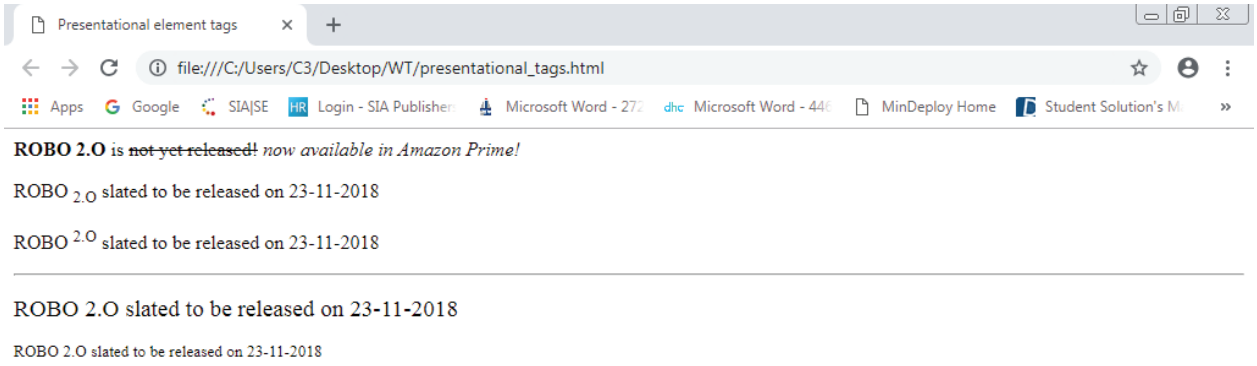
Q2. (a) Write a HTML program using presentational element tags <b>, <i>, <strike>, <sup>, <sub>, <big>, <small>, <hr>.

Answer :

Program

```
<html>
<title>Presentational element tags</title>
<body>
<p><b>ROBO 2.O</b> is <strike>not yet released!</strike><i> now available in Amazon Prime!</i></p>
<p>ROBO <sub>2.O</sub> slated to be released on 23-11-2018</p>
<p>ROBO <sup>2.O</sup> slated to be released on 23-11-2018</p>
<hr>
<p><big>ROBO 2.O slated to be released on 23-11-2018</big></p>
<p><small>ROBO 2.O slated to be released on 23-11-2018</small></p>
</body>
</html>
```

Output



(b) Write a HTML program using phrase element tags <blockquote>, <cite>, <abbr>, <acronym>, <kbd>, <address>.

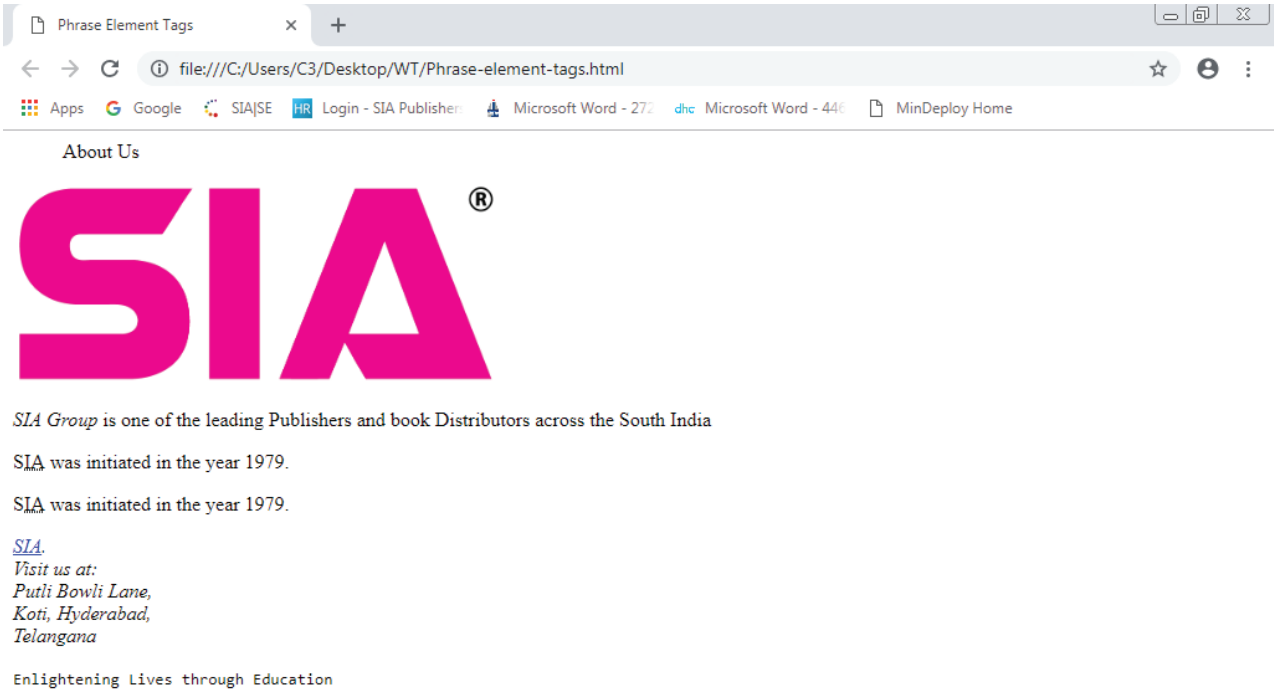
Answer :

Program

```
<html>
<title>Phrase Element Tags</title>
<body>
<blockquote cite= “http://www.siapublishers.com/”>
About Us
</blockquote>
<img src= “SIAMAINLOGO.png” alt= “Error”>
<p><cite>SIA Group</cite> is one of the leading Publishers and book Distributors across the South India </p>
<p><abbr title= “Syed Ibadat Ali”>SIA</abbr> was initiated in the year 1979.</p>
<p><acronym title= “Syed Ibadat Ali”>SIA</acronym> was initiated in the year 1979.</p>
<address><a href= “support@siapublishers.com”>SIA</a>.<br>
```

```
Visit us at:<br>
Putli Bowli Lane,<br>
Koti, Hyderabad,<br>
Telangana
</address><br>
<kbd>Enlightening Lives through Education</kbd>
</body>
</html>
```

Output



Q3. (a) Write a HTML program using different list types.

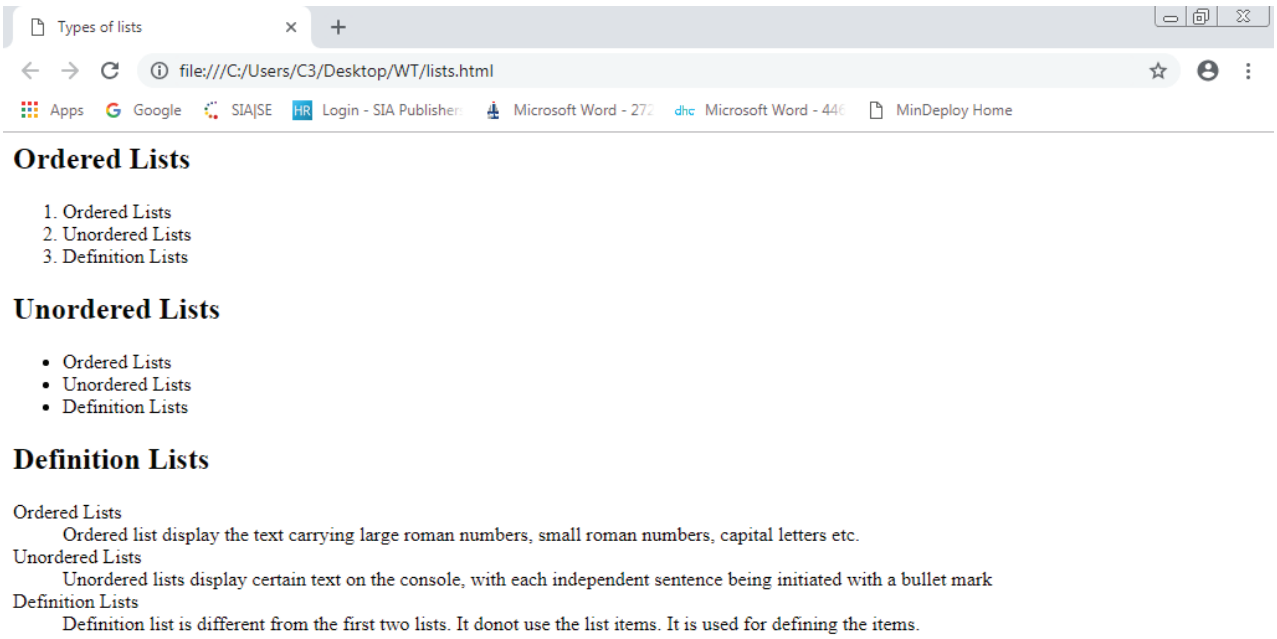
Answer :

Program

```
<html>
<title>Types of lists</title>
<body>
<h2>Ordered Lists</h2>
<ol>
<li> Ordered Lists </li>
<li> Unordered Lists </li>
<li> Definition Lists </li>
</ol>
<h2>Unordered Lists</h2>
<ul>
<li> Ordered Lists </li>
<li> Unordered Lists </li>
```

```
<li> Definition Lists </li>
</ul>
<h2>Definition Lists</h2>
<dl>
<dt>Ordered Lists</dt>
<dd> Ordered list display the text carrying large roman numbers, small roman numbers, capital letters etc.</dd>
<dt>Unordered Lists</dt>
<dd>Unordered lists display certain text on the console, with each independent sentence being initiated with a bullet mark</dd>
<dt>Definition Lists</dt>
<dd> Definition list is different from the first two lists. It donot use the list items. It is used for defining the items.</dd>
</dl>
</body>
</html>
```

Output



(b) Write a HTML page that displays ingredients and instructions to prepare a recipe.

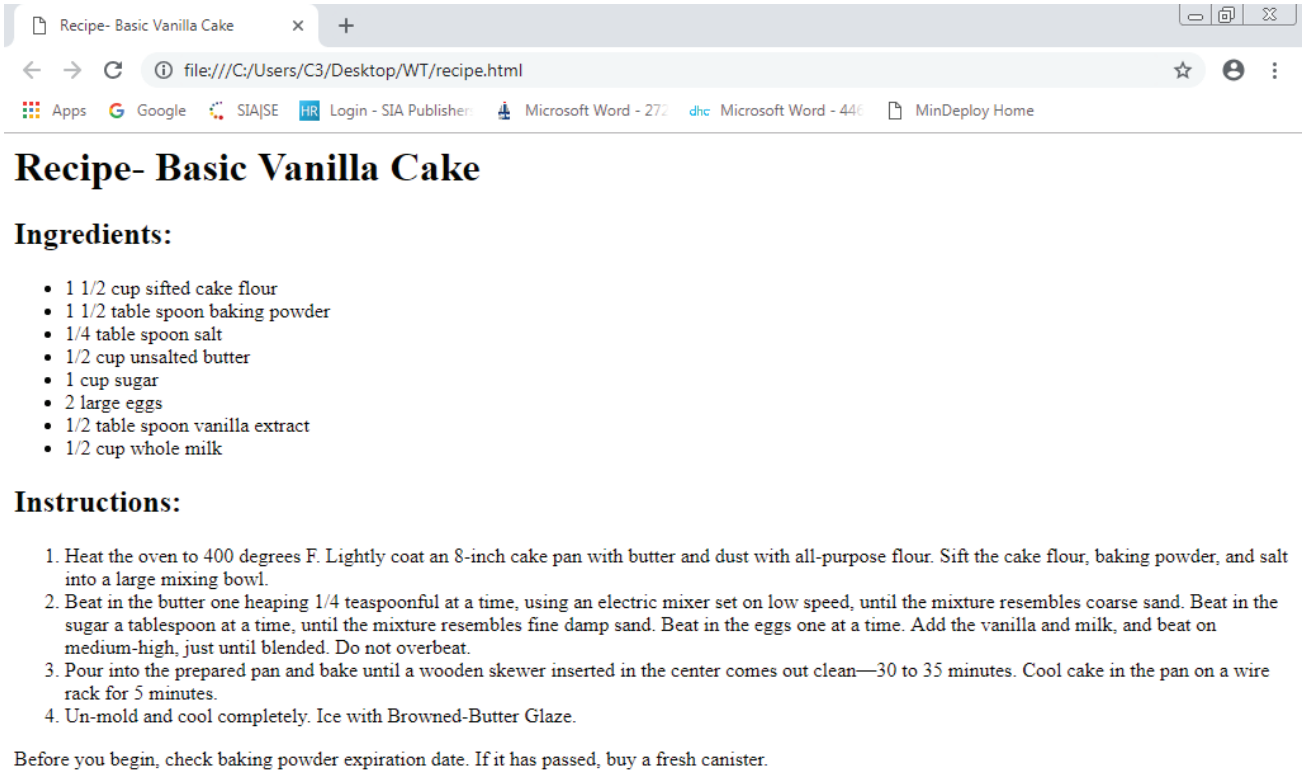
Answer :

Program

```
<html>
<head>
<title>Recipe- Basic Vanilla Cake</title>
</head>
<body>
<h1>Recipe- Basic Vanilla Cake</h1>
<h2>Ingredients:</h2>
<ul>
<li>1 1/2 cup sifted cake flour</li>
```

```
<li>1 1/2 table spoon baking powder</li>
<li>1/4 table spoon salt</li>
<li>1/2 cup unsalted butter</li>
<li>1 cup sugar</li>
<li>2 large eggs</li>
<li>1/2 table spoon vanilla extract</li>
<li>1/2 cup whole milk</li>
</ul>
<h2>Instructions:</h2>
<ol>
<li>Heat the oven to 400 degrees F. Lightly coat an 8-inch cake pan with butter and dust with all-purpose flour. Sift the cake flour, baking powder, and salt into a large mixing bowl.</li>
<li>Beat in the butter one heaping 1/4 teaspoonful at a time, using an electric mixer set on low speed, until the mixture resembles coarse sand. Beat in the sugar a tablespoon at a time, until the mixture resembles fine damp sand. Beat in the eggs one at a time. Add the vanilla and milk, and beat on medium-high, just until blended. Do not overbeat.</li>
<li>Pour into the prepared pan and bake until a wooden skewer inserted in the center comes out clean—30 to 35 minutes. Cool cake in the pan on a wire rack for 5 minutes.</li>
<li>Un-mold and cool completely. Ice with Browned-Butter Glaze.</li>
</ol>
<p>Before you begin, check baking powder expiration date. If it has passed, buy a fresh canister.</p>
</body>
</html>
```

Output



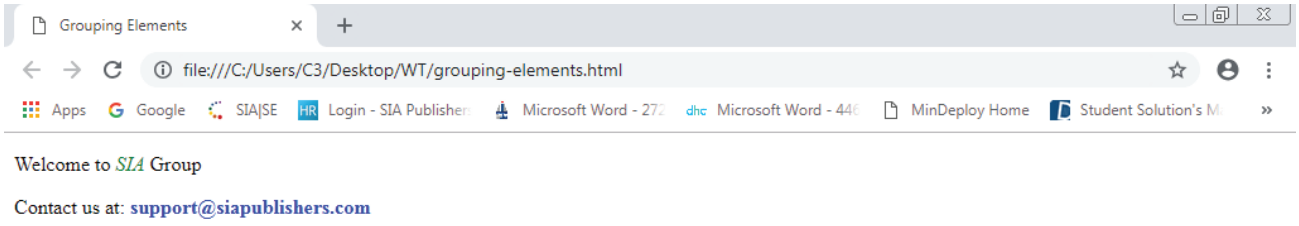
Q4. (a) Write a HTML program using grouping elements and .

Answer :

Program

```
<html>
<title>Grouping Elements</title>
<body>
<div class="Footnotes">
<p>Welcome to <i><span style = "color:green;">
SIA</span></i> Group</p>
<p>Contact us at: <b><span style = "color:blue;">
support@siapublishers.com</span></b></p>
</div>
</body>
</html>
```

Output



(b) Write a HTML Menu page for example cafe site.

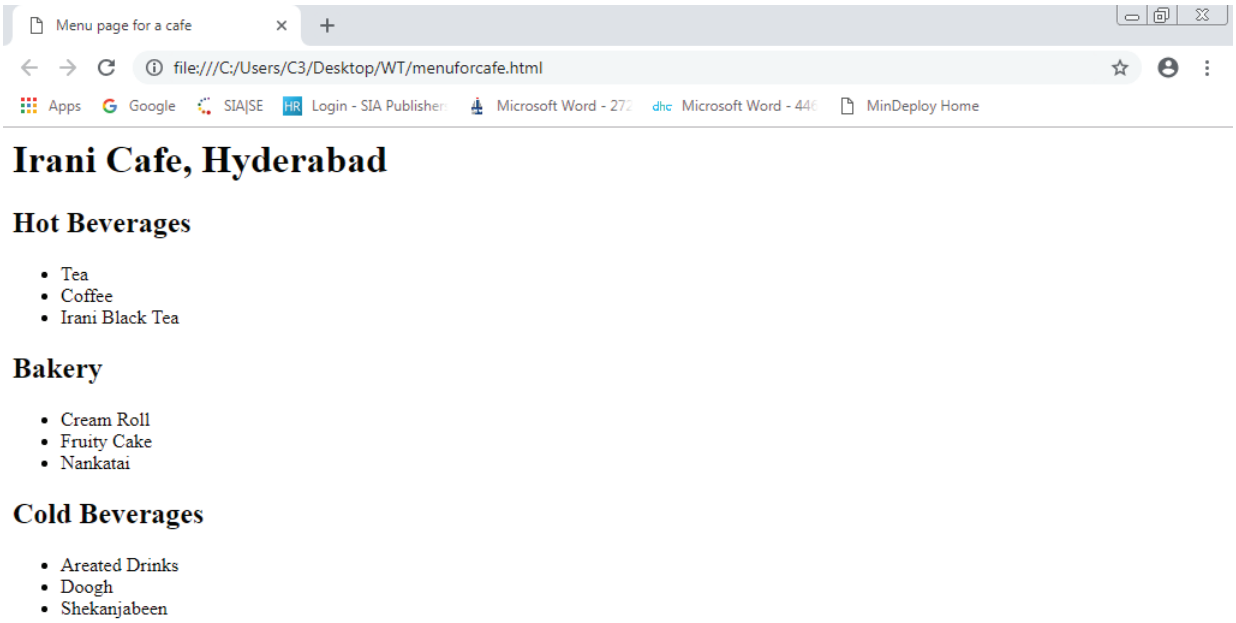
Answer :

Program

```
<html>
<head>
<title>Menu page for a cafe</title>
</head>
<body>
<h1>Irani Cafe, Hyderabad</h1>
<h2>Hot Beverages</h2>
<ul>
<li>Tea</li>
<li>Coffee</li>
<li>Irani Black Tea</li>
</ul>
<h2>Bakery</h2>
<ul>
<li>Cream Roll</li>
<li>Fruity Cake</li>
<li>Nankatai</li>
</ul>
<h2>Cold Beverages</h2>
```

```
<ul>
<li>Areated Drinks</li>
<li>Doogh</li>
<li>Shekanjabeen</li>
</ul>
</body>
</html>
```

Output



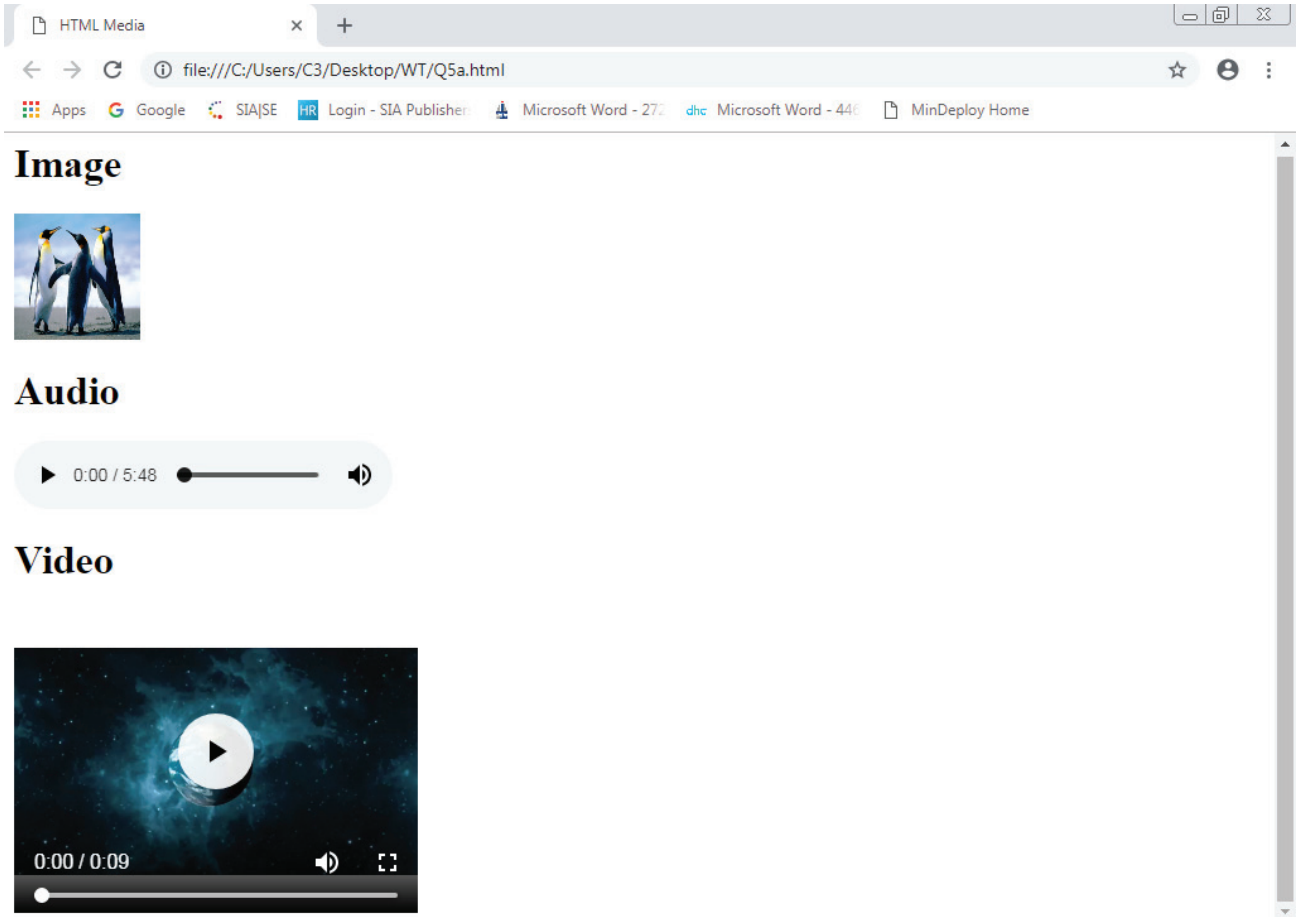
Q5. (a) Write a HTML program using images, audios, videos.

Answer :

Program

```
<html>
<head>
<title>HTML Media</title>
</head>
<body>
<h1>Image</h1>
<img src="Penguins.jpg" height="100" width= "100 "alt="error"><br>
<h1>Audio</h1>
<audio controls>
<source src= "Kalimba.mp3" type= "audio/mpeg">
</audio><br>
<h1>Video</h1>
<video width= "320" height= "240" controls>
<source src= "videoplayback.mp4" type= "video/mp4">
</video>
</body>
</html>
```

Output



(b) Write a HTML program to create your time table.

Answer :

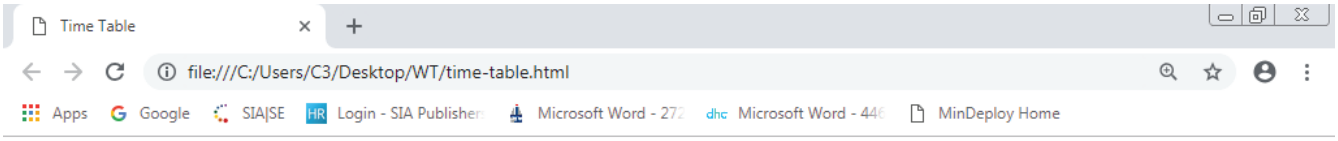
Program

```
<html>
<head>
<title>Time Table</title>
</head>
<body>
<H1>TIME TABLE</H1>
<table border= “2” cellspacing= “3” align= “center”>
<tr>
<td align=“center”>
<td>9:00-10:00
<td>10:00-11:00
<td>11:00-12:00
<td>12:00-1:00
<td>1:00-2:00
<td>2:00-3:00
<td>3:00-4:00
<td>4:00-5:00
</tr>
<tr>
```

```
<td align= “center”>MONDAY
<td align= “center”>---<td align= “center”>DBMS<br>
<td align= “center”>DE&M<br>
<td align= “center”>EM&E<br>
<td rowspan= “6”align= “center”>L<br>U<br>N<br>C<br>H
<td align= “center”>LA&VC<br>
<td align= “center”>WT<br>
<td align= “center”>Remedial class
</tr>
<tr>
<td align= “center”>TUESDAY
<td align= “center”>DE&M<br>
<td align= “center”>EM&E<br>
<td align= “center”>WT<br>
<td align= “center”>---
<td align= “center”>DBMS<BR>
<td align= “center”>LA&VC<br>
<td align= “center”>Library
</tr>
<tr>
<td align= “center”>WEDNESDAY
<td align= “center”>DBMS<br>
<td align= “center”>DE&M<BR>
<td align= “center”>WT<br>
<td align= “center”>---
<td colspan= “3” align= “center”>DBMS LAB
</tr>
<tr>
<td align= “center”>THURSDAY
<td align= “center”>LA&VC<br>
<td align= “center”>EM&E<br>
<td align= “center”>WT<BR>
<td align= “center”>DE&M
<td align= “center”>---<br>
<td align= “center”>DBMS<br>
<td align= “center”>Library
</tr>
<tr>
<td align= “center”>FRIDAY
<td align= “center”>EM&E<BR>
<td align= “center”>LA&VC<br>
<td align= “center”>EM&E<br>
<td align= “center”>---
<td align= “center”>WT<br>
<td align= “center”>---<br>
```

```
<td align= “center”>---
</tr>
<tr>
<td align= “center”>SATURDAY
<td align= “center”>WT<br>
<td colspan= “3” align= “center”>WT LAB
<td align= “center”>---<br>
<td align= “center”>DBMS<br>
<td align= “center”>Library
</tr>
</body>
</html>
```

Output



TIME TABLE

	9:00-10:00	10:00-11:00	11:00-12:00	12:00-1:00	1:00-2:00	2:00-3:00	3:00-4:00	4:00-5:00
MONDAY	---	DBMS	DE&M	EM&E		LA&VC	WT	Remedial class
TUESDAY	DE&M	EM&E	WT	---		DBMS	LA&VC	Library
WEDNESDAY	DBMS	DE&M	WT	---		DBMS LAB		
THURSDAY	LA&VC	EM&E	WT	DE&M		---	DBMS	Library
FRIDAY	EM&E	LA&VC	EM&E	---		WT	---	---
SATURDAY	WT	WT LAB				---	DBMS	Library

LUNCH

Q6. Write a HTML program to create a form using text inputs, password inputs, multiple line text input, buttons, check boxes, radio buttons, select boxes, file select boxes.

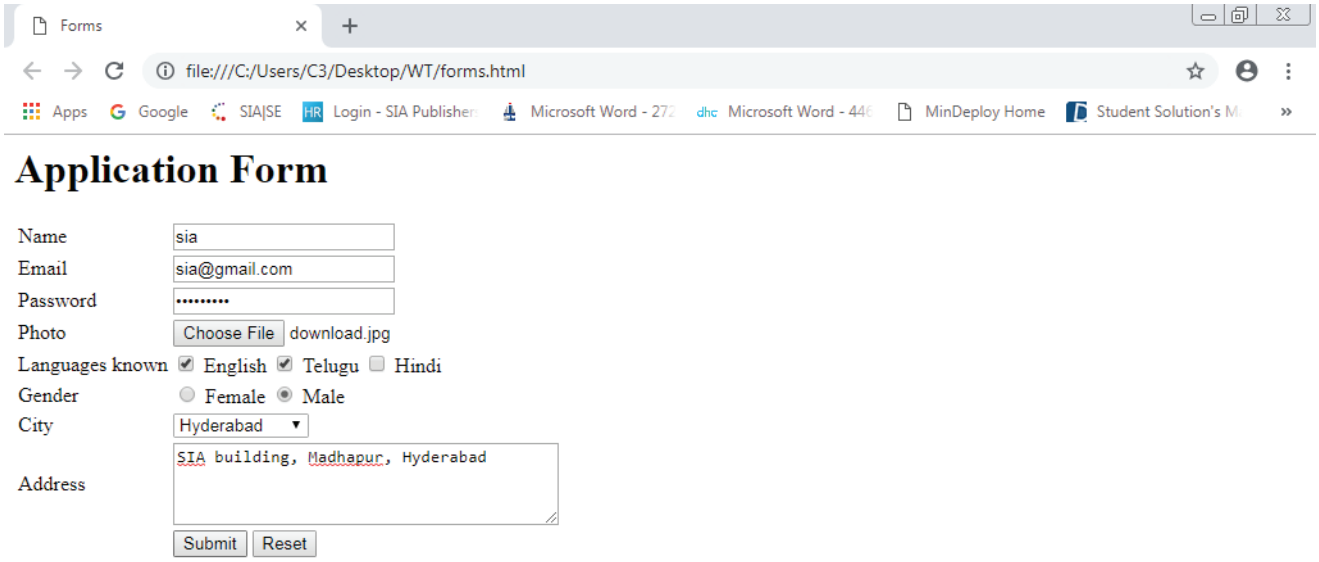
Answer :

Program

```
<html>
<title>Forms</title>
<body>
<h1>Application Form</h1>
<form>
    <table>
      <tr>
        <td>
          <label for= “name”>Name</label>
        </td>
        <td>
          <input type= “text” id= “name” name= “username”>
        </td>
      </tr>
```

```
<tr>
  <td>
    <label for= “email”>Email</label>
  </td>
  <td>
  <input type= “text” id= “email” name= “usermail”>
  </td>
</tr>
<tr>
  <td>
    <label for= “pass”>Password</label>
  </td>
  <td>
    <input type= “password” id= “pass”>
  </td>
</tr>
<tr>
  <td>
    <label for= “photo”>Photo</label>
  </td>
  <td>
    <input type= “file” id= “photo”>
  </td>
</tr>
<tr>
  <td>
    <label>Languages known</label>
  </td>
  <td>
    <label>
      <input type= “checkbox”> English</label>
    <label>
      <input type= “checkbox”> Telugu</label>
<label>
      <input type= “checkbox”> Hindi</label>
  </td>
</tr>
<tr>
  <td>
    <label>Gender</label>
  </td>
  <td>
    <label>
      <input type= “radio” value= “f” name= “gender”>
      Female</label>
      <label>
     <input value= “m” type= “radio” name= “gender”>
      Male</label>
  </td>
</tr>
<tr>
  <td>
    <label for= “city”>City</label>
  </td>
  <td>
    <select id= “city” name= “city”>
    <option>--Select City--</option>
        <option>Hyderabad</option>
        <option>Warangal</option>
        <option>Bangalore</option>
        <option>Chennai</option>
    </select>
  </td>
</tr>
<tr>
  <td>
    <label>Address</label>
  </td>
  <td>
    <textarea rows= “4” cols= “40”></textarea>
  </td>
</tr>
<tr>
  <td></td>
  <td>
    <input type= “submit” value= “Submit”>
    <input type= “reset”>
  </td>
</tr>
</table>
</form>
</body>
</html>
```

Output



Q7. Write a HTML program to create a frames and links between frames.

Answer :

Program

f.html

```
<html>
<head>
<title>Master Page</title>
</head>
<frameset cols= “50%,50%”>
<frame src= “f1.html”>
<frame src= “f2.html”>
</frameset>
</html>
```

f1.html

```
<html>
<head>
<title>PAGE1</title>
</head>
<body>
Welcome to Page 1
<br>
Click Here <a href= “f3.html”> to link to page 3 </a>
</body>
</html>
```

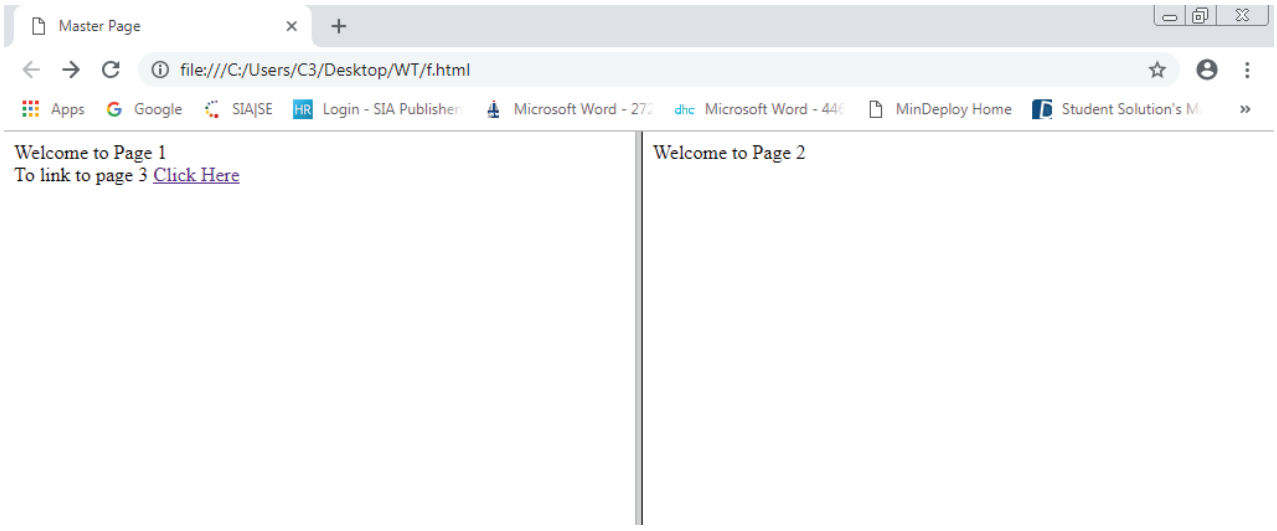
f2.htm

```
<html>
<head>
<title> PAGE2 </title>
</head>
<body>
Welcome to Page 2
</body>
</html>
```

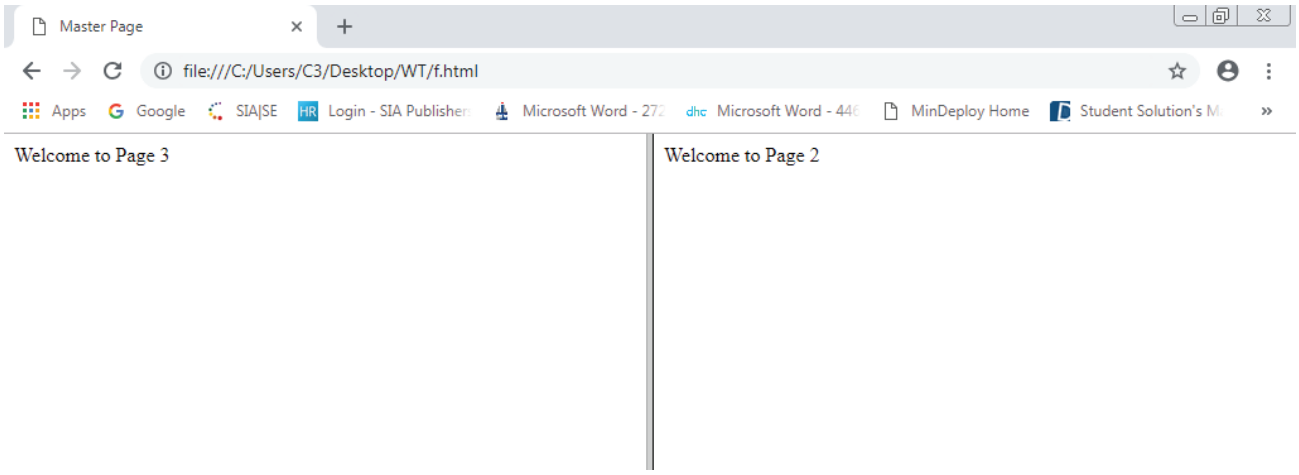
f3.html

```
<html>
<head>
<title>Page3</title>
</head>
<body>
Welcome to Page 3
</body>
</html>
```

Output



When the user clicks on the 'Click Here' then the output will be as shown in below figure,



Q8. Write a HTML program to create different types of style sheets.

Answer :

Program

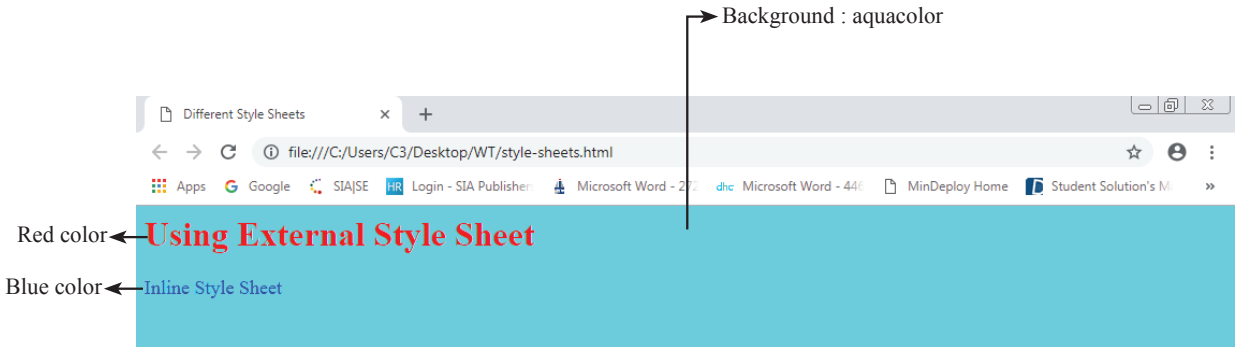
style.css

```
h1 {
    color: red;
  }
```

style-sheets.html

```
<html>
<head>
<title>Different Style Sheets</title>
<style type="text/css">
@import url("style.css");
body {
 background-color: aqua;
 font-size: 16px;
}<!--Internal Style Sheet-->
</style>
</head>
<body>
<h1>Using External Style Sheet</h1>
<p style="color:blue;font-size:18px;">Inline Style Sheet</p>
</body>
</html>
```

Output



Q9. Write a HTML program to create CSS on links, lists, tables and generated content.

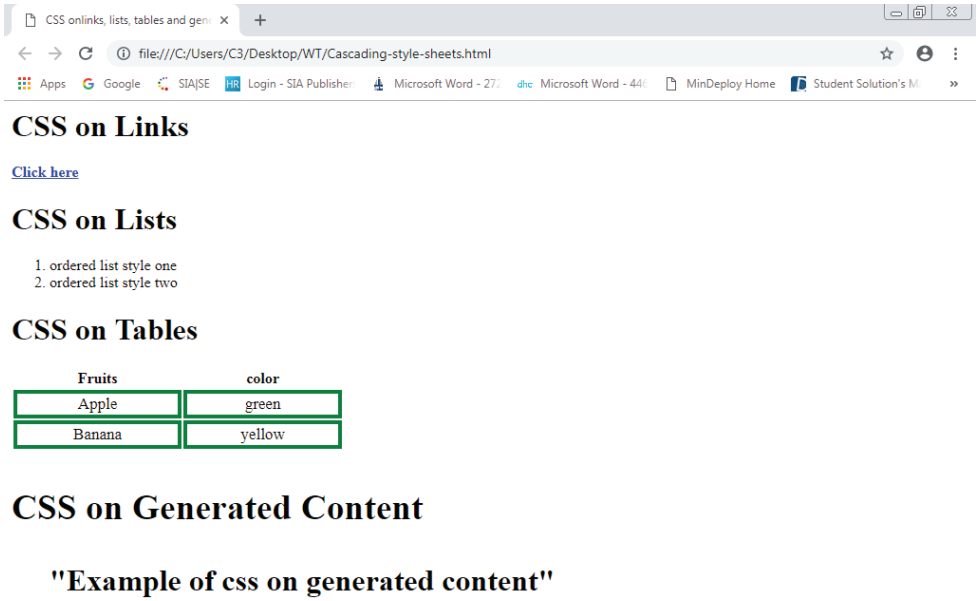
Answer :

Program

```
<html>
<title>CSS onlinks, lists, tables and generated content</title>
<head>
<style type= "text/css">
links.c { color: blue; }
li .a, b {list-style-type : decimal}
table {width : 350px ; border-collapse : separate ;}
td {font-size : 17px ; text-align : center ; border-width : 4px ; border-color : green ; border-style : solid ; border-spacing: 3pxpx ; 3px}
```

```
blockquote:before{ content: open-quote; }
blockquote:after{ content: close-quote; }
</style>
</head>
<body>
<h1>CSS on Links</h1>
<links class= “c”><p><b><a href= “default.asp” target= “_blank”>Click here</a></b></p></links>
<h1>CSS on Lists</h1>
<ol>
<li class= “a”>ordered list style one</li>
<li class= “b”>ordered list style two</li>
</ol>
<h1>CSS on Tables<h1>
<table>
<tr>
<th>Fruits</th>
<th>color</th>
</tr>
<tr>
<td>Apple</td>
<td>green</td>
</tr>
<tr>
<td>Banana</td>
<td>yellow</td>
</tr>
<table>
<h3>CSS on Generated Content</h3>
<blockquote>Example of css on generated content</blockquote>
</body>
</html>
```

Output



Q10. Write a HTML program to create your college web site using multi column layouts.

Answer :

Program

```
<html>
<head>
<title>SIA Home</title>
<meta charset= “utf-8”>
<meta name= “viewport” content= “width=device-width, initial-scale=1”>
<style>
* {
box-sizing: border-box;
}
body {
margin: 0;
}
.header {
background-color: AQUA;
padding: 9px;
font-size: 40px;
}
.topnav {
overflow: hidden;
background-color: gray;
}
.topnav a {
float: left;
display: block;
color: #f2f2f2;
text-align: center;
padding: 14px 16px;
text-decoration: none;
}
.topnav a:hover {
background-color: #ddd;
color: black;
}
.column {
float: left;
padding: 8px;
}
.column.side {
width: 25%;
background-color: silver;
}
.column.middle {
width: 50%;
}
.row:after {
content: “ ”;
display: table;
clear: both;
}
@media screen and (max-width: 600px) {
.column.side, .column.middle {
width: 100%;
}
}
.footer {
background-color: olive;
padding: 7px;
text-align: center;
}
</style>
</head>
<body>
<div class= “header”>
<img src= “SIAMAINLOGO.png” alt= “inaugral.jpg” height= “50” width= “70” <h1><b> Institute of Technology and Science</b></h1> </image>
</div>
<div class= “topnav”>
<a href= “#”>Administration</a>
<a href= “#”>Courses</a>
<a href= “#”>Examinations</a>
<a href= “#”>Affiliation</a>
<a href= “#”>Accredation</a>
```

```
<a href= “#”>About</a>
</div>
<div class= “row”>
<div class= “column side”>
<h2>Quick Links</h2>
    <a href=“#”>Annual Reports</a><br><br>
    <a href=“#”>Classwork Timetables</a><br><br>
    <a href=“#”>Holiday Notification</a><br><br>
    <a href=“#”>Student Activity center</a><br><br>
    <a href=“#”>Training & Placement</a><br><br>
    <a href=“#”>Research</a><br><br>
    <a href=“#”>NBA Accredation</a><br><br>
    <a href=“#”>Academics</a><br><br>
    <a href=“#”>Almanac</a><br><br>
    <a href=“#”>Departments</a><br><br>
    <a href=“#”>Contact Us</a>
</div>
<div class=“column middle”>
<img src=“DSC00649.jpg” alt=“inaugral.jpg” height=“400” width=“530”>
<p>The institution was started in 2017 with two B.Tech programs (viz., Computer Science Engineering and Electronics and Communication Engineering) in sprawling lush green campus of 65 acres, in the area well connected by rail and road.</p>
</div>
<div class=“column side”>
<h2>Vision</h2>
<p>To make our students technologically superior and ethically strong by providing quality education with the help of our dedicated faculty and staff and thus improve the quality of human life.</p>
<h3>Mission</h3>
<p>To provide latest technical knowledge, analytical and practical skills, managerial competence and interactive abilities to students, so that their employability is enhanced.
To provide a strong human resource base for catering to the changing needs of the Industry and Commerce.
To inculcate a sense of brotherhood and national integrity.</p>
</div>
</div>
<div class=“footer”>
<p><b>Copy Right &#169; 2018. All Rights Reserved by M.Nikhil</b></p>
</div>
</body>
</html>
```

Output



Q11. Write a HTML program to create your college web site using for mobile device.

Answer :

Program

```
<html>
<head>
<title>SIA Home</title>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<style>
@media screen and (max-width: 481px){
* {
box-sizing: border-box;
}
body {
margin: 0;
}
.header {
background-color: AQUA;
padding: 9px;
font-size: 40px;
}
```

```
.topnav {
overflow: hidden;
background-color: gray;
}
.topnav a {
float: left;
display: block;
color: #f2f2f2;
text-align: center;
padding: 14px 16px;
text-decoration: none;
}
.topnav a:hover {
background-color: #ddd;
color: black;
}
.column {
float: left;
padding: 8px;
}
.column.side {
width: 25%;
background-color: silver;
}
.column.middle {
width: 50%;
}
.row:after {
content: "";
display: table;
clear: both;
}
.column.side, .column.middle {
width: 100%;
}
.footer {
background-color: olive;
padding: 7px;
text-align: center;
}
}
</style>
</head>
<body>
<div class="header">
<img src="SIAMAINLOGO.png" alt="inaugral.jpg" height="50" width="70" <h1><b> Institute of Technology and Science</b></h1> </image>
</div>
<div class="topnav">
<a href="#">Administration</a>
<a href="#">Courses</a>
<a href="#">Examinations</a>
<a href="#">Affiliation</a>
<a href="#">Accredation</a>
<a href="#">About</a>
</div>
<div class="row">
<div class="column side">
<h2>Quick Links</h2>
    <a href="#">Annual Reports</a><br><br>
    <a href="#">Classwork Timetables</a><br><br>
    <a href="#">Holiday Notification</a><br><br>
    <a href="#">Student Activity center</a><br><br>
    <a href="#">Training & Placement</a><br><br>
    <a href="#">Research</a><br><br>
    <a href="#">NBA Accredation</a><br><br>
    <a href="#">Academics</a><br><br>
    <a href="#">Almanac</a><br><br>
    <a href="#">Departments</a><br><br>
    <a href="#">Contact Us</a>
</div>
<div class="column middle">
<img src="DSC00649.jpg" alt="inaugral.jpg" height="400" width="530">
<p>The institution was started in 2017 with two B.Tech programs (viz., Computer Science Engineering and Electronics and Communication Engineering) in sprawling lush green campus of 65 acres, in the area well connected by rail and road.</p>
</div>
<div class="column side">
<h2>Vision</h2>
<p>To make our students technologically superior and ethically strong by providing quality education with the help of our dedicated faculty and staff and thus improve the quality of human life.</p>
<h3>Mission</h3>
<p>To provide latest technical knowledge, analytical and practical skills, managerial competence and interactive abilities to students, so that their employability is enhanced.
To provide a strong human resource base for catering to the changing needs of the Industry and Commerce.
To inculcate a sense of brotherhood and national integrity.</p>
</div>
</div>
```

```
<div class="footer">
<p><b>Copy Right &#169; 2018. All Rights Reserved by M.Nikhil</b></p>
</div>
</body>
</html>
```

Output



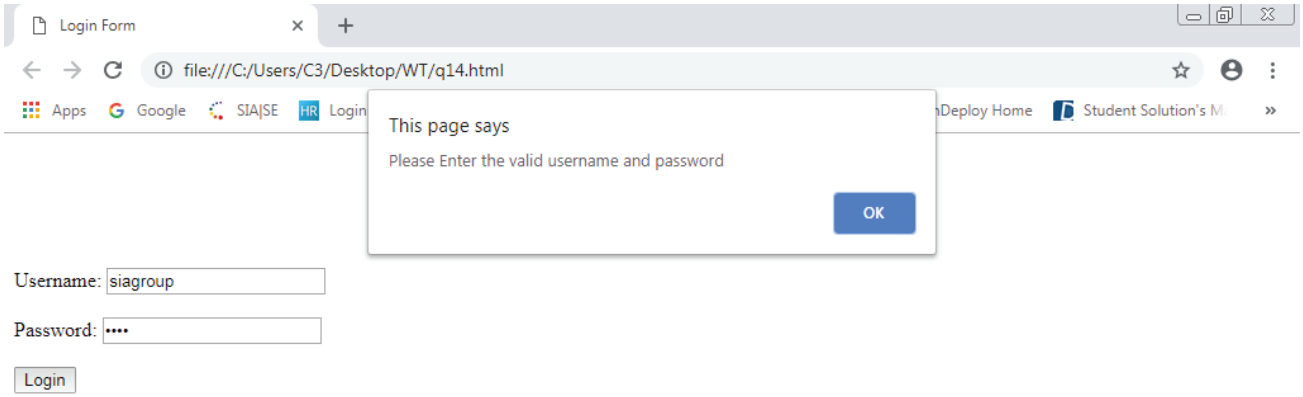
Q12. Write a HTML program to create login form and verify username and password using DOM.

Answer :

Program

```
<!DOCTYPE HTML>
<html>
<head>
<title>Login Form</title>
</head>
<body>
<h1 style= text-align: center;
font-size:40pt;
color:#00FF00;">Login Form
</h1>
<form name= "login">
Username: <input type="text" name="username"/><br><br>
Password: <input type="password" name="pswrd"/><br><br>
<input type="button" onclick="validate(this.form)" value="Login"/><br>
</form>
<script language="javascript">
function validate(form)
{
if(form.username.value == "siagroup" && form.pswrd.value == "siagrouppassword")
{
window.open('http://www.siapublishers.com/')
}
else
{
alert("Please Enter the valid username and password")
}
}
</script>
</body>
</html>
```

Output



Q13. (a) Write a JavaScript program to calculate area of rectangle using function.

Answer :

Program

```
<html>
<head>
<script type="text/JavaScript">
function Area_Rectangle(length, width) {
area =  length * width
return area
}
</script>
<title>Area of a Rectangle</title>
</head>
<body>
<form name="frmArea">
Enter the length and width of your rectangle to calculate the area:<br />
Length: <input type="text" name="txtLength" size="4" /><br />
Width: <input type="text" name="txtWidth" size="4" /><br />
<input type="button" value= "Calculate Area" onclick= "alert(Area_Rectangle(document.frmArea.txtLength.value, document.frmArea.txtWidth.value))" />
</form>
</body>
</html>
```

Output



(b) Write a JavaScript program to wish good morning, good afternoon, good evening depending on the current time.

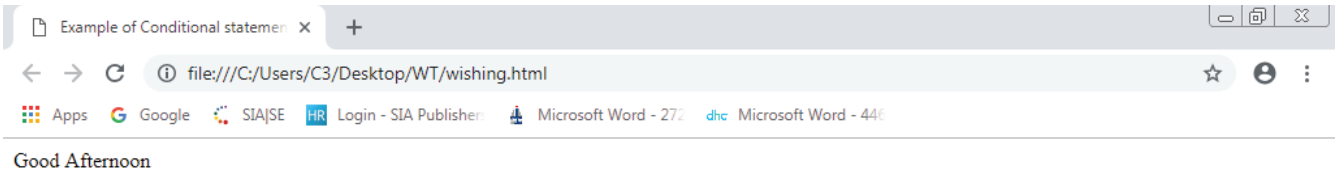
Answer :

Program

```
<html>
<head>
<title>Example of Conditional statements</title>
</head>
<body>
<script type= "text/JavaScript">
var date = new Date( );
var time = date.getHours( );
if (time < 12) {
document.write('Good Morning');
}
```

```
else if ( time >12 && time < 18) {
document.write('Good Afternoon');
}
else {
document.write('Good Evening');
}
</script>
</body>
</html>
```

Output



Q14. (a) Write a JavaScript program using switch case.

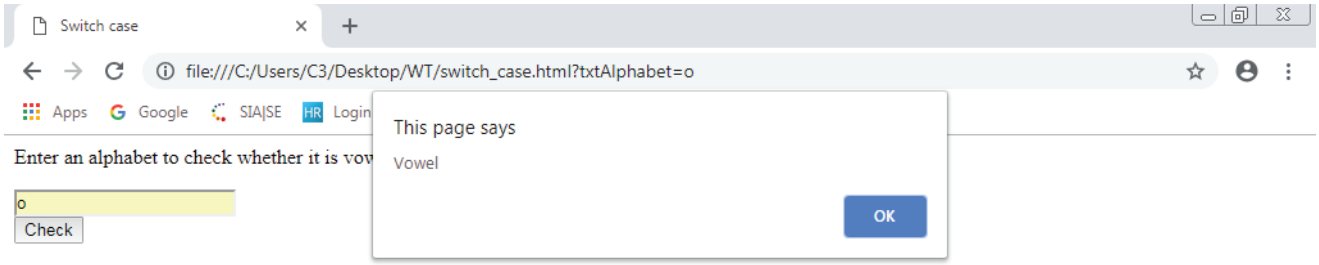
Answer :

Program

```
<html>
<head>
<title>Switch case</title>
<script type="text/JavaScript">
function checkAlphabet( ) {
switch (document.frmAlphabet.txtAlphabet.value){
case "a":
alert("Vowel")
break;
case "e":
alert("Vowel")
break;
case "i":
alert("Vowel")
break;
case "o":
alert("Vowel")
break;
case "u":
alert("Vowel")
break;
default : alert("Consonant");
}
}
</script>
</head>
<body>
```

```
<p>Enter an alphabet to check whether it is vowel or consonant:</p>
<form name= “frmAlphabet”>
<input type= “text” name= “txtAlphabet” /><br />
<input type= “button” value= “Check” onclick= “checkAlphabet()” />
</form>
</body>
</html>
```

Output



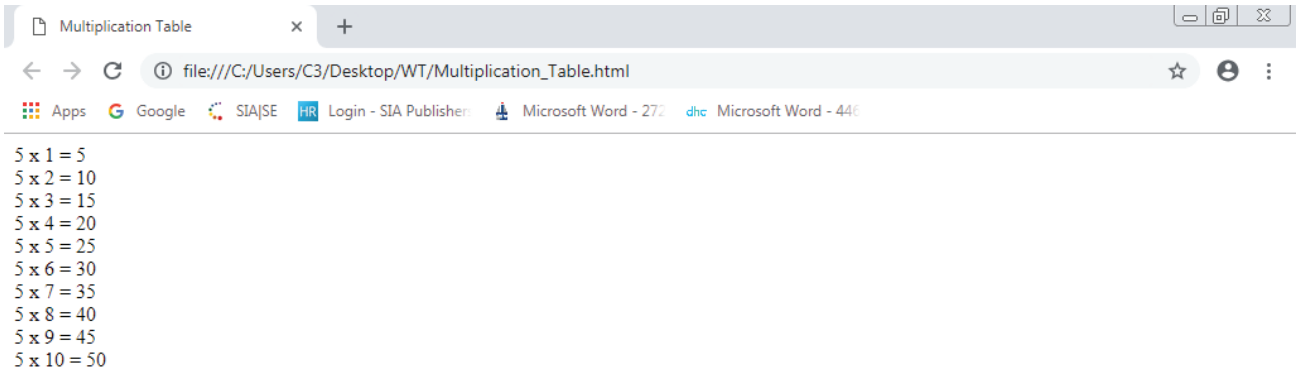
(b) Write a JavaScript program to print multiplication table of given number using loop.

Answer :

Program

```
<html>
<head>
<title>Multiplication Table</title>
</head>
<body>
<p>
<script type= “text/JavaScript”>
num = 5
i = 1
while (i <= 10) {
document.write(num + “ x “ + i +” =” + (num * i) + “<br />” );
i ++
}
</script>
</p>
</body>
</html>
```

Output



Q15. (a) Write a JavaScript programs using any 5 events.

Answer :

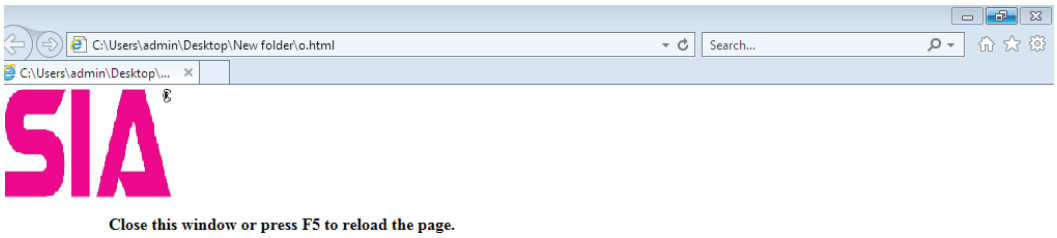
(i) onLoad Event

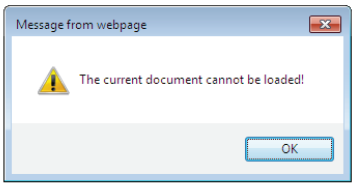
onLoad event gets invoked as soon as a web document is opened. It is handled by the onLoad event handler. It is mostly used when a webpage loads its complete content.

Program

```
<!DOCTYPE html>
<html>
<head>
<script type="text/javascript">
function dis()
{
alert ("The current document cannot be loaded!");
}
</script>
</head>
<body onUnload="dis()">
<img src="Libraries\Pictures\logo.jpg" onLoad="loadImage()" width="100" height="132">
 <b>Close this window or press F5 to reload the page.</b>
</body>
</html>
```

Output





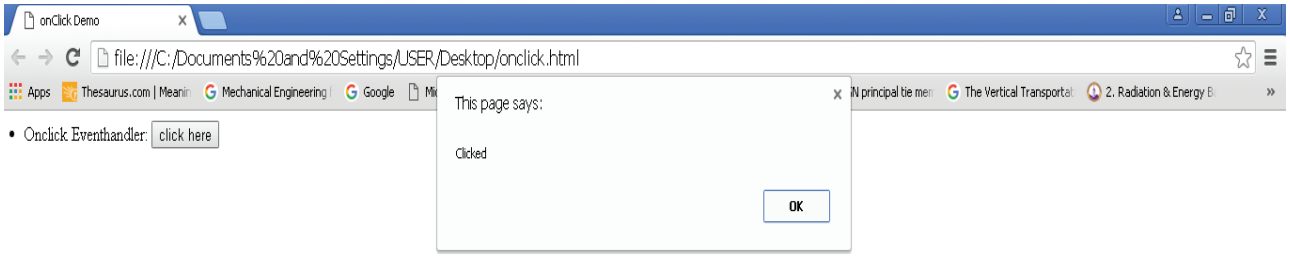
(ii) onClick Event

When element or mouse is clicked, the onClick event handler calls the javaScript function. The specific javaScript code or function is being executed on the occurrence of the click event. It returns a false value if used with check boxes, links, radio, reset and submit buttons so that the normal action that is associated with it will be cancelled.

Program

```
<!DOCTYPE html>
<html>
<body>
<title> onClick Demo </title>
<li>Onclick Eventhandler:
<input type=button value= "click here" onClick= 'alert("Clicked")'>
</body>
</html>
```

Output



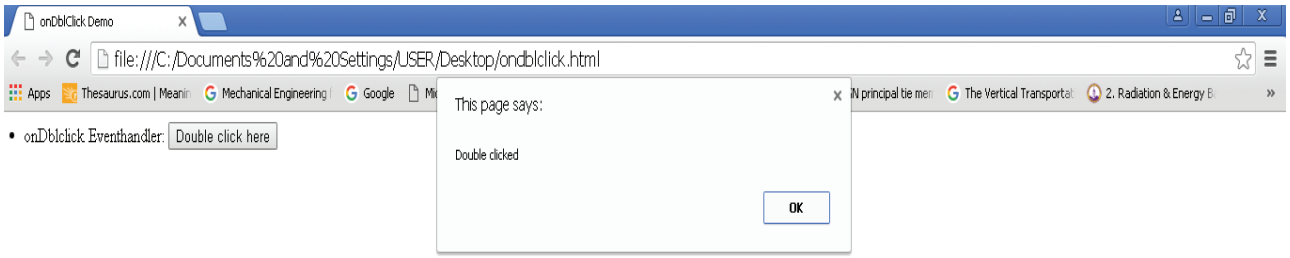
(iii) onDblClick Event

onDblClick event occurs when the user double clicks on an object. The events such as onMouseDown, onMouseUp, onDblClick will lead to this event in the same order. When onDblClick event occurs all the actions that are associated with the above events will be executed. onDblClick event is handled by the onDblClick event handler.

Program

```
<!DOCTYPE html>
<html>
<body>
<title> onDblClick Demo </title>
<li>onDblClick Eventhandler:
<input type=button value="Double click here" onDblClick='alert("Double clicked")'>
</body>
</html>
```

Output



(iv) onKeyPress Event

The onKeyPress event occurs when the user presses the key or holds down the key on key board. This event is handled by onKeyPress event handler.

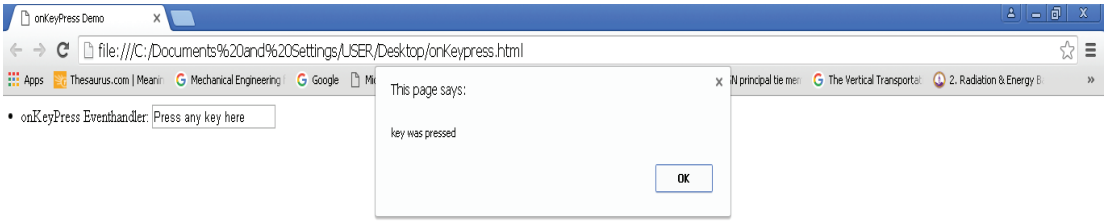
onKeyPress event invokes when the key is pressed and released over the element.

A simple example to illustrate the concept of onKeyPress event handler is given below,

Program

```
<!DOCTYPE html>
<html>
<body>
<title> onKeyPress Demo </title>
<li>onKeyPress Eventhandler:
<input type=text value="Press any key here" size=20 onKeyPress='alert ("key was pressed")'>
</body>
</html>
```

Output



(v) onFocus Event

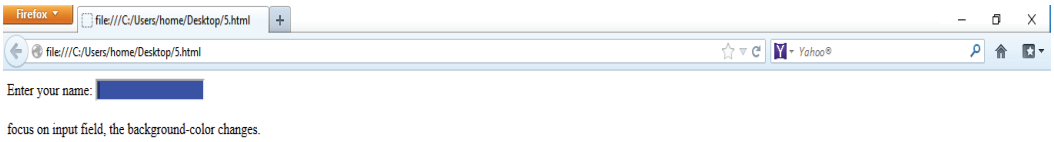
When a window frame or form element is focused on, the onFocus event handler calls the java script function. The specific java script code or function will be executed on the occurrence of a focus event. The element can be focused by clicking on it or by tabbing it.

The onFocus event handler occurs when an element gains focus.

Program

```
<html>
<body>
Enter your name: <input type="text" onfocus="func(this)">
<p>focus on input field, the background-color changes.</p>
<script language = "javascript">
function func(x)
{
x.style.background = "blue";
}
</script>
</body>
</html>
```

Output



Activate Windows
Go to Settings to activate Windows.



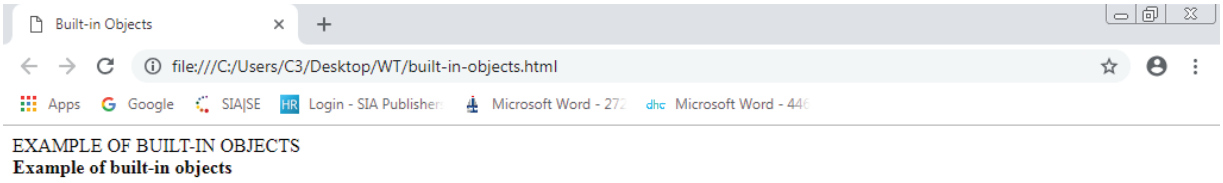
(b) Write a JavaScript program using JavaScript built-in objects.

Answer :

Program

```
<html>
<head>
<title>Built-in Objects</title>
</head>
<body>
<script type= "text/JavaScript">
myString = new String('Example of built-in objects')
Strng1 = myString.toUpperCase()
Strng2 = myString.bold()
document.write(Strng1 + "<br/>")
document.write(Strng2)
</script>
</body>
</html>
```

Output



Q16. Write a JavaScript program to create registration form and validate all fields using form validation.

Answer :

Program

form-validation.html

```
<html>
<head>
<title>Form validation</title>
<style type="text/css">
.formTitle {vertical-align:top; text-align:right;}
</style>
<script type="text/JavaScript">
function validate(form) {
var returnValue = true
var name=form.txtName.value
if (name== " ")
```

```
{
returnValue = false;
alert("You must enter a name")
document.frmRegistration.txtName.focus();
}
var email=form.txtEmail.value
var rxEmail = /^\w(\.?[\w-])*@\w(\.?[\w-])*\.[a-z]{2,6}(\.[a-z]{2})?$/i;
if (!rxEmail.test(email))
{
returnValue = false;
alert("You must enter a valid email address")
document.frmRegistration.txtEmail.focus()
}
var phone=form.txtPhone.value
if (phone==" ")
{
returnValue = false;
alert("You must enter your phone number")
document.frmRegistration.txtPhone.focus();
}
var radioChosen = false;
var radioButtons = form.radGender;
for (var i=0; i<radioButtons.length; i++) {
if (radioButtons[i].checked)
{
radioChosen=true;
}
}
if (radioChosen == false) {
returnValue = false;
alert("You must choose gender");
}
var Address=form.txtAddress.value
var AddressWords = Address.split(" ");
if (Address==" ")
{
returnValue = false;
alert("You must enter the address")
document.frmRegistration.txtAddress.focus();
}
```

```
return returnValue
}
</script>
</head>
<body>
<form name="frmRegistration" action="welcome.html" method="post" onsubmit="return validate(this);">
<h2>Registration Form</h2>
<table>
<tr>
<td class="formTitle">Name: </td>
<td><input type="text" name="txtName" size="18" /></td>
</tr>
<tr>
<td class="formTitle">Email: </td>
<td><input type="text" name="txtEmail" size="18" /></td>
</tr>
<tr>
<td class="formTitle">Phone: </td>
<td><input type="text" name="txtPhone" size="18" /></td>
</tr>
<tr>
<td class="formTitle">Gender: </td>
<td><input type="radio" name="radGender" value="Male" /> Male<br />
<input type="radio" name="radGender" value="Female" /> Female
</td>
</tr>
<tr>
<td class="formTitle">Address: </td>
<td><textarea name="txtAddress" cols="30" rows="3" /></textarea></td>
</tr>
<tr>
<td class="formTitle"></td>
<td><input type="submit" value="Register"  /></td>
</tr>
</table>
</form>
</body>
</head>
</html>
```

welcome.html

```
<html>
<head>
</head>
<body>
<h1>Registration done successfully</h1>
</body>
</html>
```

Output

When the user enters some invalid details (i.e., incorrect Email id) then the output will be as shown in the below figure,

